# GOVERNMENT OF MADHYA PRADESH

## PUBLIC WORKS DEPARTMENT





# SCHEDULE OF RATES (SOR)
# For ELECTRICAL

**In Force From January, 1st 2024**

**Issued by :-**

**Engineer-in-Chief (Building)**

**MP Public Works Department**

**BHOPAL**

# FOREWORD

The Schedule of Rates (SOR) for Electrical works published by Madhya Pradesh Public Works Department is a comprehensive document and is very useful in the execution of internal as well as external electrical and electro-mechanical works in various building projects. This Schedule of Rates is useful not only to the engineers of MPPWD but also to other government departments and undertakings that execute their work in accordance with the MPPWD Schedule of Rates.

The Schedule of Rates was last published in the year 2022. The prices of labour and materials have substantially changed since then. Taking into consideration labour rates, material cost, the Schedule of Rates has been revised. Accordingly, the electrical schedule of rates for 2024 has been prepared and issued. The items of this schedule are based on the CPWD General Specifications for Electrical Works Part-I – Internal, 2013.

In this SOR, rates have been proposed exclusive of GST, which shall be paid extra at prevailing rates as per Govt. of M.P. rules.

Due care has been taken to frame the Schedule of Rates as correctly as possible. It is, however, possible that some errors might have crept in. If an error or omission is discovered, it should be brought to the attention of this office.

In order to simplify the understanding of SOR, item names for each item and sub-item have been assigned. It is advised that the cost abstract of the estimate may include only item number, sub item number, item name, unit, rate, quantity, and amount. This will enable easy at-a-glance reading of the estimate. However, it is always required to include a full description of items in both the detailed estimate and the BOQ.

I extend my gratitude and thanks to the SOR committee members **Shri S.L. Suryavanshi** Chief Engineer (Building) Bhopal Zone, **Shri Jitendra Singh Chouhan** Superintendent Engineer (Building) E&M, **Shri Sunil Dehariya** Executive Engineer (Building) E&M, **Shri Hemant Jhariya** Assistant Engineer (Building) E&M, **Shri Aditya Prakash Singh** Sub Engineer (Building) E&M, for taking the initiative to launch the new SOR.

I also thank **Mrs. Kratika Patwa** Assistant Engineer (Building) E&M**, Mrs. Lawrene Francis** Sub Engineer (Building) E&M, **Shri Piyush Pandey** Sub Engineer (Building) E&M, **Shri Yogesh Nilajiwar** QS (Building) E&M, who worked hard to prepare this SOR under the guidance of **Shri S.L. Suryavanshi** Chief Engineer (Building) Bhopal Zone **Shri Jitendra Singh Chouhan** Superintendent Engineer (Building) E&M **without which, the revision would not have been possible.**

I am sure that this schedule of rates will be useful to all concerned.

Suggestions for improving this SOR, if any, are welcome and should be sent to **buildingsorelectrical@gmail.com** or **jpdpiuembhopal@gmail.com**.

This SOR is available on the departmental website at www.mppwd.gov.in and shall be in force w.e.f. January 01, 2024.



**Place: Bhopal**
**Dated:** 01/01/2024

**(Er. G.P. Mehra)**
**ENGINEER IN CHIEF (BUILDING)**
**MPPWD BHOPAL**

# GENERAL NOTES

1 **Interpretations**

a. The **Engineer In Chief** (Building), Public Works Department Govt. of M.P. shall be sole deciding authority as to the meaning, interpretation and implications for various provisions in this publications. His decision shall be final and binding.

b. Wherever any reference is made to any Indian Standards, it shall be taken as reference to the latest edition with all the amendments issued thereto.

c. In the event of any variation between the CPWD specifications (adopted) and the Indian Standard the former shall take precedence over the later.

d. Precedence of specification in SOR- In case of any contradiction in the provisions of the CPWD specifications (adopted) and this document, the provisions of this document would take precedence.

e. Precedence of rate in SOR- The rates given in this Schedule for all items are final, binding and conclusive, in case of doubt and printing mistakes if any, the decision of the **Engineer In Chief** (Building), PWD Govt. of M.P. shall be final and binding.

2 **Definitions:**

The following terms and expressions in the SOR shall have the meaning or implication hereby assigned to them unless otherwise specified elsewhere.

2.1 Specifications would refer to the General Specification for Electrical Works issued by Central Public Works Department (CPWD) as applicable as on date along with up to date amendments, addendums and errata.

2.2 Engineer-in-Charge: The "Engineer-in-Charge" means the engineer officer of Executive Engineer rank in Building Division/ Road (B&R) Division/ E&M Division, who shall supervise and In-charge of the works and who shall sign the contract on behalf of the Governor of M.P.

2.3 IS: The standards, specification and code of practices issued by the Bureau of Indian Standards along with up to date amendments.

2.4 Superintendent Engineer means officer in-charge of the circle.

2.5 Chief Engineer means officer in-charge of Zone.

2.6 Engineer-in-chief (Building) means head of the department, who heads Building wing of MPPWD.

2.7 Engineer-in-Chief means head of department who heads Building and Road wing of M.P. PWD.

2.8 Competent Authority i.e as per manual/ NIT / Agreement.

2.9 Site: The 'site' shall mean the land/or other places on, in, into or through which the work is to be executed under the contract or any adjacent land, path or street through which the work is to be executed under the contract, or any adjacent land, path or street which may be allotted or used for the purpose of carrying out the contract.

2.10 Best: The word 'best' when used shall mean that in the opinion of the Engineer- in-Charge, there is no superior material/ article and workmanship obtainable in the market and trade respectively. As far as possible the standard required shall be specified in preference to the word 'best'.

2.11 Department: "Department" mean Public Works Department of Govt. of M.P.

2.12 ISI, IS, BIS shall mean the relevant Indian Standards.

3 Only Copper conductor multi-strand cable with PVC Insulated FRLS having minimum oxygen index value 29% is to be used.

4 (a) The point wiring for light/fan/bell, socket outlet 6A, 16A includes switches, sockets, ceiling rose, lamp holder, conduit, copper multistrand FRLS wire with copper lugs (to be compulsorily used), fixing and drawing of all items and green colour PVC insulated multistrand FRLS copper wire as earth wire. However, switch board with sheet (modular or non-modular type) is not included in the rates of point wiring and shall be paid separately and switch board should be selected so as to accommodate at least one similar switch & socket in future.

(b) Secondary point shall mean the light point which is looped from another point (not connected directly from the switch), this secondary point shall be measured from the looping point.

5 Circuit wiring is to be done using 2.5 sq mm copper multistrand FRLS cable. The rate of the 2.5 sq mm copper multistrand FRLS earth wire (green in colour), to be run inside the same conduit, shall be payable separately as per actual measurements. Similarly, the earth wire (green in colour) for sub-main shall run inside the same conduit as the phase wire and neutral wire of the sub-main but shall be paid separately. Separate conduits should be used for point, circuit, power and sub-main wiring.

6 All non-modular and modular accessories should be made of polycarbonate material only.

7 The rates of wiring have been taken considering FRLS wire only. If FR-LSH wire are used anywhere, 3% rates may be paid additionally for related items.

8 Chapter containing overhead electrification has also been enhanced as per norms of MPSEB and B.E.E. While recording the measurement of overhead lines, sag up to 3% on DOG conductors and up to 5% on Panther conductors, shall be given separately.

9 The rates given in this SOR are based on the following classification:

**CLASSIFICATION OF POINTS**

| **Point** | **Light/Fan/Bell/2Way Points/ Socket 6A etc.** | **Power Wiring 16A** |
|---|---|---|
| Short | Up to 3m. | Up to 3m. |
| Medium | Above 3m up to 6m | Above 3m up to 6m |
| Long | Above 6m up to 10m | Above 6m up to 10m |
| Extra Long-I | - | Above 10m up to 15m |
| Extra Long-II | - | Above 15m up to 20m |
| Extra Long-III | - | Above 20m up to 25m |
| As Per Linear Measurement | Above 10m | Above 25m |

**NOTE:**

1. Material used above 10m (Point Wiring)/25m (Power Wiring) shall be paid as per actual measurements on linear basis.
2. If two or more points are controlled by single switch then 2nd & subsequent points shall be measured as secondary point from the looping point.
3. Any special wiring not covered above shall be measured on the linear basis and shall be paid as per SOR (E/M) items.

10 **CLASSIFICATION OF CONDUIT WIRING**

**Maximum number of PVC insulated 650/1100 V grade Aluminium/ Copper conductor cable conforming to IS 694:1990**

| Nominal cross sectional area of | 20 mm | | 25 mm | | 32 mm | | 38 mm | | 51 mm | | 64 mm | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S | B | S | B | S | B | S | B | S | B | S | B |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 1.5 | 5 | 4 | 10 | 8 | 18 | 12 | - | - | - | - | - | - |
| 2.5 | 5 | 3 | 8 | 6 | 12 | 10 | - | - | - | - | - | - |
| 4 | 3 | 2 | 6 | 5 | 10 | 8 | - | - | - | - | - | - |
| 6 | 2 | - | 5 | 4 | 8 | 7 | - | - | - | - | - | - |
| 10 | 2 | - | 4 | 6 | 6 | 5 | 8 | 6 | - | - | - | - |
| 16 | - | - | 2 | 2 | 3 | 3 | 6 | 5 | 10 | 7 | 12 | 8 |
| 25 | - | - | - | - | 3 | 2 | 5 | 3 | 8 | 6 | 9 | 7 |
| 35 | - | - | - | - | - | - | 3 | 2 | 6 | 5 | 8 | 6 |
| 50 | - | - | - | - | - | - | - | - | 5 | 3 | 6 | 5 |
| 70 | - | - | - | - | - | - | - | - | 4 | 3 | 5 | 4 |

**Notes:-**

1. The above table shows the maximum capacity of conduits for a simultaneous drawing in of cables.
2. The columns headed ‘S’ apply to runs of conduits which have distance not exceeding 4.25 m between draw in boxes and which do not deflect from the straight by an angle of more than 15 degrees. The columns headed ‘B’ apply to runs of conduit, which deflect from the straight by an angle of more than 15 degrees.
3. Conduit sizes are the nominal external diameters.

11 The material to be used should be in accordance with the “List of approved electrical material”. However, the Engineer-in-Charge (E&M) shall be authorized to add the material of new Make/model keeping in view the quality and the required specification. In case of non-availability of material, as specified in the ‘List of Approved electrical materials’, in any particular area at any point of time, the Engineer-in-Charge(E&M) may permit the use of different equivalent make/model. However, for this purpose the Engineer-in-Charge shall take his decision in consultation with the Competent authority.

12 **Samples approval-** sample of all materials to be used on the work shall be got approved by the contractor from the Competent Authority well in time by producing at least four sample. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new and as per the samples approved by the Competent Authority.

13 The items in different chapters of this SOR describe the basic specifications. For certain items/equipment’s, additional specifications/testing certificates etc. may be required. In such case the Engineer-in-Charge shall append these requirements as Special Conditions of the Contract after receiving due approval of the same from the Competent authority.

14 **Non SOR item in on-going agreement:** -If any new item is not included in SOR and is proposed to be executed in any on-going work, proposal with full justification thereof along with financial implication and detailed rate analysis must be submitted to Superintending Engineer of respective circle (B&R/E&M) or Chief Engineer (Building), PWD for obtaining approval as on extra item.

15 **In case** of any difference in item description and rate between BOQ and SOR, the rate and description of BOQ shall be prevail, on the other hand if there is difference in between rates only while description being the same in BOQ and SOR, the rate of SOR with up-to-date amendment shall be prevail.

16 **Non SOR items for tendering:** -If any Non S.O.R. Item is to be executed in any proposed project, rate analysis with full justification thereof must be approved by the Superintending Engineer of respective circle (B & R) or Chief Engineer (Building), PWD and shall be included in BOQ under sub head of non SOR items at the last. Tender shall be invited on amount inclusive of SOR and non SOR items and in such cases non SOR items shall become schedule (BOQ) items for that particular tender only. However it is made clear that such non SOR items shall not be assumed as sanctioned rates for other agreements.

17 The rates given are for finished item of work and all material, labour, wastage, royalties, lease rent, scaffolding, temporary work, T&P hire charges, breakage etc. are included in the rates, unless specified otherwise. Any damage to the wall surface, fittings etc. done during execution of work by the contractor shall be finished by him free of cost bringing it to the original finish condition.

18 In the event of a contradiction or conflict in the provisions of this SOR and the CPWD Specifications, the former will prevail. In situations where the provisions are not provided or silent in this SOR, the provisions of the CPWD specification shall apply. However, the Engineer-in-Charge's decision will be final and binding.

19 **Testing**:

a. All test prescribed by CPWD general specification, shall be done before, during and after installation, and shall be carried out in NABL approved lab and the result shall be submitted to the Engineer-in-Charge, in prescribed format.
b. SOR Rates including testing charges: - The rates include the element of testing of samples of various materials brought by the contractor for use in the work, as well as other necessary tests for items of work as stipulated in specifications.

20 **Acceptance at Reduced Rate:** -If any item of work is found not upto the prescribed standard but the Engineer-In-Charge is of the opinion that the same is structurally/technically adequate and can be accepted at a reduced rate, then in such cases, the Engineer-In-Charge shall have to submit proposals for appropriate reduction of rates supported by an analysis, in justification thereof, through a D.O. letter to the territorial Superintending Engineer or Chief Engineer to obtain his approval expeditiously (ordinarily within 15 days).

21 **Drawings**:

a. All works shall be carried out in accordance with the drawing approved by the competent authority and also in accordance with modification thereto as approved by the competent authority from time to time.

b. Contractor is required to submit as built drawings & SLD's of electrical wiring & conduiting along with numbers of short, medium & long points**,** after completion of work to Engineer-in-Charge (electrical) to his satisfaction.

22 The contractor is required to buy a warranty from the manufacturer or supplier of the items up to the defect liability period (DLP) mentioned in the respective contract. No extra payment shall be made to the contractor for the warranty within the duration of DLP.



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**ENGINEER IN CHIEF (BUILDING)**
**MPPWD BHOPAL**

# CONTENTS


| CHAPTER No. | DESCRIPTION | PAGE No. |
|---|---|---|
| 1 | 2 | 3 |
| 1 | WIRING IN SURFACE RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR | 1-2 |
| 2 | WIRING IN CONCEALED RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR | 3-4 |
| 3 | WIRING IN SURFACE RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR | 5-6 |
| 4 | WIRING IN CONCEALED RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR | 7-8 |
| 5 | WIRING IN SURFACE RIGID P.V.C. CASING AND CAPING WITH COPPER CONDUCTOR | 9-10 |
| 6 | POWER WIRING IN SURFACE RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR | 11 |
| 7 | POWER WIRING IN CONCEALED RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR | 12 |
| 8 | POWER WIRING IN SURFACE RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR | 13 |
| 9 | POWER WIRING IN CONCEALED RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR | 14 |
| 10 | POWER WIRING IN SURFACE RIGID P.V.C. CASING AND CAPING WITH COPPER CONDUCTOR | 15 |
| 11 | WIRING IN EXISTING CONCEALED RIGID CONDUIT SYSTEM WITH COPPER CONDUCTOR | 16 |
| 12 | WIRING IN EXISTING SURFACE RIGID CONDUIT SYSTEM WITH COPPER CONDUCTOR | 17 |
| 13 | SUPPLY AND FIXING M.S. CONDUIT FOR CONCEALED CONDUIT SYSTEM | 18 |
| 14 | SUPPLY AND FIXING M.S. CONDUIT FOR SURFACE CONDUIT SYSTEM | 19 |
| 15 | SUPPLY AND FIXING PVC CONDUIT FOR SURFACE CONDUIT SYSTEM | 20 |
| 16 | SUPPLY AND FIXING PVC. CONDUIT FOR CONCEALED CONDUIT SYSTEM | 21 |
| 17 | SUPPLY AND FIXING PVC CASING AND CAPPING ON SURFACE SYSTEM | 22 |
| 18 | M.S. BOXES | 23 |
| 19 | WIRING IN EXISTING CONDUIT/PVC CASING AND CAPPING | 24-25 |
| 20 | MAIN AND SUB MAINS IN SURFACE RIGID STEEL CONDUIT | 26-27 |
| 21 | SUB-MAIN IN CONCEALED RIGID STEEL CONDUIT IN COPPER CONDUCTOR | 28-29 |
| 22 | MAIN AND SUB-MAINS IN SURFACE RIGID PVC CONDUIT | 30-31 |
| 23 | SUBMAIN IN CONCEALED PVC CONDUIT IN COPPER CONDUCTOR | 32-33 |
| 24 | REWIRING IN EXISTING RIGID STEEL/PVC NON-METALLIC CONDUIT (Copper conductor) | 34-35 |

| | | |
|---|---|---|
| 25 | IRON CLAD/METAL CLAD SWITCHES AND SWITCH FUSES UNIT CONFORMING TO I.S. 4064 & 4047 -1967 | 36-39 |
| 26 | BUSBAR CHAMBERS AND SANDWICH BUS TRUNKING | 40-51 |
| 27 | MOULDED CASE CIRCUIT BREAKERS, ISOLATORS, MCCB (TRIPPING CHARACTERISTICS CONFORMING TO IS-8828 AND BS-3871 PART -I, II) | 52-61 |
| 28 | FIXING OF SFUs, DBs, BUSBAR, MCCBs, MCBs, ISOLATORs & RCBOs | 62-64 |
| 29 | ACCESSORIES | 65-69 |
| 30 | FANS | 70-72 |
| 31 | LUMINAIRES | 73-79 |
| 32 | MISCELLANEOUS | 80-81 |
| 33 | LABOUR RATES FOR WIRING | 82-83 |
| 34 | PAINTING | 84 |
| 35 | DISMANTLING OF CIVIL & ELECTRICAL WORKS | 85-86 |
| 36 | CIVIL WORKS | 87 |
| 37 | EARTHING AND LOOP EARTHING | 88-93 |
| 38 | TELEPHONE, LOCAL AREA NETWORK (LAN), TV CABLE | 94-100 |
| 39 | WATER PUMP AND FIRE PUMP | 101-108 |
| 40 | RACEWAY AND uPVC BUS TRUNCKING | 109-112 |
| 41 | HT AND LT CABLES | 113-130 |
| 42 | EXTERNAL ELECTRIFICATION AND OVERHEAD | 131-154 |
| 43 | INTERCOM / EPABX SYSTEMS | 155-156 |
| 44 | LIFT/ELEVATORS | 157-162 |
| 45 | MGPS | 163-171 |
| 46 | AIR CONDITIONER HVAC | 172-186 |
| 47 | UNINTERRUPTED POWER SUPPLY (UPS) | 187-190 |
| 48 | DIESEL GENERATING SET | 191-193 |
| 49 | FIRE DETECTION AND ALARM SYSTEM | 194-196 |
| 50 | CONVENTIONAL FIRE ALARM SYSTEM | 197-198 |
| 51 | AUTOMATIC FIRE DETECTION AND SUPPRESSION SYSTEM | 199-199 |
| 52 | ENERGY SAVING AND PROTECTION SOLUTIONS FOR BUILDINGS | 200-211 |
| 53 | AUDIO SPEAKERS AND AMPLIFIERS | 212-218 |
| 54 | CLOSED CIRCUIT TELEVISION (CCTV) | 219-231 |
| 55 | SOLAR WATER HEATER AND SOLAR PLANT | 232-235 |
| 56 | FACADE LIGHT | 236-241 |
| 57 | ELECTRICAL/ELECTRONIC EQUIPMENTS | 242-244 |
| NABL Test | | 245-246 |

# CHAPTER - 1

## WIRING IN SURFACE RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **1.1** | | **Point wiring surface rigid steel conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid steel conduit ISI marked of suitable size and 1.5sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit including painting, screwless connector for joints etc. as required as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **1.1.1** | **Light/fan points primary** | **Light point/fan points**. | | | |
| | 1.1.1.1 | Short | Primary short point | Each | 538.00 | 607.00 |
| | 1.1.1.2 | Medium | Primary medium point | Each | 1154.00 | 1223.00 |
| | 1.1.1.3 | Long | Primary long point | Each | 1855.00 | 1924.00 |
| | **1.1.2** | **Looping point secondary** | **(Light point / fan points) looping point** | | | |
| | 1.1.2.1 | Short | Secondary short point | Each | 269.00 | 304.00 |
| | 1.1.2.2 | Medium | Secondary medium point | Each | 577.00 | 612.00 |
| | 1.1.2.3 | Long | Secondary long point | Each | 928.00 | 962.00 |
| | **1.1.3** | **Socket 6A sep. board** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 1.1.3.1 | Short | Short point | Each | 547.00 | 730.00 |
| | 1.1.3.2 | Medium | Medium point | Each | 1199.00 | 1383.00 |
| | 1.1.3.3 | Long | Long point | Each | 1956.00 | 2139.00 |
| | **1.1.4** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 1.1.4.1 | Short | Short point | Each | 505.00 | 609.00 |
| | 1.1.4.2 | Medium | Medium point | Each | 1066.00 | 1170.00 |
| | 1.1.4.3 | Long | Long point | Each | 1702.00 | 1805.00 |
| | **1.1.5** | **Twin control point** | **Twin control light points** | | | |
| | 1.1.5.1 | Short | Short point | Each | 577.00 | 675.00 |
| | 1.1.5.2 | Medium | Medium point | Each | 1230.00 | 1328.00 |
| | 1.1.5.3 | Long | Long point | Each | 1967.00 | 2066.00 |
| **1.2** | | **Socket same board 6A** | Point wiring (excluding metallic switch box & sheet but including switches, sockets) for 3/5 pin 6 Amp. Socket outlet point with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid steel conduit ISI marked of suitable size and 1.5sq. mm. PVC insulated copper earth continuity conductor of green colour inside conduit with required materials as per specification on same board. | Each | 82.00 | 265.00 |

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **1.3** | | **Circuit wiring 2.5Sq. mm** | Wiring for circuit wiring with PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid steel conduit of ISI marked suitable size including painting etc. as required as per specification | | | |
| | 1.3.1 | 2 Wire | 2x2.5 Sq.mm. | Metre | 197.00 | 197.00 |
| | 1.3.2 | 4 Wire | 4x2.5 Sq.mm. | Metre | 292.00 | 292.00 |
| | 1.3.3 | 6 Wire | 6x2.5 Sq.mm. | Metre | 357.00 | 357.00 |
| | 1.3.4 | 8 Wire | 8x2.5 Sq.mm. | Metre | 421.00 | 421.00 |

**Note: -** Secondary point shall be measured from the looping point. Total wattage of looped points shall not exceed 200 Watts.

# CHAPTER – 2

## WIRING IN CONCEALED RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **2.1** | | **Point wiring concealed rigid Steel conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid steel conduit ISI marked of suitable size and 1.5sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit including painting, screwless connector for joints etc. as required as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **2.1.1** | **Light/fan points primary** | **Light point/fan points.** | | | |
| | 2.1.1.1 | Short | Primary short point | Each | 564.00 | 633.00 |
| | 2.1.1.2 | Medium | Primary medium point | Each | 1201.00 | 1270.00 |
| | 2.1.1.3 | Long | Primary long point | Each | 1940.00 | 2008.00 |
| | **2.1.2** | **Looping point secondary** | **(Light point / fan points) looping point** | | | |
| | 2.1.2.1 | Short | Secondary short point | Each | 282.00 | 317.00 |
| | 2.1.2.2 | Medium | Secondary medium point | Each | 601.00 | 635.00 |
| | 2.1.2.3 | Long | Secondary long point | Each | 970.00 | 1004.00 |
| | **2.1.3** | **Socket 6A sep. board** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 2.1.3.1 | Short | Short point | Each | 570.00 | 535.00 |
| | 2.1.3.2 | Medium | Medium point | Each | 1226.00 | 1209.00 |
| | 2.1.3.3 | Long | Long point | Each | 2039.00 | 2004.00 |
| | **2.1.4** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 2.1.4.1 | Short | Short point | Each | 531.00 | 484.00 |
| | 2.1.4.2 | Medium | Medium point | Each | 1112.00 | 1066.00 |
| | 2.1.4.3 | Long | Long point | Each | 1786.00 | 1740.00 |
| | **2.1.5** | **Twin control point** | **Twin control light points** | | | |
| | 2.1.5.1 | Short | Short point | Each | 614.00 | 568.00 |
| | 2.1.5.2 | Medium | Medium point | Each | 1288.00 | 1242.00 |
| | 2.1.5.3 | Long | Long point | Each | 2064.00 | 2018.00 |

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **2.2** | | **Socket same board 6A** | Point wiring (excluding metallic switch box & sheet but including switches, sockets) for 3 pin 6 Amp. Socket outlet point with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid steel conduit ISI marked of suitable size and 1.5sq. mm. PVC insulated copper earth continuity conductor of green colour inside conduit with required materials as per specification on same board. | Each | 82.00 | 265.00 |
| **2.3** | | **Circuit wiring 2.5Sq. mm** | Wiring for circuit wiring with PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid steel conduit of ISI marked suitable size including painting etc. as required as per specification | | | |
| | 2.3.1 | 2 Wire | 2x2.5 Sq.mm. | Metre | 205.00 | 205.00 |
| | 2.3.2 | 4 Wire | 4x2.5 Sq.mm. | Metre | 311.00 | 311.00 |
| | 2.3.3 | 6 Wire | 6x2.5 Sq.mm. | Metre | 378.00 | 378.00 |
| | 2.3.4 | 8 Wire | 8x2.5 Sq.mm. | Metre | 444.00 | 444.00 |

**Note: -** Secondary point shall be measured from the looping point. Total wattage of looped points shall not exceed 200 Watts.

# CHAPTER – 3

## WIRING IN SURFACE RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **3.1** | | **Point wiring surface rigid P.V.C. conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid P.V.C. conduit (MMS) ISI marked of suitable size and 1.5sq. mm. PVC FRLS insulated copper earth continuity conductor of green colour inside conduit etc. as required as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **3.1.1** | **Light/fan points primary** | **Light point/fan points.** | | | |
| | 3.1.1.1 | Short | Primary short point | Each | 338.00 | 408.00 |
| | 3.1.1.2 | Medium | Primary medium point | Each | 640.00 | 729.00 |
| | 3.1.1.3 | Long | Primary long point | Each | 1070.00 | 1141.00 |
| | **3.1.2** | **Looping point secondary** | **(Light point/fan points) looping point** | | | |
| | 3.1.2.1 | Short | Secondary short point | Each | 169.00 | 204.00 |
| | 3.1.2.2 | Medium | Secondary medium point | Each | 320.00 | 365.00 |
| | 31..2.3 | Long | Secondary long point | Each | 535.00 | 571.00 |
| | **3.1.3** | **Socket 6A sep. board** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 3.1.3.1 | Short | Short point | Each | 355.00 | 540.00 |
| | 3.1.3.2 | Medium | Medium point | Each | 729.00 | 933.00 |
| | 3.1.3.3 | Long | Long point | Each | 1194.00 | 1379.00 |
| | **3.1.4** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 3.1.4.1 | Short | Short point | Each | 307.00 | 411.00 |
| | 3.1.4.2 | Medium | Medium point | Each | 601.00 | 704.00 |
| | 3.1.4.3 | Long | Long point | Each | 918.00 | 1022.00 |
| | **3.1.5** | **Twin control point** | **Twin control light points** | | | |
| | 3.1.5.1 | Short | Short point | Each | 377.00 | 488.00 |
| | 3.1.5.2 | Medium | Medium point | Each | 745.00 | 874.00 |
| | 3.1.5.3 | Long | Long point | Each | 1183.00 | 1294.00 |
| **3.2** | | **Socket same board 6A** | Point wiring (excluding metallic switch box & sheet but including switches, sockets) for 3 pin 6 Amp. Socket outlet point with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid P.V.C. conduit (MMS) ISI marked of suitable size and 1.5sq. mm. PVC insulated copper earth continuity conductor of green colour inside conduit with required materials as per specification on same board. | Each | 82.00 | 265.00 |

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **3.3** | | **Circuit wiring 2.5Sq. mm** | Wiring for circuit wiring with PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid P.V.C. conduit (MMS) of ISI marked suitable size etc. as required as per specification | | | |
| | 3.3.1 | 2 Wire | 2x2.5 Sq.mm. | Metre | 122.00 | 122.00 |
| | 3.3.2 | 4 Wire | 4x2.5 Sq.mm. | Metre | 202.00 | 202.00 |
| | 3.3.3 | 6 Wire | 6x2.5 Sq.mm. | Metre | 267.00 | 267.00 |
| | 3.3.4 | 8 Wire | 8x2.5 Sq.mm. | Metre | 331.00 | 331.00 |

**Note: -** Secondary point shall be measured from the looping point. Total wattage of looped points shall not exceed 200 Watts.

# CHAPTER – 4

## WIRING IN CONCEALED RIGID P.V.C. CONDUIT WITH FRLS COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **4.1** | | **Point wiring concealed rigid P.V.C. conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid P.V.C. conduit (HMS) ISI marked of suitable size and 1.5sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit, screwless connector for joints etc. as required as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **4.1.1** | **Light/fan points primary** | **Light point/fan points.** | | | |
| | 4.1.1.1 | Short | Primary short point | Each | 383.00 | 451.00 |
| | 4.1.1.2 | Medium | Primary medium point | Each | 761.00 | 830.00 |
| | 4.1.1.3 | Long | Primary long point | Each | 1231.00 | 1300.00 |
| | **4.1.2** | **Looping point secondary** | **(Light point/fan points) looping point** | | | |
| | 4.1.2.1 | Short | Secondary short point | Each | 192.00 | 226.00 |
| | 4.1.2.2 | Medium | Secondary medium point | Each | 381.00 | 415.00 |
| | 4.1.2.3 | Long | Secondary long point | Each | 616.00 | 650.00 |
| | **4.1.3** | **Socket 6A sep. board** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 4.1.3.1 | Short | Short point | Each | 391.00 | 574.00 |
| | 4.1.3.2 | Medium | Medium point | Each | 807.00 | 990.00 |
| | 4.1.3.3 | Long | Long point | Each | 1332.00 | 1515.00 |
| | **4.1.4** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 4.1.4.1 | Short | Short point | Each | 352.00 | 455.00 |
| | 4.1.4.2 | Medium | Medium point | Each | 695.00 | 797.00 |
| | 4.1.4.3 | Long | Long point | Each | 1079.00 | 1183.00 |
| | **4.1.5** | **Twin control point** | **Twin control light points** | | | |
| | 4.1.5.1 | Short | Short point | Each | 432.00 | 531.00 |
| | 4.1.5.2 | Medium | Medium point | Each | 850.00 | 948.00 |
| | 4.1.5.3 | Long | Long point | Each | 1355.00 | 1453.00 |

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** | **7** |
| **4.2** | | **Socket same board 6A** | Point wiring (excluding metallic switch box & sheet but including switches, sockets) for 3 pin 6 Amp. socket outlet point with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid P.V.C. conduit (HMS) ISI marked of suitable size and 1.5sq. mm. PVC insulated copper earth continuity conductor of green colour inside conduit with required materials as per specification on same board. | Each | 82.00 | 265.00 |
| **4.3** | | **Circuit wiring 2.5Sq. mm** | Wiring for circuit wiring with PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid P.V.C. conduit (HMS) of ISI marked suitable size etc. as required as per specification | | | |
| | 4.3.1 | 2 Wire | 2x2.5 Sq.mm. | Metre | 139.00 | 139.00 |
| | 4.3.2 | 4 Wire | 4x2.5 Sq.mm. | Metre | 224.00 | 224.00 |
| | 4.3.3 | 6 Wire | 6x2.5 Sq.mm. | Metre | 288.00 | 288.00 |
| | 4.3.4 | 8 Wire | 8x2.5 Sq.mm. | Metre | 353.00 | 353.00 |

**Note: -** Secondary point shall be measured from the looping point. Total wattage of looped points shall not exceed 200 Watts.

# CHAPTER - 5

## WIRING IN SURFACE RIGID P.V.C. CASING AND CAPING WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **5.1** | | **Point wiring surface rigid P.V.C. casing & capping** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in Surface rigid P.V.C. casing and caping ISI marked of suitable size and 1.5sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside casing & capping screwless connector for joints etc. as required as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **5.1.1** | **Light/fan points primary** | **Light point/fan points.** | | | |
| | 5.1.1.1 | Short | Primary short point | Each | 303.00 | 358.00 |
| | 5.1.1.2 | Medium | Primary medium point | Each | 607.00 | 641.00 |
| | 5.1.1.3 | Long | Primary long point | Each | 932.00 | 943.00 |
| | **5.1.2** | **Looping point secondary** | **(Light point/fan points) looping point** | | | |
| | 5.1.2.1 | Short | Secondary short point | Each | 152.00 | 179.00 |
| | 5.1.2.2 | Medium | Secondary medium point | Each | 304.00 | 321.00 |
| | 5.1.2.3 | Long | Secondary long point | Each | 466.00 | 472.00 |
| | **5.1.3** | **Socket 6A sep. board** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 5.1.3.1 | Short | Short point | Each | 311.00 | 481.00 |
| | 5.1.3.2 | Medium | Medium point | Each | 652.00 | 801.00 |
| | 5.1.3.3 | Long | Long point | Each | 1033.00 | 1094.00 |
| | **5.1.4** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 5.1.4.1 | Short | Short point | Each | 270.00 | 360.00 |
| | 5.1.4.2 | Medium | Medium point | Each | 518.00 | 588.00 |
| | 5.1.4.3 | Long | Long point | Each | 780.00 | 826.00 |
| | **5.1.5** | **Twin control point** | **Twin control light points** | | | |
| | 5.1.5.1 | Short | Short point | Each | 353.00 | 437.00 |
| | 5.1.5.2 | Medium | Medium point | Each | 693.00 | 757.00 |
| | 5.1.5.3 | Long | Long point | Each | 1056.00 | 1096.00 |

| Item No. | Sub Item No. | Item Name | Description of Item<br>Wiring With 1.5sq. mm. Copper Stranded Conductor | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **5.2** | | **Socket same board 6A** | Point wiring (excluding metallic switch box & sheet but including switches, sockets) for 3 pin 6 Amp. socket outlet point with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in Surface rigid P.V.C. casing and caping ISI marked of suitable size and 1.5sq. mm. PVC insulated copper earth continuity conductor of green colour inside casing & capping with required materials as per specification on same board. | Each | 82.00 | 265.00 |
| **5.3** | | **Circuit wiring 2.5Sq. mm** | Wiring for circuit wiring with PVC insulated cable FRLS with copper multi strand conductor ISI marked in Surface rigid P.V.C. casing and caping of ISI marked suitable size etc. as required as per specification | | | |
| | 5.3.1 | 2 Wire | 2x2.5 Sq.mm. | Metre | 106.00 | 106.00 |
| | 5.3.2 | 4 Wire | 4x2.5 Sq.mm. | Metre | 174.00 | 174.00 |
| | 5.3.3 | 6 Wire | 6x2.5 Sq.mm. | Metre | 239.00 | 239.00 |
| | 5.3.4 | 8 Wire | 8x2.5 Sq.mm. | Metre | 304.00 | 304.00 |

**Note: -** Secondary point shall be measured from the looping point. Total wattage of looped points shall not exceed 200 Watts.

# CHAPTER – 6

## POWER WIRING IN SURFACE RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **6.1** | | **Power point wiring surface steel rigid conduit** | Point wiring (excluding metallic switch box & sheet) for 3 Pin 16 Amp. socket outlet point with 4sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid steel conduit ISI marked of suitable size including painting etc. with 20/16 Amp. F.T. switch &socket/S.S. combined 20/16/6 amp. of ISI marked and 4sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit, screwless connector for joints as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **6.1.1** | **Power point** | **On separate board** | | | |
| | 6.1.1.1 | Short | Short point | Each | 791.00 | 1011.00 |
| | 6.1.1.2 | Medium | Medium point | Each | 1648.00 | 1867.00 |
| | 6.1.1.3 | Long | Long point | Each | 2637.00 | 2856.00 |
| | 6.1.1.4 | Ex. long-I | Extra long -I | Each | 3809.00 | 4028.00 |
| | 6.1.1.5 | Ex. long-II | Extra long -II | Each | 5133.00 | 5353.00 |
| | 6.1.1.6 | Ex. long-III | Extra long -III | Each | 6464.00 | 6684.00 |
| **6.2** | | **Same board** | Same board switch & socket 16 Amp./S.S. combined 20/16/6 Amp | Each | 217.00 | 437.00 |

# CHAPTER – 7

## POWER WIRING IN CONCEALED RIGID STEEL CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **7.1** | | **Power point wiring concealed steel rigid conduit** | Point wiring (excluding metallic switch box & sheet) for 3 pin 16 Amp. socket outlet point with 4sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid steel conduit ISI marked of suitable size including painting etc. with 20/16 Amp. F.T. switch & socket/S.S. combined 20/16/6 Amp. of ISI marked and 4sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit, screwless connector for joints as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **7.1.1** | **Power point** | **On separate board** | | | |
| | 7.1.1.1 | Short | Short point | Each | 853.00 | 1072.00 |
| | 7.1.1.2 | Medium | Medium point | Each | 1717.00 | 1937.00 |
| | 7.1.1.3 | Long | Long point | Each | 2720.00 | 2940.00 |
| | 7.1.1.4 | Ex. long-I | Extra long -I | Each | 3961.00 | 4181.00 |
| | 7.1.1.5 | Ex. long-II | Extra long-II | Each | 5332.00 | 5551.00 |
| | 7.1.1.6 | Ex. long-III | Extra long-III | Each | 6721.00 | 6941.00 |
| **7.2** | | **Same board** | Same board switch & socket 16 Amp./S.S. combined 20/16/6 Amp | Each | 217.00 | 437.00 |

# CHAPTER – 8

## POWER WIRING IN SURFACE RIGID P.V.C.CONDUIT WITH FRLS COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **8.1** | | **Power point wiring surface rigid P.V.C. conduit** | Point wiring (excluding metallic switch box & sheet) for 3 Pin 16 Amp. socket outlet point with 4sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid P.V.C. conduit (MMS) ISI marked of suitable size etc. with 20/16 Amp. F.T. switch & socket/S.S. combined 20/16/6 Amp. of ISI marked and 4sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit, screwless connector for joints as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **8.1.1** | **Power point** | **On separate board** | | | |
| | 8.1.1.1 | Short | Short point | Each | 625.00 | 845.00 |
| | 8.1.1.2 | Medium | Medium point | Each | 1249.00 | 1468.00 |
| | 8.1.1.3 | Long | Long point | Each | 1963.00 | 2183.00 |
| | 8.1.1.4 | Ex. long-I | Extra long -I | Each | 2849.00 | 3069.00 |
| | 8.1.1.5 | Ex. long-II | Extra long -II | Each | 3825.00 | 4045.00 |
| | 8.1.1.6 | Ex. long-III | Extra long -III | Each | 4814.00 | 5033.00 |
| **8.2** | | **Same board** | Same board switch & socket 16 Amp./S.S. combined 20/16/6 Amp | Each | 217.00 | 437.00 |

# CHAPTER – 9

## POWER WIRING IN CONCEALED RIGID P.V.C.CONDUIT WITH FRLS COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **9.1** | | **Power point wiring concealed rigid P.V.C. conduit** | Point wiring (excluding metallic switch box & sheet) for 3 Pin 16 Amp. socket outlet point with 4sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in concealed rigid P.V.C. conduit (HMS) ISI marked of suitable size etc. with 20/16 Amp. F.T. switch & socket/S.S. combined 20/16/6 Amp. of ISI marked and 4sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside conduit, screwless connector for joints as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **9.1.1** | **Power point** | **On separate board** | | | |
| | 9.1.1.1 | Short | Short point | Each | 765.00 | 985.00 |
| | 9.1.1.2 | Medium | Medium point | Each | 1530.00 | 1749.00 |
| | 9.1.1.3 | Long | Long point | Each | 2393.00 | 2612.00 |
| | 9.1.1.4 | Ex. long-I | Extra long-I | Each | 3429.00 | 3648.00 |
| | 9.1.1.5 | Ex. long-II | Extra long-II | Each | 4551.00 | 4771.00 |
| | 9.1.1.6 | Ex. long-III | Extra long-III | Each | 5766.00 | 5986.00 |
| **9.2** | | **Same board** | Same board switch & socket 16 Amp./S.S. combined 20/16/6 Amp | Each | 217.00 | 437.00 |

# CHAPTER - 10

## POWER WIRING IN SURFACE RIGID P.V.C.CASING AND CAPING WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **10.1** | | **Power point wiring surface rigid P.V.C. casing & caping** | Point wiring (excluding metallic switch box & sheet) for 3 Pin 16 Amp. socket outlet point with 4sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid P.V.C. casing and caping ISI marked of suitable size etc. with 20/16 Amp. F.T. switch & socket/S.S. combined 20/16/6 Amp. of ISI marked and 4sq. mm. PVC insulated FRLS copper earth continuity conductor of green colour inside capping, screwless connector for joints as per specification for :- | | Using flush type normal accessories | Using modular accessories |
| | **10.1.1** | **Power point** | **On separate board** | | | |
| | 10.1.1.1 | Short | Short point | Each | 594.00 | 807.00 |
| | 10.1.1.2 | Medium | Medium point | Each | 1181.00 | 1383.00 |
| | 10.1.1.3 | Long | Long point | Each | 1850.00 | 2041.00 |
| | 10.1.1.4 | Ex. long-I | Extra long-I | Each | 2685.00 | 2860.00 |
| | 10.1.1.5 | Ex. long-II | Extra long-II | Each | 3603.00 | 3761.00 |
| | 10.1.1.6 | Ex. long-III | Extra long-III | Each | 4523.00 | 4664.00 |
| **10.2** | | **Same board** | Same board switch & socket 16 Amp./S.S. combined 20/16/6 Amp | Each | 217.00 | 437.00 |

# CHAPTER – 11

## WIRING IN EXISTING CONCEALED RIGID CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **11.1** | | **Point wiring In existing concealed rigid conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable FRLS with copper multi strand conductor ISI marked in existing concealed rigid conduit ISI marked of suitable size with PVC insulated FRLS copper earth continuity conductor of green colour, screwless connector for joints as per specification including earthing of regulator etc as required for :- | | Using flush type normal accessories | Using modular accessories |
| | **11.1.1** | **Light/fan points.** | **Light point/fan/ex-fan points.** | | | |
| | 11.1.1.1 | Short | Short point | Each | 214.00 | 262.00 |
| | 11.1.1.2 | Medium | Medium point | Each | 431.00 | 464.00 |
| | 11.1.1.3 | Long | Long point | Each | 683.00 | 688.00 |
| | **11.1.2** | **6A socket point** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 11.1.2.1 | Short | Short point | Each | 216.00 | 370.00 |
| | 11.1.2.2 | Medium | Medium point | Each | 434.00 | 570.00 |
| | 11.1.2.3 | Long | Long point | Each | 686.00 | 796.00 |
| | **11.1.3** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 11.1.3.1 | Short | Short point | Each | 181.00 | 266.00 |
| | 11.1.3.2 | Medium | Medium point | Each | 343.00 | 417.00 |
| | 11.1.3.3 | Long | Long point | Each | 530.00 | 583.00 |
| | **11.1.4** | **Twin control point** | **Twin control light points** | | | |
| | 11.1.4.1 | Short | Short point | Each | 260.00 | 338.00 |
| | 11.1.4.2 | Medium | Medium point | Each | 477.00 | 538.00 |
| | 11.1.4.3 | Long | Long point | Each | 711.00 | 743.00 |

# CHAPTER – 12

## WIRING IN EXISTING SURFACE RIGID CONDUIT SYSTEM WITH COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs.) | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **12.1** | | **Point wiring in existing surface rigid conduit** | Point wiring (excluding metallic switch box & sheet but including switches, sockets, lamp holders/ceiling roses etc) with 1.5sq. mm. PVC insulated cable copper stranded conductor ISI marked FRLS in existing surface rigid conduit ISI marked of suitable size with PVC insulated FRLS copper earth continuity conductor of green colour as per specification including earthing of regulator etc as required for :- | | Using flush type normal accessories | Using modular accessories |
| | **12.1.1** | **Light/fan points.** | **Light point/fan /ex-fan points.** | | | |
| | 12.1.1.1 | Short | Short point | Each | 204.00 | 272.00 |
| | 12.1.1.2 | Medium | Medium point | Each | 413.00 | 483.00 |
| | 12.1.1.3 | Long | Long point | Each | 653.00 | 720.00 |
| | **12.1.2** | **6A socket point** | **3 Pin 6 Amp socket outlet on separate board** | | | |
| | 12.1.2.1 | Short | Short point | Each | 207.00 | 384.00 |
| | 12.1.2.2 | Medium | Medium point | Each | 415.00 | 594.00 |
| | 12.1.2.3 | Long | Long point | Each | 656.00 | 828.00 |
| | **12.1.3** | **Call bell point** | **Call bell/buzzer points** | | | |
| | 12.1.3.1 | Short | Short point | Each | 173.00 | 276.00 |
| | 12.1.3.2 | Medium | Medium point | Each | 328.00 | 435.00 |
| | 12.1.3.3 | Long | Long point | Each | 507.00 | 616.00 |
| | **12.1.4** | **Twin control point** | **Twin control light points** | | | |
| | 12.1.4.1 | Short | Short point | Each | 260.00 | 344.00 |
| | 12.1.4.2 | Medium | Medium point | Each | 477.00 | 556.00 |
| | 12.1.4.3 | Long | Long point | Each | 711.00 | 773.00 |

# CHAPTER – 13

## SUPPLY AND FIXING M.S.CONDUIT FOR CONCEALED CONDUIT SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **13.1** | | **Concealed M.S. conduit H.G.** | Supplying and fixing rigid steel conduit ISI marked along with the accessories in concealed system including cutting the wall and plastering & repainting the wall with matching colour to bring in its original condition as required | | |
| | 13.1.1 | 20 mm. 16 SWG | H.G. conduit 20 mm, wall thickness 1.6mm. | Metre | 123.00 |
| | 13.1.2 | 25 mm. 16 SWG | H.G. conduit 25 mm, wall thickness 1.6mm. | Metre | 155.00 |
| | 13.1.3 | 32 mm. 16 SWG | H.G. conduit 32 mm, wall thickness 1.6mm. | Metre | 212.00 |
| | 13.1.4 | 40 mm. 14 SWG | H.G. conduit40 mm, wall thickness 2.0mm. | Metre | 339.00 |
| | 13.1.5 | 50 mm. 14 SWG | H.G. conduit 50 mm, wall thickness 2.0mm. | Metre | 440.00 |

# CHAPTER – 14

## SUPPLY AND FIXING M.S. CONDUIT FOR SURFACE CONDUIT SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **14.1** | | **Surface M.S. conduit H.G.** | Supplying and fixing rigid steel conduit ISI marked along with the accessories on surface including painting etc. as required | | |
| | 14.1.1 | 20 mm. 16 SWG | H.G. conduit 20 mm, wall thickness 1.6mm. | Metre | 121.00 |
| | 14.1.2 | 25 mm. 16 SWG | H.G. conduit 25 mm, wall thickness 1.6mm. | Metre | 151.00 |
| | 14.1.3 | 32 mm. 16 SWG | H.G. conduit 32 mm, wall thickness 1.6mm. | Metre | 212.00 |
| | 14.1.4 | 40 mm. 14 SWG | H.G. conduit 40 mm, wall thickness 2.0mm. | Metre | 338.00 |
| | 14.1.5 | 50 mm. 14 SWG | H.G. conduit 50 mm, wall thickness 2.0mm. | Metre | 441.00 |

# CHAPTER – 15

## SUPPLY AND FIXING PVC CONDUIT FOR SURFACE CONDUIT SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **15.1** | | **Surface PVC conduit MMS** | Supplying and fixing PVC conduit as required confirming to IS 9537 (Part-3), ISI marked along with accessories on surface etc. as required. | | |
| | 15.1.1 | 20 mm. MMS | PVC conduit 20 mm. (MMS) | Metre | 51.00 |
| | 15.1.2 | 25 mm. MMS | PVC conduit 25 mm. (MMS) | Metre | 67.00 |
| | 15.1.3 | 32 mm. MMS | PVC conduit 32 mm. (MMS) | Metre | 102.00 |
| | 15.1.4 | 40 mm. MMS | PVC conduit 40 mm. (MMS) | Metre | 136.00 |
| | 15.1.5 | 50 mm. MMS | PVC conduit 50 mm. (MMS) | Metre | 193.00 |

# CHAPTER – 16

## SUPPLY AND FIXING PVC. CONDUIT FOR CONCEALED CONDUIT SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **16.1** | | **Concealed PVC conduit HMS** | Supply and fixing PVC conduits required confirming to IS 9537 (Part-3), ISI marked along with the accessories in concealed system etc. as required. | | |
| | 16.1.1 | 20 mm. HMS | PVC. conduit 20 mm (HMS) | Metre | 61.00 |
| | 16.1.2 | 25 mm. HMS | PVC. conduit 25 mm (HMS) | Metre | 74.00 |
| | 16.1.3 | 32 mm. HMS | PVC. conduit 32 mm (HMS) | Metre | 115.00 |
| | 16.1.4 | 40 mm. HMS | PVC. conduit 40 mm (HMS) | Metre | 147.00 |
| | 16.1.5 | 50 mm. HMS | PVC. conduit 50 mm (HMS) | Metre | 211.00 |

# CHAPTER – 17

## SUPPLY AND FIXING PVC CASING AND CAPPING ON SURFACE SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate ( in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **17.1** | | **PVC casing and capping (double fold)** | Supplying and fixing PVC casing and capping (double fold) ISI marked along with accessories on surface etc. as required | | |
| | 17.1.1 | 20 mm x 12mm | PVC casing and capping (double fold) 20 mm x 12mm | Metre | 26.00 |
| | 17.1.2 | 25 mm x 12mm | PVC casing and capping (double fold) 25 mm x 12mm | Metre | 34.00 |
| | 17.1.3 | 32 mm x 12mm | PVC casing and capping (double fold) 32 mm x 12mm | Metre | 39.00 |
| | 17.1.4 | 38 mm x 20mm | PVC casing and capping (double fold) 38 mm x 20mm | Metre | 47.00 |
| | 17.1.5 | 50 mm x 16mm | PVC casing and capping (double fold) 50 mm x 16mm | Metre | 75.00 |

# CHAPTER – 18
# M.S.BOXES

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate ( in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **18.1** | | **Normal MS switch box (surface/concealed)** | Supplying and fixing factory fabricated hot dip galvanized switch box for normal flush type accessories of minimum 1.2 mm thickness for up to 200mmx250mm size boxes and 1.6mm minimum thickness for higher size boxes, earth terminal stud with nut and washer, including 3 mm thick synthetic phenolic resin bonded laminated sheet conforming to grade P-I of IS:2036-1974 on surface/concealed including all necessary accessories, painting etc. as required as per specification: | | |
| | 18.1.1 | MS box 100 X100 | MS box 100 mm X 100 mm X 60mm | Each | 64.00 |
| | 18.1.2 | MS box 250 X150 | MS box 200 mm X 150 mm X 60 mm | Each | 77.00 |
| | 18.1.3 | MS box 180 X100 | MS box 180 mm X 100 mm X 60 mm | Each | 57.00 |
| | 18.1.4 | MS box 125 X 300 | MS box 125 mm X 300 mm X 60 mm | Each | 73.00 |
| | 18.1.5 | MS box 200 X250 | MS box 200 mm X 250 mm X 60 mm | Each | 84.00 |
| | 18.1.6 | MS box 250 X300 | MS box 250 mm X 300 mm X 60 mm | Each | 125.00 |
| | 18.1.7 | MS box 300 X 380 | MS box 300mm X 380 mm X 60 mm | Each | 239.00 |
| **18.2** | | **Modular G.I. metal box (concealed)** | Supplying and fixing of approved make modular type G.I. metal box 1.2mm minimum thick and depth 50mm for single row boxes and 60mm depth for two/three row boxes earth terminal stud with nut and washer with modular base and cover plate including fixing in concealed etc. as required for:- | | |
| | 18.2.1 | G.I box 1 Or 2 module | 1Or 2 Module metal box | Each | 182.00 |
| | 18.2.2 | G.I box 3 module | 3 Module metal box | Each | 216.00 |
| | 18.2.3 | G.I box 4 module | 4 Module metal box | Each | 244.00 |
| | 18.2.4 | G.I box 6 module | 6 Module metal box | Each | 309.00 |
| | 18.2.5 | G.I box 8/9 module | 8/9 Module metal box | Each | 373.00 |
| | 18.2.6 | G.I box 12 module | 12 Module metal box | Each | 502.00 |
| | 18.2.7 | G.I box 16 module | 16 Module metal box | Each | 585.00 |
| | 18.2.8 | G.I box 18 module | 18 Module metal box | Each | 642.00 |
| **18.3** | | **Modular PVC box (surface)** | Supplying and fixing of approved make modular type PVC box minimum depth 40mm and minimum thickness 2mm with modular base and cover plate including fixing on surface etc. as required for:- | | |
| | 18.3.1 | PVC box 1 Or 2 module | 1 Or 2 Module PVC box | Each | 155.00 |
| | 18.3.2 | PVC box 3 module | 3 Module PVC box | Each | 177.00 |
| | 18.3.3 | PVC box 4 module | 4 Module PVC box | Each | 210.00 |
| | 18.3.4 | PVC box 6 module | 6 Module PVC box | Each | 254.00 |
| | 18.3.5 | PVC box 8/9 module | 8/9 Module PVC box | Each | 314.00 |
| | 18.3.6 | PVC box 12 module | 12 Module PVC box | Each | 402.00 |
| | 18.3.7 | PVC box 16 module | 16 Module PVC box | Each | 442.00 |

# CHAPTER – 19

## WIRING IN EXISTING CONDUIT/PVC CASING AND CAPPING

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **19.1** | | **Single core PVC insulated FRLS cable** | Supplying and drawing single core PVC insulated cable FRLS with copper multi strand conductor ISI marked in existing rigid PVC casing and capping/conduit in surface or concealed as per specification. | | |
| | **19.1.1** | **1.5 sq. mm.** | **1.5 sq. mm. wire** | | |
| | 19.1.1.1 | 1 wire | 1 X 1.5 sq. mm. | Metre | 19.00 |
| | 19.1.1.2 | 2 wire | 2 X 1.5 sq. mm. | Metre | 38.00 |
| | 19.1.1.3 | 3 wire | 3 X 1.5 sq. mm. | Metre | 56.00 |
| | 19.1.1.4 | 4 wire | 4 X 1.5 sq. mm. | Metre | 75.00 |
| | 19.1.1.5 | 5 wire | 5 X 1.5 sq. mm. | Metre | 94.00 |
| | 19.1.1.6 | 6 wire | 6 X 1.5 sq. mm. | Metre | 113.00 |
| | 19.1.1.7 | 7 wire | 7 X 1.5 sq. mm. | Metre | 132.00 |
| | 19.1.1.8 | 8 wire | 8 X 1.5 sq. mm. | Metre | 151.00 |
| | 19.1.1.9 | 9 wire | 9 X 1.5 sq. mm. | Metre | 170.00 |
| | 19.1.1.10 | 10 wire | 10 X 1.5 sq. mm. | Metre | 189.00 |
| | **19.1.2** | **2.5 sq. mm.** | **2.5 sq. mm. wire** | | |
| | 19.1.2.1 | 1 wire | 1 X 2.5 sq. mm. | Metre | 30.00 |
| | 19.1.2.2 | 2 wire | 2 X 2.5 sq. mm. | Metre | 60.00 |
| | 19.1.2.3 | 3 wire | 3 X 2.5 sq. mm. | Metre | 90.00 |
| | 19.1.2.4 | 4 wire | 4 X 2.5 sq. mm. | Metre | 120.00 |
| | 19.1.2.5 | 5 wire | 5 X 2.5 sq. mm. | Metre | 150.00 |
| | 19.1.2.6 | 6 wire | 6 X 2.5 sq. mm. | Metre | 180.00 |
| | 19.1.2.7 | 7 wire | 7 X 2.5 sq. mm. | Metre | 210.00 |
| | 19.1.2.8 | 8 wire | 8 X 2.5 sq. mm. | Metre | 240.00 |
| | 19.1.2.9 | 9 wire | 9 X 2.5 sq. mm. | Metre | 271.00 |
| | 19.1.2.10 | 10 wire | 10 X 2.5 sq. mm. | Metre | 273.00 |
| | **19.1.3** | **4.00 sq. mm.** | **4.00 sq. mm. wire** | | |
| | 19.1.3.1 | 1 wire | 1 X 4.00 sq. mm. | Metre | 45.00 |
| | 19.1.3.2 | 2 wire | 2 X 4.00 sq. mm. | Metre | 90.00 |
| | 19.1.3.3 | 3 wire | 3 X 4.00 sq. mm. | Metre | 134.00 |
| | 19.1.3.4 | 4 wire | 4 X 4.00 sq. mm. | Metre | 180.00 |
| | 19.1.3.5 | 5 wire | 5 X 4.00 sq. mm. | Metre | 224.00 |
| | 19.1.3.6 | 6 wire | 6 X 4.00 sq. mm. | Metre | 269.00 |
| | 19.1.3.7 | 7 wire | 7 X 4.00 sq. mm. | Metre | 314.00 |
| | 19.1.3.8 | 8 wire | 8 X 4.00 sq. mm. | Metre | 360.00 |
| | 19.1.3.9 | 9 wire | 9 X 4.00 sq. mm. | Metre | 405.00 |
| | 19.1.3.10 | 10 wire | 10 X 4.00 sq. mm. | Metre | 451.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | **19.1.4** | **6.00 sq. mm.** | **6.00 sq. mm. wire** | | |
| | 19.1.4.1 | 1 wire | 1 X 6.00 sq. mm. | Metre | 68.00 |
| | 19.1.4.2 | 2 wire | 2 X 6.00 sq. mm. | Metre | 135.00 |
| | 19.1.4.3 | 3 wire | 3 X 6.00 sq. mm. | Metre | 202.00 |
| | 19.1.4.4 | 4 wire | 4 X 6.00 sq. mm. | Metre | 269.00 |
| | 19.1.4.5 | 5 wire | 5 X 6.00 sq. mm. | Metre | 336.00 |
| | 19.1.4.6 | 6 wire | 6 X 6.00 sq. mm. | Metre | 404.00 |
| | 19.1.4.7 | 7 wire | 7 X 6.00 sq. mm. | Metre | 471.00 |
| | 19.1.4.8 | 8 wire | 8 X 6.00 sq. mm. | Metre | 538.00 |
| | 19.1.4.9 | 9 wire | 9 X 6.00 sq. mm. | Metre | 606.00 |
| | 19.1.4.10 | 10 wire | 10 X 6.00 sq. mm. | Metre | 673.00 |
| | **19.1.5** | **10.00 sq. mm.** | **10.00 sq. mm. cable** | | |
| | 19.1.5.1 | 1 wire | 1 X 10.00 sq. mm. | Metre | 111.00 |
| | 19.1.5.2 | 2 wire | 2 X 10.00 sq. mm. | Metre | 218.00 |
| | 19.1.5.3 | 3 wire | 3 X 10.00 sq. mm. | Metre | 324.00 |
| | 19.1.5.4 | 4 wire | 4 X 10.00 sq. mm. | Metre | 431.00 |
| | 19.1.5.5 | 5 wire | 5 X 10.00 sq. mm. | Metre | 538.00 |
| | 19.1.5.6 | 6 wire | 6 X 10.00 sq. mm. | Metre | 652.00 |
| | **19.1.6** | **16.00 sq. mm.** | **16.00 sq. mm. cable** | | |
| | 19.1.6.1 | 1 wire | 1 X 16.00 sq. mm. | Metre | 188.00 |
| | 19.1.6.2 | 2 wire | 2 X 16.00 sq. mm. | Metre | 368.00 |
| | 19.1.6.3 | 3 wire | 3 X 16.00 sq. mm. | Metre | 550.00 |
| | 19.1.6.4 | 4 wire | 4 X 16.00 sq. mm. | Metre | 730.00 |
| | 19.1.7 | 25.00 sq. mm. | **25.00 sq. mm. cable** | | |
| | 19.1.7.1 | 1 wire | 1 X 25.00 sq. mm. | Metre | 300.00 |
| | 19.1.7.2 | 2 wire | 2 X 25.00 sq. mm. | Metre | 591.00 |
| | 19.1.7.3 | 3 wire | 3 X 25.00 sq. mm. | Metre | 883.00 |
| | 19.1.7.4 | 4 wire | 4 X 25.00 sq. mm. | Metre | 1170.00 |
| | **19.1.8** | **35.00 sq. mm.** | **35.00 sq. mm. cable** | | |
| | 19.1.8.1 | 1 wire | 1 X 35.00 sq. mm. | Metre | 415.00 |
| | 19.1.8.2 | 2 wire | 2 X 35.00 sq. mm. | Metre | 822.00 |
| | 19.1.8.3 | 3 wire | 3 X 35.00 sq. mm. | Metre | 1228.00 |
| | 19.1.8.4 | 4 wire | 4 X 35.00 sq. mm. | Metre | 1631.00 |

# CHAPTER – 20

## MAIN AND SUB MAINS IN SURFACE RIGID STEEL CONDUIT

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **20.1** | | **Sub mains in surface rigid steel** | Wiring for sub-mains with PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid steel ISI marked conduit of suitable size (conduit included) including accessories long neck threaded type, connection painting etc, as required as per specification. (Refer General Notes for Classification of Conduit Wiring Point No.10) | | |
| | **20.1.1** | **2 Wire** | **2 Wire sub main** | | |
| | 20.1.1.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 208.00 |
| | 20.1.1.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 238.00 |
| | 20.1.1.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 288.00 |
| | 20.1.1.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 402.00 |
| | 20.1.1.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 538.00 |
| | 20.1.1.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 824.00 |
| | 20.1.1.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1060.00 |
| | **20.1.2** | **3 Wire** | **3 Wire sub main** | | |
| | 20.1.2.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 240.00 |
| | 20.1.2.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 285.00 |
| | 20.1.2.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 353.00 |
| | 20.1.2.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 513.00 |
| | 20.1.2.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 777.00 |
| | 20.1.2.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1113.00 |
| | 20.1.2.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1605.00 |
| | **20.1.3** | **3 1/2 Wire** | **3 1/2 Wire sub main** | | |
| | 20.1.3.1 | 3 x 2.5 + 1 x 1.5 sq. mm. cable | 3 x 2.5 + 1 x 1.5 sq. mm. cable | Metre | 260.00 |
| | 20.1.3.2 | 3 x 4.0 + 1 x 2.5 sq. mm. cable | 3 x 4.0 + 1 x 2.5 sq. mm. cable | Metre | 320.00 |
| | 20.1.3.3 | 3 x 6.0 + 1 x 4.0 sq. mm. cable | 3 x 6.0 + 1 x 4.0 sq. mm. cable | Metre | 434.00 |
| | 20.1.3.4 | 3 x 10.0 + 1 x 6.0 sq. mm. cable | 3 x 10.0 + 1 x 6.0 sq. mm. cable | Metre | 649.00 |
| | 20.1.3.5 | 3 x 16.0 + 1 x 10.0 sq. mm. cable | 3 x 16.0 + 1 x 10.0 sq. mm. cable | Metre | 895.00 |
| | 20.1.3.6 | 3 x 25.0 + 1 x 16.0 sq. mm. cable | 3 x 25.0 + 1 x 16.0 sq. mm. cable | Metre | 1427.00 |
| | 20.1.3.7 | 3 x 35.0 + 1 x 16.0 sq. mm. cable | 3 x 35.0 + 1 x 16.0 sq. mm. cable | Metre | 1784.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | **20.1.4** | **4 Wire** | **4 Wire sub main** | | |
| | 20.1.4.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 271.00 |
| | 20.1.4.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 335.00 |
| | 20.1.4.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 457.00 |
| | 20.1.4.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 631.00 |
| | 20.1.4.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 965.00 |
| | 20.1.4.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1536.00 |
| | 20.1.4.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 2009.00 |

# CHAPTER – 21

## SUB-MAIN IN CONCEALED RIGID STEEL CONDUIT IN COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 21.1 | | **Sub-mains in recessed/concealed rigid steel conduit** | Wiring for sub-mains with PVC insulated cable FRLS with copper multi strand conductor ISI marked in recessed/concealed rigid steel ISI marked conduit of suitable size (Conduit included) including accessories long neck threaded type, connection painting etc, as required as per specification. (Refer General Notes for Classification of Conduit Wiring Point No.10) | | |
| | **21.1.1** | **2 Wire** | **2 Wire sub main** | | |
| | 21.1.1.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 238.00 |
| | 21.1.1.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 268.00 |
| | 21.1.1.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 344.00 |
| | 21.1.1.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 431.00 |
| | 21.1.1.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 563.00 |
| | 21.1.1.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 852.00 |
| | 21.1.1.7 | 35.0 sq. mm. cable | 35.0 sq. mm. | Metre | 1121.00 |
| | **21.1.2** | **3 Wire** | **3 Wire sub main** | | |
| | 21.1.2.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 270.00 |
| | 21.1.2.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 315.00 |
| | 21.1.2.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 383.00 |
| | 21.1.2.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 538.00 |
| | 21.1.2.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 812.00 |
| | 21.1.2.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1161.00 |
| | 21.1.2.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1683.00 |
| | **21.1.3** | **3 1/2 Wire** | **3 1/2 Wire sub main** | | |
| | 21.1.3.1 | 3 x 2.5 + 1 x 1.5 sq. mm. cable | 3 x 2.5 + 1 x 1.5 sq. mm. cable | Metre | 290.00 |
| | 21.1.3.2 | 3 x 4.0 + 1 x 2.5 sq. mm. cable | 3 x 4.0 + 1 x 2.5 sq. mm. cable | Metre | 346.00 |
| | 21.1.3.3 | 3 x 6.0 + 1 x 4.0 sq. mm. cable | 3 x 6.0 + 1 x 4.0 sq. mm. cable | Metre | 459.00 |
| | 21.1.3.4 | 3 x 10.0 + 1 x 6.0 sq. mm. cable | 3 x 10.0 + 1 x 6.0 sq. mm. cable | Metre | 661.00 |
| | 21.1.3.5 | 3 x 16.0 + 1 x 10.0 sq. mm. cable | 3 x 16.0 + 1 x 10.0 sq. mm. cable | Metre | 923.00 |
| | 21.1.3.6 | 3 x 25.0 + 1 x 16.0 sq. mm. cable | 3 x 25.0 + 1 x 16.0 sq. mm. cable | Metre | 1490.00 |
| | 21.1.3.7 | 3 x 35.0 + 1 x 16.0 sq. mm. cable | 3 x 35.0 + 1 x 16.0 sq. mm. cable | Metre | 1872.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | **21.1.4** | **4 Wire** | **4 Wire sub main** | | |
| | 21.1.4.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 185.00 |
| | 21.1.4.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 361.00 |
| | 21.1.4.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 482.00 |
| | 21.1.4.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 702.00 |
| | 21.1.4.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 991.00 |
| | 21.1.4.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1575.00 |
| | 21.1.4.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 2097.00 |

# CHAPTER - 22

## MAIN AND SUB-MAINS IN SURFACE RIGID PVC CONDUIT

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **22.1** | | **Sub-mains in surface rigid PVC conduit (MMS)** | Wiring for sub-mains with PVC insulated cable FRLS with copper multi strand conductor ISI marked in surface rigid PVC conduit (MMS) ISI marked of suitable size (conduit included) including 2mm thick accessories, connection etc, as required as per specification. (Refer General Notes for Classification of Conduit Wiring Point No.10) | | |
| | **22.1.1** | **2 Wire** | **2 Wire sub main** | | |
| | 22.1.1.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 136.00 |
| | 22.1.1.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 167.00 |
| | 22.1.1.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 191.00 |
| | 22.1.1.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 292.00 |
| | 22.1.1.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 443.00 |
| | 22.1.1.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 693.00 |
| | 22.1.1.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 943.00 |
| | **22.1.2** | **3 Wire** | **3 Wire sub main** | | |
| | 22.1.2.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 167.00 |
| | 22.1.2.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 208.00 |
| | 22.1.2.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 241.00 |
| | 22.1.2.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 364.00 |
| | 22.1.2.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 449.00 |
| | 22.1.2.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 986.00 |
| | 22.1.2.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1364.00 |
| | **22.1.3** | **3 1/2 Wire** | **3 1/2 Wire sub main** | | |
| | 22.1.3.1 | 3 x 2.5 + 1 x 1.5 sq. mm. cable | 3 x 2.5 + 1 x 1.5 sq. mm. cable | Metre | 182.00 |
| | 22.1.3.2 | 3 x 4.0 + 1 x 2.5 sq. mm. cable | 3 x 4.0 + 1 x 2.5 sq. mm. cable | Metre | 239.00 |
| | 22.1.3.3 | 3 x 6.0 + 1 x 4.0 sq. mm. cable | 3 x 6.0 + 1 x 4.0 sq. mm. cable | Metre | 331.00 |
| | 22.1.3.4 | 3 x 10.0 + 1 x 6.0 sq. mm. cable | 3 x 10.0 + 1 x 6.0 sq. mm. cable | Metre | 519.00 |
| | 22.1.3.5 | 3 x 16.0 + 1 x 10.0 sq. mm. cable | 3 x 16.0 + 1 x 10.0 sq. mm. cable | Metre | 765.00 |
| | 22.1.3.6 | 3 x 25.0 + 1 x 16.0 sq. mm. cable | 3 x 25.0 + 1 x 16.0 sq. mm. cable | Metre | 1197.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 22.1.3.7 | 3 x 35.0 + 1 x 16.0 sq. mm. cable | 3 x 35.0 + 1 x 16.0 sq. mm. cable | Metre | 1542.00 |
| | **22.1.4** | **4 Wire** | **4 Wire sub main** | | |
| | 22.1.4.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 199.00 |
| | 22.1.4.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 257.00 |
| | 22.1.4.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 360.00 |
| | 22.1.4.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 530.00 |
| | 22.1.4.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 844.00 |
| | 22.1.4.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1318.00 |
| | 22.1.4.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1779.00 |

# CHAPTER – 23

## SUBMAIN IN CONCEALED PVC CONDUIT IN COPPER CONDUCTOR

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **23.1** | | **Sub mains in recessed/concealed rigid PVC conduit (HMS)** | Wiring for sub-mains with PVC insulated cable FRLS with copper multi strand conductor ISI marked in recessed/concealed rigid PVC conduit (HMS) ISI marked of suitable size (conduit included) including 2mm thick accessories, connection etc, as required as per specification. (Refer General Notes for Classification of Conduit Wiring Point No.10) | | |
| | **23.1.1** | **2 Wire** | **2 Wire sub main** | | |
| | 23.1.1.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 175.00 |
| | 23.1.1.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 219.00 |
| | 23.1.1.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 264.00 |
| | 23.1.1.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 348.00 |
| | 23.1.1.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 499.00 |
| | 23.1.1.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 754.00 |
| | 23.1.1.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1022.00 |
| | **23.1.2** | **3 Wire** | **3 Wire sub main** | | |
| | 23.1.2.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 205.00 |
| | 23.1.2.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 254.00 |
| | 23.1.2.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 321.00 |
| | 23.1.2.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 457.00 |
| | 23.1.2.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 698.00 |
| | 23.1.2.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1027.00 |
| | 23.1.2.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1402.00 |
| | **23.1.3** | **3 1/2 Wire** | **3 1/2 Wire sub main** | | |
| | 23.1.3.1 | 3 x 2.5 + 1 x 1.5 sq. mm. cable | 3 x 2.5 + 1 x 1.5 sq. mm. cable | Metre | 224.00 |
| | 23.1.3.2 | 3 x 4.0 + 1 x 2.5 sq. mm. cable | 3 x 4.0 + 1 x 2.5 sq. mm. cable | Metre | 280.00 |
| | 23.1.3.3 | 3 x 6.0 + 1 x 4.0 sq. mm. cable | 3 x 6.0 + 1 x 4.0 sq. mm. cable | Metre | 375.00 |
| | 23.1.3.4 | 3 x 10.0 + 1 x 6.0 sq. mm. cable | 3 x 10.0 + 1 x 6.0 sq. mm. cable | Metre | 559.00 |
| | 23.1.3.5 | 3 x 16.0 + 1 x 10.0 sq. mm. cable | 3 x 16.0 + 1 x 10.0 sq. mm. cable | Metre | 805.00 |
| | 23.1.3.6 | 3 x 25.0 + 1 x 16.0 sq. mm. cable | 3 x 25.0 + 1 x 16.0 sq. mm. cable | Metre | 1232.00 |
| | 23.1.3.7 | 3 x 35.0 + 1 x 16.0 sq. mm. cable | 3 x 35.0 + 1 x 16.0 sq. mm. cable | Metre | 1577.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | **23.1.4** | **4 Wire** | **4 Wire sub main** | | |
| | 23.1.4.1 | 2.5 sq. mm. cable | 2.5 sq. mm. cable | Metre | 235.00 |
| | 23.1.4.2 | 4.0 sq. mm. cable | 4.0 sq. mm. cable | Metre | 295.00 |
| | 23.1.4.3 | 6.0 sq. mm. cable | 6.0 sq. mm. cable | Metre | 398.00 |
| | 23.1.4.4 | 10.0 sq. mm. cable | 10.0 sq. mm. cable | Metre | 601.00 |
| | 23.1.4.5 | 16.0 sq. mm. cable | 16.0 sq. mm. cable | Metre | 873.00 |
| | 23.1.4.6 | 25.0 sq. mm. cable | 25.0 sq. mm. cable | Metre | 1341.00 |
| | 23.1.4.7 | 35.0 sq. mm. cable | 35.0 sq. mm. cable | Metre | 1802.00 |

# CHAPTER – 24

## REWIRING IN EXISTING RIGID STEEL/PVC NON METALLIC CONDUIT (Copper conductor)

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **24.1** | | **Rewiring** | Rewiring in existing surface/concealed rigid steel/PVC nonmetallic conduit with PVC insulated cable with copper multi strand conductor FRLS of ISI marked and other materials as required including replacement of wornout/damaged/missing material, painting etc. complete as per specification. This shall include all work, arrangement required for wiring but shall exclude switch, ceiling rose, fitting etc. | | |
| | **24.1.1** | **Point wiring** | **Light/fan/3 Pin 6 Amp socket outlet on separate board/call bell/buzzer points :-** | | |
| | 24.1.1.1 | Short | Short point | Each | 184.00 |
| | 24.1.1.2 | Medium | Medium point | Each | 380.00 |
| | 24.1.1.3 | Long | Long point | Each | 585.00 |
| | **24.1.2** | **Twin control point** | **Twin control light point** | | |
| | 24.1.2.1 | Short | Short point | Each | 222.00 |
| | 24.1.2.2 | Medium | Medium point | Each | 456.00 |
| | 24.1.2.3 | Long | Long point | Each | 698.00 |
| | **24.1.3** | **Circuit wiring** | **Circuit wiring** | | |
| | 24.1.3.1 | 1.5 sq. mm. | 2 X 1.5 sq. mm. | Metre | 48.00 |
| | 24.1.3.2 | 2.5 sq. mm. | 2 X 2.5 sq. mm. | Metre | 71.00 |
| **24.2** | | **6 Amp same board** | 3 Pin 6 Amp socket outlet point on same board | Each | 38.00 |
| **24.3** | | **2 Wire sub main** | **2 Wire sub main** | | |
| | 24.3.1 | 2.5 sq. mm. | 2 X 2.5 sq. mm. | Metre | 76.00 |
| | 24.3.2 | 4.0 sq. mm. | 2 X 4.00 sq. mm. | Metre | 106.00 |
| | 24.3.3 | 6.0 sq. mm. | 2 X 6.00 sq. mm. | Metre | 151.00 |
| | 24.3.4 | 10.0 sq. mm. | 2 X 10.0 sq. mm. | Metre | 236.00 |
| | 24.3.5 | 16.0 sq. mm. | 2 X 16.00 sq. mm. | Metre | 372.00 |
| **24.4** | | **3 Wire sub main** | **3 Wire sub main** | | |
| | 24.4.1 | 2.5 sq. mm. | 3 X 2.5 sq. mm. | Metre | 106.00 |
| | 24.4.2 | 4.0 sq. mm. | 3 X 4.00 sq. mm. | Metre | 151.00 |
| | 24.4.3 | 6.0 sq. mm. | 3 X 6.00 sq. mm. | Metre | 221.00 |
| | 24.4.4 | 10.0 sq. mm. | 3 X 10.0 sq. mm. | Metre | 347.00 |
| | 24.4.5 | 16.0 sq. mm. | 3 X 16.00 sq. mm. | Metre | 551.00 |
| | 24.4.6 | 25.0 sq. mm. | 3 X 25.00 sq. mm. | Metre | 881.00 |
| | 24.4.7 | 35.0 sq. mm. | 3 X 35.00 sq. mm. | Metre | 1228.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **24.5** | | **3 1/2 Wire sub main** | **3 1/2 Wire sub main** | | |
| | 24.5.1 | 2.5 sq. mm. | 3 x 2.5 + 1 x 1.5 sq. mm. | Metre | 126.00 |
| | 24.5.2 | 4.0 sq. mm. | 3 x 4.0 + 1 x 2.5 sq. mm. | Metre | 182.00 |
| | 24.5.3 | 6.0 sq. mm. | 3 x 6.0 + 1 x 4.0 sq. mm. | Metre | 267.00 |
| | 24.5.4 | 10.0 sq. mm. | 3 x 10.0 + 1 x 6.0 sq. mm. | Metre | 416.00 |
| | 24.5.5 | 16.0 sq. mm. | 3 x 16.0 + 1 x 10.0 sq. mm. | Metre | 662.00 |
| | 24.5.6 | 25.0 sq. mm. | 3 x 25.0 + 1 x 16.0 sq. mm. | Metre | 1060.00 |
| | 24.5.7 | 35.0 sq. mm. | 3 x 35.0 + 1 x 16.0 sq. mm. | Metre | 1406.00 |
| **24.6** | | **4 Wire sub main** | **4 Wire sub main** | | |
| | 24.6.1 | 2.5 sq. mm. | 4 X 2.5 sq. mm. | Metre | 137.00 |
| | 24.6.2 | 4.0 sq. mm. | 4 X 4.0 sq. mm. | Metre | 198.00 |
| | 24.6.3 | 6.0 sq. mm. | 4 X 6.0 sq. mm. | Metre | 290.00 |
| | 24.6.4 | 10.0 sq. mm. | 4 X 10.0 sq. mm. | Metre | 457.00 |
| | 24.6.5 | 16.0 sq. mm. | 4 X 16.0 sq. mm. | Metre | 730.00 |
| | 24.6.6 | 25.0 sq. mm. | 4 X 25.0 sq. mm. | Metre | 1169.00 |
| | 24.6.7 | 35.0 sq. mm. | 4 X 35.0 sq. mm. | Metre | 1631.00 |

# CHAPTER - 25

## IRON CLAD/METAL CLAD SWITCHES AND SWITCH FUSES UNIT CONFIRMING TO I.S.S. 4064 & 4047 -1967

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **25.1** | | **Iron clad/metal clad double pole switch** | Supply of I.S.I. Marked iron clad / metal clad double pole switch fuse (conduit entry ) /cable gland entry with two earthing terminals -- | | |
| | 25.1.1 | 16A | 16 Amps. 250 Volts | Each | 542.00 |
| | 25.1.2 | 32A | 32 Amps. 250 Volts | Each | 955.00 |
| **25.2** | | **Triple pole and neutral switch fuse with two earthing terminals** | Supply of I.S.I. Marked iron clad / metal clad triple pole and neutral switch fuse (conduit entry ) /cable gland entry with two earthing terminals -- | | |
| | 25.2.1 | 16A | 16 Amps. 415/500 Volts. | Each | 1190.00 |
| | 25.2.2 | 32A | 32 Amps. 415/500 Volts. | Each | 1732.00 |
| | 25.2.3 | 63A | 63 Amps. 415/500 Volts. | Each | 3769.00 |
| | 25.2.4 | 100A | 100 Amps. 415/500 Volts. | Each | 6981.00 |
| | 25.2.5 | 200A | 200 Amps. 415/500 Volts. | Each | 10902.00 |
| | 25.2.6 | 320A | 320 Amps 415/500 Volts | | 13226.00 |
| **25.3** | | **Triple pole and neutral switch fuse with two earthing terminals & without HRC fuses** | Supplying of ISI Marked iron clad / metal clad triple pole and neutral switch fuse unit with two earthing terminals & without HRC fuses -- | | |
| | 25.3.1 | 16A | 16 Amps. 415/500 Volts. | Each | 1539.00 |
| | 25.3.2 | 32A | 32 Amps. 415/500 Volts. | Each | 1940.00 |
| | 25.3.3 | 63A | 63 Amps. 415/500 Volts. | Each | 4234.00 |
| | 25.3.4 | 100A | 100 Amps. 415/500 Volts. | Each | 7929.00 |
| **25.4** | | **Change over switches (off load)** | Supplying of ISI Marked Change over switches in sheet enclosure confirming to IS : 4064 / IS : 13947 (part I & III ) with side handle operated 4 pole with two earthing terminals | | |
| | 25.4.1 | 16A (off load) | 16 Amps. 415/500 Volts. (off load ) | Each | 1577.00 |
| | 25.4.2 | 32A (off load) | 32 Amps. 415/500 Volts. (off load ) | Each | 2025.00 |
| | 25.4.3 | 63A (off load) | 63 Amps. 415/500 Volts. (off load ) | Each | 4308.00 |
| | 25.4.4 | 100A (off load) | 100 Amps. 415/500 Volts. (off load ) | Each | 8012.00 |
| | 25.4.5 | 200A (off load) | 200 Amps. 415/500 Volts. (off load ) | Each | 14156.00 |
| | 25.4.6 | 300A (off load) | 300 Amps. 415/500 Volts. (off load ) | Each | 19078.00 |
| | 25.4.7 | 400A (off load) | 400 Amps. 415/500 Volts. (off load ) | Each | 24389.00 |
| | 25.4.8 | 630A (off load) | 630 Amps. 415/500 Volts. (off load ) | Each | 41441.00 |
| | 25.4.9 | 800A (off load) | 800 Amps. 415/500 Volts. (off load ) | Each | 51672.00 |
| **25.5** | | **Change over switches (open - execution )** | Supply of ISI Marked **On-Load Change over switches** panel mouniting type confirming to IS : 13947 (part I & III ) with **front operated** 4 pole, 415 V with two earthing terminals if required. | | |
| | 25.5.1 | 63A (open execution) | 63 Amps. (open - execution ) | Each | 4122.00 |
| | 25.5.2 | 100A (open execution) | 100 Amps. (open - execution ) | Each | 5452.00 |
| | 25.5.3 | 125A (open execution) | 125 Amps. .(open - execution ) | Each | 8198.00 |
| | 25.5.4 | 160A (open execution) | 160 Amps. (open - execution ) | Each | 8972.00 |
| | 25.5.5 | 200A (open execution) | 200 Amps. (open - execution ) | Each | 10767.00 |
| | 25.5.6 | 250A (open execution) | 250 Amps. (open - execution ) | Each | 14831.00 |
| | 25.5.7 | 320A (open execution) | 320 Amps. .(open - execution ) | Each | 16796.00 |
| | 25.5.8 | 400A (open execution) | 400 Amps. .(open - execution ) | Each | 22650.00 |
| | 25.5.9 | 630A (open execution) | 630 Amps. .(open - execution ) | Each | 28902.00 |
| | 25.5.10 | 800A (open execution) | 800 Amps. .(open - execution ) | Each | 42714.00 |
| | 25.5.11 | 63A (With enclosure) | 63 Amps. (With enclosure ) | Each | 5801.00 |
| | 25.5.12 | 100A (With enclosure) | 100 Amps.(With enclosure ) | Each | 7424.00 |
| | 25.5.13 | 125A (With enclosure) | 125 Amps. .(With enclosure ) | Each | 11034.00 |
| | 25.5.14 | 160A (With enclosure) | 160 Amps. .(With enclosure ) | Each | 12095.00 |
| | 25.5.15 | 200A (With enclosure) | 200 Amps. .(With enclosure ) | Each | 13987.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 25.5.16 | 320A (With enclosure) | 320 Amps. .(With enclosure ) | Each | 20689.00 |
| | 25.5.17 | 400A (With enclosure) | 400 Amps. .(With enclosure ) | Each | 30396.00 |
| | 25.5.18 | 630A (With enclosure) | 630 Amps. .(With enclosure ) | Each | 35141.00 |
| | 25.5.19 | 800A (With enclosure) | 800 Amps. .(With enclosure ) | Each | 51513.00 |
| **25.6** | | **Automatic transfer switches (ATS)** | Supply of ISI Marked **AutomaticTransfer Switches (ATS)** confirming to IEC : 60947-1 & IEC : 60947-6-1 with automatic inbuilt time delay 4 pole, 415 V with two earthing terminals if required. | | |
| | 25.6.1 | 100A (open execution) | 100 Amp (open execution ) | Each | 31835.00 |
| | 25.6.2 | 160 Amp (open execution) | 160 Amp (open execution ) | Each | 41648.00 |
| | 25.6.3 | 200A (open execution) | 200 Amp (open execution ) | Each | 55818.00 |
| | 25.6.4 | 315 Amp (open execution) | 315 Amp (open execution ) | Each | 66067.00 |
| | 25.6.5 | 400A (open execution) | 400 Amp (open execution ) | Each | 76371.00 |
| | 25.6.6 | 630A (open execution) | 630 Amp (open execution ) | Each | 98999.00 |
| | 25.6.7 | 100A (with enclosure) | 100 Amp (with enclosure ) | Each | 35755.00 |
| | 25.6.8 | 160A (with enclosure) | 160 Amp (with enclosure ) | Each | 45568.00 |
| | 25.6.9 | 200A (with enclosure) | 200 Amp (with enclosure ) | Each | 59738.00 |
| | 25.6.10 | 315A (with enclosure) | 315 Amp (with enclosure ) | Each | 69992.00 |
| | 25.6.11 | 400A (with enclosure) | 400 Amp (with enclosure ) | Each | 83820.00 |
| | 25.6.12 | 630A (with enclosure) | 630 Amp (with enclosure ) | Each | 106447.00 |
| **25.7** | | **Two way centre off MCB changeover switch** | Supply of **Two Way Centre off MCB Changeover Switch** confirming to IS:13947 (part III ) / IEC : 60947-3, 2/4 pole ,AC 240 / 415 Volt, 50 Hz | | |
| | 25.7.1 | 25A, (DP) | 25 Amps, Double Pole (DP) | Each | 1000.00 |
| | 25.7.2 | 40A, (DP) | 40 Amps, Double Pole (DP) | Each | 1501.00 |
| | 25.7.3 | 25A, (FP) | 25 Amps, Four Pole (FP) | Each | 1910.00 |
| | 25.7.4 | 40A, (FP) | 40 Amps, Four Pole (FP) | Each | 2992.00 |
| **25.8** | | **Switch disconnector fuse (SDF), open execution, with 2 HBC fuses** | Supply of ISI Marked **Switch Disconnector Fuse (SDF), open execution** (Panel mounting cubicle type) confirming to IS:13947 (part I & III ) 2 pole , AC -23A ,50 Hz , 415 Volt **with 2 HBC fuses** complete. | | |
| | 25.8.1 | 32A | 32 Amps | Each | 1411.00 |
| | 25.8.2 | 63A | 63 Amps | Each | 2081.00 |
| | 25.8.3 | 100A | 100 Amps | Each | 3829.00 |
| | 25.8.4 | 125A | 125 Amps | Each | 4547.00 |
| **25.9** | | **Switch disconnector fuse (SDF), in sheet steel enclosure, 2 HBC fuses** | Supply of ISI Marked **Switch Disconnector Fuse (SDF), in sheet steel enclosure** (Panel mounting cubicle type) confirming to IS:13947 (part I & III ) 2 pole , AC -23A ,50 Hz , 415 Volt **with 2 HBC fuses** complete. | | |
| | 25.9.1 | 32A | 32 Amps | Each | 2042.00 |
| | 25.9.2 | 63A | 63 Amps | Each | 2537.00 |
| | 25.9.3 | 100A | 100 Amps | Each | 4556.00 |
| | 25.9.4 | 125A | 125 Amps | Each | 4620.00 |
| **25.10** | | **Switch disconnector fuse (SDF), open execution, with 3 HBC fuses** | Supply of ISI Marked **Switch Disconnector Fuse (SDF), open execution** (Panel mounting cubicle type) confirming to IS:13947 (part I & III ) 3 pole and neutral , AC -23A ,50 Hz , 415 Volt **with 3 HBC fuses** complete. | | |
| | 25.10.1 | 32A | 32 Amps | Each | 1783.00 |
| | 25.10.2 | 63A | 63 Amps | Each | 2390.00 |
| | 25.10.3 | 100A | 100 Amps | Each | 4718.00 |
| | 25.10.4 | 125A | 125 Amps | Each | 5690.00 |
| | 25.10.5 | 160A | 160 Amps | Each | 6544.00 |
| | 25.10.6 | 200A | 200 Amps | Each | 8074.00 |
| | 25.10.7 | 250A | 250 Amps | Each | 9557.00 |
| | 25.10.8 | 320A | 320 Amps | Each | 12215.00 |
| | 25.10.9 | 400A | 400 Amps | Each | 14714.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 25.10.10 | 630A | 630 Amps | Each | 23410.00 |
| **25.11** | | **Switch disconnector fuse (SDF), in sheet steel enclosure, with 3 HBC fuses** | Supply of ISI Marked **Switch Disconnector Fuse (SDF), in sheet steel enclosure (Panel mounting cubicle type)** confirming to IS:13947 (part I & III ) 3 pole and neutral , AC -23A ,50 Hz , 415 Volt **with 3 HBC fuses** complete. | | |
| | 25.11.1 | 32A | 32 Amps | Each | 3274.00 |
| | 25.11.2 | 63A | 63 Amps | Each | 3920.00 |
| | 25.11.3 | 100A | 100 Amps | Each | 6549.00 |
| | 25.11.4 | 125A | 125 Amps | Each | 7869.00 |
| | 25.11.5 | 160A | 160 Amps | Each | 8705.00 |
| | 25.11.6 | 200A | 200 Amps | Each | 10817.00 |
| | 25.11.7 | 250A | 250 Amps | Each | 12491.00 |
| | 25.11.8 | 320A | 320 Amps | Each | 15160.00 |
| | 25.11.9 | 400A | 400 Amps | Each | 17518.00 |
| | 25.11.10 | 630A | 630 Amps | Each | 27519.00 |
| **25.12** | | **Direct on-line starters** | Supply of ISI Marked **Direct On-Line Starters** confirming to IS : 13947-4 with suitable over load relay range and contactor size, 415 V , 3 -Phase with two earthing terminals | | |
| | 25.12.1 | 0.75/1.0/2.0/3.0/5.0/7.5 HP | 0.75 / 1.0 / 2.0 / 3.0 / 5.0 / 7.5 HP | Each | 1740.00 |
| | 25.12.2 | 10 HP | 10 HP | Each | 2607.00 |
| **25.13** | | **Automatic star-delta starters** | Supply of ISI Marked **Automatic Star-Delta Starters** confirming to IS : 13947-4 with suitable over load relay range and contactor size & timer 415 V , 3 -Phase with two earthing terminals | | |
| | 25.13.1 | 3 to 12.5 HP | 3 to 12.5 HP | Each | 8150.00 |
| | 25.13.2 | 15.0 HP | 15.0 HP | Each | 8009.00 |
| | 25.13.3 | 20 HP | 20 HP | Each | 9079.00 |
| | 25.13.4 | 25 HP | 25 HP | Each | 10186.00 |
| | 25.13.5 | 30 HP | 30 HP | Each | 11324.00 |
| | 25.13.6 | 35 HP | 35 HP | Each | 15702.00 |
| | 25.13.7 | 50 HP | 50 HP | Each | 24767.00 |
| | 25.13.8 | 75 HP | 75 HP | Each | 42875.00 |
| **25.14** | | **High breaking capacity (HBC) fuse links bolted connection type** | Supply & fixing of **High Breaking Capacity (HBC) Fuse** Links bolted connection type as per IS : 13703 : 1993 & IEC : 60269 :1986 with 80 kA breaking capacity of approved make | | |
| | 25.14.1 | 20A to 32A | 20 Amp. to 32 Amp | Each | 62.00 |
| | 25.14.2 | 36A to 63A | 36 Amp to 63 Amp | Each | 75.00 |
| | 25.14.3 | 80A to 125A | 80 Amp to 125 Amp | Each | 161.00 |
| | 25.14.4 | 160A to 250A | 160 Amp to 250 Amp | Each | 244.00 |
| | 25.14.5 | 300A to 315A | 300 Amp to 315 Amp | Each | 357.00 |
| | 25.14.6 | 400A to 500A | 400 Amp to 500 Amp | Each | 717.00 |
| | 25.14.7 | 630A | 630 Amp | Each | 1313.00 |
| **25.15** | | **High breaking capacity (HBC) fuse links knife (DIN) type** | Supply & fixing of **High Breaking Capacity (HBC) Fuse** Links knife (DIN) type as per IS : 13703 : 1993 & IEC : 60269 :1986 with 80 kA breaking capacity of approved make | | |
| | 25.15.1 | 20A to 100A | 20 Amp. to 100 Amp | Each | 212.00 |
| | 25.15.2 | 125A to 160A | 125 Amp to 160 Amp | Each | 275.00 |
| | 25.15.3 | 200 Amp to 315 Amp | 200 Amp to 315 Amp | Each | 487.00 |
| | 25.15.4 | 350A to 400A | 350 Amp to 400 Amp | Each | 624.00 |
| | 25.15.5 | 425A to 630A | 425 Amp to 630 Amp | Each | 946.00 |
| **25.16** | | **Porcelain rewireable fuse unit** | Supply & fixing of I.S.I.marked Porecelain Rewireable fuse unit of approved make or confirming to I.S.S. 2086-1982 | | |
| | 25.16.1 | 16A/240 Volts | 16 Amps./240 Volts | Each | 44.00 |
| | 25.16.2 | 16A/415 Volts or 32 Amp/240 Volts | 16 Amps./415 Volts or 32 Amps/240 Volts | Each | 87.00 |
| | 25.16.3 | 32A/415 Volts | 32 Amps /415 Volts | Each | 145.00 |
| | 25.16.4 | 63A/415 Volts | 63 Amps /415 Volts | Each | 309.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 25.16.5 | 100A/415 Volts | 100 Amps /415 Volts | Each | 548.00 |
| | 25.16.6 | 200A/415 Volts | 200 Amps /415 Volts | Each | 983.00 |
| | 25.16.7 | 300A/415 Volts | 300 Amps /415 Volts | Each | 1178.00 |
| **25.17** | | **DMC/Nylon fuse holder fitting** | Supply & fixing of ISI marked DMC /Nylon Fuse holder fitting / Fuse Base for knife type HBC fuse Links of approved make & confirming to IS:13703-2-1 | | |
| | 25.17.1 | 20A/415 Volts | DMC /Nylon Fuse holder fitting 20 Amps /415 Volts | Each | 111.00 |
| | 25.17.2 | 32A/415 Volts | DMC /Nylon Fuse holder fitting 32 Amps /415 Volts | Each | 179.00 |
| | 25.17.3 | 63A/415 Volts | DMC /Nylon Fuse holder fitting 63 Amps /415 Volts | Each | 221.00 |
| | 25.17.4 | 100A/415 Volts | DMC /Nylon Fuse holder fitting 100 Amps /415 Volts | Each | 553.00 |
| | 25.17.5 | 160A/415 Volts | DMC /Nylon Fuse Base 160 Amps /415 Volts | Each | 263.00 |
| | 25.17.6 | 250A/415 Volts | DMC /Nylon Fuse Base 250 Amps /415 Volts | Each | 492.00 |
| | 25.17.7 | 400A/415 Volts | DMC /Nylon Fuse Base 400 Amps /415 Volts | Each | 718.00 |
| | 25.17.8 | 630A/415 Volts | DMC /Nylon Fuse Base 630 Amps /415 Volts | Each | 950.00 |

# CHAPTER – 26

## BUSBAR CHAMBERS AND SANDWICH BUS TRUNKING

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **26.1** | | **Busbar chamber, 415 V 4 way** | Supplying of metal clad ISI marked busbar chamber made of heavy gauge iron sheet complete with copper strip and busbar supports as per specification conforming to IS -4064 | | |
| | 26.1.1 | 32 Amp | 32 Amp 415 VOLTS, 4 way | Each | 5570.00 |
| | 26.1.2 | 60/63 Amp | 60/63 Amp 415 VOLTS, 4 way | Each | 6213.00 |
| | 26.1.3 | 100 Amp | 100 Amp 415 VOLTS, 4 way | Each | 6883.00 |
| | 26.1.4 | 200 Amp | 200 Amp 415 VOLTS, 4 way | Each | 8513.00 |
| | 26.1.5 | 300 Amp | 300 Amp 415 VOLTS, 4 way | Each | 14722.00 |
| | 26.1.6 | 400 Amp | 400 Amp 415 VOLTS, 4 way | Each | 16357.00 |
| | 26.1.7 | 500 Amp | 500/630 Amp 415 VOLTS, 4 way | Each | 26169.00 |
| **26.2** | | **Outdoor aluminium sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of outdoor aluminium sandwich bus trunking 415V 4P class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 65 (with canopy) protection having external 2 runs aluminium earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.2.1 | 400 Amp, 25KA | 400 Amp 415V, 4P, 25KA | Metre | 6442.00 |
| | 26.2.2 | 500 Amp, 30KA | 500 Amp 415V 4P 4W, 30KA | Metre | 7079.00 |
| | 26.2.3 | 630 Amp, 40KA | 630 Amp 415V 4P, 40KA | Metre | 7710.00 |
| | 26.2.4 | 800 Amp, 50KA | 800 Amp 415V 4P, 50KA | Metre | 8986.00 |
| | 26.2.5 | 1000 Amp, 50KA | 1000 Amp 415V 4P, 50KA | Metre | 10277.00 |
| | 26.2.6 | 1250 Amp, 65KA | 1250 Amp 415V 4P, 65KA | Metre | 11895.00 |
| | 26.2.7 | 1600 Amp, 80KA | 1600 Amp 415V 4P, 80KA | Metre | 15302.00 |
| | 26.2.8 | 2000 Amp, 80KA | 2000 Amp 415V 4P, 80KA | Metre | 18914.00 |
| | 26.2.9 | 2250 Amp, 120KA | 2250 Amp 415V 4P, 120KA | Metre | 21441.00 |
| | 26.2.10 | 2500 Amp, 120KA | 2500 Amp 415V 4P, 120KA | Metre | 25053.00 |
| | 26.2.11 | 3200 Amp, 120KA | 3200 Amp 415V 4P, 120KA | Metre | 28286.00 |
| | 26.2.12 | 3600 Amp, 120KA | 3600 Amp 415V 4P, 120KA | Metre | 31435.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.2.13 | 4000 Amp, 120KA | 4000 Amp 415V 4P, 120KA | Metre | 36464.00 |
| **26.3** | | **Accessories for outdoor aluminium sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of accessories for outdoor aluminium sandwich bus trunking 415V 4P class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 65 (with canopy) protection having external 2 runs aluminium earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.3.1 | 90 degree bend 400 Amp | 90 degree bend 400 Amp 415V 4P, 25KA | Each | 7516.00 |
| | 26.3.2 | 90 degree bend 500 Amp | 90 degree bend 500 Amp 415V 4P 4W, 30KA | Each | 8307.00 |
| | 26.3.3 | 90 degree bend 630 Amp | 90 degree bend 630 Amp 415V 4P, 40KA | Each | 8995.00 |
| | 26.3.4 | 90 degree bend 800 Amp | 90 degree bend 800 Amp 415V 4P, 50KA | Each | 10484.00 |
| | 26.3.5 | 90 degree bend 1000 Amp | 90 degree bend 1000 Amp 415V 4P, 50KA | Each | 11989.00 |
| | 26.3.6 | 90 degree bend 1250 Amp | 90 degree bend 1250 Amp 415V 4P, 65KA | Each | 13878.00 |
| | 26.3.7 | 90 degree bend 1600 Amp | 90 degree bend 1600 Amp 415V 4P, 80KA | Each | 17853.00 |
| | 26.3.8 | 90 degree bend 2000 Amp | 90 degree bend 2000 Amp 415V 4P, 80KA | Each | 22068.00 |
| | 26.3.9 | 90 degree bend 2250 Amp | 90 degree bend 2250 Amp 415V 4P, 120KA | Each | 25014.00 |
| | 26.3.10 | 90 degree bend 2500 Amp | 90 degree bend 2500 Amp 415V 4P, 120KA | Each | 29229.00 |
| | 26.3.11 | 90 degree bend 3200 Amp | 90 degree bend 3200 Amp 415V 4P, 120KA | Each | 32993.00 |
| | 26.3.12 | 90 degree bend 3600 Amp | 90 degree bend 3600 Amp 415V 4P, 120KA | Each | 36674.00 |
| | 26.3.13 | 90 degree bend 4000 Amp | 90 degree bend 4000 Amp 415V 4P, 120KA | Each | 42541.00 |
| | 26.3.14 | Adaptor box 400 Amp | Adaptor box 400 Amp 415V, 4P, 25KA | Each | 3948.00 |
| | 26.3.15 | Adaptor box 500 Amp | Adaptor box 500 Amp 415V 4P 4W, 30KA | Each | 4422.00 |
| | 26.3.16 | Adaptor box 630 Amp | Adaptor box 630 Amp 415V 4P, 40KA | Each | 5117.00 |
| | 26.3.17 | Adaptor box 800 Amp | Adaptor box 800 Amp 415V 4P, 50KA | Each | 5283.00 |
| | 26.3.18 | Adaptor box 1000 Amp | Adaptor box1000 Amp 415V 4P, 50KA | Each | 5283.00 |
| | 26.3.19 | Adaptor box1250 Amp | Adaptor box 1250 Amp 415V 4P, 65KA | Each | 5488.00 |
| | 26.3.20 | Adaptor box1600 Amp | Adaptor box1600 Amp 415V 4P, 80KA | Each | 5765.00 |
| | 26.3.21 | Adaptor box 2000 Amp | Adaptor box 2000 Amp 415V 4P, 80KA | Each | 5922.00 |
| | 26.3.22 | Adaptor box 2250 Amp | Adaptor box 2250 Amp 415V 4P,120KA | Each | 9081.00 |
| | 26.3.23 | Adaptor box 2500 Amp | Adaptor box 2500 Amp 415V 4P,120KA | Each | 9081.00 |
| | 26.3.24 | Adaptor box 3200 Amp | Adaptor box 3200 Amp 415V 4P,120KA | Each | 9476.00 |
| | 26.3.25 | Adaptor box 3600 Amp | Adaptor box 3600 Amp 415V 4P,120KA | Each | 11845.00 |
| | 26.3.26 | Adaptor box 4000 Amp | Adaptor box 4000 Amp 415V 4P,120KA | Each | 11845.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.3.27 | Flange box 400 Amp | Flange box 400 Amp | Each | 3767.00 |
| | 26.3.28 | Flange box 500 Amp | Flange box 500 Amp | Each | 4161.00 |
| | 26.3.29 | Flange end 630 Amp | Flange end 630 Amp | Each | 5038.00 |
| | 26.3.30 | Flange end 800 Amp | Flange end 800 Amp | Each | 5370.00 |
| | 26.3.31 | Flange end 1000 Amp | Flange end 1000 Amp | Each | 5370.00 |
| | 26.3.32 | Flange end 1250 Amp | Flange end 1250 Amp | Each | 5528.00 |
| | 26.3.33 | Flange end 1600 Amp | Flange end 1600 Amp | Each | 5922.00 |
| | 26.3.34 | Flange end 2000 Amp | Flange end 2000 Amp | Each | 5922.00 |
| | 26.3.35 | Flange end 2250 Amp | Flange end 2250 Amp | Each | 8040.00 |
| | 26.3.36 | Flange end 2500 Amp | Flange end 2500 Amp | Each | 8830.00 |
| | 26.3.37 | Flange end 3200 Amp | Flange end 3200 Amp | Each | 10591.00 |
| | 26.3.38 | Flange end 3600 Amp | Flange end 3600 Amp | Each | 11788.00 |
| | 26.3.39 | Flange end 4000 Amp | Flange end 4000 Amp | Each | 13674.00 |
| | 26.3.40 | Expansion joint 400 Amp | Expansion joint 400 Amp– provision (If required) | Each | 17451.00 |
| | 26.3.41 | Expansion joint 500 Amp | Expansion joint 500 Amp– provision (If required) | Each | 17717.00 |
| | 26.3.42 | Expansion joint 630 Amp | Expansion joint 630 Amp– provision (If required) | Each | 17965.00 |
| | 26.3.43 | Expansion joint 800 Amp | Expansion joint 800 Amp– provision (If required) | Each | 18487.00 |
| | 26.3.44 | Expansion joint 1000 Amp | Expansion joint 1000 Amp– provision (If required) | Each | 19049.00 |
| | 26.3.45 | Expansion joint 1250 Amp | Expansion joint 1250 Amp– provision (If required) | Each | 19723.00 |
| | 26.3.46 | Expansion joint 1600 Amp | Expansion joint 1600 Amp– provision (If required) | Each | 24064.00 |
| | 26.3.47 | Expansion joint 2000 Amp | Expansion joint 2000 Amp– provision (If required) | Each | 25572.00 |
| | 26.3.48 | Expansion joint 2250 Amp | Expansion joint 2250 Amp– provision (If required) | Each | 46089.00 |
| | 26.3.49 | Expansion joint 2500 Amp | Expansion joint 2500 Amp– provision (If required) | Each | 48796.00 |
| | 26.3.50 | Expansion joint 3200 Amp | Expansion joint 3200 Amp– provision (If required) | Each | 51190.00 |
| | 26.3.51 | Expansion joint 3600 Amp | Expansion joint 3600 Amp– provision (If required) | Each | 53583.00 |
| | 26.3.52 | Expansion joint 4000 Amp | Expansion joint 4000 Amp– provision (If required) | Each | 57355.00 |
| | 26.3.53 | End feed unit, 400 Amp | End feed unit with provision for switchgear but without switchgear for 400 Amp | Each | 54619.00 |
| | 26.3.54 | End feed unit, 500 Amp | End feed unit with provision for switchgear but without switchgear for 500 Amp | Each | 55175.00 |
| | 26.3.55 | End feed unit, 630 Amp | End feed unit with provision for switchgear but without switchgear for 630 Amp | Each | 55701.00 |
| | 26.3.56 | End feed unit, 800 Amp | End feed unit with provision for switchgear but without switchgear for 800 Amp | Each | 56211.00 |
| | 26.3.57 | End feed unit, 1000 Amp | End feed unit with provision for switchgear but without switchgear for 1000 Amp | Each | 57839.00 |
| | 26.3.58 | End feed unit, 1250 Amp | End feed unit with provision for switchgear but without switchgear for 1250 Amp | Each | 75613.00 |
| | 26.3.59 | End feed unit, 1600 Amp | End feed unit with provision for switchgear but without switchgear for 1600 Amp | Each | 78451.00 |
| | 26.3.60 | End feed unit, 2000 Amp | End feed unit with provision for switchgear but without switchgear for 2000 Amp | Each | 81462.00 |
| | 26.3.61 | End feed unit, 2250 Amp | End feed unit with provision for switchgear but without switchgear for 2250 Amp | Each | 99993.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.3.62 | End feed unit, 2500 Amp | End feed unit with provision for switchgear but without switchgear for 2500 Amp | Each | 103002.00 |
| | 26.3.63 | End feed unit, 3200 Amp | End feed unit with provision for switchgear but without switchgear for 3200 Amp | Each | 105661.00 |
| | 26.3.64 | End feed unit, 3600 Amp | End feed unit with provision for switchgear but without switchgear for 3600 Amp | Each | 108321.00 |
| | 26.3.65 | End feed unit, 4000 Amp | End feed unit with provision for switchgear but without switchgear for 4000 Amp | Each | 112511.00 |
| | 26.3.66 | End cover for 400 Amp | End cover for 400 Amp | Each | 1489.00 |
| | 26.3.67 | End cover for 500 Amp | End cover for 500 Amp | Each | 1489.00 |
| | 26.3.68 | End cover for 630 Amp | End cover for 630 Amp | Each | 1489.00 |
| | 26.3.69 | End cover for 800 Amp | End cover for 800 Amp | Each | 1848.00 |
| | 26.3.70 | End cover for 1000 Amp | End cover for 1000 Amp | Each | 2156.00 |
| | 26.3.71 | End cover for 1250 Amp | End cover for 1250 Amp | Each | 2361.00 |
| | 26.3.72 | End cover for 1600 Amp | End cover for 1600 Amp | Each | 2464.00 |
| | 26.3.73 | End cover for 2000 Amp | End cover for 2000 Amp | Each | 2566.00 |
| | 26.3.74 | End cover for 2250 Amp | End cover for 2250 Amp | Each | 3182.00 |
| | 26.3.75 | End cover for 2500 Amp | End cover for 2500 Amp | Each | 3388.00 |
| | 26.3.76 | End cover for 3200 Amp | End cover for 3200 Amp | Each | 3685.00 |
| | 26.3.77 | End cover for 3600 Amp | End cover for 3600 Amp | Each | 3798.00 |
| | 26.3.78 | End cover for 4000 Amp | End cover for 4000 Amp | Each | 4106.00 |
| **26.4** | | **Outdoor copper sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of outdoor Copper Sandwich bus trunking 415V 4P Class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 65 (with canopy) protection having external 2 runs copper earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.4.1 | 630 Amp, 40KA | 630 Amp 415V, 4P, 40KA | Metre | 17767.00 |
| | 26.4.2 | 800 Amp, 50KA | 800 Amp 415V, 4P, 50KA | Metre | 21266.00 |
| | 26.4.3 | 1000 Amp, 65KA | 1000 Amp 415V, 4P, 65KA | Metre | 28276.00 |
| | 26.4.4 | 1250 Amp, 100KA | 1250 Amp 415V, 4P, 100KA | Metre | 38778.00 |
| | 26.4.5 | 1600 Amp, 100KA | 1600 Amp 415V, 4P, 100KA | Metre | 47548.00 |
| | 26.4.6 | 2000 Amp, 100KA | 2000 Amp 415V, 4P, 100KA | Metre | 65269.00 |
| | 26.4.7 | 2250 Amp, 100KA | 2250 Amp 415V, 4P, 100KA | Metre | 74021.00 |
| | 26.4.8 | 2500 Amp, 100KA | 2500 Amp 415V, 4P, 100KA | Metre | 84740.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.4.9 | 3200 Amp, 120KA | 3200 Amp 415V, 4P, 120KA | Metre | 93184.00 |
| | 26.4.10 | 3600 Amp, 120KA | 3600 Amp 415V, 4P, 120KA | Metre | 111004.00 |
| | 26.4.11 | 4000 Amp, 120KA | 4000 Amp 415V, 4P, 120KA | Metre | 128616.00 |
| **26.5** | | **Accessories for outdoor copper sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of accessories for outdoor copper Sandwich bus trunking 415V 4P class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 65 (with canopy) protection having external 2 runs copper earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.5.1 | 90 degree bend 630 Amp | 90 degree bend 630 Amp 415V, 4P, 40KA | Each | 21326.00 |
| | 26.5.2 | 90 degree bend 800 Amp | 90 degree bend 800 Amp 415V, 4P, 50KA | Each | 25525.00 |
| | 26.5.3 | 90 degree bend 1000 Amp | 90 degree bend 1000 Amp 415V, 4P, 65KA | Each | 33932.00 |
| | 26.5.4 | 90 degree bend 1250 Amp | 90 degree bend 1250 Amp 415V, 4P, 100KA | Each | 46538.00 |
| | 26.5.5 | 90 degree bend 1600 Amp | 90 degree bend 1600 Amp 415V, 4P, 100KA | Each | 57058.00 |
| | 26.5.6 | 90 degree bend 2000 Amp | 90 degree bend 2000 Amp 415V, 4P, 100KA | Each | 78323.00 |
| | 26.5.7 | 90 degree bend 2250 Amp | 90 degree bend 2250 Amp 415V, 4P, 100KA | Each | 88826.00 |
| | 26.5.8 | 90 degree bend 2500 Amp | 90 degree bend 2500 Amp 415V, 4P, 100KA | Each | 101687.00 |
| | 26.5.9 | 90 degree bend 3200 Amp | 90 degree bend 3200 Amp 415V, 4P, 120KA | Each | 111821.00 |
| | 26.5.10 | 90 degree bend 3600 Amp | 90 degree bend 3600 Amp 415V, 4P, 120KA | Each | 133204.00 |
| | 26.5.11 | 90 degree bend 4000 Amp | 90 degree bend 4000 Amp 415V, 4P, 120KA | Each | 154339.00 |
| | 26.5.12 | Copper flexible 400 Amp | Copper flexible 400 Amp | Each | 18646.00 |
| | 26.5.13 | Copper flexible 500 Amp | Copper flexible 500 Amp | Each | 21285.00 |
| | 26.5.14 | Copper flexible 630 Amp | Copper flexible 630 Amp | Each | 21285.00 |
| | 26.5.15 | Copper flexible 800 Amp | Copper flexible 800 Amp | Each | 23384.00 |
| | 26.5.16 | Copper flexible 1000 Amp | Copper flexible 1000 Amp | Each | 23384.00 |
| | 26.5.17 | Copper flexible 1250 Amp | Copper flexible 1250 Amp | Each | 30605.00 |
| | 26.5.18 | Copper flexible 1600 Amp | Copper flexible 1600 Amp | Each | 36317.00 |
| | 26.5.19 | Copper flexible 2000 Amp | Copper flexible 2000 Amp | Each | 59630.00 |
| | 26.5.20 | Copper flexible 2250 Amp | Copper flexible 2250 Amp | Each | 59630.00 |
| | 26.5.21 | Copper flexible 2500 Amp | Copper flexible 2500 Amp | Each | 69475.00 |
| | 26.5.22 | Copper flexible 3200 Amp | Copper flexible 3200 Amp | Each | 85497.00 |
| | 26.5.23 | Copper flexible 3600 Amp | Copper flexible 3600 Amp | Each | 105002.00 |
| | 26.5.24 | Copper flexible 4000 Amp | Copper flexible 4000 Amp | Each | 113732.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.5.25 | Expansion joint 630 Amp– provision (if required) | Expansion joint 630 Amp– provision (if required) | Each | 23653.00 |
| | 26.5.26 | Expansion joint 800 Amp | Expansion joint 800 Amp– provision (if required) | Each | 25402.00 |
| | 26.5.27 | Expansion joint 1000 Amp | Expansion joint 1000 Amp– provision (if required) | Each | 28905.00 |
| | 26.5.28 | Expansion joint 1250 Amp | Expansion joint 1250 Amp– provision (if required) | Each | 34155.00 |
| | 26.5.29 | Expansion joint 1600 Amp | Expansion joint 1600 Amp– provision (if required) | Each | 38541.00 |
| | 26.5.30 | Expansion joint 2000 Amp | Expansion joint 2000 Amp– provision (if required) | Each | 50323.00 |
| | 26.5.31 | Expansion joint 2250 Amp | Expansion joint 2250 Amp– provision (if required) | Each | 54699.00 |
| | 26.5.32 | Expansion joint 2500 Amp | Expansion joint 2500 Amp– provision (if required) | Each | 60059.00 |
| | 26.5.33 | Expansion joint 3200 Amp | Expansion joint 3200 Amp– provision (if required) | Each | 113873.00 |
| | 26.5.34 | Expansion joint 3600 Amp | Expansion joint 3600 Amp– provision (if required) | Each | 129911.00 |
| | 26.5.35 | Expansion joint 4000 Amp | Expansion joint 4000 Amp– provision (if required) | Each | 145762.00 |
| | 26.5.36 | End feed unit, 630Amp | End feed unit with provision for switchgear but without switchgear for 630 Amp | Each | 70546.00 |
| | 26.5.37 | End feed unit, 600Amp | End feed unit with provision for switchgear but without switchgear for 800 Amp | Each | 77551.00 |
| | 26.5.38 | End feed unit, 1000 Amp | End feed unit with provision for switchgear but without switchgear for 1000 Amp | Each | 88053.00 |
| | 26.5.39 | End feed unit, 1250 Amp | End feed unit with provision for switchgear but without switchgear for 1250 Amp | Each | 113248.00 |
| | 26.5.40 | End feed unit, 1600 Amp | End feed unit with provision for switchgear but without switchgear for 1600 Amp | Each | 122216.00 |
| | 26.5.41 | End feed unit, 2000 Amp | End feed unit with provision for switchgear but without switchgear for 2000 Amp | Each | 130969.00 |
| | 26.5.42 | End feed unit, 2250 Amp | End feed unit with provision for switchgear but without switchgear for 2250 Amp | Each | 139721.00 |
| | 26.5.43 | End feed unit, 2500Amp | End feed unit with provision for switchgear but without switchgear for 2500 Amp | Each | 150440.00 |
| | 26.5.44 | End feed unit, 3200Amp | End feed unit with provision for switchgear but without switchgear for 3200 Amp | Each | 175309.00 |
| | 26.5.45 | End feed unit, 3600Amp | End feed unit with provision for switchgear but without switchgear for 3600 Amp | Each | 193129.00 |
| | 26.5.46 | End feed unit, 4000Amp | End feed unit with provision for switchgear but without switchgear for 4000 Amp | Each | 210741.00 |
| | 26.5.47 | End cover for 630 Amp | End cover for 630 Amp | Each | 1489.00 |
| | 26.5.48 | End cover for 800 Amp | End cover for 800 Amp | Each | 1848.00 |
| | 26.5.49 | End cover for 1000 Amp | End cover for 1000 Amp | Each | 2156.00 |
| | 26.5.50 | End cover for 1250 Amp | End cover for 1250 Amp | Each | 2361.00 |
| | 26.5.51 | End cover for 1600 Amp | End cover for 1600 Amp | Each | 2464.00 |
| | 26.5.52 | End cover for 2000 Amp | End cover for 2000 Amp | Each | 2566.00 |
| | 26.5.53 | End cover for 2250 Amp | End cover for 2250 Amp | Each | 3182.00 |
| | 26.5.54 | End cover for 2500 Amp | End cover for 2500 Amp | Each | 3388.00 |
| | 26.5.55 | End cover for 3200 Amp | End cover for 3200 Amp | Each | 3685.00 |
| | 26.5.56 | End cover for 3600 Amp | End cover for 3600 Amp | Each | 3798.00 |
| | 26.5.57 | End cover for 4000 Amp | End cover for 4000 Amp | Each | 4106.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **26.6** | | **Indoor aluminium Sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of indoor aluminium sandwich bus trunking 415V 4P Class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 54 protection having external 2 runs aluminium earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.6.1 | 400 Amp, 25KA | 400 Amp 415V, 4P, 25KA | Metre | 5798.00 |
| | 26.6.2 | 500 Amp, 30KA | 500 Amp 415V 4P, 30KA | Metre | 6372.00 |
| | 26.6.3 | 630 Amp, 40KA | 630 Amp 415V 4P, 40KA | Metre | 6940.00 |
| | 26.6.4 | 800 Amp, 50KA | 800 Amp 415V 4P, 50KA | Metre | 8088.00 |
| | 26.6.5 | 1000 Amp, 50KA | 1000 Amp 415V 4P, 50KA | Metre | 9249.00 |
| | 26.6.6 | 1250 Amp, 65KA | 1250 Amp 415V 4P, 65KA | Metre | 10706.00 |
| | 26.6.7 | 1600 Amp, 80KA | 1600 Amp 415V 4P, 80KA | Metre | 13772.00 |
| | 26.6.8 | 2000 Amp, 80KA | 2000 Amp 415V 4P, 80KA | Metre | 17023.00 |
| | 26.6.9 | 2250 Amp, 120KA | 2250 Amp 415V 4P, 120KA | Metre | 19297.00 |
| | 26.6.10 | 2500 Amp, 120KA | 2500 Amp 415V 4P, 120KA | Metre | 22547.00 |
| | 26.6.11 | 3200 Amp, 120KA | 3200 Amp 415V 4P, 120KA | Metre | 25457.00 |
| | 26.6.12 | 3600 Amp, 120KA | 3600 Amp 415V 4P, 120KA | Metre | 28291.00 |
| | 26.6.13 | 4000 Amp, 120KA | 4000 Amp 415V 4P, 120KA | Metre | 32817.00 |
| **26.7** | | **Accessories for indoor aluminium sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of accessories for Indoor Aluminium Sandwich bus trunking 415V 4P Class F insulation (155 Deg. C) with insulation material multi layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade / Aluminum casing of 2.5mm may also be provided. IP 54 protection having external 2 runs Aluminium earthing 50% cross section area confirm to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately The rating should be as follows :- | | |
| | 26.7.1 | 90 degree bend 400 Amp | 90 degree bend 400 Amp 415V, 4P, 25KA | Each | 6764.00 |
| | 26.7.2 | 90 degree bend 500 Amp | 90 degree bend 500 Amp 415V 4P 4W, 30KA | Each | 7477.00 |
| | 26.7.3 | 90 degree bend 630 Amp | 90 degree bend 630 Amp 415V 4P, 40KA | Each | 8096.00 |
| | 26.7.4 | 90 degree bend 800 Amp | 90 degree bend 800 Amp 415V 4P, 50KA | Each | 9436.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.7.5 | 90 degree bend 1000 Amp | 90 degree bend 1000 Amp 415V 4P, 50KA | Each | 10791.00 |
| | 26.7.6 | 90 degree bend 1250 Amp | 90 degree bend 1250 Amp 415V 4P, 65KA | Each | 12491.00 |
| | 26.7.7 | 90 degree bend 1600 Amp | 90 degree bend 1600 Amp 415V 4P, 80KA | Each | 16067.00 |
| | 26.7.8 | 90 degree bend 2000 Amp | 90 degree bend 2000 Amp 415V 4P, 80KA | Each | 19861.00 |
| | 26.7.9 | 90 degree bend 2250 Amp | 90 degree bend 2250 Amp 415V 4P, 120KA | Each | 22513.00 |
| | 26.7.10 | 90 degree bend 2500 Amp | 90 degree bend 2500 Amp 415V 4P, 120KA | Each | 26306.00 |
| | 26.7.11 | 90 degree bend 3200 Amp | 90 degree bend 3200 Amp 415V 4P, 120KA | Each | 29694.00 |
| | 26.7.12 | 90 degree bend 3600 Amp | 90 degree bend 3600 Amp 415V 4P, 120KA | Each | 33007.00 |
| | 26.7.13 | 90 degree bend 4000 Amp | 90 degree bend 4000 Amp 415V 4P, 120KA | Each | 38287.00 |
| | 26.7.14 | Adaptor box 400 Amp | Adaptor box 400 Amp 415V, 4P, 25KA | Each | 3553.00 |
| | 26.7.15 | Adaptor box 500 Amp | Adaptor box 500 Amp 415V 4P 4W, 30KA | Each | 3980.00 |
| | 26.7.16 | Adaptor box 630 Amp | Adaptor box 630 Amp 415V 4P, 40KA | Each | 4605.00 |
| | 26.7.17 | Adaptor box 800 Amp | Adaptor box 800 Amp 415V  4P, 50KA | Each | 4755.00 |
| | 26.7.18 | Adaptor box 1000 Amp | Adaptor box 1000 Amp 415V 4P, 50KA | Each | 4755.00 |
| | 26.7.19 | Adaptor box 1250 Amp | Adaptor box 1250 Amp 415V 4P, 65KA | Each | 4939.00 |
| | 26.7.20 | Adaptor box 1600 Amp | Adaptor box 1600 Amp 415V 4P, 80KA | Each | 5188.00 |
| | 26.7.21 | Adaptor box 2000 Amp | Adaptor box 2000 Amp 415V 4P, 80KA | Each | 5330.00 |
| | 26.7.22 | Adaptor box 2250 Amp | Adaptor box 2250 Amp 415V 4P, 120KA | Each | 8173.00 |
| | 26.7.23 | Adaptor box 2500 Amp | Adaptor box 2500 Amp 415V 4P, 120KA | Each | 8173.00 |
| | 26.7.24 | Adaptor box 3200 Amp | Adaptor box 3200 Amp 415V 4P, 120KA | Each | 8528.00 |
| | 26.7.25 | Adaptor box 3600 Amp | Adaptor box 3600 Amp 415V 4P, 120KA | Each | 10660.00 |
| | 26.7.26 | Adaptor box 4000 Amp | Adaptor box 4000 Amp 415V 4P, 120KA | Each | 10660.00 |
| | 26.7.27 | Flange end 400 Amp | Flange end 400 Amp | Each | 3390.00 |
| | 26.7.28 | Flange end 500 Amp | Flange end 500 Amp | Each | 3745.00 |
| | 26.7.29 | Flange end 630 Amp | Flange end 630 Amp | Each | 4534.00 |
| | 26.7.30 | Flange end 800 Amp | Flange end 800 Amp | Each | 4833.00 |
| | 26.7.31 | Flange end 1000 Amp | Flange end 1000 Amp | Each | 4833.00 |
| | 26.7.32 | Flange end 1250 Amp | Flange end 1250 Amp | Each | 4975.00 |
| | 26.7.33 | Flange end 1600 Amp | Flange end 1600 Amp | Each | 5330.00 |
| | 26.7.34 | Flange end 2000 Amp | Flange end 2000 Amp | Each | 5330.00 |
| | 26.7.35 | Flange end 2250 Amp | Flange end 2250 Amp | Each | 7236.00 |
| | 26.7.36 | Flange end 2500 Amp | Flange end 2500 Amp | Each | 7949.00 |
| | 26.7.37 | Flange end 3200 Amp | Flange end 3200 Amp | Each | 9531.00 |
| | 26.7.38 | Flange end 3600 Amp | Flange end 3600 Amp | Each | 10609.00 |
| | 26.7.39 | Flange end 4000 Amp | Flange end 4000 Amp | Each | 12306.00 |
| | 26.7.40 | Expansion joint 400 Amp | Expansion joint 400 Amp– provision (if required) | Each | 15706.00 |
| | 26.7.41 | Expansion joint 500 Amp | Expansion joint 500 Amp– provision (if required) | Each | 15945.00 |
| | 26.7.42 | Expansion joint 630 Amp | Expansion joint 630 Amp– provision (if required) | Each | 16168.00 |
| | 26.7.43 | Expansion joint 800 Amp | Expansion joint 800 Amp– provision (if required) | Each | 16638.00 |
| | 26.7.44 | Expansion joint 1000 Amp | Expansion joint 1000 Amp– provision (if required) | Each | 17144.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.7.45 | Expansion joint 1250 Amp | Expansion joint 1250 Amp– provision (if required) | Each | 17750.00 |
| | 26.7.46 | Expansion joint 1600 Amp | Expansion joint 1600 Amp– provision (if required) | Each | 21657.00 |
| | 26.7.47 | Expansion joint 2000 Amp | Expansion joint 2000 Amp– provision (if required) | Each | 23015.00 |
| | 26.7.48 | Expansion joint 2250 Amp | Expansion joint 2250 Amp– provision (if required) | Each | 41479.00 |
| | 26.7.49 | Expansion joint 2500 Amp | Expansion joint 2500 Amp– provision (if required) | Each | 43916.00 |
| | 26.7.50 | Expansion joint 3200 Amp | Expansion joint 3200 Amp– provision (if required) | Each | 46071.00 |
| | 26.7.51 | Expansion joint 3600 Amp | Expansion joint 3600 Amp– provision (if required) | Each | 48225.00 |
| | 26.7.52 | Expansion joint 4000 Amp | Expansion joint 4000 Amp– provision (if required) | Each | 51619.00 |
| | 26.7.53 | End feed unit, 400 Amp | End feed unit with provision for switchgear but without switchgear for 400 Amp | Each | 49157.00 |
| | 26.7.54 | End feed unit, 500 Amp | End feed unit with provision for switchgear but without switchgear for 500 Amp | Each | 49657.00 |
| | 26.7.55 | End feed unit, 630 Amp | End feed unit with provision for switchgear but without switchgear for 630 Amp | Each | 50130.00 |
| | 26.7.56 | End feed unit, 800 Amp | End feed unit with provision for switchgear but without switchgear for 800 Amp | Each | 50590.00 |
| | 26.7.57 | End feed unit, 1000 Amp | End feed unit with provision for switchgear but without switchgear for 1000 Amp | Each | 52055.00 |
| | 26.7.58 | End feed unit, 1250 Amp | End feed unit with provision for switchgear but without switchgear for 1250 Amp | Each | 68051.00 |
| | 26.7.59 | End feed unit, 1600 Amp | End feed unit with provision for switchgear but without switchgear for 1600 Amp | Each | 70606.00 |
| | 26.7.60 | End feed unit, 2000 Amp | End feed unit with provision for switchgear but without switchgear for 2000 Amp | Each | 73316.00 |
| | 26.7.61 | End feed unit, 2250 Amp | End feed unit with provision for switchgear but without switchgear for 2250 Amp | Each | 89993.00 |
| | 26.7.62 | End feed unit, 2500 Amp | End feed unit with provision for switchgear but without switchgear for 2500 Amp | Each | 92702.00 |
| | 26.7.63 | End feed unit, 3200 Amp | End feed unit with provision for switchgear but without switchgear for 3200 Amp | Each | 95094.00 |
| | 26.7.64 | End feed unit, 3600 Amp | End feed unit with provision for switchgear but without switchgear for 3600 Amp | Each | 97488.00 |
| | 26.7.65 | End feed unit, 4000 Amp | End feed unit with provision for switchgear but without switchgear for 4000 Amp | Each | 101260.00 |
| | 26.7.66 | End cover for 400 Amp | End cover for 400 Amp | Each | 1339.00 |
| | 26.7.67 | End cover for 500 Amp | End cover for 500 Amp | Each | 1339.00 |
| | 26.7.68 | End cover for 630 Amp | End cover for 630 Amp | Each | 1339.00 |
| | 26.7.69 | End cover for 800 Amp | End cover for 800 Amp | Each | 1663.00 |
| | 26.7.70 | End cover for 1000 Amp | End cover for 1000 Amp | Each | 1940.00 |
| | 26.7.71 | End cover for 1250 Amp | End cover for 1250 Amp | Each | 2125.00 |
| | 26.7.72 | End cover for 1600 Amp | End cover for 1600 Amp | Each | 2217.00 |
| | 26.7.73 | End cover for 2000 Amp | End cover for 2000 Amp | Each | 2310.00 |
| | 26.7.74 | End cover for 2250 Amp | End cover for 2250 Amp | Each | 2864.00 |
| | 26.7.75 | End cover for 2500 Amp | End cover for 2500 Amp | Each | 3049.00 |
| | 26.7.76 | End cover for 3200 Amp | End cover for 3200 Amp | Each | 3317.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.7.77 | End cover for 3600 Amp | End cover for 3600 Amp | Each | 3418.00 |
| | 26.7.78 | End cover for 4000 Amp | End cover for 4000 Amp | Each | 3696.00 |
| **26.8** | | **Housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure** | Supplying, erection, testing and commissioning of housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided for plug-in-box draw out type, contacts shall be silver plated and spring loaded with earth link, 415V 4P, IP 54 protection conform to IEC 60439-1&2 and IEC 61439 - 1&6 the rating should be as follows :- | | |
| | 26.8.1 | Plug-in-box, 63 Amp | Plug-in-box, 4P 63 Amp with (TM MCCB) IP54 | Each | 14688.00 |
| | 26.8.2 | Plug-in-box, 125/100 Amp | Plug-in-box, 4P 125/100 Amp with (TM MCCB) IP54 | Each | 21479.00 |
| | 26.8.3 | Plug-in-box, 160 Amp | Plug-in-box, 4P 160 Amp with (TM MCCB) IP54 | Each | 26691.00 |
| | 26.8.4 | Plug-in-box, 250 Amp | Plug-in-box,.4P 250 Amp with (TM MCCB) IP54 | Each | 32218.00 |
| | 26.8.5 | Plug-in-box, 400 Amp | Plug-in-box, 4P 400 Amp with (TM MCCB) IP54 | Each | 36798.00 |
| | 26.8.6 | Plug-in-box, 630 Amp | Plug-in-box, 4P 630 Amp with (TM MCCB) IP54 | Each | 43274.00 |
| **26.9** | | **Indoor copper sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of Indoor Copper Sandwich bus trunking 415V 4P Class F insulation (155 Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 54 protection having external 2 runs copper earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows | | |
| | 26.9.1 | 630 Amp, 40KA | 630 Amp 415V, 4P, 40KA | Metre | 17175.00 |
| | 26.9.2 | 800 Amp, 50KA | 800 Amp 415V, 4P, 50KA | Metre | 20557.00 |
| | 26.9.3 | 1000 Amp, 65KA | 1000 Amp 415V, 4P, 65KA | Metre | 27334.00 |
| | 26.9.4 | 1250 Amp, 100KA | 1250 Amp 415V, 4P, 100KA | Metre | 37500.00 |
| | 26.9.5 | 1600 Amp, 100KA | 1600 Amp 415V, 4P, 100KA | Metre | 45963.00 |
| | 26.9.6 | 2000 Amp, 100KA | 2000 Amp 415V, 4P, 100KA | Metre | 63093.00 |
| | 26.9.7 | 2250 Amp, 100KA | 2250 Amp 415V, 4P, 100KA | Metre | 71554.00 |
| | 26.9.8 | 2500 Amp, 100KA | 2500 Amp 415V, 4P, 100KA | Metre | 81915.00 |
| | 26.9.9 | 3200 Amp, 120KA | 3200 Amp 415V, 4P, 120KA | Metre | 90078.00 |
| | 26.9.10 | 3600 Amp, 120KA | 3600 Amp 415V, 4P, 120KA | Metre | 107304.00 |
| | 26.9.11 | 4000 Amp, 120KA | 4000 Amp 415V, 4P, 120KA | Metre | 124328.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **26.10** | | **Accessories for indoor door copper sandwich bus trunking 415V 4P** | Supplying, erection, testing and commissioning of accessories for Indoor door copper Sandwich bus trunking 415V 4P class F insulation (155Deg. C) with insulation material multi-layer polyethylene terephthalate, housing fabricated with 1.6mm thick electro galvanized sheet steel enclosure epoxy powder coated paint finish of RAL 7032 grade/aluminum casing of 2.5mm may also be provided. IP 54 protection having external 2 runs copper earthing 50% cross section area conform to IEC 60439-1&2 and IEC 61439 - 1&6. Every joint should be provided with mono block joint and shear-off nut. The mounting bracket shall be paid separately the rating should be as follows :- | | |
| | 26.10.1 | 90 degree bend 630 Amp | 90 degree bend 630 Amp 415V, 4P, 40KA | Each | 20615.00 |
| | 26.10.2 | 90 degree bend 800 Amp | 90 degree bend 800 Amp 415V, 4P, 50KA | Each | 24674.00 |
| | 26.10.3 | 90 degree bend 1000 Amp | 90 degree bend 1000 Amp 415V, 4P, 65KA | Each | 32801.00 |
| | 26.10.4 | 90 degree bend 1250 Amp | 90 degree bend 1250 Amp 415V, 4P, 100KA | Each | 44987.00 |
| | 26.10.5 | 90 degree bend 1600 Amp | 90 degree bend 1600 Amp 415V, 4P, 100KA | Each | 55156.00 |
| | 26.10.6 | 90 degree bend 2000 Amp | 90 degree bend 2000 Amp 415V, 4P, 100KA | Each | 75712.00 |
| | 26.10.7 | 90 degree bend 2250 Amp | 90 degree bend 2250 Amp 415V, 4P, 100KA | Each | 85865.00 |
| | 26.10.8 | 90 degree bend 2500 Amp | 90 degree bend 2500 Amp 415V, 4P, 100KA | Each | 98298.00 |
| | 26.10.9 | 90 degree bend 3200 Amp | 90 degree bend 3200 Amp 415V, 4P, 120KA | Each | 108094.00 |
| | 26.10.10 | 90 degree bend 3600 Amp | 90 degree bend 3600 Amp 415V, 4P, 120KA | Each | 128764.00 |
| | 26.10.11 | 90 degree bend 4000 Amp | 90 degree bend 4000 Amp 415V, 4P, 120KA | Each | 149194.00 |
| | 26.10.12 | Copper flexible 400 Amp | Copper flexible 400 Amp | Each | 18646.00 |
| | 26.10.13 | Copper flexible 500 Amp | Copper flexible 500 Amp | Each | 21285.00 |
| | 26.10.14 | Copper flexible 630 Amp | Copper flexible 630 Amp | Each | 21285.00 |
| | 26.10.15 | Copper flexible 800 Amp | Copper flexible 800 Amp | Each | 23384.00 |
| | 26.10.16 | Copper flexible 1000 Amp | Copper flexible 1000 Amp | Each | 23384.00 |
| | 26.10.17 | Copper flexible 1250 Amp | Copper flexible 1250 Amp | Each | 30605.00 |
| | 26.10.18 | Copper flexible 1600 Amp | Copper flexible 1600 Amp | Each | 36317.00 |
| | 26.10.19 | Copper flexible 2000 Amp | Copper flexible 2000 Amp | Each | 59630.00 |
| | 26.10.20 | Copper flexible 2250 Amp | Copper flexible 2250 Amp | Each | 59630.00 |
| | 26.10.21 | Copper flexible 2500 Amp | Copper flexible 2500 Amp | Each | 69475.00 |
| | 26.10.22 | Copper flexible 3200 Amp | Copper flexible 3200 Amp | Each | 85497.00 |
| | 26.10.23 | Copper Flexible 3600 Amp | Copper flexible 3600 Amp | Each | 105002.00 |
| | 26.10.24 | Copper flexible 4000 Amp | Copper flexible 4000 Amp | Each | 113732.00 |
| | 26.10.25 | Expansion joint 630 Amp | Expansion joint 630 Amp– provision (if required) | Each | 22864.00 |
| | 26.10.26 | Expansion joint 800 Amp | Expansion joint 800 Amp– provision (if required) | Each | 24555.00 |
| | 26.10.27 | Expansion joint 1000 Amp | Expansion joint 1000 Amp– provision (if required) | Each | 27941.00 |
| | 26.10.28 | Expansion joint 1250 Amp | Expansion joint 1250 Amp– provision (if required) | Each | 33017.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 26.10.29 | Expansion joint 1600 Amp | Expansion joint 1600 Amp– provision (if required) | Each | 37256.00 |
| | 26.10.30 | Expansion joint 2000 Amp | Expansion joint 2000 Amp– provision (if required) | Each | 48645.00 |
| | 26.10.31 | Expansion joint 2250 Amp | Expansion joint 2250 Amp– provision (if required) | Each | 52876.00 |
| | 26.10.32 | Expansion joint 2500 Amp | Expansion joint 2500 Amp– provision (if required) | Each | 58057.00 |
| | 26.10.33 | Expansion joint 3200 Amp | Expansion joint 3200 Amp– provision (if required) | Each | 110078.00 |
| | 26.10.34 | Expansion joint 3600 Amp | Expansion joint 3600 Amp– provision (if required) | Each | 125580.00 |
| | 26.10.35 | Expansion joint 4000 Amp | Expansion joint 4000 Amp– provision (if required) | Each | 140903.00 |
| | 26.10.36 | End feed unit, 630 Amp | End feed unit with provision for switchgear but without switchgear for 630 Amp | Each | 68195.00 |
| | 26.10.37 | End feed unit, 800 Amp | End feed unit with provision for switchgear but without switchgear for 800 Amp | Each | 74966.00 |
| | 26.10.38 | End feed unit, 1000 Amp | End feed unit with provision for switchgear but without switchgear for 1000 Amp | Each | 85118.00 |
| | 26.10.39 | End feed unit, 1250 Amp | End feed unit with provision for switchgear but without switchgear for 1250 Amp | Each | 109473.00 |
| | 26.10.40 | End feed unit, 1600 Amp | End feed unit with provision for switchgear but without switchgear for 1600 Amp | Each | 118142.00 |
| | 26.10.41 | End feed unit, 2000 Amp | End feed unit with provision for switchgear but without switchgear for 2000 Amp | Each | 126603.00 |
| | 26.10.42 | End feed unit, 2250 Amp | End feed unit with provision for switchgear but without switchgear for 2250 Amp | Each | 135064.00 |
| | 26.10.43 | End feed unit, 2500 Amp | End feed unit with provision for switchgear but without switchgear for 2500 Amp | Each | 145425.00 |
| | 26.10.44 | End feed unit, 3200 Amp | End feed unit with provision for switchgear but without switchgear for 3200 Amp | Each | 169465.00 |
| | 26.10.45 | End feed unit, 3600 Amp | End feed unit with provision for switchgear but without switchgear for 3600 Amp | Each | 186691.00 |
| | 26.10.46 | End feed unit, 4000 Amp | End feed unit with provision for switchgear but without switchgear for 4000 Amp | Each | 203716.00 |
| | 26.10.47 | End cover for 630 Amp | End cover for 630 Amp | Each | 1439.00 |
| | 26.10.48 | End cover for 800 Amp | End cover for 800 Amp | Each | 1786.00 |
| | 26.10.49 | End cover for 1000 Amp | End cover for 1000 Amp | Each | 2084.00 |
| | 26.10.50 | End cover for 1250 Amp | End cover for 1250 Amp | Each | 2282.00 |
| | 26.10.51 | End cover for 1600 Amp | End cover for 1600 Amp | Each | 2382.00 |
| | 26.10.52 | End cover for 2000 Amp | End cover for 2000 Amp | Each | 2481.00 |
| | 26.10.53 | End cover for 2250 Amp | End cover for 2250 Amp | Each | 3076.00 |
| | 26.10.54 | End cover for 2500 Amp | End cover for 2500 Amp | Each | 3275.00 |
| | 26.10.55 | End cover for 3200 Amp | End cover for 3200 Amp | Each | 3562.00 |
| | 26.10.56 | End cover for 3600 Amp | End cover for 3600 Amp | Each | 3672.00 |
| | 26.10.57 | End cover for 4000 Amp | End cover for 4000 Amp | Each | 3969.00 |

# CHAPTER – 27

## MOULDED CASE CIRCUIT BREAKERS, ISOLATORS, MCB (TRIPPING CHARACTERISTICS CONFORMING TO IS-8828 AND BS-3871 PART -I, II)

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.1** | | **MCCB 3Ph, 3 Pole, 50Hz, 415V** | Supply of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase,3 pole, 50 Hz, 415 Volts, AC supply with respective interrupting capacity (KA) at 415 Volts cited against their range standard conforming to IS – 8828 | | |
| | 27.1.1 | MCCB upto 63,10kA, 3P | MCCB current rating – **upto 63 Amp**, 10kA, 3P 415V AC | Each | 2721.00 |
| | 27.1.2 | MCCB 80 to 100A,10kA, 3P | MCCB current rating **-80 to 100Amp**, 10kA, 3P 415V AC | Each | 3256.00 |
| | 27.1.3 | MCCB 25 to 100A, 25kA, 3P | MCCB 25kA current rating, **3P -25 to 100Amp** & 80% -100% adjustable | Each | 7369.00 |
| | 27.1.4 | MCCB 125A, 25kA, 3P | MCCB 25kA current rating, **3P -125Amp** & 80% -100% adjustable | Each | 9953.00 |
| | 27.1.5 | MCCB 160A, 25kA, 3P | MCCB 25kA current rating, **3P -160Amp** & 80% -100% adjustable | Each | 12701.00 |
| | 27.1.6 | MCCB 200A, 25kA, 3P | MCCB 25kA current rating, **3P -200Amp** & 80% -100% adjustable | Each | 16061.00 |
| | 27.1.7 | MCCB 160A, 36kA, 3P | MCCB 36kA current rating, **3P -160Amp** &adjustable: 80% -100% thermal | Each | 14499.00 |
| | 27.1.8 | MCCB 200A, 36kA, 3P | MCCB 36kA current rating, 3P, **3P -200Amp** &adjustable: 80% -100% thermal | Each | 17555.00 |
| | 27.1.9 | MCCB 250A, 36kA, 3P | MCCB 36kA current rating, **3P -250Amp** &adjustable: 80% -100% thermal | Each | 21243.00 |
| | 27.1.10 | MCCB 315A, 36kA, 3P | MCCB 36kA current rating, **3P -315Amp** &adjustable: 80% -100% thermal | Each | 23636.00 |
| | 27.1.11 | MCCB 400A, 36kA, 3P | MCCB 36kA current rating, **3P -400Amp** &adjustable: 80% -100% thermal | Each | 24413.00 |
| **27.2** | | **MCCB 3Ph, 3 Pole, 50Hz, 415V (with Thermal - Magnetic Release)** | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase,3 pole, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply **with thermal - magnetic release** having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | 27.2.1 | MCCB 63 to 100A, 25kA, 3P | MCCB 25kA current rating - **63 to100Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 7538.00 |
| | 27.2.2 | MCCB 125A, 25kA, 3P | MCCB 25kA current rating - **125Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 10160.00 |
| | 27.2.3 | MCCB 160A, 25kA, 3P | MCCB 25kA current rating - **160Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 13327.00 |
| | 27.2.4 | MCCB 200A, 25kA, 3P | MCCB 25kA current rating - **200Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 17154.00 |
| | 27.2.5 | MCCB 160A, 36kA, 3P | MCCB 36kA current rating - **160Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 15400.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 27.2.6 | MCCB 200A, 36kA, 3P | MCCB 36kA current rating - **200Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 19291.00 |
| | 27.2.7 | MCCB 250A, 36kA, 3P | MCCB 36kA current rating - **250Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 22839.00 |
| | 27.2.8 | MCCB 315A, 36kA, 3P | MCCB 36kA current rating - **315Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 25946.00 |
| | 27.2.9 | MCCB 400A, 36kA, 3P | MCCB 36kA current rating - **400Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 26837.00 |
| | 27.2.10 | MCCB 500A, 36kA, 3P | MCCB 36kA current rating - **500Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 3 Pole | Each | 44709.00 |
| **27.3** | | **MCCB 3Ph & N, 4Pole, 50Hz, 415V (thermal - magnetic release)** | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase & N, **4 pole**, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply **with thermal-magnetic release** having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | 27.3.1 | MCCB 63 to 100A, 25kA, 4P | MCCB 25kA current rating - **63 to100Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 9781.00 |
| | 27.3.2 | MCCB 125A, 25kA, 4P | MCCB 25kA current rating - **125Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 11593.00 |
| | 27.3.3 | MCCB 160A, 25kA, 4P | MCCB 25kA current rating - **160Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 15835.00 |
| | 27.3.4 | MCCB 200A, 25kA, 4P | MCCB 25kA current rating - **200 Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 19695.00 |
| | 27.3.5 | MCCB 160A, 36kA, 4P | MCCB 36kA current rating - **160 Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 18923.00 |
| | 27.3.6 | MCCB 200A, 36kA, 4P | MCCB 36kA current rating - **200Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 21950.00 |
| | 27.3.7 | MCCB 250A, 36kA, 4P | MCCB 36kA current rating - **250Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 27693.00 |
| | 27.3.8 | MCCB 315A, 36kA, 4P | MCCB 36kA current rating - **315Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 30528.00 |
| | 27.3.9 | MCCB 400A, 36kA, 4P | MCCB 36kA current rating - **400Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 31907.00 |
| | 27.3.10 | MCCB 500A, 36kA, 4P | MCCB 36kA current rating - **500Amp** & adjustable overload setting 80% -100% of In, adjustable short circuit setting 6-10 In 4 Pole | Each | 53891.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.4** | | **MCCB 3Ph, 3Pole, 50Hz, 415V (Micro-processor release)** | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase,**3 pole**, 50 Hz, 415 Volts, (Ics=100%Icu) AC supply **with Micro-Processor release** having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | 27.4.1 | MCCB 40 to 100A, 25kA, 3P | MCCB 25kA current rating **- 40 & 100Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 18389.00 |
| | 27.4.2 | MCCB 160A, 25kA, 3P | MCCB 25kA current rating - **160Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 23257.00 |
| | 27.4.3 | MCCB 250A, 25kA, 3P | MCCB 25kA current rating - **250Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 29146.00 |
| | 27.4.4 | MCCB 40 to 100A, 36kA, 3P | MCCB 36kA current rating - **40 & 100Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 18716.00 |
| | 27.4.5 | MCCB 160A, 36kA, 3P | MCCB 36kA current rating - **160Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 21383.00 |
| | 27.4.6 | MCCB 250A, 36kA, 3P | MCCB 36kA current rating - **250Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 26116.00 |
| | 27.4.7 | MCCB 400A, 36kA, 3P | MCCB 36kA current rating - **400Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 37640.00 |
| | 27.4.8 | MCCB 630A, 36kA, 3P | MCCB 36kA current rating - **630Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 47274.00 |
| | 27.4.9 | MCCB 800A, 50kA, 3P | MCCB 50kA current rating - **800Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 72254.00 |
| | 27.4.10 | MCCB 1250A, 50kA, 3P | MCCB 50kA current rating - **1250Amp** 3 Pole &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 232253.00 |
| **27.5** | | **(MCCB) suitable for 3 phase & N, 4 pole, AC supply with micro-processor release** | Supplying of ISI marked and approved make of moulded case circuit breaker (MCCB) suitable for 3 phase & N, **4 pole**,50 Hz, 415 Volts, (Ics=100%Icu) AC supply **with microprocessor release** with inbuilt instantaneous protection & variable trip class 7.2 Ir & 6 Ir having respective breaking capacity (KA) at 415 Volts cited against their range standard conforming to IS/IEC 60947-2 | | |
| | 27.5.1 | MCCB, 25kA, 40 & 100 Amps, 4P | MCCB 25kA current rating - **40 & 100 Amps 4 Pole** & micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 21576.00 |
| | 27.5.2 | MCCB, 25kA, 160 Amps, 4P | MCCB 25kA current rating - **160 Amps 4 Pole** & micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 26273.00 |
| | 27.5.3 | MCCB, 25kA, 250 Amps, 4P | MCCB 25kA current rating - **250 Amps 4 Pole** & micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 38922.00 |
| | 27.5.4 | MCCB, 36kA, 40 & 100 Amps, 4P | MCCB 36kA current rating - **40 & 100 Amps 4 Pole** & Micro-Processor Release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 20196.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 27.5.5 | MCCB, 36kA, 160 Amps, 4P | MCCB 36kA current rating - **160 Amps 4 Pole** &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 26747.00 |
| | 27.5.6 | MCCB, 36kA, 250 Amps, 4P | MCCB 36kA current rating - **250 Amps 4 Pole** &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 39272.00 |
| | 27.5.7 | MCCB, 36kA, 400 Amps, 4P | MCCB 36kA current rating - **400 Amps 4 Pole** &micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. adjustable | Each | 47659.00 |
| | 27.5.8 | MCCB, 36kA, 630 Amps, 4P | MCCB 36kA current rating - **630 Amps 4 Pole** & micro-processor release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | Each | 70459.00 |
| | 27.5.9 | MCCB, 50kA, 800 Amps, 4P | MCCB 50kA current rating - **800 Amps 4 Pole** & Micro-Processor Release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | Each | 90200.00 |
| | 27.5.10 | MCCB, 50kA, 1250 Amps, 4P | MCCB 50kA current rating - **1250 Amps 4 Pole** & Micro-Processor Release, over load 40% -100% x In. & short circuit 1.5 - 8 x Ir. Adjustable. | Each | 315130.00 |
| **27.6** | | **MCCB 3Ph, 3/4Pole, 50Hz, 415V (without protection release)** | Supplying of approved make moulded case circuit breaker (MCCB**) without protection release** suitable for 3 phase**, 3/4 pole**, 50 Hz, 415 Volts, AC supply standard conforming to IS – 8828. | | |
| | 27.6.1 | MCCB 200A 3Pole | MCCB switch disconnector rating -**200Amp 3 Pole** without protection release. | Each | 9411.00 |
| | 27.6.2 | MCCB 400A 3Pole | MCCB switch disconnector rating -**400Amp 3 Pole** without protection release. | Each | 13560.00 |
| | 27.6.3 | MCCB 600A 3Pole | MCCB switch disconnector rating - **600Amp 3 Pole** without protection release. | Each | 16437.00 |
| | 27.6.4 | MCCB 200A 4Pole | MCCB switch disconnector rating - **200Amp 4 Pole** without protection Release. | Each | 11877.00 |
| | 27.6.5 | MCCB 400A 4Pole | MCCB switch disconnector rating - **400Amp 4 Pole** without protection release. | Each | 16942.00 |
| | 27.6.6 | MCCB 600A 4Pole | MCCB switch disconnector rating - **600Amp 4 Pole** without protection Release. | Each | 22030.00 |
| **27.7** | | **Air circuit breaker (ACB) (Microprocessor Release)** | Supply of ISI marked and approved make of air circuit breaker (ACB) with microprocessor release with overload, short circuit, neutral protection, instantaneous, ground fault protection (LSING) Ics=Icu=Icw for 1sec for total selectivity, ambient temperature 50deg.C, In with breaking capacity of 50 kA/60kA/75kA, 4 pole, 415 Volts conforming to IS/IEC : 60947-2 | | |
| | 27.7.1 | 630A, Fixed 50kA | 630Amp, manual, fixed 50kA | Each | 152207.00 |
| | 27.7.2 | 800A, Fixed 50kA | 800Amp, manual, fixed 50kA | Each | 152207.00 |
| | 27.7.3 | 1000A, Fixed 50kA | 1000Amp, manual, fixed 50kA | Each | 156559.00 |
| | 27.7.4 | 1250A, Fixed 50kA | 1250Amp, manual, fixed 50kA | Each | 190353.00 |
| | 27.7.5 | 1600A, Fixed 50kA | 1600Amp, manual, fixed 50kA | Each | 215711.00 |
| | 27.7.6 | 630A, Drawout 50kA | 630Amp, manual, drawout 50kA | Each | 233563.00 |
| | 27.7.7 | 800A, Drawout 50kA | 800Amp, manual, drawout 50kA | Each | 237659.00 |
| | 27.7.8 | 1000A, Drawout 50kA | 1000Amp, manual, drawout 50kA | Each | 274243.00 |
| | 27.7.9 | 1250A, Drawout 50kA | 1250Amp, manual, drawout 50kA | Each | 295733.00 |
| | 27.7.10 | 1600A, Drawout 50kA | 1600Amp, manual, drawout 50kA | Each | 365740.00 |
| | 27.7.11 | 2000A, Drawout 50kA | 2000Amp, manual, drawout 50kA | Each | 432582.00 |
| | 27.7.12 | 2500A, Drawout 60kA | 2500Amp, manual, drawout 60kA | Each | 508365.00 |
| | 27.7.13 | 3200A, Drawout 75kA | 3200Amp, manual, drawout 75kA | Each | 548886.00 |
| | 27.7.14 | 4000A, Drawout 75kA | 4000Amp, manual, drawout 75kA | Each | 607592.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.8** | | **MCB 'C' Series, 240/415V, 50 Cycle, 10kA/15kA** | Supply of ISI marked and accepted standard of miniature circuit breaker (MCB) of **'C' series** with short circuit indication, suitable for 240/415 Volts,50 Cycle, 10 kA/15 kA Value AC supply conforming to IS : 8828 : 1996, IEC : 60898 :2002 & 60947-2 but without enclosures :- | | |
| | **27.8.1** | **Single pole** | **Single pole** | | |
| | 27.8.1.1 | MCB SP 0.5A to 5A, 10kA | MCB SP 0.5Amp to 5Amp 10kA rating | Each | 439.00 |
| | 27.8.1.2 | MCB SP 6A to 32A, 10kA | MCB SP 6Amp to 32Amp 10kA rating | Each | 263.00 |
| | 27.8.1.3 | MCB SP 40A, 10kA | MCB SP For 40Amp 10kA rating | Each | 611.00 |
| | 27.8.1.4 | MCB SP 50A to 63A, 10kA | MCB SP 50Amp to 63Amp 10kA rating | Each | 651.00 |
| | 27.8.1.5 | MCB SP 80A, 15kA | MCB SP 80Amp 15kA rating | Each | 2409.00 |
| | 27.8.1.6 | MCB SP 100A, 15kA | MCB SP 100Amp 15kA rating | Each | 2701.00 |
| | 27.8.1.7 | MCB SP 125A, 15kA | MCB SP 125Amp 15kA rating | Each | 2921.00 |
| | **27.8.2** | **Double Pole** | **Double pole** | | |
| | 27.8.2.1 | MCB DP 0.5A to 5A, 10kA | MCB DP 0.5Amp to 5Amp 10kA rating | Each | 1243.00 |
| | 27.8.2.2 | MCB DP 6A to 32A, 10kA | MCB DP 6Amp to 32Amp 10kA rating | Each | 887.00 |
| | 27.8.2.3 | MCB DP 40A, 10kA | MCB DP For 40Amp 10kA rating | Each | 1329.00 |
| | 27.8.2.4 | MCB DP 50A to 63A, 10kA | MCB DP 50Amp to 63Amp 10kA rating | Each | 1478.00 |
| | 27.8.2.5 | MCB DP 80A, 15kA | MCB DP 80Amp 15kA rating | Each | 5344.00 |
| | 27.8.2.6 | MCB DP 100A, 15kA | MCB DP 100Amp 15kA rating | Each | 5709.00 |
| | 27.8.2.7 | MCB DP 125A, 15kA | MCB DP 125Amp 15kA rating | Each | 6274.00 |
| | **27.8.3** | **Triple pole** | **For Triple pole** | | |
| | 27.8.3.1 | MCB TP 0.5A to 5A, 10kA | MCB TP 0.5Amp to 5Amp 10kA rating | Each | 1756.00 |
| | 27.8.3.2 | MCB TP 6A to 32A, 10kA | MCB TP 6Amp to 32Amp 10kA rating | Each | 1411.00 |
| | 27.8.3.3 | MCB TP 40A, 10kA | MCB TP For 40Amp 10kA rating | Each | 2099.00 |
| | 27.8.3.4 | MCB TP 50A to 63A, 10kA | MCB TP 50Amp to 63Amp 10kA rating | Each | 2157.00 |
| | 27.8.3.5 | MCB TP 80A, 15kA | MCB TP 80Amp 15kA rating | Each | 8436.00 |
| | 27.8.3.6 | MCB TP 100A, 15kA | MCB TP 100Amp 15kA rating | Each | 8981.00 |
| | 27.8.3.7 | MCB TP 125A, 15kA | MCB TP 125Amp 15kA rating | Each | 9976.00 |
| | **27.8.4** | **Four pole** | **Four pole** | | |
| | 27.8.4.1 | MCB FP 0.5A to 5A, 10kA | MCB FP 0.5Amp to 5Amp 10kA rating | Each | 2399.00 |
| | 27.8.4.2 | MCB FP 6A to 32A, 10kA | MCB FP 6Amp to 32Amp 10kA rating | Each | 1965.00 |
| | 27.8.4.3 | MCB FP 40A, 10kA | MCB FP For 40Amp 10kA rating | Each | 2714.00 |
| | 27.8.4.4 | MCB FP 50A to 63A, 10kA | MCB FP 50Amp to 63 Amp 10kA rating | Each | 2795.00 |
| | 27.8.4.5 | MCB FP 80A, 15kA | MCB FP 80Amp 15kA rating | Each | 10806.00 |
| | 27.8.4.6 | MCB FP 100A, 15kA | MCB FP 100Amp 15kA rating | Each | 11264.00 |
| | 27.8.4.7 | MCB FP 125A, 15kA | MCB FP 125Amp 15kA rating | Each | 12724.00 |
| **27.9** | | **MCB 'D' Series, 240/415V, 50 Cycle, 10kA/15kA** | Supply of ISI marked and accepted standard of miniature circuit breaker (MCB) of **'D' series** with short circuit indication, suitable for 240/415 Volts,50 Cycle, 10 kA/15 kA Value AC supply conforming to IS : 8828 : 1996, IEC : 60898 :2002 & 60947-2 but without enclosures :- | | |
| | **27.9.1** | **Single pole** | **Single pole** | | |
| | 27.9.1.1 | MCB SP 0.5A to 5A, 10kA | MCB SP 0.5Amp to 5Amp 10kA rating | Each | 481.00 |
| | 27.9.1.2 | MCB SP 6A to 32A, 10kA | MCB SP 6Amp to 32Amp 10kA rating | Each | 383.00 |
| | 27.9.1.3 | MCB SP 40A, 10kA | MCB SP For 40Amp 10kA rating | Each | 691.00 |
| | 27.9.1.4 | MCB SP 50A to 63A, 10kA | MCB SP 50Amp to 63Amp 10kA rating | Each | 767.00 |
| | **27.9.2** | **Double Pole** | **Double pole** | | |
| | 27.9.2.1 | MCB DP 0.5A to 5A, 10kA | MCB DP 0.5Amp to 5Amp 10kA rating | Each | 1296.00 |
| | 27.9.2.2 | MCB DP 6A to 32A, 10kA | MCB DP 6Amp to 32Amp 10kA rating | Each | 1038.00 |
| | 27.9.2.3 | MCB DP 40A, 10kA | MCB DP For 40Amp 10kA rating | Each | 1453.00 |
| | 27.9.2.4 | MCB DP 50A to 63A, 10kA | MCB DP 50Amp to 63Amp 10kA rating | Each | 1555.00 |
| | **27.9.3** | **Triple pole** | **For Triple pole** | | |
| | 27.9.3.1 | MCB TP 0.5A to 5A, 10kA | MCB TP 0.5Amp to 5Amp 10kA rating | Each | 1913.00 |
| | 27.9.3.2 | MCB TP 6A to 32A, 10kA | MCB TP 6Amp to 32Amp 10kA rating | Each | 1519.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 27.9.3.3 | MCB TP 40A, 10kA | MCB TP For 40Amp 10kA rating | Each | 2306.00 |
| | 27.9.3.4 | MCB TP 50A to 63A, 10kA | MCB TP 50Amp to 63Amp 10kA rating | Each | 2382.00 |
| | **27.9.4** | **Four pole** | **Four pole** | | |
| | 27.9.4.1 | MCB FP 0.5A to 5A, 10kA | MCB FP 0.5Amp to 5Amp 10kA rating | Each | 2525.00 |
| | 27.9.4.2 | MCB FP 6A to 32A, 10kA | MCB FP 6Amp to 32Amp 10kA rating | Each | 2275.00 |
| | 27.9.4.3 | MCB FP 40A, 10kA | MCB FP For 40Amp 10kA rating | Each | 2855.00 |
| | 27.9.4.4 | MCB FP 50A to 63A, 10kA | MCB FP 50Amp to 63 Amp 10kA rating | Each | 3073.00 |
| **27.10** | | **MCB Isolators** | Supplying of MCB Isolators suitable for 240/415 Volts, 50 Hz AC supply with kA value rating 10 kA of approved make conforming to IS:13947-Part III :1993 & IEC:60947- 3:2001 (without enclosures ) :- | | |
| | **27.10.1** | **Single pole** | **Single pole** | | |
| | 27.10.1.1 | Upto 40A | Current rating upto 40Amp | Each | 410.00 |
| | 27.10.1.2 | Upto 63A | Current rating upto 63Amp | Each | 451.00 |
| | **27.10.2** | **Double pole** | **Double pole** | | |
| | 27.10.2.1 | Upto 40A | Current rating upto 40Amp | Each | 575.00 |
| | 27.10.2.2 | Upto 63A | Current rating upto 63Amp | Each | 733.00 |
| | **27.10.3** | **Triple pole** | **Triple pole** | | |
| | 27.10.3.1 | Upto 40A | Current rating upto 40Amp | Each | 1103.00 |
| | 27.10.3.2 | Upto 63A | Current rating upto 63Amp | Each | 1236.00 |
| | 27.10.3.3 | Upto 80A | Current rating upto 80Amp | Each | 1386.00 |
| | 27.10.3.4 | Upto 100A | Current rating upto 100Amp | Each | 1455.00 |
| | **27.10.4** | **Four pole** | **Four pole** | | |
| | 27.10.4.1 | Upto 40A | Current rating upto 40Amp | Each | 1097.00 |
| | 27.10.4.2 | Upto 63A | Current rating upto 63Amp | Each | 1398.00 |
| | 27.10.4.3 | Upto 80A | Current rating upto 80Amp | Each | 1665.00 |
| | 27.10.4.4 | Upto 100A | Current rating upto 100Amp | Each | 1747.00 |
| **27.11** | | **SPN MCB DB** | Supply of approved make powder coated sheet steel encloser SPN MCB DB inclusive of busbar, neutral bar, earth bar, connection copper wire not less than 16 sq.mm & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator)- | | |
| | 27.11.1 | 2 Way single door | 2 Way single door | Each | 447.00 |
| | 27.11.2 | 4 Way single door | 4 Way single door | Each | 1097.00 |
| | 27.11.3 | 4 Way double door | 4 Way double door IP 43 protection | Each | 1699.00 |
| | 27.11.4 | 6 Way double door | 6 Way double door IP 43 protection | Each | 1846.00 |
| | 27.11.5 | 8 Way double door | 8 Way double door IP 43 protection | Each | 2127.00 |
| | 27.11.6 | 12 Way double door | 12 Way double door IP 43 protection | Each | 2738.00 |
| | 27.11.7 | 16 Way double door | 16 Way double door IP 43 protection | Each | 3415.00 |
| **27.12** | | **TPN MCB DB double door** | Supplying of approved make TPN MCB DB metal double door IP 43 protection with provision for FP MCB/ Isolator/ RCCB/ RCBO as incomer and SP MCBs as outgoing inclusive of busbar, neutral bar, earth bar, connection copper wire not less than 16 sq.mm & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator): | | |
| | 27.12.1 | 4 Way (4+12) | 4 Way (4+12) | Each | 5126.00 |
| | 27.12.2 | 4 Way (8+12) | 4 Way (8+12) | Each | 4444.00 |
| | 27.12.3 | 6 Way (8+18) | 6 Way (8+18) | Each | 5526.00 |
| | 27.12.4 | 8 Way (8+24) | 8 Way (8+24) | Each | 6716.00 |
| | 27.12.5 | 12 Way (8+36) | 12 Way (8+36) | Each | 9519.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.13** | | **Vertical TPN MCB DB double door** | Supplying of approved make vertical TPN MCB DB metal double door IP 43 protection with provision for FP MCB/Isolator/RCCB/RCBO as incomer and SP/TP MCBs as outgoing inclusive of busbar, neutral bar, earth bar, & two earth terminals etc. complete as per IS:13032(exclusive of MCB & isolator) | | |
| | 27.13.1 | 4 Way (8+12) | 4 Way (8+12) | Each | 13577.00 |
| | 27.13.2 | 6 Way (8+18) | 6 Way (8+18) | Each | 15436.00 |
| | 27.13.3 | 8 Way (8+24) | 8 Way (8+24) | Each | 17263.00 |
| | 27.13.4 | 12 Way (8+36) | 12 Way (8+36) | Each | 23179.00 |
| **27.14** | | **Vertical TPN MCB DB, with provision for MCCB upto 160A TP/FP 36kA, double door** | Supplying of approved make, powder coated metal double door vertical TPN MCB DB IP 43 protection with provision for MCCB upto160A TP/FP 36kA as incomer and space for SP/TP MCBs as outgoing (without MCCB/MCBs) inclusive of busbar & connections etc | | |
| | 27.14.1 | 4 Way without MCCB | 4 Way without MCCB | Each | 13822.00 |
| | 27.14.2 | 6 Way without MCCB | 6 Way without MCCB | Each | 15671.00 |
| | 27.14.3 | 8 Way without MCCB | 8 Way without MCCB | Each | 15989.00 |
| | 27.14.4 | 12 Way without MCCB | 12 Way without MCCB | Each | 21911.00 |
| **27.15** | | **Vertical TPN MCB DB, with provision for MCCB upto 250A TP/FP 50kA, double door** | Supply of approved make, powder coated metal double door vertical TPN MCB DB IP 43 protection with provision for MCCB upto 250A TP/FP 50kA as incomer and space for SP/TP MCBs as outgoing inclusive of busbar, neutral bar, earth bar & two earth terminals etc. complete as per IEC:61439-1 & 3(without MCCB/MCBs) | | |
| | 27.15.1 | 4 Way without MCCB | 4 Way without MCCB | Each | 16825.00 |
| | 27.15.2 | 6 Way without MCCB | 6 Way without MCCB | Each | 20725.00 |
| | 27.15.3 | 8 Way without MCCB | 8 Way without MCCB | Each | 22840.00 |
| | 27.15.4 | 12 Way without MCCB | 12 Way without MCCB | Each | 27727.00 |
| **27.16** | | **Vertical TPN MCCB sub-main DB, with provision for MCCB upto 630A, 36kA/50kA** | Supply of approved make, powder coated metal double door vertical TPN MCCB sub-main DB IP 43 protection with provision for MCCB upto 630A, 36kA/50kA, TP as incomer and space for TP MCCBs as outgoing inclusive of tin-plated electrolytic copper busbar 250A/400A/630A as per incoming MCCB rating, neutral bar 100% of phase, earth bar 50% of phase & two earth terminals etc. complete as per IEC:61439-2(without MCCBs) | | |
| | 27.16.1 | 4 Way, I/C 250A, O/G 100A | 4 Way without MCCB I/C 250Amp, O/G 100Amp | Each | 38547.00 |
| | 27.16.2 | 8 Way, I/C 250A, O/G 100A | 8 Way without MCCB I/C 250Amp, O/G 100Amp | Each | 46657.00 |
| | 27.16.3 | 12 Way, I/C 250A, O/G 100A | 12 Way without MCCB I/C 250Amp, O/G 100Amp | Each | 59846.00 |
| | 27.16.4 | 16 Way, I/C 250A, O/G 100A | 16 Way without MCCB I/C 250Amp, O/G 100Amp | Each | 69987.00 |
| | 27.16.5 | 4 Way, I/C 400A, O/G 100A | 4 Way without MCCB I/C 400Amp, O/G 100Amp | Each | 43614.00 |
| | 27.16.6 | 8 Way, I/C 400A, O/G 100A | 8 Way without MCCB I/C 400Amp, O/G 100Amp | Each | 49703.00 |
| | 27.16.7 | 12 Way, I/C 400A, O/G 100A | 12 Way without MCCB I/C 400Amp, O/G 100Amp | Each | 63899.00 |
| | 27.16.8 | 16 Way, I/C 400A, O/G 100A | 16 Way without MCCB I/C 400Amp, O/G 100Amp | Each | 76076.00 |
| | 27.16.9 | 8 Way, I/C 630A, O/G 100A | 8 Way without MCCB I/C 630Amp, O/G 100Amp | Each | 78098.00 |
| | 27.16.10 | 12 Way, I/C 630A, O/G 100A | 12 Way without MCCB I/C 630Amp, O/G 100Amp | Each | 105483.00 |
| | 27.16.11 | 16 Way, I/C 630A, O/G 100A | 16 Way without MCCB I/C 630Amp, O/G 100Amp | Each | 115626.00 |
| | 27.16.12 | 4 Way, I/C 630A, O/G 250A | 4 Way without MCCB I/C 630Amp, O/G 250Amp | Each | 76076.00 |
| | 27.16.13 | 8 Way, I/C 630A, O/G 250A | 8 Way without MCCB I/C 630Amp, O/G 250Amp | Each | 85198.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.17** | | **Vertical TPN MCCB sub-main DB accessories** | Supply & fixing of double door vertical TPN MCCB sub-main DB accessories as per OEM of approved make powder coated metal box for metering, side cabling box and corner box on surface/concealed complete as required mentioned below :-. | | |
| | 27.17.1 | Metering ext. box | Metering extension box 250Amp/400Amp | Each | 6692.00 |
| | 27.17.2 | Metering ext. box | Metering extension box 630Amp | Each | 7106.00 |
| | 27.17.3 | Bottom cabling box | Bottom cabling box 250Amp/400Amp | Each | 6269.00 |
| | 27.17.4 | Bottom cabling box | Bottom cabling box 630Amp | Each | 6692.00 |
| | 27.17.5 | Corner box | Corner box 250Amp/400Amp/630Amp | Each | 2097.00 |
| | 27.17.6 | Mounting base | Mounting base 250Amp/400Amp/630Amp | Each | 3142.00 |
| | **27.17.7** | **Side cabling box 250A-100A** | **Side cabling box 250Amp - 100Amp left/right side for -** | | |
| | 27.17.7.1 | 4 Way | 4 Way | Each | 6070.00 |
| | 27.17.7.2 | 6 Way | 6 Way | Each | 7843.00 |
| | 27.17.7.3 | 8 Way | 8 Way | Each | 9405.00 |
| | 27.17.7.4 | 12 Way | 12 Way | Each | 10971.00 |
| | **27.17.8** | **Side cabling box 400A-100A** | **Side cabling box 400Amp - 100Amp left/right side for -** | | |
| | 27.17.8.1 | 4 Way | 4 Way | Each | 9405.00 |
| | 27.17.8.2 | 6 Way | 6 Way | Each | 10971.00 |
| | 27.17.8.3 | 8 Way | 8 Way | Each | 12019.00 |
| | 27.17.8.4 | 12 Way | 12 Way | Each | 13584.00 |
| | **27.17.9** | **Side cabling box 630A-2500A + 100A** | **Side cabling box 630Amp - 250Amp +100Amp left/right side for -** | | |
| | 27.17.9.1 | 4 Way/2 + 2 | 4 Way/2 + 2 | Each | 10762.00 |
| | 27.17.9.2 | 6 Way/2 + 6 | 6 Way/2 + 6 | Each | 12019.00 |
| | 27.17.9.3 | 8 Way/2 + 10 | 8 Way/2 + 10 | Each | 13998.00 |
| | 27.17.9.4 | 12 Way/2 + 14 | 12 Way/2 + 14 | Each | 15466.00 |
| **27.18** | | **Industrial type metal plug & socket DBs, SPN sheet encloser** | Supplying & fixing of approved make Industrial type metal plug & socket DBs (without MCB) SPN sheet encloser (dust protected) inclusive of 2 pole and earth metal plug and socket and space to incorporate SP MCB complete as per specification as required. | | |
| | 27.18.1 | 10A | 10 Amp | Each | 1037.00 |
| | 27.18.2 | 20A | 20 Amp | Each | 1095.00 |
| **27.19** | | **Industrial type metal plug & socket DBs, TPN sheet encloser** | Supplying & fixing of ISI marked Industrial type metal plug socket D.B.s (without MCBs) for TPN encloser three pole and earth metal plug and socket including space to incorporate TP MCB complete as per specification as required. | | |
| | 27.19.1 | 20A | 20 Amp | Each | 2249.00 |
| | 27.19.2 | 30/32A | 30/32 Amp | Each | 2543.00 |
| **27.20** | | **Industrial type metal plug & socket, 3 Pin (2P+E)** | Supply & fixing of approved make Industrial type metal plug & socket, 3 pin (2P + E) conforming to IEC : 60309 -1&3 complete as per specification | | |
| | 27.20.1 | 10A Plug only | 10Amp Plug only | Each | 153.00 |
| | 27.20.2 | 10A socket only | 10Amp socket only | Each | 153.00 |
| | 27.20.3 | 20A Plug only | 20Amp Plug only | Each | 179.00 |
| | 27.20.4 | 20A socket only | 20Amp socket only | Each | 194.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.21** | | **RCBOs** | Supplying and installing of RCBOs (Residual current circuit breaker with overload and short circuit protection) ISI marked complete as per I.S. standard conforming to IEC:61009-1 & IS:12640-2, 240/415V 50 Hz with 10 kA short circuit withstand capacity for earth leakage, overload & short circuit protection including connection in existing enclosure in approved manner as per specification. | | |
| | **27.21.1** | **DP (4 Module)** | **DP (4 module)** | | |
| | 27.21.1.1 | 2 Pole 6A- 25A, 30mA | 2 pole 6Amp to 25Amp, 30 mA sensitivity | Each | 4491.00 |
| | 27.21.1.2 | 2 Pole 32A, 30mA | 2 pole 32Amp, 30mA sensitivity | Each | 4586.00 |
| | 27.21.1.3 | 2 Pole 40A, 30mA | 2 pole 40Amp, 30mA sensitivity | Each | 5010.00 |
| | 27.21.1.4 | 2 Pole 63A, 30mA | 2 pole 63Amp, 30mA sensitivity | Each | 6372.00 |
| | 27.21.1.5 | 2 Pole 6A- 25A, 100mA | 2 pole 6Amp to 25Amp, 100mA sensitivity | Each | 4603.00 |
| | 27.21.1.6 | 2 Pole 32A, 100mA | 2 pole 32Amp, 100mA sensitivity | Each | 4666.00 |
| | 27.21.1.7 | 2 Pole 40A, 100mA | 2 pole 40Amp, 100mA sensitivity | Each | 5081.00 |
| | 27.21.1.8 | 2 Pole 63A, 100mA | 2 pole 63Amp, 100mA sensitivity | Each | 6475.00 |
| | **27.21.2** | **FP (8 Module)** | **FP (8module)** | | |
| | 27.21.2.1 | 4 Pole 16A, 30mA | 4 pole 16Amp, 30mA sensitivity | Each | 6173.00 |
| | 27.21.2.2 | 4 Pole 25A-32A, 30mA | 4 pole 25-32Amp, 30mA sensitivity | Each | 6366.00 |
| | 27.21.2.3 | 4 Pole 40A, 30mA | 4 pole 40Amp, 30mA sensitivity | Each | 7077.00 |
| | 27.21.2.4 | 4 Pole 63A, 30mA | 4 pole 63Amp, 30mA sensitivity | Each | 7908.00 |
| | 27.21.2.5 | 4 Pole 16A, 100/300mA | 4 pole 16Amp, 100/300mA sensitivity | Each | 6356.00 |
| | 27.21.2.6 | 4 Pole 25A-32A, 100A/300mA | 4 pole 25-32Amp, 100/300mA sensitivity | Each | 6476.00 |
| | 27.21.2.7 | 4 Pole 40A, 100/300mA | 4 pole 40Amp, 100/300mA sensitivity | Each | 7180.00 |
| | 27.21.2.8 | 4 Pole 63A, 100/300mA | 4 pole 63Amp, 100/300mA sensitivity | Each | 7926.00 |
| **27.22** | | **SPDs**, Type 2 maximum discharge current (Imax) upto70kA &type 1+2 (Imax) upto100kA | Supplying and installing of SPDs (surge protection device) ISI marked complete as per I.S. standard conforming to IEC:61543 - 11 for direct & indirect lightning surges protection type 2 maximum discharge current (Imax) up to 70kA &type 1+2 (Imax) up to 100KA including connection in existing enclosure in approved manner as per specification. | | |
| | 27.22.1 | Type 2, 40kA, 1P+N | Type 2 : 40kA, 1P+N protection against indirect lightning & switching surges | Each | 6642.00 |
| | 27.22.2 | Type 2, 70kA, 1P+N | Type 2 : 70kA, 1P+N protection against indirect lightning & switching surges | Each | 10630.00 |
| | 27.22.3 | Type 2, 40kA, 3P+N | Type 2 : 40kA, 3P+N protection against indirect lightning & switching surges | Each | 13282.00 |
| | 27.22.4 | Type 2, 70kA, 3P+N | Type 2 : 70kA, 3P+N protection against indirect lightning & switching surges | Each | 17713.00 |
| | 27.22.5 | Type 1+2, 50kA, 1P+N | Type 1+2 : 50kA, 1P+N protection against direct & indirect lightning surges | Each | 9561.00 |
| | 27.22.6 | Type 1+2, 100kA, 1P+N | Type 1+2 : 100kA, 1P+N protection against direct & indirect lightning surges | Each | 10630.00 |
| | 27.22.7 | Type 1+2, 50kA, 3P+N | Type 1+2 : 50kA, 3P+N protection against direct & indirect lightning surges | Each | 18770.00 |
| | 27.22.8 | Type 1+2, 100kA, 3P+N | Type 1+2 : 100kA, 3P+N protection against direct & indirect lightning surges | Each | 19477.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **27.23** | | **Steel enclosure, neutral bar suitable for MCB/RCBOs/MCCB** | Supply & fixing of approved make accepted standard sheet steel enclosure (dust protected) duly powder coated paint neutral bar suitable for MCB/RCBOs/MCCB of capacity as required etc. complete as per specification (exclusive of M.C.B./ MCCB/RCBOs) | | |
| | **27.23.1** | **For MCB** | **For MCB** | | |
| | 27.23.1.1 | SP enclosure | SP enclosure | Each | 285.00 |
| | 27.23.1.2 | DP/TP/FP enclosure | DP/TP/FP enclosure | Each | 341.00 |
| | 27.23.1.3 | FP RCBO enclosure | FP RCBO enclosure | Each | 602.00 |
| | **27.23.2** | **For MCCB** | **For MCCB** | | |
| | 27.23.2.1 | 63A/100A, 3P/4P enclosure | 63Amp/100Amp, 3P/4P enclosure | Each | 1225.00 |
| | 27.23.2.2 | 160A, 3P/4P enclosure | 160Amp, 3P/4P enclosure | Each | 2285.00 |
| | 27.23.2.3 | 250A, 3P/4P enclosure | 250Amp, 3P/4P enclosure | Each | 4058.00 |
| | 27.23.2.4 | 630A, 3P/4P enclosure | 630Amp, 3P/4P enclosure | Each | 6149.00 |
| **27.24** | | **Cable end box** | Supplying and fixing cable end box (loose wire box) suitable for triple pole and neutral, sheet steel, MCB/MCCB distribution board, 415 V, on surface/recess, complete with testing and commissioning etc. as required. | | |
| | 27.24.1 | TPN DB MCB | TPN DB MCB SP O/G, (V) SP/TP O/G | Each | 696.00 |
| | 27.24.2 | TPN DB MCCB | TPN DB MCCB (V) SP/TP O/G | Each | 2461.00 |

# CHAPTER - 28

## FIXING OF SFUs, DBs, BUSBAR, MCCBs, MCBs, ISOLATORs & RCBOs

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **28.1** | | **Fixing of metal clad iron clad enclosure on 25 X 5 mm flat** | Fixing of metal clad iron clad enclosure on 25 X 5 mm flat iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| | 28.1.1 | 15/16A SP/DP switch/SPN | 15/16Amp SP/DP switch/SPN, DB up to 4 way | Each | 476.00 |
| | 28.1.2 | 32A DP switch/SPN | 32Amp DP switch/SPN DB up to 8 way/16 Amp TPN DB up to 3 way per pole or 16 Amp TP or TPN switch/16 Amp 415Volt Isolator | Each | 475.00 |
| | 28.1.3 | 32A TP or TPN Switch | 32 Amp TP or TPN switch/32 Amp 415Volt Isolator/16 Amp TPN DB up to 6 way per pole/32 Amp 415Volt bus bar 4 way | Each | 477.00 |
| **28.2** | | **Fixing of metal clad, 25x25x5 mm angle** | Fixing of metal clad enclosure on 25X 25 X 5 mm angle iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| | 28.2.1 | Fixing of 63A TP or TPN | Fixing of 63Amp TP or TPN main switches/DB 63Amp 240/415 Volt 4 to 8 way or 63 Amp change over switch/63Amp 415Volt Isolator/30 Amp 415Volt TPN DB up to 6 way per pole and bus bar 4 way | Each | 484.00 |
| | 28.2.2 | Fixing of 100A TP or TPN | Fixing of 100Amp TP or TPN main switch or 100Amp change over switch/100 Amp 415Volt Isolators and bus bar 4 way | Each | 510.00 |
| | 28.2.3 | Fixing of 200A TP or TPN | Fixing of 200 Amp TP or TPN main switch or 200 Amp change over switch/200 Amp 415 Volt Isolators and busbar 4 way | Each | 521.00 |
| **28.3** | | **Fixing of metal clad, 35x35x5 mm angle** | Fixing of metal clad enclosure on 35X 35 X 5 mm angle iron clamp including supplying and fixing of clamp as required duly embedded in masonry including cable connection and other required material as per specification :- | | |
| | 28.3.1 | Fixing of 300/400A TP or TPN main switch or change over switch | Fixing of 300/400Amp TP or TPN main switch or 300/400Amp change over switch/300/400Amp 415 Volt Isolators and busbar 4 way | Each | 459.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 28.3.2 | Fixing of 500/600A TP Or TPN main switch or change over switch | Fixing of 500/600Amp TP or TPN main switch or 500/600Amp change over switch 500/600Amp 415 Volt Isolators and busbar 4 way | Each | 468.00 |
| **28.4** | | **Fixing of MCB/MCCB/ Isolator, (labour only)** | Fixing of MCB/MCCB/Isolator in sheet steel enclosure as required as per accepted practice, including mounting on busbar and cable connection etc. complete (labour only) | | |
| | 28.4.1 | MCB/Isolator SP/DP | MCB/Isolator SP/DP | Each | 22.00 |
| | 28.4.2 | MCB/MCCB Isolator TP/TPN/FP | MCB/MCCB Isolator TP/TPN/FP | Each | 28.00 |
| **28.5** | | **Labour charges** | Labour charges for fixing sheet steel enclosure, **MCB DB surface mounting type**, as per accepted practice on **25x5mm** flat iron clamp duly embedded in wall, cable connection etc. complete. | | |
| | | For Item | For item no :- | | |
| | 28.5.1 | Item SOR no. | 27.11.1 to.27.11.4 | Each | 517.00 |
| | 28.5.2 | Item SOR no. | 27.11.5 & 27.12.1 to 27.12.2 | Each | 545.00 |
| | 28.5.3 | Item SOR no. | 27.12.3 to 27.12.4 & 27.13.1 | Each | 572.00 |
| **28.6** | | **Labour charges** | Labour charges for fixing sheet steel enclosures, **MCB DB surface mounting type**, as per accepted practice on **25x25x 5 mm** angle iron clamp, including supplying and fixing of clamp duly embedded in wall, cable connection etc. complete :- | | |
| | | For Item | For item no :- | | |
| | 28.6.1 | Item SOR no. | 27.11.6 & 27.11.7; 27.12.5 ; 27.13.2 to 27.13.4 & 27.14.1 to 27.14.4 ; 27.15.1 to 27.15.4 | Each | 414.00 |
| **28.7** | | **Labour charges** | Labour charges for fixing sheet steel enclosures, **MCCB DB surface mounting type**, as per accepted practice on **35x35x 5 mm** angle iron clamp, including supplying and fixing of clamp duly embedded in wall, cable connection etc. complete :- | | |
| | 28.7.1 | Item SOR no. | 27.16.1 & 27.16.2 ; 27.16.5 & 27.16.6 | Each | 3100.00 |
| | 28.7.2 | Item SOR no. | 27.16.3 & 27.16.4 ; 27.16.7 to 27.16.9, 27.16.12, 27.16.13 | Each | 3318.00 |
| | 28.7.3 | Item SOR no. | 27.16.10 to 27.16.11 | Each | 3991.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **28.8** | | **Labour charges** | Labour charges for fixing sheet steel enclosures, **MCB/MCCBDB flush mounting type**, as per accepted practice, duly embedded and end plate completely flushed in wall, cable connection etc. complete :- | | |
| | | For item | For item no :- | | |
| | 28.8.1 | Item SOR no. | 27.11.1 to 27.11.7 ; 27.12.1 to 27.12.3 & 27.13.1 | Each | 268.00 |
| | 28.8.2 | Item SOR no. | 27.12.4 & 27.12.5 ; 27.13.2 to 27.13.4 ; 27.14.1 to 27.14.4 & 27.15.1 to 27.15.4 | Each | 297.00 |
| | 28.8.3 | Item SOR no. | 27.16.1 & 27.16.2 ; 27.16.5 & 27.16.6 | Each | 2788.00 |
| | 28.8.4 | Item SOR no. | 27.16.3 & 27.16.4 ; 27.16.7 to 27.16.9, 27.16.12, 27.16.13 | Each | 3527.00 |
| | 28.8.5 | Item SOR no. | 27.16.10 to 27.16.11 | Each | 3532.00 |

# CHAPTER - 29

# ACCESSORIES

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **29.1** | | **Pendent Holder** | Supply and fixing as per specification pendent holder of approved make including flexible cord up to 3 metre length with necessary materials complete | Each | 27.00 |
| **29.2** | | **Polycarbonate batten holder/angle holder** | Supply and fixing as per specification polycarbonate batten holder/angle holder of approved make with necessary materials complete | Each | 29.00 |
| **29.3** | | **Call bell** | Supply and fixing as per specification call bell/buzzer of approved make with necessary materials complete. | | |
| | 29.3.1 | Gong bell | Gong bell | Each | 96.00 |
| | 29.3.2 | Buzzer | Buzzer | Each | 82.00 |
| | 29.3.3 | Ding dong bell | Ding dong bell | Each | 185.00 |
| | 29.3.4 | Multi sound bell | Multi sound bell | Each | 205.00 |
| | 29.3.5 | Remote/cordless bell | Remote/cordless bell | Each | 969.00 |
| **29.4** | | **modular call bell indicator** | Supply and fixing as per specification modular call bell indicator 230Volt A.C. including reset and buzzer of approved make with necessary materials complete. | | |
| | 29.4.1 | Bell Indicator | Bell indicator with reset(1 Module) 250Volt | Each | 177.00 |
| | 29.4.2 | Buzzer modular | Buzzer modular (2 Module) | Each | 209.00 |
| **29.5** | | AUDIO CABLE | AUDIO CABLES | | |
| | 29.5.1 | **Speaker wire** | Supply and fixing of PVC insulated copper flexible speaker wire twin FRLS 1100 Volts as per IS:694-1990 of approved make | | |
| | 29.5.1.1 | Twin core 0.75 sq.mm. | (24/0.2 mm) Twin core 0.75 sq.mm. | Metre | 16.00 |
| | 29.5.1.2 | Twin core 1.00 sq.mm. | (32/0.2 mm) Twin core 1.00 sq.mm. | Metre | 20.00 |
| | 29.5.1.3 | Twin core 1.50 sq.mm. | (48/0.20 mm) Twin core 1.50 sq.mm. | Metre | 58.00 |
| | 29.5.1.4 | Twin core 2.50 sq.mm. | (80/0.20 mm) Twin core 2.50 sq.mm. | Metre | 99.00 |
| | 29.5.2 | Microphone Cable | Microphone Cable | | |
| | 29.5.2.1 | 20 AWG | Supply and fixing of 20 AWG (16/0.20mm) x 2 core ATC copper conductor, PE insulated, core twisted in a circular form, Inner sheathed with moisture Proof Polymer, braid shielded (80%) with ATC copper, outer Jacketed with Matt finish Rugged Flexible PVC | Metre | 65.00 |
| | 29.5.2.2 | 18 AWG | Supply and fixing of 18 AWG (24/0.20mm) x 2 core ATC copper conductor, PE insulated, core twisted in a circular form, Inner sheathed with moisture Proof Polymer, braid shielded (80%) with ATC copper, outer Jacketed with Matt finish Rugged Flexible PVC. | Metre | 83.00 |
| | 29.5.3 | DIGITAL SNAKE CABLE | DIGITAL SNAKE CABLE | | |
| | 29.5.3.1 | 8 Pair Digital Snake Cable | Supply and fixing 8 Pair Digital Snake Cable – 24 AWG (19/0.12mm) x 2 core ABC copper conductor, PE insulated, core twisted in a circular form, Mylar Polyester Taped, ABC Spiral shielded + Aluminum foil shielded, PVC Inner Sheathed, Matt Finish High Flex Special PVC Overall Sheathed Snake Cable | Metre | 293.00 |
| | 29.5.3.2 | 16 Pair Digital Snake Cable | Supply and fixing 16 Pair Digital Snake Cable – 24 AWG (19/0.12mm) x 2 core ABC copper conductor, PE insulated, core twisted in circular form, Mylar Polyester Taped, ABC Spiral shielded + Aluminum foil shielded, PVC Inner Sheathed, Matt Finish High Flex Special PVC Overall Sheathed Snake Cable | Metre | 571.00 |
| **29.6** | | **Multi core flexible cables** | Supply and fixing of multi core round FRLSH PVC insulated copper (flexible) conductor & PVC sheathed cables 1100Volts as per IS:694-1990 of approved make | | |
| | 29.6.1 | Three core 0.5 sq.mm. | (16/0.2 mm)Three core 0.5 sq.mm. | Metre | 20.00 |
| | 29.6.2 | Three core 0.75 sq.mm. | (24/0.2 mm) Three core 0.75 sq.mm. | Metre | 25.00 |
| | 29.6.3 | Three core 1.5 sq.mm. | (30/0.25 mm) Three core 1.5 sq.mm. | Metre | 50.00 |
| | 29.6.4 | Three core 2.5 sq.mm. | (50/0.25 mm) Three core 2.5 sq.mm. | Metre | 86.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 29.6.5 | Three core 4.0 sq.mm. | (56/0.3 mm) Three core 4.0 sq.mm. | Metre | 132.00 |
| **29.7** | | **LED lamp** | Supply and fixing in position LED lamp of approved make | | |
| | 29.7.1 | 0.5W 250Volt | 0.5 Watt250Volt of required colour lamp | Each | 50.00 |
| | 29.7.2 | 3W 250Volt | 3 Watt250Volt | Each | 80.00 |
| | 29.7.3 | 5W 250Volt | 5 Watt250Volt | Each | 98.00 |
| | 29.7.4 | 7W 250Volt | 7 Watt250Volt | Each | 108.00 |
| | 29.7.5 | 9W 250Volt | 9 Watt250Volt | Each | 110.00 |
| | 29.7.6 | 12W 250Volt | 12 Watt250Volt | Each | 191.00 |
| | 29.7.7 | 15W 250Volt | 15 Watt250Volt | Each | 223.00 |
| | 29.7.8 | 18W 250Volt | 18 Watt250Volt | Each | 303.00 |
| | 29.7.9 | 25/27W 250Volt | 25/27 Watt 250Volt | Each | 722.00 |
| | 29.7.10 | 35/40W 250Volt | 35/40 Watt 250Volt | Each | 757.00 |
| | 29.7.11 | 45W 250Volt | 45 Watt 250Volt | Each | 742.00 |
| | 29.7.12 | 50W 250Volt | 50 Watt 250Volt | Each | 1186.00 |
| **29.8** | | **Rechargeable Inverter LED Bulb** | Supply and fixing of rechargeable inverter LED Bulb, reduces to 40-50% (emergency application) with Lithium Ion battery 2600mAh, backup upto 5 hrs. having color temp upto 6500K, IP-20 rating, THD (Total harmonic distortion) <20% | | |
| | 29.8.1 | 7W, Rechargeable LED Bulb | 7W, Rechargeable Inverter LED Bulb | Each | 622.00 |
| | 29.8.2 | 9W, Rechargeable LED Bulb | 9W, Rechargeable Inverter LED Bulb | Each | 676.00 |
| **29.9** | | **Switch** | Supply and fixing as per specification Switch of approved make ISI marked with necessary material complete. | | |
| | 29.9.1 | 6A SP, (piano key type) | 6Amp SP 250 Volt deluxe (piano key type) | Each | 18.00 |
| | 29.9.2 | 16/20A, (piano key type) | 16/20Amp SP 250 Volt deluxe (piano key type) | Each | 68.00 |
| | 29.9.3 | 6A SP, Two way F/T | 6Amp SP 250 Volt two way F/T | Each | 21.00 |
| | 29.9.4 | 6/10A SP, Modular switch | 6/10Amp SP 250 Volt modular switch | Each | 48.00 |
| | 29.9.5 | 16/20A SP Modular switch | 16/20Amp SP 250 Volt modular switch | Each | 89.00 |
| | 29.9.6 | 10A SP Modular switch | 10Amp SP 250 Volt modular switch 2 way | Each | 77.00 |
| | 29.9.7 | 32 Amps, Modular switch | 32 Amps D.P.250 Volt modular switch with indicator LED 2 Module | Each | 223.00 |
| | 29.9.8 | Blank plate 1 module | Blank plate 1 module | Each | 14.00 |
| | 29.9.9 | Surface switch 6A | Surface switch 6Amp | Each | 24.00 |
| **29.10** | | **Socket** | Supply and fixing as per specification socket outlet of approved make & ISI marked with necessary material complete | | |
| | 29.10.1 | 6A, 3 Pin flush Type | 6Amp 250 Volt 3 Pin flush type | Each | 50.00 |
| | 2910.2 | 16A, 3 Pin flush Type | 16Amp 250 Volt 3 Pin flush type | Each | 54.00 |
| | 29.10.3 | 10A, 3 Pin modular type | 10Amp 250 Volt 3 Pin modular type | Each | 98.00 |
| | 29.10.4 | 10/16/20A 3 Pin modular type | 10/16/20Amp 250 Volt 3 Pin modular type | Each | 140.00 |
| **29.11** | | **6A, Bell push switch** | Supply and fixing as per specification 6Amp 250Volt bell push switch of approved make ISI marked with necessary material complete. | | |
| | 29.11.1 | Modular switch bell | 6/10Amp SP 250 Volt modular switch bell push | Each | 78.00 |
| | 29.11.2 | Flush type bell | Flush type bell push | Each | 32.00 |
| | 29.11.3 | Surface bell | Surface bell push 6A | Each | 41.00 |
| **29.12** | | **Switch/socket** | Supply and fixing as per specification switch socket combined of approved make & ISI marked with necessary material complete | | |
| | 29.12.1 | 3/6 Pin flush type | 20/10Amp 250Volt 3/6 Pin flush type | Each | 74.00 |
| **29.13** | | **Ceiling rose** | Supply and fixing as per specification polycarbonate ceiling rose 3 plate of approved make ISI marked with necessary material complete. | Each | 37.00 |
| **29.14** | | **3 Pin plug top** | Supply and fixing of approved make 3 pin plug top ISI marked | | |
| | 29.14.1 | 6A 3 Pin top | 6Amp 250Volt 3 Pin top | Each | 58.00 |
| | 29.14.2 | 16A 3 Pin top | 16Amp 250Volt 3 Pin top | Each | 92.00 |
| **29.15** | | **Danger board** | Supply and fixing as per specification caution/danger board of approved make with necessary material complete. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 29.15.1 | 200 X 150 mm, For MV/LT | Small size 200 X 150 mm for MV/LT | Each | 97.00 |
| | 29.15.2 | 250 X 200 mm, For | Large size 250 X 200 mm for | Each | 155.00 |
| **29.16** | | **Condenser 440V** | Supply and fixing of MPP condenser 440 Volt single phase of approved make as required as per specification | | |
| | 29.16.1 | 2.5 mfd 440V | Condenser 2.5 mfd 440Volt | Each | 53.00 |
| | 29.16.2 | 3.15 mfd 440V | Condenser 3.15 mfd 440Volt | Each | 61.00 |
| | 29.16.3 | 4.00 mfd 440V | Condenser 4.00 mfd 440Volt | Each | 82.00 |
| | 29.16.4 | 6.00 mfd 440V | Condenser 6.00 mfd 440Volt | Each | 90.00 |
| | 29.16.5 | 36.00 mfd 440V | Condenser 36.00 mfd 440Volt | Each | 425.00 |
| | 29.16.6 | 45.00 mfd 440V | Condenser 45.00 mfd 440Volt | Each | 456.00 |
| | 29.16.7 | 50.00 mfd 440V | Condenser 50.00 mfd 440Volt | Each | 502.00 |
| | 29.16.8 | 60.00 mfd 440V | Condenser 60.00 mfd 440Volt | Each | 552.00 |
| **29.17** | | **Condenser 250V** | Supply and fixing of MPP condenser 250Volt single phase of approved make as required as per specification | | |
| | 29.17.1 | 4 mfd 250V | Condenser 4 mfd 250Volt | Each | 56.00 |
| | 29.17.2 | 8 mfd 250V | Condenser 8 mfd 250Volt | Each | 69.00 |
| | 29.17.3 | 10 mfd 250V | Condenser 10 mfd 250Volt | Each | 100.00 |
| | 29.17.4 | 15 mfd 250V | Condenser 15 mfd 250Volt | Each | 137.00 |
| | 29.17.5 | 20 mfd 250V | Condenser 20 mfd 250Volt | Each | 153.00 |
| | 29.17.6 | 33 mfd 250V | Condenser 33 mfd 250Volt | Each | 222.00 |
| | 29.17.7 | 42 mfd 250V | Condenser 42 mfd 250Volt | Each | 273.00 |
| **29.18** | | **APP heavy duty capacitor 440V, 3 Ph.** | Supply and fixing of mixed die-electric ultra-heavy duty APP capacitor 440Volt, 3 Ph., peak inrush current upto 500 x In, over current upto 3 x In(normal current), operating losses total not more than 0.35 W/kVAR, operating life not less than 300000 hrs, switching 20000 operation/year, as per IS:13340-1993, IS:13341-1992, IS:13585-1994, of approved make as required as per specification | | |
| | 29.18.1 | 5 - 25 kVAR | 5 to 25 kVAR bank | Per kVAR | 849.00 |
| **29.19** | | **MPP H heavy duty capacitor 415V, 3 Ph.** | Supply and fixing of MPP - H heavy duty capacitor 415 Volt, 3 Ph., peak inrush current upto 250 x In, over current upto 1.8 x In(normal current), operating losses total not more than 0.35 W/kVAR, switching 8000 operation/year, as per IS:13340-1993, IS:13341-1992 of approved make as required as per specification | | |
| | 29.19.1 | 1 - 4 kVAR | 1 to 4 kVAR bank | Per kVAR | 309.00 |
| | 29.19.2 | 5 - 25 kVAR | 5 to 25 kVAR bank | Per kVAR | 325.00 |
| **29.20** | | **MH lamp, Single Ended Type** | Supply and fixing of MH lamp single ended type ISI marked of approved make as required as per specification | | |
| | 29.20.1 | 70 Watt | 70 Watt | Each | 1092.00 |
| | 29.20.2 | 150 Watt | 150 Watt | Each | 1182.00 |
| | 29.20.3 | 250 Watt | 250 Watt | Each | 1455.00 |
| | 29.20.4 | 400 Watt | 400 Watt | Each | 1523.00 |
| **29.21** | | **MH lamp, Double Ended Type** | Supply and fixing of MH lamp double ended type ISI marked of approved make as required as per specification | | |
| | 29.21.1 | 70 Watt | 70 Watt | Each | 843.00 |
| | 29.21.2 | 150 Watt | 150 Watt | Each | 891.00 |
| **29.22** | | **Ball bearing** | Supply and fixing of ball bearing of approved make as required as per specification | | |
| | 29.22.1 | 6201 | 6201 | Each | 155.00 |
| | 29.22.2 | 6202 | 6202 | Each | 165.00 |
| | 29.22.3 | 6203 | 6203 | Each | 176.00 |
| | 29.22.4 | 6204 | 6204 | Each | 207.00 |
| **29.23** | | **Fan Regulator** | Supply and fixing of approved make flush type step type electronic fan regulator including connection etc. as required on existing board | | |
| | 29.23.1 | 100 Watt | 100 Watt | Each | 169.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **29.24** | | **Electronic dimmer/regulator** | Supply and fixing of approved make electronic dimmer/regulator including connection etc. as required on existing board | | |
| | 29.24.1 | 450 Watt | 450 Watt | Each | 146.00 |
| | 29.24.2 | 650 Watt | 650 Watt | Each | 255.00 |
| **29.25** | | **Modular electronic fan regulator** | Supply and fixing of approved make step type modular electronic fan regulator including connection etc. as required on existing board | | |
| | 29.25.1 | 1 Module | 100 Watt (1 module) | Each | 315.00 |
| | 29.25.2 | 2 Module | 100 Watt(2 module) | Each | 484.00 |
| **29.26** | | **Modular electronic dimmer/regulator** | Supply and fixing of approved make step type modular electronic dimmer including connection etc. as required on existing board | | |
| | 29.26.1 | 2 Module | 650 Watt(2 module) | Each | 303.00 |
| **29.27** | | **LT panel** | Supply, fixing, testing & commissioning wall/floor mounted LT panel primer coated with powder coated paint & provided with required gasket for dust/vermin proof with degree of protection IP42 suitable for 415Volt 3 phase, 50 Hz, 4 wire system fabricated out of CRCA sheet upto 2 mm thick (1.6 mm for doors) duly compartmentalized for incomer, bus section, outgoings, cable alleys & CT, PT Ampere Metre, Volt Metre, selector switches, frequency Metre, phase indicating lamp, Energy Metre complete including cost of busbar supports, detachable cable gland plates, 2 earthing terminals, internal wiring & fixing of separately supplied MCBs, MCCBs, ACB, panel mounted changeover switch/SFUs, etc. as required but excluding cost of busbar strips, Ampere Metre, Volt Metre, selector switch as per approved design & specification | Kg | 397.00 |
| **29.28** | | **LT panel accessories** | Supply and fixing of LT panel accessories of approved make in existing LT panel including connections etc. as required as per specification. | | |
| | 29.28.1 | Digital Ampere Metre | Digital Ampere Metre with CTs with selector switch | P.SET | 2125.00 |
| | 29.28.2 | Digital Volt Metre | Digital Volt Metre with selector switch & HRC fuse | P.SET | 2398.00 |
| | 29.28.3 | Frequency Metre | Frequency Metre | Each | 1435.00 |
| | 29.28.4 | Multifunction Meter | Multifunction Meter | Each | 2587.00 |
| | 29.28.5 | Copper bus bar strips | Copper bus bar strips with PVC sleeves | Kg | 788.00 |
| | 29.28.6 | Aluminium bus bar strips | Aluminium bus bar strips with PVC sleeves | Kg | 284.00 |
| | 29.28.7 | LED lamp indicator | LED lamp indicator | Each | 255.00 |
| **29.29** | | **Capacitor Duty Contactor** | Supplying and fixing of capacitor duty contactor of approved make in existing power factor/LT panel including connections etc.as required as per specification. | | |
| | 29.29.1 | Upto 3 kVAR | Upto 3 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 1431.00 |
| | 29.29.2 | For 5 kVAR | For 5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 1495.00 |
| | 29.29.3 | For 8.5 kVAR | For 8.5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 1596.00 |
| | 29.29.4 | For 10 kVAR | For 10 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 1721.00 |
| | 29.29.5 | For 12.5 kVAR | For 12.5 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 1831.00 |
| | 29.29.6 | For 15 kVAR | For 15 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 2512.00 |
| | 29.29.7 | For 20 kVAR | For 20 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 2948.00 |
| | 29.29.8 | For 25 kVAR | For 25 KVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 3261.00 |
| | 29.29.9 | For 30 kVAR | For 30 kVAR 440V,3Ph,50Hz, Aux contact-1NO | Each | 4353.00 |
| **29.30** | | **APFC relay** | Supplying and fixing of APFC relay 1A/5 A site selectable, measurement of individual current and voltage harmonic (THD), capacitor failure indication, In-built temperature sensor of approved make in existing power factor Panel including connections etc. as required as per specification as mentioned below : | | |
| | 29.30.1 | 8 Step relay | 8 Step APFC relay (144 x144mm) | Each | 13351.00 |
| | 29.30.2 | 12 Step relay | Automatic power factor relay 12 Steps 230V 144 x 144mm | Each | 17505.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **29.31** | | **CT** | Supplying and fixing of CT 1A/5A as per site selectable busbar dimension of approved make in existing main panel for power factor Panel including connections etc. as required as per specification. | Each | 1664.00 |
| **29.32** | | **Analog time switch for automatic ON - OFF street light** | Supplying and fixing of analog time switch for automatic ON - OFF street light of approved make in existing LT panel including connections etc. as required as per specification | Each | 2689.00 |
| **29.33** | | **Contractor 2 Pole rating 240/415V** | Supplying and fixing of contractor 2 Pole rating 240/415V :for ON - OFF street light of approved make in existing LT panel including connections etc. as required as per specification | | |
| | 29.33.1 | 20A | 20A | Each | 559.00 |
| | 29.33.2 | 25A | 25A | Each | 666.00 |
| | 29.33.3 | 40A | 40A | Each | 832.00 |

# CHAPTER - 30

## FANS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **30.1** | | **Electric ceiling fan** | Supplying, erection and testing of approved make electric Ceiling fan of double ball bearing complete with standard down rod, canopy, hanging shackle, Aluminium blades, without regulator, A.C. 230-250 volts including connections with all necessary material complete of approved as required confirming to IS :374/2019 with upto date amendments. | | |
| | 30.1.1 | 900 mm | Ceiling fan - 900 mm sweep | Each | 1898.00 |
| | 30.1.2 | 1200 mm sweep | Ceiling Fan (ISI marked)-1200 mm Sweep, with minimum air delivery 210 $m^3$/min and service value ≥ 4.00 as per BEE.. | Each | 2292.00 |
| | 30.1.3 | 1400 mm sweep | Ceiling Fan ( ISI marked )-1400 mm Sweep, minimum air delivery 245 $m^3$/min and service value ≥ 4.10 as per BEE. | Each | 2540.00 |
| | 30.1.4 | 1200 mm sweep Ornamental fan | Ceiling Fan Ornamental 1200 mm Sweep, with minimum air delivery 210 $m^3$/min and service value ≥ 4.00 as per BEE. | Each | 3012.00 |
| **30.2** | | **Wall mounting fan** | Supplying, erection and testing of approved make wall mounting fan oscillating type with base, blades, guard, speed regulator etc. AC 230-250 volts with connections and including raw bolt/anchor hole fastener etc. complete finished and as required. | | |
| | 30.2.1 | 300 mm | 300 mm sweep | Each | 2478.00 |
| | 30.2.2 | 400 mm | 400 mm sweep | Each | 2810.00 |
| **30.3** | | **Cabin fan** | Supplying, erection and testing of approved make cabin fan oscillating type with base, blades, guard, speed regulator etc. AC 230-250 volts with connections and including raw bolt/anchor hole fastener etc. complete finished and as required. | | |
| | 30.3.1 | 300 mm | 300 mm sweep | Each | 2479.00 |
| **30.4** | | **Fresh air fan** | Supplying, erection and testing of approved make 'Fresh air fan' AC 230-250 volts with connection and including frame bolt/anchor hole fasteners etc. complete finished and as required. | | |
| | 30.4.1 | 150 mm | 150 mm sweep bush type | Each | 1466.00 |
| | 30.4.2 | 225 mm | 225 mm sweep bush type | Each | 1497.00 |
| | 30.4.3 | 300 mm | 300 mm sweep bush type | Each | 1698.00 |
| | 30.4.4 | 225 mm | 225 mm sweep double ball bearing | Each | 1817.00 |
| | 30.4.5 | 300 mm | 300 mm sweep double ball bearing | Each | 1973.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **30.5** | | **Deluxe fresh air fan** | Supplying and fixing of deluxe fresh air fan with louvers (ventilating fan) with self-closing louvers of decorative PVC blades mounting square frame of approved make complete with all necessary material as required | | |
| | 30.5.1 | 150 mm | 150mm RPM1300/1400 | Each | 1406.00 |
| | 30.5.2 | 200 mm | 200mm RPM1300/1400 | Each | 1654.00 |
| | 30.5.3 | 250 mm | 250mm RPM1300/1400 | Each | 1875.00 |
| | 30.5.4 | 300 mm | 300mm RPM1300/1400 | Each | 2004.00 |
| **30.6** | | **Exhaust fan heavy duty** | Supplying, erecting and testing of approved make exhaust fan heavy duty with mounting frame, blades AC 230-250 complete connection and including, frame bolt/anchor hole fasteners etc. complete finished and as required. | | |
| | 30.6.1 | 300 mm | 300mm sweep RPM 900 / 1400 | Each | 2571.00 |
| | 30.6.2 | 380 mm | 380mm sweep RPM 900 /1400 | Each | 3271.00 |
| | 30.6.3 | 450 mm | 450mm sweep RPM 900 / 1400 | Each | 4259.00 |
| | 30.6.4 | 600 mm | 600mm sweep RPM 900 | Each | 8702.00 |
| **30.7** | | **Inline fan** | Supplying, installation/fixing, and testing of approved make exhaust/fresh Air inline fan Ac 220-240V, 50 Hz size 150mm (6 inches) 23-25 watts including frame bolt/ anchor hole fastners etc. having fully copper winding motor, silent operation and delightful aesthetics complete including connections. | | |
| | 30.7.1 | 150mm (white) | 150mm (white) | Each | 1874.00 |
| | 30.7.2 | 150mm(SS Finished) | 150mm (SS Finished) | Each | 2048.00 |
| **30.8** | | **Pedestal fan** | Supplying and testing of approved make Pedestal fan A.C. 230-250 volts with blades speed regulator 'pedestal etc duly wired with all necessary material complete as required | | |
| | 30.8.1 | 400 mm | 400 mm Sweep | Each | 2511.00 |
| **30.9** | | **Air circulator fan** | Supplying, erection and testing of approved make air circulator fan A.C. 230-250 volts with blades speed regulator etc duly wired with all necessary material complete as required | | |
| | 30.9.1 | Air circulator wall mounting fan | Air circulator wall mounting fan | | |
| | 30.9.1.1 | 450 mm | 450 mm sweep | Each | 5149.00 |
| | 30.9.1.2 | 600 mm | 600 mm sweep | Each | 5888.00 |
| | 30.9.2 | Air circulator pedestal fan | Air circulator pedestal fan | | |
| | 30.9.2.1 | 450 mm | 450 mm sweep | Each | 5741.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 30.9.2.2 | 600 mm | 600 mm sweep | Each | 6651.00 |
| **30.10** | | **Cooler fan (replacement)** | Supplying and replacement of cooler fan 500 mm (20") ISI marked with all necessary material including connection complete of approved as required | Each | 1633.00 |
| **30.11** | | **Cooler pump (replacement)** | Supply and replacement of ISI marked cooler submersible water pump complete with all necessary material including connection of approved make as required | Each | 381.00 |
| **30.12** | | **M.S. angle** | Supplying and fixing of cooler/A.C. M.S. angle iron stand including flat iron supports, grouting in wall if required, welding, painting with primer and two coat of superior quality paint complete. | Kg | 92.00 |
| **30.13** | | **Down rod for ceiling fan** | Supply and fixing M.S. down rod 14 SWG (2mm) thick for ceiling fan including threading, painting, making holes etc. as required complete.<br>Note: For new works difference of installed & standard down rod length shall be paid. | Metre | 114.00 |

# CHAPTER - 31

# LUMINAIRES

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **31.1** | | **CFL Roadway lighting** | Supplying, fixing & testing of approved make of non - integral type CFL roadway lighting consisting of powder coated sheet steel housing with epoxy white powder coated CRCA sheet steel gear tray, clear acrylic cover complete with all accessories (without lamp) including fixing on existing G.I. pipe, pole with connection and all required material suitable for - | | |
| | 31.1.1 | 1 x 85W Retrofit CFL/LED | For 1 x 85W Retrofit CFL/LED (Without control gear) | Each | 1666.00 |
| **31.2** | | **Post top lantern** | Supply, fixing & testing of approved make globe type polycarbonate post top lantern, consisting of cast aluminium spigot complete with all accessories, lamp holder, (without lamp) including fixing on gate pillars/exiting G.I. pipe, pole with connection & all required materials on existing required size of G.I. pipe, suitable for CFL/LED lamp for mentioned below sizes :- | | |
| | 31.2.1 | 200 mm (8") | 200 mm (8") | Each | 569.00 |
| | 31.2.2 | 250 mm (10") | 250 mm (10") | Each | 791.00 |
| | 31.2.3 | 300 mm (12") | 300 mm (12") | Each | 971.00 |
| **31.3** | | **LED tube rod fixture** | Supply and fixing Led tube rod comprising of LED tube with non-integral/integral driver, upto 6500K color temp having 40000 burning hrs life with minimum @ L 70, system lumen output should be minimum with system efficacy> 100 lm/Watt. LED driver PF> 0.95 & THD < 20%. The colour rendering index of LED light should be more than 70. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.3.1 | Tube light LED 1 X 8/10W | Tube light LED 1 X 8/10Watt Integral i/c batten aluminium body, PC diffuser. | Each | 502.00 |
| | 31.3.2 | Tube light LED 1 X 12W | Tube light LED 1 X 12Watt Integral i/c batten aluminium body, PC diffuser. | Each | 605.00 |
| | 31.3.3 | Tube light LED 1 X 18W | Tube light LED 1 X 18Watt, Integral i/c batten aluminium body, PC diffuser. | Each | 713.00 |
| | 31.3.4 | Tube light LED 1 X 20/22W | Tube light LED 1 X 20/22Watt, Integral i/c batten aluminium body, PC diffuser. | Each | 782.00 |
| | 31.3.5 | LED retrofit glass tube | LED retrofit glass tube i/c driver only 20/22Watt,color temp 3000 - 6000K without box suitable for replacing existing 36/40Watt FTL | Each | 763.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **31.4** | | **LED Panel (recessed mounting)** | Supply and fixing of recessed mounting type Led light fixture, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF> 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.4.1 | 2' X 2', 36W, 3000-6500k | LED luminiare 2' X 2', 36Watt, color temp 3000-6500k as required. | Each | 3103.00 |
| | 31.4.2 | 2' X 2', 42/45W, 3000-6500k | LED luminiare 2' X 2', 42/45Watt, color temp 3000-6500k as required. | Each | 4215.00 |
| **31.5** | | **LED light (surface mounting)** | Supply and fixing of surface mounting type LED light fixture, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver, PF> 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.5.1 | 2' X 2', 36W, 3000-6500k | LED luminiare 2' X 2', 36Watt, color temp 3000-6500k as required. | Each | 4384.00 |
| | 31.5.2 | 2' X 2', 42/45W, 3000-6500k | LED luminiare 2' X 2', 42/45Watt, color temp 3000-6500k as required. | Each | 4472.00 |
| **31.6** | | **LED down lighter (recessed mounting)** | Supply and fixing recessed mounting LED down lighter, LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF > 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium/CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job.. | | |
| | 31.6.1 | 3W | 3 Watt, color temp 3000-6500k as required. | Each | 426.00 |
| | 31.6.2 | 5/6W | 5/6 Watt, color temp 3000-6500k as required. | Each | 719.00 |
| | 31.6.3 | 7/8/10W | 7/8/10 Watt, color temp 3000-6500k as required. | Each | 761.00 |
| | 31.6.4 | 12W | 12 Watt, color temp 3000-6500k as required. | Each | 846.00 |
| | 31.6.5 | 14/15W | 14/15 Watt, color temp 3000-6500k as required. | Each | 1150.00 |
| | 31.6.6 | 18/20W | 18/20Watt, color temp 3000-6500k as required. | Each | 1651.00 |
| | 31.6.7 | 22/24W | 22/24Watt, color temp 3000-6500k as required. | Each | 2377.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **31.7** | | **LED down lighter (surface mounting)** | Supply and fixing surface mounting LED down lighter, LED of 1 to 3Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100 lm/Watt. LED driver PF > 0.95, THD < 20% & surge protection 4KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium/CRCA powder coated frame with glare free diffused polycarbonate cover. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job.. | | |
| | 31.7.1 | 3W | 3 Watt, color temp 3000-6500k as required. | Each | 687.00 |
| | 31.7.2 | 5/6W | 5/6 Watt, color temp 3000-6500k as required. | Each | 704.00 |
| | 31.7.3 | 7/8/10W | 7/8/10 Watt, color temp 3000-6500k as required. | Each | 726.00 |
| | 31.7.4 | 14/15W | 14/15 Watt, color temp 3000-6500k as required. | Each | 1400.00 |
| | 31.7.5 | 18/20W | 18/20 Watt, color temp 3000-6500k as required. | Each | 1783.00 |
| | 31.7.6 | 21/22/24W | 21/22/24Watt, color temp 3000-6500k as required. | Each | 2809.00 |
| **31.8** | | **COB LIGHT** | Supply and fixing of recessed mounting LED swivel type COB downlighter, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 70. Housing made of pressure die cast aluminium / CRCA powder coated frame with glare free diffused polycorbonate cover. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| | 31.8.1 | COB LIGHT | 3W | Each | 1228.00 |
| | 31.8.2 | COB LIGHT | 6/7W | Each | 1229.00 |
| | 31.8.3 | COB LIGHT | 9W | Each | 1544.00 |
| | 31.8.4 | COB LIGHT | 12/15W | Each | 2022.00 |
| | 31.8.5 | COB LIGHT | 18/20W | Each | 2925.00 |
| | 31.8.6 | COB LIGHT | 22/24W | Each | 3758.00 |
| **31.9** | | **Linear profile suspended LED light** | Supply and fixing of Linear profile suspended LED lumanaire, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 80. Housing made of pressure die cast aluminium / CRCA powder coated frame with high transmission diffuser. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c mounting arrangment, connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| | 31.9.1 | | 32/40W | Each | 8712.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 31.9.2 | | 72/80W | Each | 17316.00 |
| **31.10** | | **Linear profile recessed LED light** | Supply and fixing of Linear profile suspended LED lumanaire, having color temp upto 6500K & having 20000 burning hrs life with minimum @ L 70, system lumen output should be minimum with efficacy>80lm/W. LED driver PF > 0.9, THD < 20% & surge protection 5KV. The colour rendering index of LED light should be more than 80. Housing made of pressure die cast aluminium / CRCA powder coated frame with high transmission diffuser. Submission LM 80-08 Form LED Source Manufacturer & LM79-08 / IS16106 from NABL approved lab. Manufacturer manadatory. i/c mounting arrangment, connection wire, testing etc. to complete the job. Guarantee as per tender agreement condition. | | |
| | 31.10.1 | | 32/40W | Each | 7478.00 |
| | 31.10.2 | | 72/80W | Each | 14146.00 |
| **31.11** | | **Post top lantern LED fitting** | Supply and fixing integral post top lantern LED fitting comprises of copper dust finish cast aluminium spigot and spun aluminium canopy fixed with opal polycarbonate, pipe arrangement for vertical mounting, open construction driver and accessories are wired upto terminal block. LED of 1 to 3 Watt each assembled on single MCPCB, having color temp upto 6500K & having 50000 burning hrs. life with minimum @ L 70, system lumen output should be minimum with efficacy>100lm/Watt. LED driver PF > 0.95 & surge protection 10KV. The colour rendering index of LED light should be more than 70. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.11.1 | 40W | 40 Watt LED, color temp 3000-6500k as required. | Each | 7768.00 |
| | 31.11.2 | 25W | 25 Watt LED, color temp 3000-6500k as required. | Each | 4106.00 |
| **31.12** | | **Street light with high power LED system** | Supply and fixing street light with high power LED of 3 to 6 W each on existing bracket assembled on single MCPCB, system lumens output with efficacy>120 lm/W. luminiare having color temp upto 6500K & 50000 burning hrs. life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF> 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass/Lens type, IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.12.1 | 15-20W | 15Watt to 20Watt, color temp 3000-6500k as required. | Each | 2316.00 |
| | 31.12.2 | 24-25W | 24Watt to 25Watt, color temp 3000-6500k as required. | Each | 2737.00 |
| **31.13** | | **Street light with high power LED system** | Supply and fixing street light with high power LED of 3 to 6 Watt each on existing bracket assembled on single MCPCB and additional unique peanut lens on each LED, system lumens output with efficacy>120 lm/Watt. luminiare having color temp upto 6500K & 50000 hrs. burning life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF> 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass, IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347 i/c connection wire, testing etc. to complete the job. | | |
| | 31.13.1 | 40W | 40Watt to 48Watt, color temp 3000-6500k as required. | Each | 6105.00 |
| | 31.13.2 | 60W | 60Watt, color temp 3000-6500k as required. | Each | 9791.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 31.13.3 | 90W | 90Watt, color temp 3000-6500k as required. | Each | 14173.00 |
| | 31.13.4 | 120W | 120Watt, color temp 3000-6500k as required. | Each | 21183.00 |
| | 31.13.5 | 150W | 150Watt, color temp 3000-6500k as required. | Each | 22936.00 |
| | 31.13.6 | 180W/177Watt | 180W/177Watt, color temp 3000-6500k as required. | Each | 33452.00 |
| | 31.13.7 | 250/240W | 250/240Watt, color temp 3000-6500k as required. | Each | 34766.00 |
| **31.14** | | **LED flood light** | Supplying and fixing flood light with high power LED of 3 to 6 Watt each assembled on single MCPCB and additional unique peanut lens on each LED, system lumens output with efficacy>120 lm/Watt. luminiare having color temp upto 6500K & 50000 hrs. burning life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare comprises of driver, PF > 0.95 & surge protection 10KV. Housing made of pressure die cast aluminium with heat resistant flat glass, IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.14.1 | 10W | 10Watt, color temp 3000-6500k as required. | Each | 1285.00 |
| | 31.14.2 | 30W | 30Watt, color temp 3000-6500k as required. | Each | 5178.00 |
| | 31.14.3 | 60W | 60Watt, color temp 3000-6500k as required. | Each | 7872.00 |
| | 31.14.4 | 90/100W | 90/100Watt, color temp 3000-6500k as required. | Each | 11553.00 |
| | 31.14.5 | 120W | 120Watt, color temp 3000-6500k as required. | Each | 16022.00 |
| | 31.14.6 | 150W | 150Watt, color temp 3000-6500k as required. | Each | 21572.00 |
| | 31.14.7 | 180W | 180Watt, color temp 3000-6500k as required. | Each | 27268.00 |
| | 31.14.8 | 250W | 250Watt, color temp 3000-6500k as required. | Each | 34279.00 |
| | 31.14.9 | 300W | 300Watt, color temp 3000-6500k as required. | Each | 34717.00 |
| **31.15** | | **Aviation light LED** | Supply, Installation testing and commissioning of red colour aviation light LED integral type comprising of aluminium powder coated housing with polycarbonate enclosure IP 65 on existing pipe/support including connection, testing etc. as required. Guarantee as per tender agreement condition. | | |
| | 31.15.1 | 20 watt(300 LED) | 20 watt (300 LED) | Each | 3958.00 |
| **31.16** | | **Bollard** | Supply, fixing & testing of approved make of LED integral type bollard cylindrical/square shape housing cast aluminium dome shape top cover fixed with cylindrical shape acrylic cover having base plate with holes for direct mounting complete with all accessories including preparation of foundation, fixing, connection etc. as required. LED of 1 to 3 Watt each assembled on single MCPCB, system lumens output with efficacy>90 lm/W. luminaire having color temp upto 6500K & 50000 burning hrs. life with minimum @ L 70, The colour rendering index of LED light should be more than 70. Luminiare, comprises of driver PF> 0.95 & surge protection 10KV. IP65 protection. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.16.1 | 8-10W cylindrical shape | Bollard LED 8-10 Watt cylindrical shape height > 700mm, color temp 3000-6000k | Each | 4614.00 |
| | 31.16.2 | 8-10W mushroom shape | Bollard LED 8-10 Watt mushroom shape height > 600mm, color temp 3000-6000k | Each | 4344.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **31.17** | | **LED strip** | Supplying, fixing & testing of approved make flexible LED strip comprising of 60 LED per meter W/O driver for direct mounting complete with all accessories including connection lead, fixing, connection as required. Submission LM 79-08/IS16106 (2012), IEC60598, IEC61347i/c connection wire, testing etc. to complete the job. | | |
| | 31.17.1 | 24/25W | Flexible LED strip 24Watt/25Watt - 5 M indoor cove light W/O Driver IP 20 -33 | Each | 1236.00 |
| | 31.17.2 | 72/75W | Flexible LED strip 72Watt/75Watt - 5 M indoor light W/O driver IP 20 - 33 | Each | 1859.00 |
| | 31.17.3 | 24/25W | Flexible LED strip 24Watt/25Watt - 5 M outdoor cove light W/O driver IP65 | Each | 1994.00 |
| | 31.17.4 | 72/75W | Flexible LED strip 72Watt/75Watt - 5 M outdoor light W/O driver IP65 | Each | 2805.00 |
| **31.18** | | **Power supply for LED strip** | Supplying, fixing & testing of approved make Power supply for LED strip | | |
| | 31.18.1 | 25/30W | Power supply 25Watt/30Watt for LED strip IP25/IP33 | Each | 533.00 |
| | 31.18.2 | 48W | Power supply 48Watt for LED strip IP33 | Each | 755.00 |
| | 31.18.3 | 72/75W | Power supply 72Watt/75Watt for LED strip IP25/IP33 | Each | 1020.00 |
| | 31.18.4 | 25/30W | Power supply 25Watt/30Watt for LED strip IP65 | Each | 1331.00 |
| | 31.18.5 | 48W | Power supply 48Watt for LED strip IP65 | Each | 1598.00 |
| | 31.18.6 | 72/75W | Power supply 72Watt/75Watt for LED strip IP65 | Each | 3905.00 |
| **31.19** | | **LED Bulk Head** | Supplying, fixing and testing of approved make LED bulkhead ABS body, PC diffuser including fixing on wall as required, with necessary material complete | | |
| | 31.19.1 | 6W | 6W IP 54 | Each | 1484.00 |
| | 31.19.2 | 10W | 10W IP 55 | Each | 1805.00 |
| **31.20** | | **Integral step/skirt LED light** | Supply, fixing and testing of approved make integral step/skirt LED light suitable for 3 module box and plate including fixing on wall as required, with necessary material to complete the job. Guarantee as per tender agreement condition. | Each | 676.00 |
| **31.21** | | **LED Fire signage, single side hanged** | Supplying and fixing LED fire signage i/c 6V. 9AH, lithium battery sealed 2hrs backup single side, hanged/fixed on wall or ceiling etc. as required with necessary material to complete the job. Guarantee as per tender agreement condition. | Each | 5953.00 |
| **31.22** | | **LED Fire signage, double side hanged** | Supplying and fixing LED fire signage i/c 6V. 9AH, lithium battery sealed 2hrs backup double side hanged/fixed on wall or ceiling etc. as required with necessary material to complete the job. Guarantee as per tender agreement condition. | Each | 7093.00 |
| **31.23** | | **Fixing of street light fitting** | Fixing of street light bracket of required angular shape and size pipe bracket on wall/pole including M.S. clamp not less than 40X5mm flat, nut, bolt 12mm of required length with necessary material including connection 3 core flexible wire etc. complete as required excluding cost of pipe paid separately. | Each | 498.00 |
| **31.24** | | **Panel LED driver** | Supplying Fixing and testing of approved and suitable make LED driver for an existing LED luminary panel. | | |
| | 31.24.1 | 1'x1' 24W | LED Luminiare 1'x1', 24W Panel Driver | Each | 558.00 |
| | 31.24.2 | 2'x2', 36W | LED Luminiare 2'x2', 36W Panel Driver | Each | 670.00 |
| **31.25** | | **Street Light LED Driver** | Supplying fixing and testing of approved and suitable make LED driver for an existing LED street light. | | |
| | 31.25.1 | 24-25W | 24Watt to 25Watt, color temp 3000-6500k | Each | 537.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 31.25.2 | 40W | 40Watt to 48Watt, color temp 3000-6500k | Each | 903.00 |
| | 31.25.3 | 60W | 60Watt, color temp 3000-6500k | Each | 1135.00 |
| | 31.25.4 | 90W | 90Watt, color temp 3000-6500k | Each | 1683.00 |
| | 31.25.5 | 120W | 120Watt, color temp 3000-6500k | Each | 2279.00 |
| | 31.25.6 | 150W | 150Watt, color temp 3000-6500k | Each | 2582.00 |
| **31.26** | | **Flood Light LED Driver** | Supplying, fixing and testing of approved and suitable make LED driver for an existing LED Flood light. | | |
| | 31.26.1 | 60W | 60Watt, color temp 3000-6500k | Each | 808.00 |
| | 31.26.2 | 90W | 90Watt, color temp 3000-6500k | Each | 1420.00 |
| | 31.26.3 | 120W | 120Watt, color temp 3000-6500k | Each | 1546.00 |
| | 31.26.4 | 150W | 150Watt, color temp 3000-6500k | Each | 2046.00 |
| **31.27** | | **Decorative Street Light** | Supply & Fixing of Decorative LED Street Light of 1 to 3 watt with each , system lumens output with Efficacy>=135lm/W(@6500k), It's body made of Die cast aluminium using pressure die-casting process. Hole for coupling on pipes Ø 60 mm, for lateral installation. Silicon gasket enusures the IP protection. IP Rating - IP66. Power LED Module with high efficacy LED's on single Printed Circuit Board with metal core plate. Colour Rendering Index: Ra > 70. Electronic Power Supply for LED Module, which offers Protection against Short Circuit, Over- Voltage & Over- Current, with in-built surge protection upto 4kv (an additional surge protection device of 10kV also provided in-built the fixture). PF >0.95 and THD<10%. Lens cover in PC (IK09). Estimated LED Lifetime is L70 @ 1,00,000 hours minimum. LM79-80 report from LED manufaturer should be submitted. Safety test report as per IEC 60598-2-3; IEC 60598-1 / IS 10322 IEC 61347 i/c connection wire , testing etc-Supporting Test Report from NABL approved lab, with 5 year on site warranty. | | |
| | 31.27.1 | 40 W | 40Watt, Color Temp 3000-6500k as required | Each | 6578.00 |
| | 31.27.2 | 60 W | 60Watt, Color Temp 3000-6500k as required | Each | 9927.00 |
| | 31.27.3 | 80/90 W | 80/90Watt, Color Temp 3000-6500k as required | Each | 14113.00 |
| | 31.27.4 | 100 W | 100Watt, Color Temp 3000-6500k as required | Each | 15548.00 |
| | 31.27.5 | 120W | 120Watt, Color Temp 3000-6500k as required | Each | 21289.00 |
| | 31.27.6 | 150 W | 150Watt, Color Temp 3000-6500k as required | Each | 23083.00 |
| | 31.27.7 | 180 W | 180Watt, Color Temp 3000-6500k as required | Each | 29900.00 |
| | 31.27.8 | 200 W | 200Watt, Color Temp 3000-6500k as required | Each | 30259.00 |
| | 31.27.9 | 250 W | 250Watt, Color Temp 3000-6500k as required | Each | 34684.00 |

# CHAPTER - 32

# MISCELLANEOUS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **32.1** | | **Fixing of fluorescent tube on wall ceiling** | Fixing of fluorescent tube fitting on wall-ceiling on T.W. /PVC block with necessary material including connection 3 core flexible wire etc. complete as required. | Each | 100.00 |
| **32.2** | | **Fixing of fluorescent tube with pendent** | Fixing of fluorescent tube fitting with pendent arrangement of two pieces of approved chrome plated metallic chain with necessary material including connection 3 core flexible wire etc. complete as required. | | |
| | 32.2.1 | Pendent upto1.5 Metre | For pendent length upto 1.5 Metre | Each | 162.00 |
| | 32.2.2 | Excess of 1.5 Metre | Add additional pendent length in excess of 1.5 Metre | Metre | 99.00 |
| **32.3** | | **Fixing for ceiling fan clamp** | Providing and fixing circular/hexagonal cast iron or M.S. sheet box for ceiling fan clamp of internal dia 140mm, 73mm height, top lid of 1.5mm thick M.S. sheet with its top surface hacked for proper bonding, top lid shall be screwed into the cast iron/M.S. sheet box by mean of 3.3mm dia. round headed screws, one lock at the corners. Clamp shall be made of 12mm dia M.S. bar bent to shape as per standard drawing. | Each | 90.00 |
| **32.4** | | **Fixing fan clamp of 12 mm dia steel bar** | Supplying and fixing fan clamp of 12 mm dia steel bar as per specification complete. | | |
| | 32.4.1 | For stone slab | For stone slab roof "J" Type hook | Each | 58.00 |
| | 32.4.2 | For RCC slab/beam | For R.C.C. slabs/beams | Each | 58.00 |
| | 32.4.3 | Existing RCC slab/beam | For existing R.C.C. slabs/beams | Each | 151.00 |
| **32.5** | | **Fan clamp on existing T.W. /M.S. structure beam** | Supplying and fixing fan clamp with hook on existing T.W./M.S. structure beam trusses with required size flat iron etc. as per specification complete :- | Each | 146.00 |
| **32.6** | | **Erection of fan** | Erection of ceiling fan in position with regulator on existing switch board including wiring, testing etc. with necessary material complete as required. | Each | 95.00 |
| **32.7** | | **Erection of fan regulator** | Erection of fan regulator in position with necessary material connection etc. complete as required. | Each | 37.00 |
| **32.8** | | **Erection of exhaust fan** | Erection of exhaust fan in position with required PVC sheathed cable, frame bolts etc. complete including connections. | Each | 171.00 |
| **32.9** | | **Numbering of ceiling fan** | Numbering of ceiling fans or fitting with approved paint including all labour material etc. as required. | | |
| | 32.9.1 | Letter upto 5 CM | Letter upto 5 CM height | Each | 8.00 |
| | 32.9.2 | Above 7CM to 15 CM | Letter above 7 CM but upto 15 CM height | Each | 11.00 |
| **32.10** | | **Fixing of T.W. board** | Providing and fixing T.W. board of suitable size (maximum 12" X 18") for fixing of house service Metre with necessary materials and connection testing etc. complete as per specification : | | |
| | 32.10.1 | 1 Ph Metre | Single phase Metre | Each | 151.00 |
| | 32.10.2 | 3 Ph Metre | Three phase Metre | Each | 195.00 |
| **32.11** | | **Fixing of wood wool/khas pads** | Providing & fixing of wood wool/khas pads of normal size desert/room cooler panels set of 3 nos including fixing wire mesh on pads. | Each | 189.00 |
| **32.12** | | **Servicing of cooler fan motor, pump** | Servicing & overhauling of cooler fan motor, pump including replacement of distribution, delivery pipes, PVC elbows, painting etc. as reqd. complete with necessary material & labour. | Each | 291.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **32.13** | | **Rewinding of fans** | Rewinding of fans by copper enameled wire of suitable gauge including all required material, insulation, varnishing, connection leads, oiling .greasing etc complete with testing including dismantling & refitting of fan at site. Scrap old material is not returnable to the department. | | |
| | 32.13.1 | Ceiling(old)/exhaust fan | Ceiling fan 1400 mm sweep old model/exhaust fan 450 mm sweep | Each | 1064.00 |
| | 32.13.2 | Ceiling(new)/exhaust fan | Ceiling fan 1400 mm sweep new model/exhaust fan 380 mm sweep | Each | 685.00 |
| | 32.13.3 | Ceiling(old)/exhaust fan | Ceiling fan 1200 mm sweep old model/cooler fan 500 mm (20")sweep | Each | 807.00 |
| | 32.13.4 | Ceiling(new)/exhaust fan | Ceiling fan 1200 mm sweep new model/exhaust fan 300 mm sweep | Each | 637.00 |
| | 32.13.5 | Fresh air fan/wall bracket/cabin fan | Fresh air fan up to 300 mm sweep/wall bracket/cabin fan 400 mm sweep | Each | 507.00 |
| **32.14** | | **Servicing of fans** | Cleaning, overhauling, servicing of ceiling/exhaust/wall bracket fan by applying grease & oil, including testing with mounting/hanging arrangement including replacement of safety pins, check nuts, lock nut, nut bolts etc. making perfect alignment & refitting at site as required. | Each | 166.00 |
| **32.15** | | **GI Sheet 22 SWG exhaust fan cowl** | Supply and fixing of GI sheet 22 SWG exhaust fan cowl including grouting bolt, nut, & finishing etc on existing hole to complete the job. | sq. Metre | 1700.00 |
| **32.16** | | **GI sheet 24 SWG exhaust fan cowl** | Supply and fixing of GI sheet 24 SWG exhaust fan louvers including grouting bolt, nut, & finishing etc. on existing hole to complete the job as mentioned below :- | | |
| | 32.16.1 | 300mm | Sweep 300mm | Each | 953.00 |
| | 32.16.2 | 380mm | Sweep 380mm | Each | 1369.00 |
| | 32.16.3 | 450mm | Sweep 450mm | Each | 2069.00 |
| | 32.16.4 | 600mm | Sweep 600mm | Each | 2745.00 |
| **32.17** | | **Elect. Sanitary napkin incinerator** | Supply, Installation, Testing and Commissioning of Electrical Napkin Incinerator Wall mounted factory fabricated with CRCA sheet steel of 1.2mm thickness duly powder coated suitable for 230V 50Hz AC Supply with auto cut-off feature having insulation of thermal ceramic sheet with glass wool of 4-5mm thick with MCB protection & aprox.2 meter flexible flue gas pipe with clamp. | | |
| | 32.17.1 | 30 Per Day | Sanitary Napkin Incinerator( Capacity 30 per Day) | Each | 30874.00 |
| | 32.17.2 | 50 Per Day | Sanitary Napkin Incinerator( Capacity 50 per Day) | Each | 38666.00 |
| | 32.17.3 | 100 Per Day | Sanitary Napkin Incinerator( Capacity 100 per Day) | Each | 45666.00 |
| | 32.17.4 | 200 Per Day | Sanitary Napkin Incinerator( Capacity 200 per Day) | Each | 54270.00 |

# CHAPTER - 33

## LABOUR RATES FOR WIRING

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **33.1** | | **Point wiring for surface rigid steel** | Labor rate for point wiring a light/fan/twin control light/socket out let on separate board/call bell point in (open) surface rigid steel conduit system. | Each | 261.00 |
| **33.2** | | **Point wiring for recessed/concealed rigid steel** | Labor rate for point wiring a light fan/twin control light/socket out let on separate board/call bell point in recessed/concealed rigid steel conduit system. | Each | 330.00 |
| **33.3** | | **Point wiring for surface rigid PVC conduit/casing and capping system.** | Labour rate for point wiring a light/fan/twin control light/socket out let on separate board/call bell point in (open) surface rigid PVC conduit/casing and capping system. | Each | 166.00 |
| **33.4** | | **Point wiring for surface rigid PVC conduit/casing and capping system.** | Labour rate for point wiring a light/fan/twin control light/socket out let on separate board/call bell point in recessed/concealed rigid PVC conduit system. | Each | 233.00 |
| **33.5** | | **Point wiring for surface rigid steel/PVC conduit/casing and capping** | Labour rate for point wiring a power point in surface rigid steel/PVC conduit/casing and capping system including labour for fixing earth continuity conductor. | Each | 383.00 |
| **33.6** | | **Power point wiring for recessed/concealed rigid steel/PVC conduit** | Labour rate for point wiring a power point in recessed/concealed rigid steel/PVC conduit system including labour for fixing earth continuity conductor. | Each | 473.00 |
| **33.7** | | **Fixing rigid steel/PVC conduit** | Labour rate for fixing rigid steel/PVC conduit along with accessories in surface including painting etc. as required. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 33.7.1 | 20mm | 20 mm conduit | Meter | 10.00 |
| | 33.7.2 | 25mm & 32mm | 25 mm & 32 mm conduit | Meter | 14.00 |
| | 33.7.3 | 40mm & 50mm | 40 mm & 50 mm conduit | Meter | 20.00 |
| **33.8** | | **Rewiring in surface conduit** | Labor rate for rewiring a light/fan/twin control light/socket out let on separate board/call bell point in rigid steel/rigid PVC nonmetallic conduit/PVC casing and capping in (open) surface conduit wiring system. | Each | 90.00 |
| **33.9** | | **Rewiring in recessed/concealed conduit** | Labor rate for rewiring a light/fan/twin control light/socket out let on separate board/call bell point in rigid steel/rigid PVC nonmetallic conduit in recessed/concealed conduit wiring system. | Each | 72.00 |
| **33.10** | | **Fixing rigid steel/PVC conduit** | Labour rate for fixing rigid steel/PVC conduit with accessories as per specification in concealed as required. | | |
| | 33.10.1 | 20mm | 20 mm conduit | Meter | 13.00 |
| | 33.10.2 | 25mm & 32mm | 25 mm & 32 mm conduit | Meter | 16.00 |
| | 33.10.3 | 40mm & 50mm | 40 mm & 50 mm conduit | Meter | 23.00 |
| **33.11** | | **Drawing cables in existing rigid steel conduit** | Labour rate for drawing cables in existing rigid steel conduit (surface or recessed) | | |
| | 33.11.1 | 2 Wires | "A" 2 Wires -- | | |
| | 33.11.1.1 | 2.5/4/6 sq mm | 2.5 sq. mm./4 sq. mm./6 sq. mm. | Meter | 2.00 |
| | 33.11.1.2 | 10/16/25 sq mm | 10 sq. mm./16 sq. mm./25 sq. mm. | Meter | 12.00 |
| | 33.11.1.3 | 35 sq mm | 35 sq. mm. | Meter | 15.00 |
| | 33.11.2 | 3 & 4 Wires | **"B" 3 Wires & 4 wires** | | |
| | 33.11.2.1 | 2.5/4/6 sq mm | 2.5 sq. mm./4 sq. mm./6 sq. mm. | Meter | 5.00 |
| | 33.11.2.2 | 10/16/25 sq mm | 10 sq. mm./16 sq. mm./25 sq. mm. | Meter | 15.00 |
| | 33.11.2.3 | 35 sq mm | 35 sq. mm. | Meter | 18.00 |

**NOTE**

1. The socket out let point on the same board shall be treated as 1/2 point in the respective system of wiring

2. Labor rate for point are inclusive of labour for fitting all accessories and fitting such as switch, ceiling rose sockets, brass lamp holders etc. as required

3. Labor rate for each circuit shall be same as one point in respective system of wiring.

# CHAPTER - 34

# PAINTING

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **34.1** | | **Painting of ceiling fan** | Painting of ceiling fan including making surface clean and spray painting of fan Motor, blades, etc. with two coat of required shade complete with material and labour including dismantling and errection, preparation of surface | Each | 278.00 |
| **34.2** | | **Painting of desert cooler** | Painting of desert cooler including making surface even, prima paint coat from inside & outside and spray painting of body, fan Motor, blades, etc. with two coat of required shade complete with material and labour. | Each | 508.00 |
| **34.3** | | **Painting of water cooler** | Painting of water cooler including scraping of old paint making surface even, prima paint coat and spray painting of body, etc. with two coat of required shade complete with material and labour. | Each | 726.00 |
| **34.4** | | **Painting of poles** | Painting of poles | | |
| | 34.4.1 | Removing of old paint | Removing of old paint from steel and other metallic surface and making surface even. | Per sq meter | 57.00 |
| | 34.4.2 | Prima coat with red oxide zinc chromide paint | Prima coat over new steel and other metallic surface with red oxide zinc chromide paint complete with supply of material labour etc. including preparation of surface. | Per sq meter | 76.00 |
| | 34.4.3 | Prima coat with aluminium paint | Prima coat with aluminium paint, brushing complete with material and labour including preparation of surface | Per sq meter | 96.00 |

# CHAPTER - 35

## DISMANTLING OF CIVIL & ELECTRICAL WORKS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **35.1** | | **Dismantling of point/circuit wiring accessories** | Dismantling of existing light/fan/ bell/socket out let point on separate board/wiring and circuit with insulated wire in casing/conduit/ batten complete with fitting and accessories | Per point | 16.00 |
| **35.2** | | **Making the site clear of the dismantled installation** | Making the site clear of the dismantled installation by refilling the hole with cement mortar & finished with paint gitties and replacing to match the colour& wall ceiling. | Per point | 27.00 |
| **35.3** | | **Dismantling of existing socket** | Dismantling of existing socket outlet point on same board in any system of wiring | Per point | 8.00 |
| **35.4** | | **Dismantling the existing armoured/unarmoured cable** | Dismantling the existing armoured/unarmoured cable or 2/3/4 wires of submain, main, in any system of wiring including recoiling. | Metre | 8.00 |
| **35.5** | | **Dismantling of existing underground cable** | Dismantling of existing underground cable safely with its accessories and refilling the trench to ground level including recoiling of dismantled cable. | Metre | 29.00 |
| **35.6** | | **Dismantling of boards or angle/flat and making site clear** | Dismantling of D.F. board, D.P. switch, T.P. & T.P.N. switch or D.B. of any size complete with board or angle/flat iron frame and making site clear including the refilling hole & repainting to match colour of wall. | Each | 30.00 |
| **35.7** | | **Dismantling of boards/switches** | Dismantling of D.F. board, D.P. switch, T.P. & T.P.N. switch or D.B. of any size only | Each | 10.00 |
| **35.8** | | **Dismantling of fans** | Dismantling the existing ceiling/exhaust/cabin fan with complete accessories and making the site clear. | Each | 37.00 |
| **35.9** | | **Dismantling of poles** | Dismantling of existing steel pole, R.S. Joist, girder with bracket clamp insulator or stay from the cement concrete foundation and making the site clear by refilling the pit to ground level i/c transportation to required destination within township. | Each | 953.00 |
| **35.10** | | **Dismantling of overhead conductor** | Dismantling of existing overhead ACSR, AAAC, GI, conductor and making the site clear i/c transportation to required destination within township. | Metre | 3.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **35.11** | | **Dismantling of water tight fluorescent tube** | Dismantling of existing water tight fluorescent tube fitting with making the site clear. | Each | 20.00 |
| **35.12** | | **Dismantling of water tight fittings S.V./M.H./L.E.D. lamp** | Dismantling of existing water tight fitting S.V/M.H/L.E.D lamp fitting with extension bracket, clamp from pole or wall with making the site clear. | Each | 35.00 |
| **35.13** | | **Dismantling of geyser or storage type water heater** | Dismantling of existing geyser or storage type water heater and making the site good by colour wash etc. as required. | Each | 41.00 |
| **35.14** | | **Civil work for exhaust fan** | Providing recess in brick wall suitable for erecting of exhaust fan up to 450 sweep complete with grouting of nut bolts, plastering and colour wash to match the colour of wall. | | |
| | 35.14.1 | Upto 9" | For wall thickness upto 9" | Each | 404.00 |
| | 35.14.2 | Upto 15" | For wall thickness upto 15" | Each | 560.00 |
| | 35.14.3 | Stone wall thickness upto 15" | For the stone wall thickness upto 15" | Each | 793.00 |

# CHAPTER - 36

# CIVIL WORKS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| 36 | **Note:** Items regarding civil works shall be taken and executed on the basis of rates of respective current civil MPPWD S.O.R. | | | | |

# CHAPTER - 37

# EARTHING AND LOOP EARTHING

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **37.1** | | **With G.I. earth pipe** | Earthing with G.I. earth pipe 4.5 metre long and 40 mm dia with masonry encloser in cement mortar, cover plate having locking arrangement on the top etc. (but without charcoal or coke and salt) complete as required | Each | 3312.00 |
| **37.2** | | **Add extra for using salt and charcoal/coke for pipe earth electrode** | Add extra for using salt and charcoal/coke for pipe earth electrode as required including excavation & refilling | | |
| | 37.2.1 | Excavation | Excavation 4.5 m by manual labour | Each | 1942.00 |
| | 37.2.2 | Excavation | Excavation by making hole with auger | Each | 1622.00 |
| **37.3** | | **Earthing with G.I. earth plate 600x600x6 mm** | Earthing with G.I. earth plate 600mm X 600mm X 6mm thick including accessories and providing masonry encloser in cement mortar, cover plate having locking arrangement on the top and G.I. watering pipe 20mm dia 2.7 metre long etc. (but without charcoal or coke and salt) complete as required | Each | 7487.00 |
| **37.4** | | **Earthing with copper earth plate 600x600x3 mm** | Earthing with copper earth plate 600mm X 600mm X 3mm thick including accessories and providing masonry encloser in cement mortar, cover plate having locking arrangement on the top and G.I. watering pipe 20mm dia 2.7 metre long etc. (but without charcoal or coke and salt) complete as required | Each | 16341.00 |
| **37.5** | | **Add extra for using salt and charcoal/ coke for G.I /copper plate (murrum)** | Add extra for using salt and charcoal/coke for G.I. plate or copper plate earth electrode as required including excavation and refilling the pit in all type of soil, murrum | Each | 1655.00 |
| | 37.5.1 | Add extra for using salt and charcoal/coke for G.I/copper plate (black soil) | Add extra for using salt and charcoal/coke for G.I. plate or copper plate earth electrode as required including excavation in hard rock and refilling the pit with black soil. | Each | 2783.00 |
| **37.6** | | **GI wire conductor earth electrode** | Supply and laying 4.0mm (8 SWG) dia. G.I. wire at 0.5 metre below ground level as conductor earth electrode including jointing etc. as required. | Metre | 21.00 |
| **37.7** | | **Cu wire conductor earth electrode** | Supply and laying 4mm dia. copper wire at 0.5 metre below ground level as conductor earth electrode including soldering etc. as required. | Metre | 114.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **37.8** | | **G.I. wire conductor earth electrode** | Supply and laying 4.87mm 6 SWG G.I. wire at 0.5 metre below ground level as conductor earth electrode including jointing etc. as required. | Metre | 25.00 |
| **37.9** | | **25x5 mm G.I. strip** | Supply and laying 25mm X 5mm G.I. strip at 0.5 metre below ground level as strip earth electrode including jointing etc. as required. | Metre | 130.00 |
| **37.10** | | **25x6 mm G.I. strip** | Supply and laying 25mm X 6mm G.I. strip at 0.5 metre below ground level/surface as strip earth electrode including jointing etc. as required. | Metre | 147.00 |
| **37.11** | | **25x5 mm Cu strip** | Providing and fixing 25mm X 5mm copper strip in 40 mm dia G.I. Pipe from earth electrode as required | Metre | 1490.00 |
| **37.12** | | **25x5 G.I. strip** | Providing and fixing 25mm X 5mm G.I. strip in 40 mm dia G.I. pipe from earth electrode as required | Metre | 928.00 |
| **37.13** | | **Earth connections from earth electrode with 4.00 mm dia G.I. wire** | Providing and laying earth connections from earth electrode with 4.00 mm dia G.I. wire in 15mm dia G.I. Pipe from earth electrode as required. | Metre | 222.00 |
| **37.14** | | **Earth connections from earth electrode with 4.00 mm dia cu. wire** | Providing and laying earth connections from earth electrode with 4.00 mm dia copper wire in 15mm dia G.I. Pipe from earth electrode as required. | Metre | 308.00 |
| **37.15** | | **25x5 mm Cu strip on surface** | Providing and fixing 25mm X 5mm copper strip on surface or in recessed for connection etc. as required | Metre | 1137.00 |
| **37.16** | | **25x5 mm G.I. strip on surface** | Providing and fixing 25mm X 5mm G.I. strip on surface or in recessed for connection etc. as required | Metre | 130.00 |
| **37.17** | | **6 SWG dia G.I. wire on surface/recessed** | Providing and fixing 4.0mm (8 SWG) dia G.I. wire on surface or in recessed for loop earthing as required | Metre | 25.00 |
| **37.18** | | **Loop earthing** | Providing and fixing 4.87mm (6 SWG) dia. G.I. wire on surface or in recessed for loop earthing as required | Metre | 30.00 |
| **37.19** | | **4 mm cu. wire on surface/recessed** | Providing and fixing 4.00mm copper wire on surface or in recessed for loop earthing as required | Metre | 127.00 |
| **37.20** | | **50x6 mm G.I. strip earth electrode** | Supply and laying 50mm X 6mm G.I. strip at 0.5 metre below ground level/surface as strip earth electrode including jointing etc. as required. | Metre | 319.00 |
| **37.21** | | **Lightning conductor** | **Lightning conductor** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 37.21.1 | **18mm dia 500mm long bright nickel plated copper tube** | Providing and fixing of **lightning conductor** finial made of 18mm dia 500mm long bright nickel plated **copper tube** tapered having single prong at top with bright nickel plated brass base M10 for connection and intersection of flat or round conductor including holes, clamp complete as required. | Each | 3947.00 |
| **37.22** | | **18mm dia 500mm long bright nickel plated GI tube** | Providing and fixing of **lightning conductor** finial made of 18mm dia 500mm long bright nickel plated **GI tube** tapered having single prong at top with bright nickel plated GI base M10 for connection and intersection of flat or round conductor including holes, clamp complete as required. | Each | 1285.00 |
| **37.23** | | **Fixing of lightning conductor, 25 mm dia 300 mm long cu./G.I. tube** | Fixing of lightning conductor finial made of 25mm dia 300mm long copper tube/G.I. tube having single prong at top with base plate etc. complete as required. | Each | 535.00 |
| **37.24** | | **Riveting/sweating and soldering** | Riveting/sweating and soldering of copper/G.I. Tape (with another copper/G.I. Tape, base of finial or any other metallic object) as required | Each | 107.00 |
| **37.25** | | **Cu. tape 20x3 mm (for horizontal runs)** | Providing and fixing copper tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for horizontal runs) | Metre | 567.00 |
| **37.26** | | **Cu. tape 20x3 mm (for vertical runs)** | Providing and fixing copper tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for vertical runs) | Metre | 591.00 |
| **37.27** | | **G.I. tape 20x3 mm (for horizontal runs)** | Providing and fixing G.I. tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for horizontal runs) | Metre | 74.00 |
| **37.28** | | **G.I. tape 20x3 mm (for vertical runs)** | Providing and fixing G.I. tape 20mm X 3mm thick on parapet and surface of wall of lightning conductor as required (for vertical runs) | Metre | 85.00 |
| **37.29** | | **Fixing cu./G.I. tape 20x3 mm (for horizontal runs)** | Fixing of copper/G.I. tape 20mm X 3mm thick on parapet and surface of wall for lightning conductor as required (for horizontal runs) | Metre | 23.00 |
| **37.30** | | **Fixing cu./G.I. tape 20x3 mm (for vertical runs)** | Fixing of copper/G.I. tape 20mm X 3mm thick on parapet and surface of wall for lightning conductor as required (for vertical runs) | Metre | 27.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **37.31** | | **Testing joint 20x3 mm cu. strip** | Providing and fixing "Testing Joint" made by 20mm X 3mm thick copper strip 125mm long with 4 Nos of tinned Brass bolts, Nut, check nuts and spring washers etc. complete as required. | Each | 525.00 |
| **37.32** | | **Cu. tape 32x6 mm** | Providing and laying copper tape 32mm X 6mm thick for earth electrode directly in ground as required | Metre | 1686.00 |
| **37.33** | | **G.I. tape 38x6 mm** | Providing and laying G.I. tape 38mm X 6mm thick for earth electrode directly in ground as required | Metre | 222.00 |
| **37.34** | | **Laying cu./G.I. tape 32x6mm** | Laying of copper/G.I. tape 32mm X 6mm thick for earth electrode directly in ground as required | Metre | 23.00 |
| **37.35** | | **Earthing** | **Earthing For Domestic Electrical Installation** | | |
| | 37.35.1 | **Earthing 16 mm dia 2.5 metre long galvanized iron or steel rod electrode** | Providing and fixing earthing arrangement with 16mm dia 2.5 metre long **galvanized iron or steel rod** electrode including packing of charcoal powder and salt as per specification watering pipe 19mm dia G.I. connection, cover plate having locking arrangement on the top etc. complete with refilling the pit as required, including excavation of earth pit in soil/ soft rock. | Each | 3318.00 |
| | 37.35.2 | **Earthing 38.1 mm dia 2.5 metre long galvanized iron pipe electrode** | Providing and fixing earthing arrangement with 38.1mm dia 2.5 metre long **galvanized iron pipe** electrode including packing of charcoal powder and salt as per specification watering pipe 19mm dia G.I. connection, cover plate having locking arrangement on the top etc. complete with refilling the pit as required, including excavation of earth pit in soil / soft rock | Each | 3670.00 |
| **37.36** | | **G.I. earth lead** | Supply and laying G.I. earth lead duly installed on wall or in G.I. Pipe including connection etc. complete as required. | | |
| | 37.36.1 | 13 sq. mm. | 13 sq. mm. ( 4 mm dia ) | Metre | 26.00 |
| | 37.36.2 | 25 sq. mm. | 25 sq. mm. ( 5.6 mm dia ) | Metre | 38.00 |
| | 37.36.3 | 40 sq. mm. | 40 sq. mm. ( 7.1 mm dia ) | Metre | 48.00 |
| **37.37** | | **G.I. earth conductor** | **G.I. or steel earth continuity conductor** | | |
| | 37.37.1 | 6 sq. mm. | 6 sq. mm. | Metre | 22.00 |
| | 37.37.2 | 10 sq. mm. | 10 sq. mm. | Metre | 29.00 |
| **37.38** | | **Earth conductor of 4 sq. mm. al.** | Providing and laying earth continuity conductor of 4 sq. mm. aluminium along the run of final sub circuit in PVC conduit in an approved manner. | Metre | 28.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **37.39** | | **PVC insulated cu. cable earth continuity conductor** | Supply and drawing PVC insulated copper cable (earth continuity conductor) of green colour FRLS with copper multi strand conductor ISI mark in existing conduit along with other wires as required as per specification : | | |
| | 37.39.1 | 4 sq. mm. | 4 sq. mm. | Metre | 52.00 |
| | 37.39.2 | 6 sq. mm. | 6 sq. mm. | Metre | 71.00 |
| | 37.39.3 | 10 sq. mm. | 10 sq. mm. | Metre | 102.00 |
| | 37.39.4 | 16 sq. mm. | 16 sq. mm. | Metre | 156.00 |
| | 37.39.5 | 25 sq. mm. | 25 sq. mm. | Metre | 234.00 |
| **37.40** | | **Safe earthing electrode consisting pipe-in-pipe technology** | Supply and erection of approved make safe earthing electrode consisting pipe-in-pipe technology as per IS 3043-1987 made of corrosion free G.I. pipes with constant ohmic value surrounded by highly conductive compound with high charge dissipation suitable for effective and maintenance free earthing in earth pit of minimum bore dia. 150mm size, as mentioned below : | | |
| | 37.40.1 | 2 Metre G.I. pipe, outer dia 50 mm | With 2 metre G.I. pipes having outer pipe dia of 50mm having 80-200 micron galvanizing, Inner pipe dia. of 25mm having 200-250 micron galvanizing, connection terminal dia. of 12mm in nominal soil with 25 kg (one Bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | Each | 8972.00 |
| | 37.40.2 | 3 Metre G.I. pipe, outer dia 50 mm | With 3 metre G.I. pipes having outer pipe dia of 50mm having 80-200 micron galvanizing, Inner pipe dia. of 25mm having 200-250 Micron galvanizing, connection terminal dia. of 12mm in nominal soil with 50 kg (two bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | Each | 13140.00 |
| | 37.40.3 | 2 Metre pipe, outer dia 80 mm | With 2 metre pipe of 80mm outer dia., 40mm inner dia and 14mm terminal dia. In soft/hard rock/marshy soil with 25 kgs. (one bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | Each | 18262.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 37.40.4 | 3 Metre pipe, outer dia 80 mm | With 3 metre pipe of 80mm outer dia., 40mm inner dia and 14mm terminal dia. In soft/hard rock/marshy soil with 50 kgs. (two bag) back filling bentonite compound UL listed and confirm to relevant International/Indian standards around electrode upto ground level. | Each | 21419.00 |
| **37.41** | | **Back filling bentonite compound** | Providing back filling bentonite compound suitable for safe earthing device. The product should be UL listed and confirm to relevant International/Indian standards. | Kg | 134.00 |
| **37.42** | | **Polycarbonate heavy duty pit cover** | Supply and installation of polycarbonate heavy duty pit cover tested for 5000Kg SWL for earth electrode at ground level size 300x300x155 mm with lockable lid. | Each | 1501.00 |

# CHAPTER - 38

# TELEPHONE, LOCAL AREA NETWORK (LAN), TV CABLE

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **38.1** | | **Co-axial T.V. cable** | Supplying and drawing co-axial T.V. cable RG-6 Grade, 0.7 mm solid copper conductor PE insulated, shielded with fine tined copper braid and protected with PVC sheathed in existing surface/concealed, steel/PVC conduit as required. | Metre | 30.00 |
| **38.2** | | **Telephone cable, 0.5 mm PVC insulated cu. conductor** | Supplying and drawing following pair, 0.5 mm PVC insulated copper conductor unarmoured telephone cable in existing surface/concealed, steel/PVC conduit as required. | | |
| | 38.2.1 | 0.5 mm, 1 Pair | Telephone wire 0.5 mm, 1 Pair | Metre | 19.00 |
| | 38.2.2 | 0.5 mm, 2 Pair | Telephone wire 0.5 mm, 2 Pair | Metre | 36.00 |
| | 38.2.3 | 0.5 mm, 4 Pair | Telephone wire 0.5 mm, 4 Pair | Metre | 48.00 |
| | 38.2.4 | 0.5 mm, 10 Pair | Telephone wire 0.5 mm, 10 Pair | Metre | 97.00 |
| **38.3** | | **Telephone cable, 0.5 mm solid annealed cu., jelly filled** | Supply installation, testing and commissioning of 0.5 mm size solid annealed copper, jelly filled telephone cables, polythene insulated, petroleum jelly filled with poly-al laminate moisture barrier, polythene sheathed, armoured polythene jacketed telephone cables as per DOT specifications of following sizes | | |
| | 38.3.1 | 10 Pair | 10 Pair | Metre | 173.00 |
| | 38.3.2 | 20 Pair | 20 Pair | Metre | 248.00 |
| | 38.3.3 | 50 Pair | 50 Pair | Metre | 560.00 |
| **38.4** | | **Telephone tag block** | Supply, installation testing and commissioning of telephone tag block krone connector with enclosure complete as mentioned below :- | | |
| | 38.4.1 | 10 Pair | 10 Pair | each | 693.00 |
| | 38.4.2 | 20 Pair | 20 Pair | each | 1009.00 |
| | 38.4.3 | 30 Pair | 30 Pair | each | 1271.00 |
| | 38.4.4 | 50 Pair | 50 Pair | each | 2424.00 |
| | 38.4.5 | 100 Pair | 100 Pair | each | 3571.00 |
| | 38.4.6 | 200 Pair | 200 Pair | each | 7483.00 |
| **38.5** | | **RJ-11 telephone jack** | Supply, fixing and testing RJ - 11 telephone jack single with shutter F.T | each | 82.00 |
| **38.6** | | **RJ-11 telephone jack modular** | Supply, fixing and testing of RJ 11 telephone jack modular (1 Module) | each | 174.00 |
| **38.7** | | **T.V. outlet F.T.** | Supply, fixing and testing of T.V. Outlet F.T | each | 82.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **38.8** | | **T.V. outlet modular** | Supply, fixing and testing of T.V. outlet modular (1Module) | each | 175.00 |
| **38.9** | | **RJ-45 computer jack (1Module)** | Supply, installation testing and commissioning RJ - 45 computer jack cat 6 with shutter modular (1Module) | | |
| | 38.9.1 | Cat 6 with shutter modular | RJ - 45 computer socket cat 6 with shutter modular (1Module) | each | 288.00 |
| | 38.9.2 | Cat 6 tool less connection | RJ - 45 computer socket cat 6 tool less connection (1Module) | each | 443.00 |
| **38.10** | | **RJ-45 computer jack (2Module)** | Supply, installation testing and commissioning RJ - 45 computer jack cat 6 with shutter modular (2Module) | each | 689.00 |
| **38.11** | | **4 Pair un-shielded twisted pair (UTP)** | Supplying and drawing 4 pair un-shielded twisted pair (UTP), 0.5 mm PVC insulated copper conductor un-armoured Cat-6 computer cable FRLS in existing surface/concealed, steel/PVC. conduit/ casing-n-capping as required. | Metre | 56.00 |
| **38.12** | | **4 Pair shielded twisted pair (STP)** | Supplying and drawing 4 pair shielded twisted pair (STP), 0.5 mm PVC insulated copper conductor un-armoured Cat-6 computer cable FRLS in existing surface/concealed, steel/PVC. conduit/ casing-n-capping as required. | Metre | 104.00 |
| **38.13** | | **Patch cord** | Supply, installation testing and commissioning of patch cord as mentioned below :- | | |
| | 38.13.1 | CAT 6 RJ45, UTP 1metre | CAT 6 RJ45 patch cord UTP 1 metre | each | 436.00 |
| | 38.13.2 | CAT 6 RJ45, UTP 2metre | CAT 6 RJ45 patch cord UTP 2 metre | each | 548.00 |
| | 38.13.3 | CAT 6 RJ45, UTP 3metre | CAT 6 RJ45 patch cord UTP 3 metre | each | 601.00 |
| | 38.13.4 | CAT 6 RJ45, UTP 3metre | CAT 6 RJ45 patch cord UTP 5 metre | each | 657.00 |
| **38.14** | | **HDMI connector &braided cord** | Supply, installation testing and commissioning of High-Speed HDMI corrosion-resistant connector & braided cable, Supports Ethernet, 3D, 4K video, computers and other HDMI-enabled devices Black as required to complete the job as mentioned below :- | | |
| | 38.14.1 | 1 Metre, HDMI cord | 1 Metre, HDMI cord | each | 1058.00 |
| | 38.14.2 | 2 Metre, HDMI cord | 2 Metre, HDMI cord | each | 1801.00 |
| | 38.14.3 | 5 Metre, HDMI cord | 5 Metre, HDMI cord | each | 3825.00 |
| | 38.14.4 | 10 Metre, HDMI cord | 10 Metre, HDMI cord | each | 6590.00 |
| | 38.14.5 | 15 Metre, HDMI cord | 15 Metre, HDMI cord | each | 10702.00 |
| | 38.14.6 | 20 Metre, HDMI cord. | 20 Metre, HDMI cord. | each | 15752.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **38.15** | | **Patch panel, RJ 45** | Supply, installation testing and commissioning of 19" modular patch - 1 U patch panel 24, RJ 45 cat-6 connector | each | 11792.00 |
| **38.16** | | **Patch panel, RJ 11** | Supply, installation testing and commissioning of 19" modular telephone patch panel 24 - 1 U with RJ 11 connector | each | 11444.00 |
| **38.17** | | **Rack** | Supply, installation testing and commissioning of wall mounting rack, for computer switches/Patch Panel etc complete as mentioned below :- | | |
| | 38.17.1 | 6U cabinet | 6U cabinet with min. Dimensions 366X600X600, wall mounting with 6 point 5 Amp. socket power supply strip, and fan, cable manager, tray as per Site requirement | each | 9753.00 |
| | 38.17.2 | 9U cabinet | 9U cabinet with min. Dimension 488X600X600, wall mounting with 6 point 5 Amp. socket power supply strip, and fan, cable manager, tray as per Site requirement | each | 13087.00 |
| | 38.17.3 | 12U cabinet | 12U cabinet with min. Dimension 590X600X600, wall mounting with 6 point 5 Amp. socket power supply strip, and fan, cable manager, tray as per Site requirement | each | 18211.00 |
| | 38.17.4 | 15U cabinet | 15U cabinet with min. Dimension 615X600X500, wall mounting with 6 point 5 Amp. socket power supply strip, and fan, cable manager, tray as per Site requirement | each | 22967.00 |
| **38.18** | | **Floor mounting rack, base frame** | Supply installation, testing and commissioning of floor mounting rack, base frame, top & bottom, front glass door with perforation, rear vented door, lockable & detachable side panels, 2 pairs 19" rail with depth members. fixed shelf, complete as mentioned below :- | | |
| | 38.18.1 | 22U cabinet | 22U cabinet with min. Dimensions 1150X600X600, floor mounting with power supply and fan, tray. Equipment support angle, vertical cable manager, 12 point 5Amp. vertical power strip, earthing continuity kit, castors (1 set of 4 nos.), captive H/W set of 10, fan tray with 4 fan, for computer . | each | 47900.00 |
| | 38.18.2 | 30U cabinet | 30U cabinet 1350X600X600, floor mounting with power supply and fan, tray. Equipment support angle, vertical cable manager, 12 point 5Amp. vertical power strip, earthing continuity kit, castors (1 set of 4 nos.), captive H/W set of 10, fan tray with 4 fan, for computer | each | 59409.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 38.18.3 | 36U cabinet | 36U cabinet 1600X600X800, floor mounting with power supply and Fan, tray, equipment support angle, vertical cable manager, 12 point 5Amp. vertical power strip, earthing continuity kit, castors (1 set of 4 nos.), captive H/W set of 10, fan tray with 4 fan, for computer | each | 67089.00 |
| | 38.18.4 | 42U cabinet | 42U cabinet 1900X600X1000, floor mounting with power supply and fan, tray, equipment support angle, vertical cable manager, 12 point 5Amp. vertical power strip, earthing continuity kit, castors (1 set of 4 nos.), captive H/W set of 10, fan tray with 4 fan, for computer | each | 87147.00 |
| **38.19** | | | **CAT6A/OFC Networking (10G Support)** | | |
| | 38.19.1 | CAT6A Cables | | | |
| | 38.19.1.1 | CAT6A U/FTP Cable | Supply, Installation, lying, drawing of 4 Pair CAT6A U/FTP Cable having 23AWG condutor, LSZH Compliant with ETL & TIA/EIA- 568-C.2, Coupling Attenuation should be 45dB min @30-100 MHz 40-20 Log (f/100) @ 100-500 MHz, Should be certified by ETL/UL in existing conduit / wiring system as direction of EIC. | Metre | 178.00 |
| | 38.19.1.2 | CAT6A SFTP | Supply, Installation, lying, drawing of 4 Pair CAT6A SFTP Cable having 23AWG condutor, Voltage rating (Vdc): <80, 1-250 MHz: 100 +/- 15 Ohm, 250-500 MHz: 100 +/- 22 Ohm, ETL/UL in existing conduit / wiring system as direction of EIC. | Metre | 214.00 |
| | 38.19.2 | CAT6A Patch Panel | | | |
| | 38.19.2.1 | CAT6 A 24 PORT PATCH PANEL | Supply, Installation, Commissioning of 24 Port Patch Panel loaded with Category-6A I/O shielded Jacks complying as per the ANSI/TIA-568C.2 Patch Panel, 19" Rack Mount, 1U Height, I/O should be Spring-loaded shutter to protects from dust and contaminants, Should be certified by ETL/UL | Each | 25901.00 |
| | 38.19.2.2 | CAT6 A 48 PORT PATCH PANEL | Supply, Installation, Commissioning of 48 Port Patch Panel loaded with Category-6A I/O shielded Jacks complying as per the ANSI/TIA-568C.2 Patch Panel, 19" Rack Mount, 1U Height, I/O should be Spring-loaded shutter to protects from dust and contaminants, Should be certified by ETL/UL | Each | 50097.00 |
| | 38.19.3 | CAT6 A SHIELDED TWISTED PAIR PATCH CORDS | CAT6 A SHIELDED TWISTED PAIR PATCH CORDS | | |
| | 38.19.3.1 | 1 Mtr Patch Cord | SITC of 1 Mtr CAT6 A SHIELDED TWISTED PAIR PATCH CORDS ETL/UL. | Each | 709.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 38.19.3.2 | 2 Mtr Patch Cord | SITC of 2 Mtr CAT6 A SHIELDED TWISTED PAIR PATCH CORDS ETL/UL. | Each | 858.00 |
| | 38.19.3.3 | 3 Mtr Patch Cord | SITC of 3 Mtr CAT6 A SHIELDED TWISTED PAIR PATCH CORDS ETL/UL. | Each | 1063.00 |
| | 38.19.3.4 | 5 Mtr Patch Cord | SITC of 5 Mtr CAT6 A SHIELDED TWISTED PAIR PATCH CORDS ETL/UL. | Each | 1454.00 |
| | 38.19.4 | CAT6 A Computer Jack | | | |
| | 38.19.4.1 | CAT6A I/O PVC | CAT6A RJ - 45 computer socketwith shutter modular (1Module) Category-6A as per ANSI/TIA-568-C.2, I/O should be Spring-loaded shutter or cap to protects from dust and contaminants, Wire AccoSModation: 22-24 AWG solid, Should be certified by ETL/UL | Each | 1199.00 |
| | 38.19.4.2 | Cat-6A Sheilded Connectors/ RJ45 Plug (Tool Less ) | CAT6A RJ - 45 computer socket , RJ-45 meets TIA-568.2-D Category 6A component performance, shutter modular (1Module) IP 20 rated,Should Supports 10GB networks, I/O should be Spring-loaded shutter or cap to protects from dust and contaminants, Wire AccoSModation: 23-24 AWG solid, Should be certified by ETL/UL | Each | 1295.00 |
| | 38.19.5 | Optical fiber cable | | | |
| | 38.19.5.1 | 6 Core OS-2 SM Cable | SITC of 6 Core OFC Cable with Armored Uni-Tube, 9/125µm,OS 2 Type, . Complying to ISO/IEC 11801, EN50173, ANSI/TIA 568-C.3, Telcordia GR-20, Uni-tube construction,Water Blocking Thixotropic Gel (Tube) Petroleum Jelly (Interstices) in indoor / outdoor ducts, direct burial and backbone cabling | Metre | 114.00 |
| | 38.19.5.2 | 12 Core OS-2 SM Cable | SITC of 12 Core OFC Cable with Armored Uni-Tube, 9/125µm,OS 2 Type, . Complying to ISO/IEC 11801, EN50173, ANSI/TIA 568-C.3, Telcordia GR-20, Uni-tube construction,Water Blocking Thixotropic Gel (Tube) Petroleum Jelly (Interstices) in indoor / outdoor ducts, direct burial and backbone cabling | Metre | 148.00 |
| | 38.19.6 | Optical fiber Patch Cord | | | |
| | 38.19.6.1 | 1 Mtr OFC Patch Cord | SITC of 1 Mtr OFC SM Patch Cord with 9/125 µm, OS2 , Duplex Zipcord , SC-LC/LC-LC / SC-SC connecctor , 900µ tight buffer, ISO/IEC 1108:2008, ANSI/TIA/ EIA 568.C.3, ANSI/TIA/EIA-492, TELECORDIA GR-409, ICEA-596 | Each | 2998.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 38.19.6.2 | 2 Mtr OFC Patch Cord | SITC of 2Mtr OFC SM Patch Cord with 9/125 µm, OS2 , Duplex Zipcord , SC-LC/LC-LC / SC-SC connecctor , 900µ tight buffer, ISO/IEC 1108:2008, ANSI/TIA/ EIA 568.C.3, ANSI/TIA/EIA-492, TELECORDIA GR-409, ICEA-597 | Each | 3140.00 |
| | 38.19.6.3 | 3 Mtr OFC Patch Cord | SITC of 3 Mtr OFC SM Patch Cord with 9/125 µm, OS2 , Duplex Zipcord , SC-LC/LC-LC / SC-SC connecctor , 900µ tight buffer, ISO/IEC 1108:2008, ANSI/TIA/ EIA 568.C.3, ANSI/TIA/EIA-492, TELECORDIA GR-409, ICEA-598 | Each | 4035.00 |
| | 38.19.6.4 | 5 Mtr OFC Patch Cord | SITC of 5 Mtr OFC SM Patch Cord with 9/125 µm, OS2 , Duplex Zipcord , SC-LC/LC-LC / SC-SC connecctor , 900µ tight buffer, ISO/IEC 1108:2008, ANSI/TIA/ EIA 568.C.3, ANSI/TIA/EIA-492, TELECORDIA GR-409, ICEA-599 | Each | 5016.00 |
| | 38.19.7 | Fiber Termination LIU | | | |
| | 38.19.7.1 | 6 Port LIU | SITC, Splicing, Termination of 6 Port Full loaded LIU with SC-LC/LC-LC / SC-SC adopter, Splicing tray, Pigtails, Termination, 1U 19” / ETSI versions available, Slidding Drawer for easy access of back side of connector, Fiber guides, radius controls & secure tie downs provided | Each | 21726.00 |
| | 38.19.7.2 | 12 Port LIU | SITC, Splicing, Termination of 12 Port Full loaded LIU with SC-LC/LC-LC / SC-SC adopter, Splicing tray, Pigtails, Termination, 1U 19” / ETSI versions available, Slidding Drawer for easy access of back side of connector, Fiber guides, radius controls & secure tie downs provided | Each | 28141.00 |
| | 38.19.7.3 | 24 Port LIU | SITC, Splicing, Termination of 24 Port Full loaded LIU with SC-LC/LC-LC / SC-SC adopter, Splicing tray, Pigtails, Termination, 1U 19” / ETSI versions available, Slidding Drawer for easy access of back side of connector, Fiber guides, radius controls & secure tie downs provided | Each | 38986.00 |
| | 38.19.7.4 | 48 Port LIU | SITC, Splicing, Termination of 48 Port Full loaded LIU with SC-LC/LC-LC / SC-SC adopter, Splicing tray, Pigtails, Termination, 1U 19” / ETSI versions available, Slidding Drawer for easy access of back side of connector, Fiber guides, radius controls & secure tie downs provided | Each | 53127.00 |
| **38.20** | | | SITC OF Gigabit Unmanaged Desktop Switch | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 38.20.1 | 5-Port Gigabit Switch | 5-Port Gigabit Unmanaged Desktop Switch. Fast and Reliable Networking. 1000Base-T Interface : 5. Auto MDI/MDIX .Switching Capacity : 10 Gbps. Operation Temperature 0 ~ 40 °C. Power Input AC : 100 ~ 240 V. | Each | 3139.00 |
| | 38.20.2 | 8-Port Gigabit Switch | 8-Port Gigabit Unmanaged Desktop Switch. Fast and Reliable Networking. 1000Base-T Interface : 8. Auto MDI/MDIX .Switching Capacity : 16 Gbps. Operation Temperature 0 ~ 40 °C. Power Input AC : 100 ~ 240 V. | Each | 4424.00 |
| | 38.20.3 | 16-Port Gigabit Switch | 16-Port Gigabit Unmanaged Desktop Switch. 1000Base-T Interface : 16. Auto MDI/MDIX .Switching Capacity : 32 Gbps. Operation Temperature 0 ~ 40 °C. Power Input AC : 100 ~ 240 V. | Each | 11517.00 |
| | 38.20.4 | 24-Port Gigabit Switch | 24-Port Gigabit Unmanaged Desktop Switch. Full wire-speed data transmission. Energy-efficient design. 1000Base-T Interface : 24. Auto MDI/MDIX .Switching Capacity : 48 Gbps. Operation Temperature 0 ~ 40 °C. Power Input AC : 100 ~ 240 V. | Each | 15279.00 |

## CHAPTER -39

## WATER PUMP AND FIRE PUMP

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.1** | | **1 Phase deep well submersible pump stainless steel body** | Supplying &installation of required capacity of single phase, 50 Hz, 220V, deep well submersible pump stainless steel body, as per IS of approved make, suitable for 4"/6 " tube well with control panel starter suitable for submersible pump with dry run protection, connections, including clamp, bore cap etc. as required as per specifications but excluding pipe, SS/Nylon wire rope and connection cable. | | |
| | 39.1.1 | 0.5 HP, 6-7 stages | **0.5 HP** with 6-7 stages, head metre 46-13, discharge LPM 10-55 | Each | 18696.00 |
| | 39.1.2 | 1 HP, 7-8 stages | **1 HP** with 7-8 stages, head metre 61-18, discharge LPM 25-90 | Each | 20201.00 |
| | 39.1.3 | 1 HP, 10-11 stages | **1 HP** with 10-11 stages, head metre 74-21, discharge LPM 15-55 | Each | 21716.00 |
| | 39.1.4 | 1 HP, 12-14 stages | **1 HP** with 12-14 stages, head metre 91-28, discharge LPM 10-45 | Each | 25246.00 |
| | 39.1.5 | 1 HP, 16-25 stages | **1 HP** with 16-25 stages, head metre 91-28, discharge LPM 10-45 | Each | 27891.00 |
| | 39.1.6 | 1.5 HP, 7-8 stages | **1.5 HP** with 7-8 stages, head metre 56-21, discharge LPM 40-125 | Each | 27160.00 |
| | 39.1.7 | 1.5 HP, 10-11 stages | **1.5 HP** with 10-11 stages, head metre 84-26, discharge LPM 25-90 | Each | 27390.00 |
| | 39.1.8 | 1.5 HP, 16-17 stages | **1.5 HP** with 16-17 stages, head metre 114-33, discharge LPM 15-55 | Each | 30150.00 |
| | 39.1.9 | 1.5 HP, 18-20 stages | **1.5 HP** with 18-20 stages, head metre 130-41, discharge LPM 10-45 | Each | 30556.00 |
| | 39.1.10 | 1.5 HP, 21-30 stages | **1.5 HP** with 21-30 stages, head metre 130-41, discharge LPM 10-45 | Each | 34076.00 |
| | 39.1.11 | 2 HP, 7-8 stages | **2 HP** with 7-8 stages, head metre 56-16, discharge LPM 65-205 | Each | 31199.00 |
| | 39.1.12 | 2 HP, 20-22 stages | **2 HP** with 20-22 stages, head metre 147-42, discharge LPM 15-55 | Each | 35906.00 |
| | 39.1.13 | 2 HP, 25 stages | **2 HP** with 25 stages, head metre 163-58, discharge LPM 10-45 | Each | 35854.00 |
| | 39.1.14 | 3 HP, 10-12 stages | **3 HP** with 10-12 stages, head metre 84-23, discharge LPM 65-205 | Each | 37891.00 |
| | 39.1.15 | 3 HP, 15-17 stages | **3 HP** with 15-17 stages, head metre 119-45, discharge LPM 40-125 | Each | 42129.00 |
| | 39.1.16 | 3 HP, 20-35 stages | **3 HP** with 20-35 stages, head metre 167-51, discharge LPM 25-90 | Each | 52131.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.2** | | **3 Phase deep well submersible pump steel body** | Supplying &installation of required capacity of three phase, 50 Hz,415V, **deep well submersible pump** steel body, as per IS of approved make, suitable upto 6 " tube well with control panel starter suitable for submersible pump with dry run protection, single phase preventer, connections, including clamp, bore cap etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.2.1 | 1 HP, 12-25 stages | **1 HP** with 12-25 stages, head metre 91-28, discharge LPM 10-45 | Each | 20061.00 |
| | 39.2.2 | 1.5 HP, 16-17 stages | **1.5 HP** with 16-17 stages, head metre 114-33, discharge LPM 15-55 | Each | 30036.00 |
| | 39.2.3 | 1.5 HP, 18-30 stages | **1.5 HP** with 18-30 stages, head metre 130-41, discharge LPM 10-45 | Each | 37862.00 |
| | 39.2.4 | 3 HP, 15-17 stages | **3 HP** with 15-17 stages, head metre 119-45, discharge LPM 40-125 | Each | 43906.00 |
| | 39.2.5 | 3 HP, 20-35 stages | **3 HP** with 20-35 stages, head metre 167-51, discharge LPM 25-90 | Each | 51697.00 |
| | 39.2.6 | 3 HP, upto 6 stages | **3 HP** with upto 6 stages, head metre 55-7, discharge LPM 60-510 | Each | 52959.00 |
| | 39.2.7 | 5 HP, upto 5 stages | **5 HP** with upto 5 stages, head metre 55-13, discharge LPM 120-510 | Each | 55145.00 |
| | 39.2.8 | 5 HP, 8-11 stages | **5 HP** with 8-11 stages, head metre 101-32, discharge LPM 60-270 | Each | 65323.00 |
| | 39.2.9 | 7.5 HP, 13-14 stages | **7.5 HP** with 13-14 stages, head metre 129-52, discharge LPM 60-270 | Each | 71984.00 |
| | 39.2.10 | 7.5 HP, 15 stages | **7.5 HP** with 15 stages, head metre 138-60, discharge LPM 60-270 | Each | 81412.00 |
| | 39.2.11 | 10 HP, 16 stages | **10 HP** with 16 stages, head metre 147-64, discharge LPM 60-270 | Each | 91240.00 |
| | 39.2.12 | 10 HP, 20 stages | **10 HP** with 20 stages, head metre 184-80, discharge LPM 60-270 | Each | 98847.00 |
| **39.3** | | **Submersible flat cable 3 core cu. wire** | Supplying and laying of submersible flat cable ISI marked 3 core copper wire of suitable size with proper clamping of approved make. | | |
| | 39.3.1 | 2.5 sq.mm | 2.5 sq.mm multi strand | Metre | 139.00 |
| | 39.3.2 | 4.0 sq.mm | 4.0 sq.mm multi strand | Metre | 205.00 |
| | 39.3.3 | 6.0 sq.mm | 6.0 sq.mm multi strand | Metre | 308.00 |
| | 39.3.4 | 10.0 sq.mm | 10.0 sq.mm multi strand | Metre | 471.00 |
| **39.4** | | **HDPE pipe** | Supplying and laying of approved make HDPE pipe as per IS:4984 and IS:14333 with latest amendment for installation of submersible pump with necessary connection, socket/couplings, tees, etc. as required (PE-63 & working pressure 6 kgf/sq cm) | | |
| | 39.4.1 | 25 mm | 25 mm Outer Dia. | Metre | 77.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 39.4.2 | 32 mm | 32 mm Outer Dia. | Metre | 120.00 |
| | 39.4.3 | 40 mm | 40 mm Outer Dia. | Metre | 182.00 |
| | 39.4.4 | 50 mm | 50 mm Outer Dia. | Metre | 279.00 |
| **39.5** | | **Nylon rope 12 mm thick** | Supplying and laying of approved make nylon rope 12 mm thick complete with binding for support of pump and motor. | Metre | 76.00 |
| **39.6** | | **Stainless steel wire rope 6 mm thick** | Supplying and laying of approved make stainless steel wire rope 6 mm thick complete with binding for support of pump and motor. | Metre | 163.00 |
| **39.7** | | **1 Phase centrifugal mono-block pump** | Supplying &installation of approved make required capacity single phase, 50 Hz,220V, centrifugal mono-block pump self-priming, connections, base channel foundation etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.7.1 | 0.5 HP head metre 6-26 | 0.5 HP head metre 6-26, discharge LPM 34-6 | Each | 10237.00 |
| | 39.7.2 | 1 HP head metre 6-42 | 1 HP head metre 6-42, discharge LPM 55-5 | Each | 11820.00 |
| **39.8** | | **1 Phase centrifugal mono-block pump** | Supplying &installation of approved make required capacity single phase, 50 Hz,240V, centrifugal mono-block pump, with starter, connections, base channel foundation etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.8.1 | 1.5 HP head metre 9-15 | **1.5 HP** head metre 9-15, discharge LPM 420-180 | Each | 23728.00 |
| | 39.8.2 | 2.0 HP head metre 12-20 | **2.0 HP** head metre 12-20, discharge LPM 540-180 | Each | 27997.00 |
| **39.9** | | **3 Phase centrifugal mono-block pump** | Supplying & installation of approved make required capacity 3 phase, 50 Hz, 415V, centrifugal mono-block pump, with starter i/c single phase preventer, connections, base channel foundation etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.9.1 | 3.0 HP head metre 6-15 | **3.0 HP** head metre 6-15, discharge LPM 1200-1000 | Each | 41197.00 |
| | 39.9.2 | 5.0 HP head metre 15-30 | **5.0 HP** head metre 15-30, discharge LPM 825-250 | Each | 53001.00 |
| | 39.9.3 | 7.5 HP head metre 24-37 | **7.5 HP** head metre 24-37, discharge LPM 780-320 | Each | 65259.00 |
| | 39.9.4 | 10.0 HP head metre 10-36 | **10.0 HP** head metre 10-36, discharge LPM 2400-1200 | Each | 73281.00 |
| | 39.9.5 | 15.0 HP head metre 20-36 | **15.0 HP** head metre 20-36, discharge LPM 2400-1200 | Each | 102282.00 |
| | 39.9.6 | 20.0 HP head metre 32-52 | **20.0 HP** head metre 32-52, discharge LPM 2400-1200 | Each | 125833.00 |
| | 39.9.7 | 25.0 HP head metre 48-60 | **25.0 HP** head metre 48-60, discharge LPM 1500-800 | Each | 148837.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.10** | | **3 Phase open well submersible pump** | Supplying &installation of approved make required capacity 3 phase, 50 Hz, 415V, open well submersible pump, with control panel starter with dry run protection, single phase preventer, connections, etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.10.1 | 3.0 HP head metre 12-36 | **3.0 HP** head metre 12-36, discharge LPM 528-120 | Each | 40286.00 |
| | 39.10.2 | 5.0 HP head metre 12-28 | **5.0 HP** head metre 12-28, discharge LPM 1200-390 | Each | 47724.00 |
| | 39.10.3 | 7.5 HP head metre12-32 | **7.5 HP** head metre 12-32, discharge LPM 1620-420 | Each | 55638.00 |
| | 39.10.4 | 10.0 HP head metre 18-40 | **10.0 HP** head metre 18-40, discharge LPM 1620-570 | Each | 75781.00 |
| **39.11** | | **1 Phase open well submersible pump** | Supplying & installation of approved make required capacity single phase, 50 Hz, 230V, open well submersible pump, with control panel starter with dry run protection, connections, etc. as required as per specifications but excluding pipe and connection cable. | | |
| | 39.11.1 | 0.5 HP head metre 6-20 | 0.5 HP head metre 6-20, discharge LPM 2-61, W/o Starter | Each | 14059.00 |
| | 39.11.2 | 1.0 HP head metre 6-30 | 1.0 HP head metre 6-30, discharge LPM 144-27, W/o Starter | Each | 20267.00 |
| | 39.11.3 | 1.5 HP head metre 22-38 | 1.5 HP head metre 22-38, discharge LPM 120-36 | Each | 28958.00 |
| | 39.11.4 | 2.0 HP head metre 14-24 | 2.0 HP head metre 14-24, discharge LPM 342-138 | Each | 33235.00 |
| **39.12** | | Pressure Booster Pump | Supplying & installation of approved make Pressure Booster Pump having stainless steel impeller with tank & controller running on single phase AC 220V and 50Hz , horizontal and vertical type as required as per specifications but excluding pipe and connection cable. | | |
| | 39.12.1 | 0.5/0.6 HP | 0.5/0.6 HP, Head Range 10-40Meter, Discharge 77-12 LPM | Each | 34692.00 |
| | 39.12.2 | 0.75/0.8 HP | 0.75/0.8 HP, Head Range 15-40 Meter, Discharge 70-12 LPM | Each | 34835.00 |
| | 39.12.3 | 1 HP | 1 HP, Head Range 15-50 Meter, Discharge 79-13 LPM | Each | 36288.00 |
| | 39.12.4 | 1.5 HP | 1.5 HP, Head Range 15-54 Meter, Discharge 125-18 LPM | Each | 41848.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.13** | | **Horizontally mounted coupled motor (Eff-1) 60hp** | Supplying, installation, testing & commissioning of approved make horizontally mounted coupled motor (Eff-1) 60hp, 2900 RPM, 3 Ph 415 V with mechanical seal pump delivery of 2280 litres per minute against total head of 70 meter so as to ensure a minimum pressure of 3.5 Kg/Cm2 at the highest. The impeller shall be closed type and accurately balanced dynamically. The pump shall be of cast iron, impeller shall be of bronze and shaft shall be of stainless steel, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) i/c coupling, coupling guard, base channel foundation &matching flanges etc. as required as per specifications. The fire pump set as per NFPA –20 specifications. | Each | 381195.00 |
| **39.14** | | **Horizontally mounted coupled motor (Eff-1) 75hp** | Supplying, installation, testing & commissioning of approved make horizontally mounted coupled motor (Eff-1) 75hp, 2900 RPM, 3 Ph 415 V with mechanical seal pump delivery of 2850 litres per minute against total head of 70 meter so as to ensure a minimum pressure of 3.5 Kg/Cm2 at the highest. The impeller shall be closed type and accurately balanced dynamically. The pump shall be of cast iron, impeller shall be of bronze and shaft shall be of stainless steel, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) i/c coupling, coupling guard, base channel foundation & matching flanges etc. as required as per specifications. The fire pump set as per NFPA –20 specifications. | Each | 390087.00 |
| **39.15** | | **Horizontally mounted coupled motor (Eff-1) 150hp** | Supplying, installation, testing & commissioning of approved make horizontally mounted coupled motor (Eff-1) 150hp, 2900 RPM, 3 Ph 415 V with mechanical seal pump delivery of 4550 litres per minute against total head of 88 meter so as to ensure a minimum pressure of 3.5 Kg/Cm2 at the highest. The impeller shall be closed type and accurately balanced dynamically. The pump shall be of cast iron, impeller shall be of bronze and shaft shall be of stainless steel, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) i/c coupling, coupling Guard, base channel foundation &matching flanges etc. as required as per specifications. The fire pump set as per NFPA –20 specifications. | Each | 436825.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.16** | | **Jockey pump horizontally mounted coupled motor (Eff-2) 5.5hp** | Supplying, installation, testing & commissioning of approved make jockey pump horizontally mounted coupled motor (Eff-2) 5.5hp, 2900 RPM, 3 Ph 415 V with mechanical seal pump, delivery of 180 litres per minute against total head of 70 / 88 meter so as to ensure a minimum pressure of 3.5 Kg/Cm2 at the highest. The impeller shall be closed type and accurately balanced dynamically. The pump shall be of cast iron, impeller shall be of bronze and shaft shall be of stainless steel. i/c coupling, coupling guard, base channel foundation & matching flanges etc. as required as per specifications. | Each | 166401.00 |
| **39.17** | | **Diesel engine driven firefighting pump water cooled 65 BHP** | Supplying &installation of approved make diesel engine driven firefighting pump water cooled 65 BHP 2150 RPM engine, fuel tank 200 Ltr (empty), accessories, matching flanges, base frame, ARV , fuel gauge, exide12 Volt battery & charger, lugs & cable between battery & charger. The pump capacity of 2280 litres per minute at total head 70 meters, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) as required. All above equipment's are mounted on common base plate. The fire pump set as per NFPA –20 specifications. | Each | 621842.00 |
| **39.18** | | **Diesel engine driven firefighting pump water cooled 105 BHP** | Supplying &installation of approved make diesel engine driven firefighting pump water cooled 105 BHP 2150 RPM engine, fuel tank 200 litre (empty), accessories, matching flanges, base frame, ARV, fuel gauge, exide 12 Volt battery & charger, lugs & cable between battery & charger. The pump capacity of 2850 litres per minute at total head 70 meters, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) as required. All above equipment's are mounted on common base plate. The fire pump set as per NFPA –20 specifications. | Each | 631922.00 |
| **39.19** | | **Diesel engine driven firefighting pump water cooled 170 BHP** | Supplying &installation of approved make diesel engine driven firefighting pump water cooled 170 BHP 2150 RPM engine, fuel tank 200 litre (empty), accessories , matching flanges, base frame ,ARV, fuel gauge, exide 12 Volt battery & charger, lugs & cable between battery & charger. The pump capacity of 4550 litres per minute at total head 88 meters, automatic air release valve, pressure gauges (combination suction gauge, 3½" dial type with ¼" cock and lever handle - 1 no, discharge gauge, 3½" dial type with ¼" cock and lever handle - 1 no.) as required. All above equipment's are mounted on common base plate. The fire pump set as per NFPA –20 specifications. | Each | 836636.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.20** | | **Booster pump mono block horizontally mounted with motor (Eff-2) 7.5hp** | Supplying, installation, testing & commissioning of approved make booster pump mono-block horizontally mounted with motor (Eff-2) 7.5hp, 2900 RPM, 3 Ph 415 V with mechanical seal pump, delivery of 600 LPM at 35 MWC against total head of 32 meter. The impeller shall be closed type and accurately balanced dynamically i/c base channel foundation etc. as required as per specifications. | Each | 35000.00 |
| **39.21** | | **Fire pumps electrical panel :** | Fire pumps electrical panel : | | |
| | 39.21.1 | **Cubical type floor mounted electrical control panel** | Supplying, installation, testing & commissioning of cubical type floor mounted electrical control panel of 2.0 mm thick CRCA sheet suitable for 380-415 Volt, 50 Hz, 3 Ph supply, NEMA type 2 (IP 31) drip proof metal free standing enclosure for firefighting pumps with suitable star-delta starter including indicating lamps, ammeter, voltmeter, selector switches for each, incoming and outgoing feeder, auxiliary switches, dry suction cut off, single phase preventer, alu. bus bar etc. The panel shall be provided with interlocking arrangement between electric motor driven and diesel engine driven pumps so that only one pump can operate at one time and in case of any failure in electric motor driven pump the diesel engine driven pump shall operate automatically. The installation shall include all control wring, accommodation for battery charger etc. and connections as required. Incoming :<br>1.No. 250 Amp 4 Pole MCCB, 35 KA<br>1 Set 300 Amp, 500V capacity 4 Nos. aluminium bus bar with epoxy insulator, heat shrink sleeve, required hole , nut bolt etc.<br>1 No. 96x 96 mm 0 to 500 Volt voltmeter selector switch and HRC fuse with base.<br>1 No. 96x 96 mm 0 to 300 Amp ammeter, selector switch and CT<br>1Set –3 nos. indicating lamp with HRC fuse and base<br>Out goings :<br>2 Nos. 200 Amp 4 Pole MCCB,25 KA<br>2 Nos. 63 Amp 4 pole MCCB,10 KA.<br>1 No. fully automatic star delta starter suitable for main pump motor I/C CT, ammeter with selector switch & control with pressure switch, pump ON indicating lamp.<br>1 No. DOL starter suitable for jockey pump motor I/C CT, ammeter with selector switch. & control with pressure switch complete with auto/manual selector switch inter connecting wring, control wiring from pressure switch to panel, pump ON indicating lamp. etc. complete as required. | Each | 299686.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **39.22** | | **Pressure switch** | Supplying &installation of approved make pressure switch with 15 mm dia brass check nut for automatic operation on required pressure including necessary wiring up to control panel. | Each | 2500.00 |
| **39.23** | | **Pressure gauges** | Supplying &installation of pressure gauges with 15 mm dia brass check nut & capable of showing pressure reading from 0. to 30 kg. | Each | 1650.00 |
| **39.24** | | **Fire panel** | Supply installation, testing and commissioning of approved make required fire panel for manual operated fire hooters etc. as required as per specifications but excluding pipe and connection cable. | Each | 12200.00 |
| **39.25** | | **DOL starter 7.5 hp** | Supply installation, testing and commissioning of approved make DOL starter 7.5 hp, 3 phase, 50 Hz, 415V, i/c single phase preventer, incoming MCB 32A 4P, 10KA , in outdoor type enclosure, connections, as required as per specifications. | Each | 5000.00 |

# CHAPTER-40

# RACEWAY AND uPVC BUS TRUNCKING

| Item No. | Sub Item | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **40.1** | | **G.I. floor raceway** | Supply, installation, testing and commissioning of G.I. floor raceway of mention below sizes 1/3 compartments including all necessary civil works. | | |
| | 40.1.1 | 75x 25 x 1.2 mm | 75 x 25 x 1.2 mm | Metre | 885.00 |
| | 40.1.2 | 100 x 25 x 1.2 mm | 100 x 25 x 1.2 mm | Metre | 1018.00 |
| | 40.1.3 | 150 x 25 x 1.2 mm | 150 x 25 x 1.2 mm | Metre | 1319.00 |
| | 40.1.4 | 225 x 25 x 1.2 mm | 225 x 25 x 1.2 mm | Metre | 1927.00 |
| | 40.1.5 | 300 x 25 x 1.2 mm | 300 x 25 x 1.2 mm | Metre | 2327.00 |
| | 40.1.6 | 75 x 38 x 1.2 mm | 75 x 38 x 1.2 mm | Metre | 1006.00 |
| | 40.1.7 | 100 x 38 x 1.2 mm | 100 x 38 x 1.2 mm | Metre | 1165.00 |
| | 40.1.8 | 150 x 38 x 1.2 mm | 150 x 38 x 1.2 mm | Metre | 1479.00 |
| | 40.1.9 | 225 x 38 x 1.2 mm | 225 x 38 x 1.2 mm | Metre | 2111.00 |
| | 40.1.10 | 300 x 38 x 1.2 mm | 300 x 38 x 1.2 mm | Metre | 2532.00 |
| **40.2** | | **G.I. junction box** | Supply, installation, testing and commissioning of G.I. junction box for duct entry 1/3 compartments junction box (frame & trap) including all necessary civil works. | | |
| | 40.2.1 | 150 x 150 x 65 - 90 mm | 150 x 150 x 65 - 90 mm | Each | 1528.00 |
| | 40.2.2 | 225 x 225 x 65 - 90mm | 225 x 225 x 65 - 90mm | Each | 1886.00 |
| | 40.2.3 | 300 x 300 x 65 - 90mm | 300 x 300 x 65 - 90mm | Each | 2522.00 |
| | 40.2.4 | 400 x 400 x 65 - 90mm | 400 x 400 x 65 - 90mm | Each | 3730.00 |
| **40.3** | | **G.I. cross over/junction box** | Supply, installation, testing and commissioning of G.I. cross over/junction box for duct entry 1/3 compartments junction box c/w fly overs (frame & trap) including all necessary civil works. | | |
| | 40.3.1 | 225 x 225 x 65 - 90mm | 225 x 225 x 65 - 90mm | Each | 2067.00 |
| | 40.3.2 | 300 x 300 x 65 - 90mm | 300 x 300 x 65 - 90mm | Each | 2747.00 |
| | 40.3.3 | 400 x 400 x 65 - 90mm | 400 x 400 x 65 - 90mm | Each | 4100.00 |

| Item No. | Sub Item | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **40.4** | | **G.I. coupler for under floor M-Tracks 25mm duct** | Supply, installation, testing and commissioning of G.I. coupler for under floor M-Tracks 25mm duct size wherever required. | Each | 70.00 |
| **40.5** | | **G.I. coupler for under floor M-Tracks 38mm duct** | Supply, installation, testing and commissioning of G.I. coupler for under floor M-Tracks 38mm duct size wherever required. | Each | 90.00 |
| **40.6** | | **G.I. fixing bracket** | Supply, installation, testing and commissioning of G.I. fixing bracket wherever required for under floor M-Tracks of duct size as mentioned below :- | | |
| | 40.6.1 | 75 X 25 mm | 75 X 25 mm | Each | 38.00 |
| | 40.6.2 | 100 X 25 mm | 100 X 25 mm | Each | 44.00 |
| | 40.6.3 | 150 X 25 mm | 150 X 25 mm | Each | 54.00 |
| | 40.6.4 | 225 X 25 mm | 225 X 25 mm | Each | 71.00 |
| | 40.6.5 | 300 X 25 mm | 300 X 25 mm | Each | 93.00 |
| **40.7** | | **G.I. fixing bracket** | Supply, installation, testing and commissioning of G.I. fixing bracket wherever required for under floor M-Tracks of duct size as mentioned below :- | | |
| | 40.7.1 | 75 X 38 mm | 75 X 38 mm | Each | 43.00 |
| | 40.7.2 | 100 X 38 mm | 100 X 38 mm | Each | 50.00 |
| | 40.7.3 | 150 X 38 mm | 150 X 38 mm | Each | 58.00 |
| | 40.7.4 | 225 X 38 mm | 225 X 38 mm | Each | 87.00 |
| | 40.7.5 | 300 X 38 mm | 300 X 38 mm | Each | 105.00 |
| **40.8** | | **G.I. vertical access box for raceway** | Supply, installation, testing and commissioning of G.I. vertical access box for raceway as mentioned below including all necessary civil works. | | |
| | 40.8.1 | 225 mm width | 225 mm width raceway | Each | 1375.00 |
| | 40.8.2 | 300 mm width | 300 mm width raceway | Each | 1580.00 |
| **40.9** | | **BUS U-PVC Trunking** | Supply & fixing of BUS U-PVC trunking without its allied accessories as reqd. | | |
| | 40.9.1 | 85 x 50 mm. | 85 x 50 mm. | Metre | 493.00 |
| | 40.9.2 | 100/105 x 50 mm. | 100/105 x 50 mm. | Metre | 601.00 |
| | 40.9.3 | 150 x 50 mm. | 150 x 50 mm. | Metre | 985.00 |

| Item No. | Sub Item | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 40.9.4 | 195 x 50 mm. | 195 x 50 mm. | Metre | 1288.00 |
| **40.10** | | **Flexible cover** | Supply & fixing flexible cover of following width on the existing size BUS U-PVC trunking as mentioned below as required. | | |
| | 40.10.1 | 40 mm width | 40 mm width. | Metre | 219.00 |
| | 40.10.2 | 65 mm width | 65 mm width. | Metre | 216.00 |
| | 40.10.3 | 85 mm width | 85 mm width. | Metre | 244.00 |
| | 40.10.4 | 130 mm width | 130 mm width. | Metre | 507.00 |
| | 40.10.5 | 180 mm width | 180 mm width. | Metre | 642.00 |
| **40.11** | | **Partition on the existing size BUS U-PVC trunking** | Supply & fixing of partition on the existing size BUS U-PVC trunking for mentioned below size as reqd. | | |
| | 40.11.1 | Separation partition | Separation partition for 80/105/150/195 X 50mm. | Metre | 155.00 |
| | 40.11.2 | Clip on separation partition | Clip on separation partition for 105/150/195 X 50mm. | Metre | 224.00 |
| **40.12** | | **Accessories for BUS UPVC trunking** | Supply & fixing of following accessories suitable for BUS UPVC trunking compete etc. as reqd. | | |
| | 40.12.1 | Base joint | Base joint | Each | 54.00 |
| **40.13** | | **Cover joints** | Supply & fixing of body joint for aligning during installation accessories suitable for mentioned below size BUS UPVC trunking compete as reqd. | | |
| | 40.13.1 | 40 mm width | Cover joints for 40 mm width | Each | 102.00 |
| | 40.13.2 | 65 mm width | Cover joints for 65 mm width | Each | 89.00 |
| | 40.13.3 | 85 mm width | Cover joints for 85 mm width | Each | 132.00 |
| | 40.13.4 | 130 mm width | Cover joints for 130 mm width | Each | 215.00 |
| | 40.13.5 | 180 mm width | Cover joints for 180 mm width | Each | 200.00 |
| **40.14** | | **BUS U-PVC trunking internal & external angles accessories** | Supply and fixing of following sizes of BUS U-PVC trunking internal & external angles accessories at desired locations etc as reqd. | | |
| | 40.14.1 | Internal angles from 80-100 degree | Internal angles-adjustable from 80-100 degree for 80/105/150/195 X 50mm. | Each | 319.00 |

| Item No. | Sub Item | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 40.14.2 | External Angles From 80-100 Degree | External angles-adjustable from 80 to 120 degree for 80/105/150/195 X 50mm. | Each | 321.00 |
| | 40.14.3 | Flat angles-90 degree for 80 x 50mm. | Flat angles-90 degree for 80 X 50mm. | Each | 400.00 |
| | 40.14.4 | Flat angles-90 degree for 105 x 50mm. | Flat angles-90 degree for 105 X 50mm. | Each | 417.00 |
| | 40.14.5 | Flat angles-90 degree for 150 x 50mm. | Flat angles-90 degree for 150 X 50mm. | Each | 1363.00 |
| | 40.14.6 | Flat angles-90 degree for 195 x 50mm. | Flat angles-90 degree for 195 X 50mm. | Each | 1754.00 |
| **40.15** | | **BUS U-PVC trunking end cap left and right side** | Supply and fixing of following sizes of BUS U-PVC trunking end cap left and right side at desired locations as reqd. | | |
| | 40.15.1 | End cap 85 x 50 mm | End cap 85 x 50 mm | Each | 123.00 |
| | 40.15.2 | End cap 105 x 50 mm | End cap 105 x 50 mm | Each | 114.00 |
| | 40.15.3 | End cap 150 x 50 mm | End cap 150 x 50 mm | Each | 214.00 |
| | 40.15.4 | End cap 195 x 50 mm | End cap 195 x 50 mm | Each | 251.00 |
| **40.16** | | **PVC frame plate for 65 mm cover** | Supply and fixing of following sizes of PVC frame plate for 65 mm cover on existing BUS U-PVC trunking at desired locations i/c cutting of existing flexible cover, partition etc as reqd. | | |
| | 40.16.1 | 2 Module | 2 Module | Each | 116.00 |
| | 40.16.2 | 4 Module | 4 Module | Each | 161.00 |
| | 40.16.3 | 6 Module | 6 Module | Each | 189.00 |

# CHAPTER -41
# HT AND LT CABLES

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.1** | | **XLPE insulated power cable, al. conductor** | Supply of XLPE insulated power cable (conforming IS-7098 Part-I) 1100 Volt grade, 1 core/2 core/3½ core/4 core ISI marked with alu. stranded /solid conductor | | |
| | **41.1.1** | **Unarmoured1 core** | **Unarmoured 1 core** | | |
| | 41.1.1.1 | 16 sq. mm. | 16 sq. mm. | Metre | 46.00 |
| | 41.1.1.2 | 25 sq. mm. | 25 sq. mm. | Metre | 69.00 |
| | 41.1.1.3 | 35 sq. mm. | 35 sq. mm. | Metre | 85.00 |
| | 41.1.1.4 | 50 sq. mm. | 50 sq. mm. | Metre | 109.00 |
| | 41.1.1.5 | 70 sq. mm. | 70 sq. mm. | Metre | 143.00 |
| | 41.1.1.6 | 95 sq. mm. | 95 sq. mm. | Metre | 187.00 |
| | 41.1.1.7 | 120 sq. mm. | 120 sq. mm. | Metre | 223.00 |
| | 41.1.1.8 | 150 sq. mm. | 150 sq. mm. | Metre | 266.00 |
| | 41.1.1.9 | 300 Sq. Mm. | 300 Sq. Mm. | Metre | 483.00 |
| | **41.1.2** | **Armoured 1 core** | **Armoured 1 core** | | |
| | 41.1.2.1 | 16 sq. mm. | 16 sq. mm. | Metre | 82.00 |
| | 41.1.2.2 | 25 sq. mm. | 25 sq. mm. | Metre | 105.00 |
| | 41.1.2.3 | 35 sq. mm. | 35 sq. mm. | Metre | 124.00 |
| | 41.1.2.4 | 50 sq. mm. | 50 sq. mm. | Metre | 149.00 |
| | 41.1.2.5 | 70 sq. mm. | 70 sq. mm. | Metre | 194.00 |
| | 41.1.2.6 | 95 sq. mm. | 95 sq. mm. | Metre | 212.00 |
| | 41.1.2.7 | 120 sq. mm. | 120 sq. mm. | Metre | 256.00 |
| | 41.1.2.8 | 150 sq. mm. | 150 sq. mm. | Metre | 302.00 |
| | 41.1.2.9 | 300 sq. mm. | 300 sq. mm. | Metre | 572.00 |
| | **41.1.3** | **Unarmoured 2 core** | **Unarmoured 2 core** | | |
| | 41.1.3.1 | 2.5 sq. mm. | 2.5 sq. mm. | Metre | 42.00 |
| | 41.1.3.2 | 4 sq. mm. | 4 sq. mm. | Metre | 49.00 |
| | 41.1.3.3 | 6 sq. mm. | 6 sq. mm. | Metre | 63.00 |
| | 41.1.3.4 | 10 sq. mm. | 10 sq. mm. | Metre | 87.00 |
| | 41.1.3.5 | 16 sq. mm. | 16 sq. mm. | Metre | 104.00 |
| | **41.1.4** | **Armoured 2 core** | **Armoured 2 core** | | |
| | 41.1.4.1 | 2.5 sq. mm. | 2.5 sq. mm. | Metre | 73.00 |
| | 41.1.4.2 | 4 sq. mm. | 4 sq. mm. | Metre | 80.00 |
| | 41.1.4.3 | 6 sq. mm. | 6 sq. mm. | Metre | 100.00 |
| | 41.1.4.4 | 10 sq. mm. | 10 sq. mm. | Metre | 121.00 |
| | 41.1.4.5 | 16 sq. mm. | 16 sq. mm. | Metre | 131.00 |
| | **41.1.5** | **Unarmoured 3 core** | **Unarmoured 3 core** | | |
| | 41.1.5.1 | 6 sq. mm. | 6 sq. mm. | Metre | 78.00 |
| | 41.1.5.2 | 10 sq. mm. | 10 sq. mm. | Metre | 104.00 |
| | 41.1.5.3 | 16 sq. mm. | 16 sq. mm. | Metre | 116.00 |
| | 41.1.5.4 | 25 sq. mm | 25 sq. mm | Metre | 167.00 |
| | 41.1.5.5 | 35 sq. mm | 35 sq. mm | Metre | 228.00 |
| | 41.1.5.6 | 50 sq. mm | 50 sq. mm | Metre | 285.00 |
| | 41.1.5.7 | 70 sq. mm | 70 sq. mm | Metre | 401.00 |
| | 41.1.5.8 | 95 sq. mm. | 95 sq. mm. | Metre | 497.00 |
| | **41.1.6** | **Armoured 3 core** | **Armoured 3 core** | | |
| | 41.1.6.1 | 6 sq. mm. | 6 sq. mm. | Metre | 112.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.1.6.2 | 10 sq. mm. | 10 sq. mm. | Metre | 142.00 |
| | 41.1.6.3 | 16 sq. mm. | 16 sq. mm. | Metre | 146.00 |
| | 41.1.6.4 | 25 sq. mm | 25 sq. mm | Metre | 192.00 |
| | 41.1.6.5 | 35 sq. mm | 35 sq. mm | Metre | 252.00 |
| | 41.1.6.6 | 50 sq. mm | 50 sq. mm | Metre | 326.00 |
| | 41.1.6.7 | 70 sq. mm | 70 sq. mm | Metre | 433.00 |
| | 41.1.6.8 | 95 sq. mm. | 95 sq. mm. | Metre | 540.00 |
| | **41.1.7** | **3½ core unarmoured** | **3½ core unarmoured** | | |
| | 41.1.7.1 | 25 sq. mm | 25 sq. mm | Metre | 210.00 |
| | 41.1.7.2 | 35 sq. mm | 35 sq. mm | Metre | 250.00 |
| | 41.1.7.3 | 50 sq. mm | 50 sq. mm | Metre | 338.00 |
| | 41.1.7.4 | 70 sq. mm | 70 sq. mm | Metre | 458.00 |
| | 41.1.7.5 | 95 sq. mm. | 95 sq. mm. | Metre | 577.00 |
| | 41.1.7.6 | 120 sq. mm. | 120 sq. mm. | Metre | 739.00 |
| | 41.1.7.7 | 150 sq. mm. | 150 sq. mm. | Metre | 866.00 |
| | 41.1.7.8 | 185 sq. mm. | 185 sq. mm. | Metre | 1100.00 |
| | 41.1.7.9 | 240 sq. mm. | 240 sq. mm. | Metre | 1327.00 |
| | 41.1.7.10 | 300 sq. mm. | 300 sq. mm. | Metre | 1697.00 |
| | 41.1.7.11 | 400 sq. mm. | 400 sq. mm. | Metre | 2142.00 |
| | **41.1.8** | **3½ core armoured** | **3½ core armoured** | | |
| | 41.1.8.1 | 25 sq. mm | 25 sq. mm | Metre | 231.00 |
| | 41.1.8.2 | 35 sq. mm | 35 sq. mm | Metre | 280.00 |
| | 41.1.8.3 | 50 sq. mm | 50 sq. mm | Metre | 373.00 |
| | 41.1.8.4 | 70 sq. mm | 70 sq. mm | Metre | 500.00 |
| | 41.1.8.5 | 95 sq. mm. | 95 sq. mm. | Metre | 625.00 |
| | 41.1.8.6 | 120 sq. mm. | 120 sq. mm. | Metre | 780.00 |
| | 41.1.8.7 | 150 sq. mm. | 150 sq. mm. | Metre | 904.00 |
| | 41.1.8.8 | 185 sq. mm. | 185 sq. mm. | Metre | 1150.00 |
| | 41.1.8.9 | 240 sq. mm. | 240 sq. mm. | Metre | 1460.00 |
| | 41.1.8.10 | 300 sq. mm. | 300 sq. mm. | Metre | 1792.00 |
| | 41.1.8.11 | 400 sq. mm. | 400 sq. mm. | Metre | 2288.00 |
| | **41.1.9** | **4 core unarmoured** | **4 core unarmoured** | | |
| | 41.1.9.1 | 6 sq. mm. | 6 sq. mm. | Metre | 92.00 |
| | 41.1.9.2 | 10 sq. mm. | 10 sq. mm. | Metre | 119.00 |
| | 41.1.9.3 | 16 sq. mm. | 16 sq. mm. | Metre | 148.00 |
| | 41.1.9.4 | 25 sq. mm | 25 sq. mm | Metre | 221.00 |
| | 41.1.9.5 | 35 sq. mm | 35 sq. mm | Metre | 278.00 |
| | 41.1.9.6 | 50 sq. mm | 50 sq. mm | Metre | 372.00 |
| | 41.1.9.7 | 70 sq. mm | 70 sq. mm | Metre | 527.00 |
| | 41.1.9.8 | 95 sq. mm. | 95 sq. mm. | Metre | 627.00 |
| | 41.1.9.9 | 120 sq. mm. | 120 sq. mm. | Metre | 755.00 |
| | 41.1.9.10 | 150 sq. mm. | 150 sq. mm. | Metre | 955.00 |
| | 41.1.9.11 | 185 sq. mm. | 185 sq. mm. | Metre | 1172.00 |
| | 41.1.9.12 | 240 sq. mm. | 240 sq. mm. | Metre | 1536.00 |
| | 41.1.9.13 | 300 sq. mm. | 300 sq. mm. | Metre | 1883.00 |
| | 41.1.9.14 | 400 sq. mm. | 400 sq. mm. | Metre | 2474.00 |
| | **41.1.10** | **4 core armoured** | **4 core armoured** | | |
| | 41.1.10.1 | 6 sq. mm. | 6 sq. mm. | Metre | 151.00 |
| | 41.1.10.2 | 10 sq. mm. | 10 sq. mm. | Metre | 181.00 |
| | 41.1.10.3 | 16 sq. mm. | 16 sq. mm. | Metre | 218.00 |
| | 41.1.10.4 | 25 sq. mm | 25 sq. mm | Metre | 279.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.1.10.5 | 35 sq. mm | 35 sq. mm | Metre | 358.00 |
| | 41.1.10.6 | 50 sq. mm | 50 sq. mm | Metre | 474.00 |
| | 41.1.10.7 | 70 sq. mm | 70 sq. mm | Metre | 646.00 |
| | 41.1.10.8 | 95 sq. mm. | 95 sq. mm. | Metre | 791.00 |
| | 41.1.10.9 | 120 sq. mm. | 120 sq. mm. | Metre | 985.00 |
| | 41.1.10.10 | 150 sq. mm. | 150 sq. mm. | Metre | 1175.00 |
| | 41.1.10.11 | 185 sq. mm. | 185 sq. mm. | Metre | 1483.00 |
| | 41.1.10.12 | 240 sq. mm. | 240 sq. mm. | Metre | 1895.00 |
| | 41.1.10.13 | 300 sq. mm. | 300 sq. mm. | Metre | 2435.00 |
| | 41.1.10.14 | 400 sq. mm. | 400 sq. mm. | Metre | 3017.00 |
| **41.2** | | **High-tension XLPE cable, 1 core armoured al. conductor** | Supply of approved high-tension XLPE cable (conforming IS-7098/II/85) as per ISI standard single core armoured with alu. solid/stranded conductor ISI marked as required | | |
| | **41.2.1** | **Single core XLPE cable 11 Kv grade** | **Single core XLPE cable 11 Kv grade** | | |
| | 41.2.1.1 | 50 sq. mm | 50 sq. mm | Metre | 302.00 |
| | 41.2.1.2 | 70 sq. mm | 70 sq. mm | Metre | 354.00 |
| | 41.2.1.3 | 95 sq. mm. | 95 sq. mm. | Metre | 477.00 |
| | 41.2.1.4 | 120 sq. mm. | 120 sq. mm. | Metre | 544.00 |
| | 41.2.1.5 | 150 sq. mm. | 150 sq. mm. | Metre | 592.00 |
| | 41.2.1.6 | 185 sq. mm. | 185 sq. mm. | Metre | 696.00 |
| | 41.2.1.7 | 240 sq. mm. | 240 sq. mm. | Metre | 774.00 |
| | 41.2.1.8 | 300 sq. mm. | 300 sq. mm. | Metre | 885.00 |
| | **41.2.2** | **Single core XLPE cable 33 Kv grade** | **Single core XLPE cable 33 Kv grade** | | |
| | 41.2.2.1 | 95 sq. mm. | 95 sq. mm. | Metre | 675.00 |
| | 41.2.2.2 | 120 sq. mm. | 120 sq. mm. | Metre | 800.00 |
| | 41.2.2.3 | 150 sq. mm. | 150 sq. mm. | Metre | 920.00 |
| | 41.2.2.4 | 185 sq. mm. | 185 sq. mm. | Metre | 1013.00 |
| | 41.2.2.5 | 240 sq. mm. | 240 sq. mm. | Metre | 1158.00 |
| | 41.2.2.6 | 300 sq. mm. | 300 sq. mm. | Metre | 1321.00 |
| **41.3** | | **High-tension XLPE cable, 3 core armoured al. conductor** | Supply of approved high-tension XLPE cable (conforming IS-7098/II/85) as per ISI standard 3 core armoured with alu. solid/stranded conductor ISI marked as required | | |
| | **41.3.1** | **3 core XLPE cable 11 Kv grade** | **3 core XLPE cable 11 Kv grade** | | |
| | 41.3.1.1 | 50 sq. mm | 50 sq. mm | Metre | 786.00 |
| | 41.3.1.2 | 70 sq. mm | 70 sq. mm | Metre | 1016.00 |
| | 41.3.1.3 | 95 sq. mm. | 95 sq. mm. | Metre | 1197.00 |
| | 41.3.1.4 | 120 sq. mm. | 120 sq. mm. | Metre | 1392.00 |
| | 41.3.1.5 | 150 sq. mm. | 150 sq. mm. | Metre | 1560.00 |
| | 41.3.1.6 | 185 sq. mm. | 185 sq. mm. | Metre | 1782.00 |
| | 41.3.1.7 | 240 sq. mm. | 240 sq. mm. | Metre | 2118.00 |
| | 41.3.1.8 | 300 sq. mm. | 300 sq. mm. | Metre | 2470.00 |
| | **41.3.2** | **3 core XLPE cable 33 Kv grade** | **3 core XLPE cable 33 Kv grade** | | |
| | 41.3.2.1 | 95 sq. mm. | 95 sq. mm. | Metre | 1672.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.3.2.2 | 120 sq. mm. | 120 sq. mm. | Metre | 1927.00 |
| | 41.3.2.3 | 150 sq. mm. | 150 sq. mm. | Metre | 2418.00 |
| | 41.3.2.4 | 185 sq. mm. | 185 sq. mm. | Metre | 2675.00 |
| | 41.3.2.5 | 240 sq. mm. | 240 sq. mm. | Metre | 3050.00 |
| | 41.3.2.6 | 300 sq. mm. | 300 sq. mm. | Metre | 3400.00 |
| **41.4** | | **XLPE insulated FRLS heavy duty power cable** | Supply of XLPE insulated FRLS heavy duty power cable conforming IS-7098 (Part-1) 1988, 1100 Volt grade, 2/3/4/8/10/12 core ISI marked with copper stranded/solid conductor. | | |
| | **41.4.1** | **Armoured 2 core** | **Armoured 2 core** | | |
| | 41.4.1.1 | 2.5 sq. mm. | 2.5 sq. mm. | Metre | 136.00 |
| | 41.4.1.2 | 4 sq. mm. | 4 sq. mm. | Metre | 183.00 |
| | 41.4.1.3 | 6 sq. mm. | 6 sq. mm. | Metre | 241.00 |
| | 41.4.1.4 | 10 sq. mm. | 10 sq. mm. | Metre | 369.00 |
| | **41.4.2** | **Armoured 3 core** | **Armoured 3 core** | | |
| | 41.4.2.1 | 2.5 sq. mm. | 2.5 sq. mm. | Metre | 168.00 |
| | 41.4.2.2 | 4 sq. mm. | 4 sq. mm. | Metre | 229.00 |
| | 41.4.2.3 | 6 sq. mm. | 6 sq. mm. | Metre | 314.00 |
| | 41.4.2.4 | 10 sq. mm. | 10 sq. mm. | Metre | 504.00 |
| | 41.4.2.5 | 16 sq. mm. | 16 sq. mm. | Metre | 723.00 |
| | 41.4.2.6 | 25 sq. mm. | 25 sq. mm. | Metre | 1144.00 |
| | **41.4.3** | **Armoured 3½ core** | **Armoured 3½ core** | | |
| | 41.4.3.1 | 25 Sq. mm | 25 Sq. mm (10 Sq. mm 1/2 core). | Metre | 1338.00 |
| | 41.4.3.2 | 35 Sq. mm | 35 Sq. mm (16 Sq. mm 1/2 core) | Metre | 1759.00 |
| | 41.4.3.3 | 50 Sq. mm | 50 Sq. mm (25 Sq. mm 1/2 core) | Metre | 2469.00 |
| | 41.4.3.4 | 70 Sq. mm | 70 Sq. mm (35 Sq. mm 1/2 core) | Metre | 3317.00 |
| | 41.4.3.5 | 95 Sq. mm | 95 Sq. mm (50 Sq. mm 1/2 core) | Metre | 4816.00 |
| | 41.4.3.6 | 120 Sq. mm | 120 Sq. mm (70 Sq. mm 1/2 core) | Metre | 6190.00 |
| | **41.4.4** | **Armoured 4 core** | **Armoured 4 core** | | |
| | 41.4.4.1 | 1.5 sq. mm. | 1.5 sq. mm. | Metre | 148.00 |
| | 41.4.4.2 | 2.5 sq. mm. | 2.5 sq. mm. | Metre | 217.00 |
| | 41.4.4.3 | 4 sq. mm. | 4 sq. mm. | Metre | 302.00 |
| | 41.4.4.4 | 6 sq. mm. | 6 sq. mm. | Metre | 430.00 |
| | 41.4.4.5 | 10 sq. mm. | 10 sq. mm. | Metre | 625.00 |
| | 41.4.4.6 | 16 sq. mm. | 16 sq. mm. | Metre | 909.00 |
| | 41.4.4.7 | 25 sq. mm. | 25 sq. mm. | Metre | 1455.00 |
| | **41.4.5** | **Armoured 8 core** | **Armoured 8 core** | | |
| | 41.4.5.1 | 2.5 Sq. Mm. | 2.5 Sq. Mm. | Metre | 394.00 |
| | **41.4.6** | **Armoured 10 core** | **Armoured 10 core** | | |
| | 41.4.6.1 | 2.5 Sq. Mm. | 2.5 Sq. Mm. | Metre | 469.00 |
| | **41.4.7** | **Armoured 12 core** | **Armoured 12 core** | | |
| | 41.4.7.1 | 2.5 Sq. Mm. | 2.5 Sq. Mm. | Metre | 540.00 |
| | **41.4.8** | **Unarmoured 2 core** | **Unarmoured 2 core** | | |
| | 41.4.8.1 | 2.5 sq. Mm. | 2.5 sq. Mm. | Metre | 108.00 |
| | **41.4.9** | **Unarmoured 4 core** | **Unarmoured 4 core** | | |
| | 41.4.9.1 | 2.5 sq. Mm. | 2.5 sq. Mm. | Metre | 187.00 |
| | **41.4.10** | **Unarmoured 8 core** | **Unarmoured 8 core** | | |
| | 41.4.10.1 | 2.5 sq. Mm. | 2.5 sq. Mm. | Metre | 357.00 |
| | **41.4.11** | **Unarmoured 12 core** | **Unarmoured 12 core** | | |
| | 41.4.11.1 | 2.5 sq. Mm. | 2.5 sq. Mm. | Metre | 490.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.5** | | **Gland** | **Brass compression gland** | | |
| | **41.5.1** | **Gland for P.V.C. insulated armoured cable** | Supplying and fixing heavy duty cable gland for P.V.C. insulated armoured cable with brass washer, rubber ring complete erected with cable and lead connection etc. as per specification complete. | | |
| | 41.5.1.1 | 22 mm | Gland size 22mm suitable for cable 2,3,3½ & 4 x upto 6 sq. mm | Each | 79.00 |
| | 41.5.1.2 | 22 mm | Gland size 22mm suitable for cable 2/3, 3½, 2/4 x 10 sq. mm or 2x 16 sq. mm | Each | 97.00 |
| | 41.5.1.3 | 28 mm | Gland size 28mm for 3/4 x 16 sq. mm | Each | 138.00 |
| | 41.5.1.4 | 32 mm | Gland size 32 mm for 2/3, 3½, 4 x 25 sq. mm OR 2/3, 3½ x 35 sq. mm or 2/3/3½ x 50 sq. mm. | Each | 158.00 |
| | 41.5.1.5 | 38 mm | Gland size 38 mm 3½ x 70 sq. mm, 3x95 sq. mm | Each | 219.00 |
| | 41.5.1.6 | 45 mm | Gland size 45 mm 3/3½ x 120 sq. mm, 3½ x 95 sq. mm, 3 x 150 sq. mm. | Each | 260.00 |
| | 41.5.1.7 | 50 mm | Gland size 50 mm ,3½ x 150 sq. mm,3 x 185 sq. mm | Each | 290.00 |
| | 41.5.1.8 | 57 mm | Gland size 57 mm, 3 x 225 sq. mm, 3 ½ x 185 sq. mm | Each | 440.00 |
| | 41.5.1.9 | 70 mm | Gland size 70 mm, 3 x 240 sq. mm, 3½ x 300 sq. mm | Each | 605.00 |
| | 41.5.1.10 | 82 mm | Gland size 82 mm 3½ x 400 sq. mm | Each | 864.00 |
| **41.6** | | **Ferrules** | Supplying and fixing ferrules as per IS - specification suitable for following size of cable for circuit identification including connection as required complete | | |
| | | For conductor size- | For conductor size- | | |
| | 41.6.1 | 2.5-6.00 sq. mm | 2.5 to 6.00 sq. mm | Each | 3.00 |
| | 41.6.2 | 10.00 sq. mm | 10.00 sq. mm | Each | 4.00 |
| | 41.6.3 | 16.00 sq. mm | 16.00 sq. mm | Each | 4.00 |
| | 41.6.4 | 25.00 sq. mm | 25.00 sq. mm | Each | 6.00 |
| | 41.6.5 | 35.00 sq. mm | 35.00 sq. mm | Each | 7.00 |
| | 41.6.6 | 50.00 sq. mm | 50.00 sq. mm | Each | 11.00 |
| | 41.6.7 | 70.00 sq. mm | 70.00 sq. mm | Each | 14.00 |
| | 41.6.8 | 95.00 sq. mm | 95.00 sq. mm | Each | 16.00 |
| | 41.6.9 | 120.00 sq. mm | 120.00 sq. mm | Each | 18.00 |
| | 41.6.10 | 150.00 sq. mm | 150.00 sq. mm | Each | 25.00 |
| | 41.6.11 | 185.00 sq. mm | 185.00 sq. mm | Each | 32.00 |
| | 41.6.12 | 240.00 sq. mm | 240.00 sq. mm | Each | 46.00 |
| | 41.6.13 | 300.00 sq. mm | 300.00 sq. mm | Each | 57.00 |
| | 41.6.14 | 400.00 sq. mm | 400.00 sq. mm | Each | 70.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.7** | | **Crimping type alum. lugs** | Supplying and fixing crimping type alum. lugs as per I.S.S. specification suitable for following size of cable with alu./copper solid/stranded conductor evenly crimped with high/pressure tool and connected to switch gear/bus/M.C.C.B./M.C.B. etc. as required complete. | | |
| | | For conductor size- | For conductor size- | | |
| | 41.7.1 | 6 -16 sq. mm | 6mm to 16 sq. mm | Each | 8.00 |
| | 41.7.2 | 25 sq. mm | 25 sq. mm | Each | 14.00 |
| | 41.7.3 | 35 sq. mm | 35 sq. mm | Each | 22.00 |
| | 41.7.4 | 50 sq. mm | 50 sq. mm | Each | 22.00 |
| | 41.7.5 | 70 sq. mm | 70 sq. mm | Each | 35.00 |
| | 41.7.6 | 95 sq. mm. | 95 sq. mm. | Each | 48.00 |
| | 41.7.7 | 120 sq. mm. | 120 sq. mm. | Each | 58.00 |
| | 41.7.8 | 150 sq. mm. | 150 sq. mm. | Each | 66.00 |
| | 41.7.9 | 180 sq. mm | 180 sq. mm | Each | 76.00 |
| | 41.7.10 | 240 sq. mm. | 240 sq. mm. | Each | 116.00 |
| | 41.7.11 | 300 sq. mm. | 300 sq. mm. | Each | 138.00 |
| | 41.7.12 | 400 sq. mm. | 400 sq. mm. | Each | 213.00 |
| **41.8** | | **Pin terminal lugs** | Supplying & fixing pin terminal lugs as per ISS specification suitable for cable evenly cramped with high pressure tool & connection to switch gear terminal as required complete. | | |
| | **41.8.1** | **Aluminum lugs pin type** | **Aluminum lugs pin type** | | |
| | 41.8.1.1 | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | Each | 19.00 |
| | 41.8.1.2 | Upto 25 sq. mm conductor | Upto 25 sq. mm conductor size | Each | 22.00 |
| | 41.8.1.3 | 35 sq. mm conductor | 35 sq. mm conductor size | Each | 26.00 |
| | **41.8.2** | **Copper lugs pin type** | **Copper lugs pin type** | | |
| | 41.8.2.1 | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | Each | 30.00 |
| | 41.8.2.2 | Upto 25 sq. mm conductor | Upto 25 sq. mm conductor size | Each | 42.00 |
| | 41.8.2.3 | 35 sq. mm conductor | 35 sq. mm conductor size | Each | 50.00 |
| | **41.8.3** | **Aluminum lugs tube type** | **Aluminum lugs tube type** | | |
| | 41.8.3.1 | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | Each | 6.00 |
| | 41.8.3.2 | 25 sq. mm conductor | 25 sq. mm conductor size | Each | 10.00 |
| | 41.8.3.3 | 35 sq. mm conductor | 35 sq. mm conductor size | Each | 13.00 |
| | 41.8.3.4 | 50 sq. mm. conductor | 50 sq. mm. conductor size | Each | 17.00 |
| | **41.8.4** | **Copper lugs tube type** | **Copper lugs tube type** | | |
| | 41.8.4.1 | Upto 16 sq. mm conductor | Upto 16 sq. mm conductor size | Each | 13.00 |
| | 41.8.4.2 | 25 sq. mm conductor | 25 sq. mm conductor size | Each | 22.00 |
| | 41.8.4.3 | 35 sq. mm conductor | 35 sq. mm conductor size | Each | 32.00 |
| | 41.8.4.4 | 50 sq. mm. conductor | 50 sq. mm. conductor size | Each | 43.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.9** | | **Cable end termination with heat shrinkable jointing kit** | Providing &making cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 33 KV/11 KV single core XLPE alum. conductor cable as required as per specification and as per accepted standard including connection testing complete. | | |
| | **41.9.1** | **11 KV XLPE single core** | **Heat shrinkable joint, kit 11 KV XLPE single core** | | |
| | 41.9.1.1 | 11 KV XLPE cable O.D. termination | Heat shrinkable Jointing kit 11 KV XLPE cable O.D. termination | | |
| | 41.9.1.1.1 | 1 x 16-50 sq. mm. | 1 x 16-50 sq. mm. | Each | 3214.00 |
| | 41.9.1.1.2 | 1 x 70-120 sq. mm. | 1 x 70-120 sq. mm. | Each | 3511.00 |
| | 41.9.1.1.3 | 1 x 150-300 sq. mm. | 1 x 150-300 sq. mm. | Each | 3960.00 |
| | 41.9.1.1.4 | 1 x 400-630 sq. mm. | 1 x 400-630 sq. mm. | Each | 5714.00 |
| | **41.9.1.2** | **11 KV XLPE cable I.D. termination** | **Heat shrinkable jointing kit 11 KV XLPE cable I.D. termination** | | |
| | 41.9.1.2.1 | 1 x 16-50 sq. mm. | 1 x 16-50 sq. mm. | Each | 2940.00 |
| | 41.9.1.2.2 | 1 x 70-120 sq. mm. | 1 x 70-120 sq. mm. | Each | 3177.00 |
| | 41.9.1.2.3 | 1 x 150-300 sq. mm. | 1 x 150-300 sq. mm. | Each | 3523.00 |
| | 41.9.1.2.4 | 1 x 400-630 sq. mm. | 1 x 400-630 sq. mm. | Each | 4893.00 |
| | **41.9.1.3** | **11 KV XLPE cable straight through termination** | **Heat shrinkable jointing kit 11 KV XLPE cable straight through termination** | | |
| | 41.9.1.3.1 | 1 x 16-35 sq. mm. | 1 x 16-35 sq. mm. | Each | 9790.00 |
| | 41.9.1.3.2 | 1 x 50-95 sq. mm. | 1 x 50-95 sq. mm. | Each | 10701.00 |
| | 41.9.1.3.3 | 1 x 120-185 sq. mm. | 1 x 120-185 sq. mm. | Each | 11221.00 |
| | 41.9.1.3.4 | 1 x 300-400 sq. mm. | 1 x 300-400 sq. mm. | Each | 12674.00 |
| | 41.9.1.3.5 | 1 x 400-500 sq. mm. | 1 x 400-500 sq. mm. | Each | 13921.00 |
| | **41.9.2** | **33 KV XLPE single core** | **Heat shrinkable joint, kit 33 KV XLPE single core** | | |
| | **41.9.2.1** | **33 KV XLPE cable O.D. termination** | **Heat shrinkable Jointing kit 33 KV XLPE cable O.D. termination** | | |
| | 41.9.2.1.1 | 1 x 25-50 sq. mm. | 1 x 25-50 sq. mm. | Each | 5372.00 |
| | 41.9.2.1.2 | 1 x 70-185 sq. mm. | 1 x 70-185 sq. mm. | Each | 5809.00 |
| | 41.9.2.1.3 | 1 x 240-400 sq. mm. | 1 x 240-400 sq. mm. | Each | 7519.00 |
| | **41.9.2.2** | **33 KV XLPE cable I.D. termination** | **Heat shrinkable jointing kit 33 KV XLPE cable I.D. termination** | | |
| | 41.9.2.2.1 | 1 x 25-50 sq. mm. | 1 x 25-50 sq. mm. | Each | 4853.00 |
| | 41.9.2.2.2 | 1 x 70-185 sq. mm. | 1 x 70-185 sq. mm. | Each | 5320.00 |
| | 41.9.2.2.3 | 1 x 240-400 sq. mm. | 1 x 240-400 sq. mm. | Each | 7018.00 |
| | **41.9.2.3** | **33 KV XLPE cable straight through termination** | **Heat shrinkable jointing kit 33 KV XLPE cable straight through termination** | | |
| | 41.9.2.3.1 | 1 x 35-70 sq. mm. | 1 x 35-70 sq. mm. | Each | 16877.00 |
| | 41.9.2.3.2 | 1 x 95-150 sq. mm. | 1 x 95-150 sq. mm. | Each | 18791.00 |
| | 41.9.2.3.3 | 1 x 185-300 sq. mm. | 1 x 185-300 sq. mm. | Each | 20726.00 |
| | 41.9.2.3.4 | 1 x 400 sq. mm. | 1 x 400 sq. mm. | Each | 24469.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.10** | | **Cable end termination with heat shrinkable jointing kit** | Providing & Making cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 33 KV/11 KV 3 core XLPE alum. conductor cable as required as per specification and as per accepted standard including connection testing complete. | | |
| | **41.10.1** | **11 KV XLPE, 3 core** | **Heat shrinkable joint, kit 11 KV XLPE, 3 core** | | |
| | **41.10.1.1** | **11 KV XLPE cable O.D. termination** | **Heat shrinkable Jointing kit 11 KV XLPE cable O.D. termination** | | |
| | 41.10.1.1.1 | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | Each | 9194.00 |
| | 41.10.1.1.2 | 3 x 70-120 sq. mm. | 3 x 70-120 sq. mm. | Each | 9488.00 |
| | 41.10.1.1.3 | 3 x 150-225 sq. mm. | 3 x 150-225 sq. mm. | Each | 10963.00 |
| | 41.10.1.1.4 | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | Each | 11389.00 |
| | **41.10.1.2** | **11 KV XLPE cable I.D. termination** | **Heat shrinkable jointing kit 11 KV XLPE cable I.D. termination** | | |
| | 41.10.1.2.1 | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | Each | 8342.00 |
| | 41.10.1.2.2 | 3 x 70-120 sq. mm. | 3 x 70-120 sq. mm. | Each | 8431.00 |
| | 41.10.1.2.3 | 3 x 150-225 sq. mm. | 3 x 150-225 sq. mm. | Each | 9734.00 |
| | 41.10.1.2.4 | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | Each | 9470.00 |
| | **41.10.1.3** | **11 KV XLPE cable straight joint** | **Heat shrinkable jointing kit 11 KV XLPE cable straight joint** | | |
| | 41.10.1.3.1 | 3 x 16-50 sq. mm. | 3 x 16-50 sq. mm. | Each | 18520.00 |
| | 41.10.1.3.2 | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | Each | 19986.00 |
| | 41.10.1.3.3 | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | Each | 21585.00 |
| | 41.10.1.3.4 | 3 x 240-300 sq. mm. | 3 x 240-300 sq. mm. | Each | 23710.00 |
| | 41.10.1.3.5 | 3 x 400-500 sq. mm. | 3 x 400-500 sq. mm. | Each | 29159.00 |
| | **41.10.2** | **33 KV XLPE, 3 core** | **Heat shrinkable joint, kit 33 KV XLPE, 3 core** | | |
| | **41.10.2.1** | **33 KV XLPE cable O.D. termination** | **Heat shrinkable jointing kit 33 KV XLPE cable O.D. termination** | | |
| | 41.10.2.1.1 | 3 x 35-50 sq. mm. | 3 x 35-50 sq. mm. | Each | 15549.00 |
| | 41.10.2.1.2 | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | Each | 16311.00 |
| | 41.10.2.1.3 | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | Each | 16600.00 |
| | 41.10.2.1.4 | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | Each | 20808.00 |
| | **41.10.2.2** | **33 KV XLPE cable I.D. termination** | **Heat shrinkable Jointing kit 33 KV XLPE cable I.D. termination** | | |
| | 41.10.2.2.1 | 3 x 35-50 sq. mm. | 3 x 35-50 sq. mm. | Each | 13614.00 |
| | 41.10.2.2.2 | 3 x 70-95 sq. mm. | 3 x 70-95 sq. mm. | Each | 14302.00 |
| | 41.10.2.2.3 | 3 x 120-185 sq. mm. | 3 x 120-185 sq. mm. | Each | 15295.00 |
| | 41.10.2.2.4 | 3 x 240-400 sq. mm. | 3 x 240-400 sq. mm. | Each | 18182.00 |
| | **41.10.2.3** | **33 KV XLPE cable straight joint** | **Heat shrinkable Jointing kit 33 KV XLPE cable straight joint** | | |
| | 41.10.2.3.1 | 3 x 35-70 sq. mm. | 3 x 35-70 sq. mm. | Each | 36816.00 |
| | 41.10.2.3.2 | 3 x 95-150 sq. mm. | 3 x 95-150 sq. mm. | Each | 39937.00 |
| | 41.10.2.3.3 | 3 x 185 sq. mm. | 3 x 185 sq. mm. | Each | 44644.00 |
| | 41.10.2.3.4 | 3 x 240 sq. mm. | 3 x 240 sq. mm. | Each | 46543.00 |
| | 41.10.2.3.5 | 3 x 300 sq. mm. | 3 x 300 sq. mm. | Each | 47749.00 |
| | 41.10.2.3.6 | 3 x 400 sq. mm. | 3 x 400 sq. mm. | Each | 53732.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.11** | | **Cable straight joint/end termination with heat shrinkable jointing kit** | Supply and making cable straight joint/end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for 1.1 KV XLPE/HD alum. conductor cable as required as per specification and as per accepted standard including connection testing complete for (I.D../O.D.) termination. | | |
| | **41.11.1** | **1.1 KV XLPE/HD cable end termination jointing kit** | **Heat shrinkable jointing kit 1.1 KV XLPE/HD cable end termination jointing kit (I.D./O.D.)** | | |
| | 41.11.1.1 | 16-35 sq. mm. 1 core | 16-35 sq. mm. 1 core | Each | 446.00 |
| | 41.11.1.2 | 50-95 sq. mm. 1 core | 50-95 sq. mm. 1 core | Each | 497.00 |
| | 41.11.1.3 | 120-185 sq. mm. 1 core | 120-185 sq. mm. 1 core | Each | 595.00 |
| | 41.11.1.4 | 225-300 sq. mm. 1 core | 225-300 sq. mm. 1 core | Each | 991.00 |
| | 41.11.1.5 | 400-500 sq. mm. 1 core | 400-500 sq. mm. 1 core | Each | 1477.00 |
| | **41.11.2** | **1.1 KV XLPE/HD cable straight through jointing kit** | **Heat shrinkable jointing kit 1.1 KV XLPE/HD cable straight through jointing kit (I.D./O.D.)** | | |
| | 41.11.2.1 | 16-35 sq. mm. 1 core | 16-35 sq. mm. 1 core | Each | 462.00 |
| | 41.11.2.2 | 50-95 sq. mm. 1 core | 50-95 sq. mm. 1 core | Each | 559.00 |
| | 41.11.2.3 | 120-185 sq. mm. 1 core | 120-185 sq. mm. 1 core | Each | 768.00 |
| | 41.11.2.4 | 225-300 sq. mm. 1 core | 225-300 sq. mm. 1 core | Each | 1073.00 |
| | 41.11.2.5 | 400-500 sq. mm. 1 core | 400-500 sq. mm. 1 core | Each | 1663.00 |
| | **41.11.3** | **1.1 KV XLPE/HD cable I.D./O.D. end termination** | **Heat shrinkable Jointing kit 1.1 KV XLPE/HD cable I.D./O.D. end termination** | | |
| | 41.11.3.1 | 4-16 sq. mm. 3½/4 core | 4-16 sq. mm. 3½/4 core | Each | 3009.00 |
| | 41.11.3.2 | 25-50 sq. mm. 3½/4 core | 25-50 sq. mm. 3½/4 core | Each | 3677.00 |
| | 41.11.3.3 | 70-150 sq. mm. 3½/4 core | 70-150 sq. mm. 3½/4 core | Each | 3840.00 |
| | 41.11.3.4 | 185-300 sq. mm. 3½/4 core | 185-300 sq. mm. 3½/4 core | Each | 6329.00 |
| | 41.11.3.5 | 400-500 sq. mm. 3½/4 core | 400-500 sq. mm. 3½/4 core | Each | 8447.00 |
| | **41.11.4** | **1.1 KV XLPE/HD cable straight through jointing kit** | **Heat shrinkable jointing kit 1.1 KV XLPE/HD cable straight through jointing kit (I.D../O.D.)** | | |
| | 41.11.4.1 | 10-16 sq. mm. 3½/4core | 10-16 sq. mm. 3½/4core | Each | 1991.00 |
| | 41.11.4.2 | 25-50 sq. mm. 3½/4core | 25-50 sq. mm. 3½/4core | Each | 2505.00 |
| | 41.11.4.3 | 70 -150 sq. mm. 3½/4core | 70 -150 sq. mm. 3½/4core | Each | 3495.00 |
| | 41.11.4.4 | 185-240 sq. mm. 3½/4core | 185-240 sq. mm. 3½/4core | Each | 5822.00 |
| | 41.11.4.5 | 300-400 sq. mm. 3½/4core | 300-400 sq. mm. 3½/4core | Each | 6496.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.12** | | **G.I. pipe for protection of underground cable** | Supplying & installing G.I. pipe for protection of underground cable fixed on wall/support/in trench/fixed between two rigid existing support of wall/beam for erection of ceiling Fan/down rod for stiff pendent for light luminaries /fan/protective for earthing, lightening conductor down strip/overhead service line/for submersible cable or centrifugal pump for water supply with necessary iron clamp coupler, bend, tee, elbow, nuts and bolts etc. complete in an approved manner as required to complete excluding cost of excavation/dismantling & other finished masonry Item complete. | | |
| | **41.12.1** | **For 'B' class pipe** | **For 'B' class pipe ISI marked (IS-1161-68)** | | |
| | 41.12.1.1 | 15.00mm | 15.00mm | Metre | 167.00 |
| | 41.12.1.2 | 20.00mm | 20.00mm | Metre | 201.00 |
| | 41.12.1.3 | 25.00mm | 25.00mm | Metre | 291.00 |
| | 41.12.1.4 | 32.00mm | 32.00mm | Metre | 368.00 |
| | 41.12.1.5 | 40.00mm | 40.00mm | Metre | 423.00 |
| | 41.12.1.6 | 50.00mm | 50.00mm | Metre | 550.00 |
| | 41.12.1.7 | 65.00mm | 65.00mm | Metre | 693.00 |
| | 41.12.1.8 | 80.00mm | 80.00mm | Metre | 913.00 |
| | 41.12.1.9 | 100.00mm | 100.00mm | Metre | 1256.00 |
| | 41.12.1.10 | 150.00mm | 150.00mm | Metre | 2062.00 |
| | **41.12.2** | **For 'A' class G.I. pipe** | **For 'A' class G.I. pipe ISI marked(IS-1161-69)** | | |
| | 41.12.2.1 | 15.00mm | 15.00mm | Metre | 142.00 |
| | 41.12.2.2 | 19/20mm | 19/20mm | Metre | 194.00 |
| | 41.12.2.3 | 25.00mm | 25.00mm | Metre | 256.00 |
| | 41.12.2.4 | 32.00mm | 32.00mm | Metre | 331.00 |
| | 41.12.2.5 | 40.00mm | 40.00mm | Metre | 390.00 |
| | 41.12.2.6 | 50.00mm | 50.00mm | Metre | 503.00 |
| | 41.12.2.7 | 65.00mm | 65.00mm | Metre | 661.00 |
| | 41.12.2.8 | 80.00mm | 80.00mm | Metre | 790.00 |
| | 41.12.2.9 | 100.00mm | 100.00mm | Metre | 1151.00 |
| **41.13** | | **Rigid PVC single socket and, non-metallic pipe** | Supply and laying rigid PVC single socket and, non-metallic pipe ISI marked (IS-4985-2000) 10 Kg/sq.cm class 5 for protection of underground cable/or for water pump delivery side, for water supply complete with necessary clamp coupler, band, tee, elbow etc. in an approved manner excluding cost of excavation etc. | | |
| | 41.13.1 | 50.00mm O.D | 50.00mm O.D | Metre | 163.00 |
| | 41.13.2 | 75.00mm O.D | 75.00mm O.D | Metre | 357.00 |
| | 41.13.3 | 110.00mm O.D | 110.00mm O.D | Metre | 753.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.13.4 | 160.00mm O.D | 160.00mm O.D | Metre | 1617.00 |
| | 41.13.5 | 200.00mm O.D | 200.00mm O.D | Metre | 2585.00 |
| **41.14** | | **Double wall corrugated pipes (DWC) of HDPE** | Supplying and installing double wall corrugated pipes (DWC) of HDPE (IS 14930 Part II -marked) for cable laid underground with necessary material and at required depth upto 90cm. below road/ground surface, excluding excavation, back filling with excavated material, ramming and making the surface good. | | |
| | 41.14.1 | 50.00mm outside dia. | 50.00mm outside dia. | Metre | 133.00 |
| | 41.14.2 | 63.00mm outside dia. | 63.00mm outside dia. | Metre | 175.00 |
| | 41.14.3 | 90.00mm outside dia. | 90.00mm outside dia. | Metre | 214.00 |
| | 41.14.4 | 110.00mm outside dia. | 110.00mm outside dia. | Metre | 252.00 |
| | 41.14.5 | 120.00mm outside dia. | 120.00mm outside dia. | Metre | 279.00 |
| | 41.14.6 | 180.00mm outside dia. | 180.00mm outside dia. | Metre | 390.00 |
| | 41.14.7 | 210.00mm outside dia. | 210.00mm outside dia. | Metre | 532.00 |
| **41.15** | | **Cable laying** | **Cable laying** | | |
| | 41.15.1 | Laying, 1.1KV grade of size not exceeding 25 sq. mm | Laying of one number armoured / unarmoured power cable 1.1KV grade of size not exceeding 25 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, sand cushioning, protective covering and refilling the pit etc. as required. | Metre | 230.00 |
| | 41.15.2 | Laying, 1.1.KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm | Laying of one number armoured / unarmoured power cable 1.1.KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, sand cushioning, protective covering and refilling the pit etc. as required. | Metre | 246.00 |
| | 41.15.3 | Laying, 1.1.KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm | Laying of one number armoured / unarmoured cable 1.1.KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm direct in ground including excavation in all soil/ murrum /soft rock, and sand cushioning protective covering and refilling the pit etc. as required. | Metre | 266.00 |
| | 41.15.4 | Laying, 1.1KV grade of size not exceeding 25sq. mm, additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size not exceeding 25sq. mm direct in ground in the same trench in one tier horizontal formation including excavation in all soil/murrum/soft rock, sand cushioning protective covering and refilling the pit etc. as required. | Metre | 136.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.15.5 | Laying, 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm, additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the pit etc. as required. | Metre | 146.00 |
| | 41.15.6 | Laying, 1.1KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm., additional cable | Laying of one number additional armoured / unarmoured power cable 1.1KV grade of size exceeding 120 sq. mm. but not exceeding 400 sq. mm. direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the pit etc. as required. | Metre | 159.00 |
| | 41.15.7 | Laying, 1:1 KV grade of size not exceeding 25 sq. mm in the existing item | Laying of one number armoured / unarmoured power cable 1:1 KV grade of size not exceeding 25 sq. mm in the existing RCC hume/stone ware/G.I. pipe/ DWC Pipe/ surface in existing trench as required. | Metre | 28.00 |
| | 41.15.8 | Laying, 1.1 KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm. in the existing item | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm. in the existing RCC Hume /Stone ware/G.I. Pipe/ DWC Pipe/ surface in existing trench as required. | Metre | 35.00 |
| | 41.15.9 | Laying, 1.1 KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm. in the existing item | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm. in the existing RCC Hume /Stone ware/G.I. Pipe/ DWC Pipe/ surface in existing trench as required. | Metre | 54.00 |
| | 41.15.10 | Laying, 1.1 KV grade of size not exceeding 25 sq. mm in the existing masonry open duct | Laying of one number armoured / unarmoured power cable 1.1 KV grade of size not exceeding 25 sq. mm in the existing masonry open duct as required. | Metre | 23.00 |
| | 41.15.11 | Laying, 1.1KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm | Laying of one number armoured/unarmoured power cable 1.1KV grade of size exceeding 25sq. mm but not exceeding 120 sq. mm in the existing masonry open duct as required. | Metre | 36.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.15.12 | Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm | Laying of one number armoured/unarmoured power cable 1.1KV grade of size exceeding 120sq. mm but not exceeding 400 sq. mm in the existing masonry open duct as required. | Metre | 50.00 |
| | 41.15.13 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required. | Metre | 126.00 |
| | 41.15.14 | Laying, 33KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade 33KV of size not exceeding 300 sq. mm direct in ground including excavation sand cushioning, excluding protective covering and refilling the trench etc. as required | Metre | 125.00 |
| | 41.15.15 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm in the existing RCC/Hume/Stoneware/G.I. Pipe/ DWC Pipe/surface in existing trench as required. | Metre | 28.00 |
| | 41.15.16 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. | Laying of one number single core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | Metre | 33.00 |
| | 41.15.17 | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm in the existing RCC/Hume/Stoneware/G.I. Pipe/ DWC Pipe/surface in existing trench as required. | Metre | 34.00 |
| | 41.15.18 | Laying, 33KV grade of size not exceeding 300 sq. mm. | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | Metre | 38.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.15.19 | Laying, 11KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 11KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | Metre | 92.00 |
| | 41.15.20 | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number single core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | Metre | 91.00 |
| | 41.15.21 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. | Laying of one number 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required. | Metre | 221.00 |
| | 41.15.22 | Laying, 33KV of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable of grade 33KV of size not exceeding 300 sq. mm direct in ground including excavation in all soil/murrum/soft rock, sand cushioning, excluding protective covering and refilling the trench etc. as required | Metre | 230.00 |
| | 41.15.23 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number additional 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm direct in ground in the same trench in one tier horizontal formation including sand cushioning, excluding protective covering and refilling the trench as required. | Metre | 144.00 |
| | 41.15.24 | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number additional 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm direct in ground in the same trench in one tier horizontal formation including sand cushioning, excluding protective covering and refilling the trench etc. as required. | Metre | 143.00 |
| | 41.15.25 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm in the existing RCC/hume/stoneware/G.I. pipe/ DWC Pipe/surface in existing trench as required. | Metre | 50.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 41.15.26 | Laying, 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. | Laying of one number 3 core H.T armoured/unarmoured power cable of grade exceeding 1.1KV but not exceeding 11KV of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | Metre | 49.00 |
| | 41.15.27 | Laying, 11KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 11KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | Metre | 150.00 |
| | 41.15.28 | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm in the existing RCC/hume/stoneware/G.I. pipe/ DWC Pipe/surface in existing trench as required. | Metre | 50.00 |
| | 41.15.29 | Laying, 33KV grade of size not exceeding 300 sq. mm. | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm. in the existing masonry open duct as required. | Metre | 49.00 |
| | 41.15.30 | Laying, 33KV grade of size not exceeding 300 sq. mm | Laying of one number 3 core H.T armoured/unarmoured power cable 33KV grade of size not exceeding 300 sq. mm on wall/pole etc. as required. | Metre | 152.00 |
| | 41.15.31 | Laying, 1.1KV grade of size not exceeding 25 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size not exceeding 25 sq. mm on surface in existing trench etc. as required. | Metre | 25.00 |
| | 41.15.32 | Laying,1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm on surface in existing trench etc. as required. | Metre | 33.00 |
| | 41.15.33 | Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm on surface in existing trench etc. as required. | Metre | 41.00 |
| **41.16** | | **Excavation of the trenches in hard rock** | Excavation of the trenches in hard rock not exceeding 1.5 metre in width and lift upto 1.5 metre including getting out the excavated metal and disposal of surplus excavated soil/metal as directed within the lead of 50 metre (without blasting) | Cu. Mtr. | 691.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.17** | | **Laying, 1.1KV grade of size not exceeding 25 sq. mm** | Laying of one number armoured/unarmoured cable 1.1KV grade of size not exceeding 25 sq. mm on wall/truss with approved type of iron clamp etc. as required. | Metre | 25.00 |
| **41.18** | | **Laying, 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm** | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 25 sq. mm but not exceeding 120 sq. mm on wall/truss with approved type of iron clamp etc. as required. | Metre | 71.00 |
| **41.19** | | **Laying, 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm** | Laying of one number armoured/unarmoured cable 1.1KV grade of size exceeding 120 sq. mm but not exceeding 400 sq. mm on wall/truss with approved type of iron clamp etc. as required. | Metre | 112.00 |
| **41.20** | | **Laying of underground cable** | Laying of underground cable armoured./unarmoured as per specification in air suspended on existing approved type of iron clamp, G.I wire 7/1.6mm & shackle insulator/insulated strip complete. | | |
| | 41.20.1 | 16 sq. mm. | 1/2/3/4 Core cable upto 16 sq. mm | Metre | 25.00 |
| | 41.20.2 | 25-120 sq. mm. | 1/2/3/3½/4 Core cable 25 sq. mm to 120 sq. mm | Metre | 40.00 |
| | 41.20.3 | 150 sq. mm. & above | 1/2/3/3½/4 Core cable 150 sq. mm and above | Metre | 66.00 |
| **41.21** | | **Laying, cable cover/flag stone over** | Laying of cement concrete or approved type of cable cover/flag stone over L.T./H.T. cable & refill trench as per specification. | Metre | 16.00 |
| **41.22** | | **cable cover flag stone 200mm width & 32-50mm thick** | Supply of approved type cable cover flag stone 200mm width & 32mm to 50mm thick of required length inclusive of transport as required. | Metre | 71.00 |
| **4.23** | | **Cable route marker round/square** | Supply &fixing cable route marker round/square size 100mm, 5mm. thick, G.I. Plate duly bolted/welded on 35mm x 35mm x 5mm angle iron 60cm long including excavation, grouting in cement concrete 1:2:4 (1cement and 2 coarse sand: 4 graded stone aggregate 20mm nominal size) foundation of size 20cm x 20cm x 30cm duly embedded in ground 30cm depth or as required. | Each | 299.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.24** | | **Cable tray** | Supply and erection of hot dip G.I. cable tray perforation not more than 17.5% for specific dimensions along with tees, bends. The cable tray shall be hang from ceiling/ fixed to wall with necessary angle/flat iron/hanging rod, for ceiling suspensions, clamp, anchor fastener, nuts, bolts, washers, not more than 1.0 mtr. apart complete as per specification to complete the job. The tray shall be as follows :- | | |
| | 41.24.1 | 100x50x1.6mm thick | 100 x 50 x 1.6mm thick | Metre | 586.00 |
| | 41.24.2 | 150x50x1.6mm thick | 150 x 50 x 1.6mm thick | Metre | 699.00 |
| | 41.24.3 | 300x50x1.6mm thick | 300 x 50 x 1.6mm thick | Metre | 1027.00 |
| | 41.24.4 | 450x50x2mm thick | 450 x 50 x 2 mm thick | Metre | 1547.00 |
| | 41.24.5 | 600x50x 2 mm thick | 600 x 50 x 2 mm thick | Metre | 1909.00 |
| | 41.24.6 | 600x50x 2 mm thick | 600 x 50 x 2 mm thick | Metre | 1908.00 |
| **41.25** | | **FRP/ GRP Cable Tray (Ladder Type)** | Supply and Erecting of Ladder type Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) cable tray of specific dimensions. The cable tray shall be hung from ceiling/ fixed to wall with necessary angle/ flat iron ceiling suspensors, clamps, nuts, bolts, washers complete with required tees, bends etc. as per specification to complete the job. The tray shall be as follows :- | | |
| | 41.25.1 | 150mm | 150mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | Metre | 1228.00 |
| | 41.25.2 | 300mm | 300mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | Metre | 1371.00 |
| | 41.25.3 | 450mm | 450mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | Metre | 1535.00 |
| | 41.25.4 | 600mm | 600mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | Metre | 1695.00 |
| | 41.25.5 | 750mm | 750 mm width with straight coupler plate size: 200x70x4mm & 8 sets of SS 304 Hardware size: M6x20mm | Metre | 1878.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **41.26** | | **FRP/ GRP Cable Tray (Perforated Type)** | Supply and Erecting of Perforated type Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) cable tray of specific dimensions. The cable tray shall be hung from ceiling/ fixed to wall with necessary angle/ flat iron ceiling suspensors, clamps, nuts, bolts, washers complete with required tees, bends etc. as per specification to complete the job. The tray shall be as follows :- | | |
| | 41.26.1 | 50mm | 50 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | Metre | 819.00 |
| | 41.26.2 | 100mm | 100 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | Metre | 875.00 |
| | 41.26.3 | 150mm | 150 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | Metre | 1018.00 |
| | 41.26.4 | 200mm | 200 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | Metre | 1357.00 |
| | 41.26.5 | 300mm | 300 mm width with straight coupler plate size: 200x25x4mm & 8 sets of SS 304 Hardware size: M5x20mm | Metre | 1757.00 |
| **41.27** | | **FRP/GRP Flat Cover** | Supply and Fixing of Fiber Glass Reinforced Polymer (FRP) / Glass Fiber Reinforced Polymer (GRP) flat cover of minimum thickness 3mm as per required size of cable tray | | |
| | 41.27.1 | 50mm | Flat Cover of 50 mm width | Metre | 129.00 |
| | 41.27.2 | 100mm | Flat Cover of 100 mm width | Metre | 211.00 |
| | 41.27.3 | 150mm | Flat Cover of 150 mm width | Metre | 289.00 |
| | 41.27.4 | 200mm | Flat Cover of 200 mm width | Metre | 379.00 |
| | 41.27.5 | 300mm | Flat Cover of 300 mm width | Metre | 527.00 |
| | 41.27.6 | 450mm | Flat Cover of 450 mm width | Metre | 813.00 |
| | 41.27.7 | 600mm | Flat Cover of 600 mm width | Metre | 1040.00 |
| | 41.27.8 | 750mm | Flat Cover of 750 mm width | Metre | 1817.00 |
| **41.28** | | **Fire Survival Cable** | Supply of XLPE insulated **1.5 sq. mm**. Armoured 2 core **FRLSZH fire alarm cable (RED)** Fire Survival Cable conforming IS-7098 (Part-1) 1988, 1100 Volt grade, 2/3/4/8/10/12 core ISI marked with copper stranded/solid conductor. | Metre | 154.00 |

## CHAPTER-42 (External)

## EXTERNAL ELECTRIFICATION AND OVERHEAD

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.1** | | **Rail pole** | Supply of support for overhead line rail pole of I.S. standard including base plate, welding, drilling of required hole etc. complete as required. | | |
| | 42.1.1 | 60 kg/metre | Rail pole standard weight 60 kg/metre | Metre | 4712.00 |
| | 42.1.2 | 52 kg/metre | Rail pole standard weight 52 kg/metre | Metre | 4063.00 |
| **42.2** | | **RS Joist/H-Beam** | Supply of support for overhead line RS joist of I.S. standard including drilling of required hole etc. complete as required. | | |
| | 42.2.1 | 100x200-25.4 kg/metre | R.S. Joist 100x200 - 25.4 kg/metre | Metre | 1874.00 |
| | 42.2.2 | 175x85 - 19.3 kg/metre | R.S. Joist 175x85 - 19.3 kg/metre | Metre | 1467.00 |
| | 42.2.3 | 150x100 - 17 kg/metre | R.S. Joist 150x100 - 17 kg/metre | Metre | 1290.00 |
| | 42.2.4 | 150x80 - 14.9 kg/metre | R.S. Joist 150x80 - 14.9 kg/metre | Metre | 1256.00 |
| | 42.2.5 | 125x75 - 13 kg/metre | R.S. Joist 125x75 - 13 kg/metre | Metre | 1007.00 |
| | 42.2.6 | H - Beam 152x152mm | H - Beam 152x152mm, Std weight 37.1 Kg/Mtr | Metre | 2843.00 |
| **42.3** | | **Steel tubular pole** | Supply of steel tubular pole swaged type as per IS:2713-1980 complete with base plate and top Canopy | | |
| | 42.3.1 | SP-2 - 7.00 metre | 410 SP-2 - 7.00 metre | Each | 7799.00 |
| | 42.3.2 | SP-5 - 7.50 metre | 410 SP-5 - 7.50 metre | Each | 8302.00 |
| | 42.3.3 | SP-8 - 7.50 metre | 410 SP-8 - 7.50 metre | Each | 11025.00 |
| | 42.3.4 | SP-11 - 8.00 metre | 410 SP-11 - 8.00 metre | Each | 8701.00 |
| | 42.3.5 | SP-14 - 8.00 metre | 410 SP-14 - 8.00 metre | Each | 11911.00 |
| | 42.3.6 | SP-17 - 8.50 metre | 410 SP-17 - 8.50 metre | Each | 9356.00 |
| | 42.3.7 | SP-20 - 8.50 metre | 410 SP-20 - 8.50 metre | Each | 12140.00 |
| | 42.3.8 | SP-23 - 8.50 metre | 410 SP-23 - 8.50 metre | Each | 15852.00 |
| | 42.3.9 | SP-26 - 9.00 metre | 410 SP-26 - 9.00 metre | Each | 9933.00 |
| | 42.3.10 | SP-29 - 9.00 metre | 410 SP-29 - 9.00 metre | Each | 13005.00 |
| | 42.3.11 | SP-32 - 9.00 metre | 410 SP-32 - 9.00 metre | Each | 16340.00 |
| | 42.3.12 | SP-35 - 9.50 metre | 410 SP-35 - 9.50 metre | Each | 13872.00 |
| | 42.3.13 | SP-38 - 9.50 metre | 410 SP-38 - 9.50 metre | Each | 17102.00 |
| | 42.3.14 | SP-41 - 10.00 metre | 410 SP-41 - 10.00 metre | Each | 14168.00 |
| | 42.3.15 | SP-44 - 10.00 metre | 410 SP-44 - 10.00 metre | Each | 17623.00 |
| | 42.3.16 | SP-47 - 10.00 metre | 410 SP-47 - 10.00 metre | Each | 23251.00 |
| | 42.3.17 | SP-50 - 11.00 metre | 410 SP-50 - 11.00 metre | Each | 15284.00 |
| | 42.3.18 | SP-53 - 11.00 metre | 410 SP-53 - 11.00 metre | Each | 19204.00 |
| | 42.3.19 | SP-56 - 11.00 metre | 410 SP-56 - 11.00 metre | Each | 25335.00 |
| | 42.3.20 | SP-59 - 12.00 metre | 410 SP-59 - 12.00 metre | Each | 20686.00 |
| | 42.3.21 | SP-62 - 12.00 metre | 410 SP-62 - 12.00 metre | Each | 27363.00 |
| | 42.3.22 | SP-65 - 12.00 metre | 410 SP-65 - 12.00 metre | Each | 32227.00 |
| | 42.3.23 | SP-68 - 13.00 metre | 410 SP-68 - 13.00 metre | Each | 28934.00 |
| | 42.3.24 | SP-71 - 13.00 metre | 410 SP-71 - 13.00 metre | Each | 33847.00 |
| | 42.3.25 | SP-73 - 14.50 metre | 410 SP-73 - 14.50 metre | Each | 34642.00 |
| | 42.3.26 | SP-76 - 14.50 metre | 410 SP-76 - 14.50 metre | Each | 40968.00 |
| | 42.3.27 | SP-77 - 16.00 metre | 410 SP-77 - 16.00 metre | Each | 27112.00 |
| | 42.3.28 | SP-80 - 16.00 metre | 410 SP-80 - 16.00 metre | Each | 28220.00 |
| | 42.3.29 | SP-10 (Bend type) 8.00 metre | 410 SP-10 (Bend type) 8.00 metre | Each | 7514.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.3.30 | SP-16 (Bend type) 8.50 metre | 410 SP-16 (Bend type) 8.50 metre | Each | 8067.00 |
| | 42.3.31 | SP-25 (Bend type) 9.00 metre | 410 SP-25 (Bend type) 9.00 metre | Each | 8338.00 |
| | 42.3.32 | SP-34 (Bend type) 9.50 metre | 410 SP-34 (Bend type) 9.50 metre | Each | 13236.00 |
| | 42.3.33 | SP-40 (Bend type) 10.00 metre | 410 SP-40 (Bend type) 10.00 metre | Each | 13884.00 |
| | 42.3.34 | SP-49 (Bend type) 11.00 metre | 410 SP-49 (Bend type) 11.00 metre | Each | 14617.00 |
| **42.4** | | **PCC pole** | Supplying of pre stressed cement concrete (PCC) pole conforming to IS: 785 at work site including required transportation. | | |
| | 42.4.1 | 8 meter | PCC pole 8 meter Long 140 kg | Each | 3731.00 |
| | 42.4.2 | 9.1 meter | PCC pole 9.1 meter Long 280 kg | Each | 5099.00 |
| **42.5** | | **AAAC conductors** | Supplying and drawing all aluminium alloy conductor (AAAC) of approved make conforming to IS 398-1979 Pt. IV, including binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required including clearing of obstacles (if any) | | |
| | 42.5.1 | Weasel | AAAC 0.03 sq inch (34 sq.mm Al. EQ.)- (Weasel) | Km. | 39115.00 |
| | 42.5.2 | Rabbit | AAAC 0.05 sq inch (55 sq.mm Al. EQ.)- (Rabbit) | Km. | 61509.00 |
| | 42.5.3 | Raccoon | AAAC 0.075 sq inch (80 sq.mm Al. EQ.)- (Raccoon) | Km. | 79735.00 |
| | 42.5.4 | Dog | AAAC 0.1 sq inch (100 sq.mm Al. EQ.)- (Dog) | Km. | 104339.00 |
| | 42.5.5 | Panther | AAAC 0.2 sq inch (232 sq.mm Al. EQ.)- (Panther) | Km. | 250382.00 |
| **42.6** | | **ACSR conductor** | Supplying and drawing of stranded aluminium conductor steel reinforced (ACSR) conforming to IS:398-1976 of approved make, stringing, making off complete with binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required and clearing of obstacles (if any) etc. | | |
| | 42.6.1 | Squirrel | ACSR 0.02 sq.inch (20 sq.mm Al. EQ.)- (Squirrel) | Km. | 27345.00 |
| | 42.6.2 | Weasel | ACSR 0.03 sq inch (34 sq.mm Al. EQ.)- (Weasel) | Km. | 38982.00 |
| | 42.6.3 | Rabbit | ACSR 0.05 sq inch (55 sq.mm Al. EQ.)- (Rabbit) | Km. | 64148.00 |
| | 42.6.4 | Raccoon | ACSR 0.075 sq inch (80 sq.mm Al. EQ.)- (Raccoon) | Km. | 88530.00 |
| | 42.6.5 | Dog | ACSR 0.1 sq inch (100 sq.mm Al. EQ.)- (Dog) | Km. | 119836.00 |
| | 42.6.6 | Panther | ACSR 0.2 sq inch (261 sq.mm Al. EQ.)- (Panther) | Km. | 341832.00 |
| **42.7** | | **Covered aluminium conductor** | Supplying, drawing and testing of XLPE insulated covered aluminium conductor of approved make, stringing, making off complete with binding at existing insulator, jointing, jumpering, tearing off, connecting etc. as required and clearing of obstacles (if any) etc. | | |
| | 42.7.1 | 70 sqmm | 70 Sqmm XLPE 11KV grade | Km. | 541000.00 |
| | 42.7.2 | 99 Sqmm | 99 Sqmm XLPE 33KV grade | Km. | 679000.00 |
| **42.8** | | **Guard insulator or split insulator** | Supplying and fixing guard insulator or split insulator complete: | | |
| | 42.8.1 | 25x25 mm | 25x25 mm | Pair | 42.00 |
| | 42.8.2 | 37x37 mm | 37x37 mm | Pair | 48.00 |
| | 42.8.3 | 50x50 mm | 50x50 mm | Pair | 60.00 |
| **42.9** | | **Guard wire/earth wire/bearer wire 13 sq mm** | Supplying and drawing guard wire/earth wire/bearer wire 13 sq mm (4mm. dia) G.I. including stringing, binding at existing insulators or brackets, jointing, jumpering, connecting & cradle etc. as required and clearing of obstacles (if any) | Km. | 15675.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.10** | | **Guard wire/earth wire/bearer wire 18.62 sq mm** | Supplying and drawing guard wire/earth wire/bearer wire 18.62 sq mm (4.87mm. dia/6swg) G.I. including stringing, binding at existing insulators or brackets, jointing, jumpering, connecting & cradle etc. as required and clearing of obstacles (if any) | Km. | 18761.00 |
| **42.11** | | **G.I. Bearer wire 7/1.6mm** | Supplying and drawing G.I. bearer wire 7/1.6mm including stringing, binding at existing brackets/pole, etc. for cable support in air as required and clearing of obstacles (if any) | PM | 23.00 |
| **42.12** | | **Guards for existing overhead lines** | Supplying and erection of guards for existing overhead lines as required including split insulator required as per specification: | | |
| | 42.12.1 | Cradle type | Cradle type | Each | 38.00 |
| | 42.12.2 | Hexagonal type | Hexagonal type | Each | 30.00 |
| | 42.12.3 | Ring type | Ring type | Each | 27.00 |
| **42.13** | | **Water tight terminal box (M.S. Sheet)** | Supplying and erection of water tight terminal box made of 16 SWG M.S. Sheet (1.6mm thick) i/c rubber gasket of required size suitable for kit kat/MCB/switch including 2 nos. pole clamp of M.S. flat iron size 32x3mm, nut and bolts, The box shall be powder coated excluding the cost of the kit kat/MCB/Switch 4 Pole epoxy connector nut & bolt type, 30Amp, 500Volt complete as per specification as mentioned below. | | |
| | 42.13.1 | 20x15x10 Cms. | 20x15x10 Cms. | Each | 910.00 |
| | 42.13.2 | 30x20x10 Cms. | 30x20x10 Cms. | Each | 1158.00 |
| | 42.13.3 | 4 Pole epoxy connector | Supplying and fixing of 4 Pole epoxy connector nut & bolt type, 30Amp. 500Volt. in existing box as required as per specification: | Each | 195.00 |
| **42.14** | | **Water tight terminal box (Polycarbonate)** | Supplying and erection of water tight terminal box made of PC-GFS (polycarbonate),IP 66 /IP 67 (cable entry systems LES) in accordance with IEC 60670-22, resistant to weather influences (UV due to solar radiation, protected against rainwater, temperature resistant, impact resistant, etc. insulation voltage 1000 V a.c./d.c., wall thickness of the bottom part 2 / 2.7 mm of required size suitable for kit kat/MCB/switch including 2 nos. pole clamp of M.S. flat iron size 32x3mm, nut and bolts, excluding the cost of the kit kat/MCB/Switch, 4 Pole epoxy connector nut & bolt type, 30Amp, 500Volt complete as per specification as mentioned below. | | |
| | 42.14.1 | 104 x 104 x 70 mm | 104 x 104 x 70 mm | Each | 452.00 |
| | 42.14.2 | 130 x 180 x 77 mm | 130 x 180 x 77 mm | Each | 889.00 |
| | 42.14.3 | 155 x 210 x 92 mm | 155 x 210 x 92 mm | Each | 1384.00 |
| | 42.14.4 | 205 x 255 x 112 mm | 205 x 255 x 112 mm | Each | 2163.00 |
| **42.15** | | **Cable Gland** | Supplying and erection of cable glands of Polyamide material for knockouts as required as mentioned below: | | |
| | 42.15.1 | M20 | M20 | Each | 47.00 |
| | 42.15.2 | M25 | M25 | Each | 71.00 |
| | 42.15.3 | M32 | M32 | Each | 100.00 |
| **42.16** | | **MCB/Connector mounting** | Supplying and fixing of MCB/Connector mounting Corrosion proof mild steel DIN Rail of thickness 2.5mm to 4mm in existing box as required as per specification: | Per Metre | 107.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.17** | | **Single pole connector DIN Rail mounted** | Supply and fixing of single pole connector DIN Rail mounted Insulating body made from Polyamide 6.6 conforms to UL 94 standard, IEC 60947-7-1, 1000Volt in existing box as required as per specification: | | |
| | 42.17.1 | Upto 4sq mm cable | Suitable for upto 4sq mm cable | Each | 52.00 |
| | 42.17.2 | For 6sqmm-16sq mm cable | Suitable for 6sqmm-16sq mm cable | Each | 68.00 |
| **42.18** | | **Terminal set polyamide** | Supplying and fixing of terminal set polyamide with copper connector 5 pole including fixing arrangement for below mentioned size: | | |
| | 42.18.1 | 2.5 to 4 sq mm | Terminal set 2.5 to 4 sq mm | Each | 90.00 |
| | 42.18.2 | 6 sq mm | Terminal set 6 sq mm | Each | 208.00 |
| **42.19** | | **Erection of poles, length 7 metres but not exceeding 10 metres** | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole of length 7 metres but not exceeding 10 metres in cement concrete 1:3:6 (1 cement : 3 core sand: 6 graded stone aggregate 40mm nominal size) base, foundation, muffing including excavation and back refilling etc. as required. | Each | 4074.00 |
| **42.20** | | **Erection of poles, length exceeding 10 metres but not exceeding 15 metres** | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole of length exceeding 10 metres but not exceeding 15 metres in cement concrete M-10 (1 cement:3 core sand: 6 graded stone aggregate 40mm nominal size) base, foundation, muffing including excavation and back refilling etc. as required. | Each | 5289.00 |
| **42.21** | | **Erection of R.C.C./P.C.C. pole** | Erection of R.C.C./P.C.C. pole of length 8 meters but not exceeding 11 meters in brick ballast and ramming the foundation including excavation and refilling etc.as required. | Each | 2282.00 |
| **42.22** | | **Erection of metallic street light pole length upto 2 metres** | Erection of metallic street light pole length upto 2 metres in cement concrete M-10 (1cement:3coarse sand:6 graded stone aggregate 40mm nominal size) foundation with cement concrete collar of size 0.4 metres dia x 0.3 metres height to accommodate the looping type cable end box including excavation and refilling etc. as required. | Each | 1980.00 |
| **42.23** | | **Erection of metallic street light pole length exceeding 3 metres but not exceeding 6 metres** | Erection of metallic street light pole length 3 metres but not exceeding 6 metres in cement concrete M-10 (1cement:3coarse sand:6 graded stone aggregate 40mm nominal size) foundation with cement concrete collar of size 0.4 metres dia x 0.5 metres height to accommodate the looping type cable end box including excavation and refilling etc. as required. | Each | 2421.00 |
| **42.24** | | **Erection of pole strut in cement concrete 1:3:6** | Erection of RS Joist pole/'H' beam/steel tubular/steel rail pole strut in cement concrete 1:3:6 (1cement:3coarse sand: 6 aggregate 40mm nominal size) foundation including excavation and refilling and secured with holding clamp, nuts and bolts, etc. as required. | Each | 3854.00 |
| **42.25** | | **Erection of RCC/PCC pole strut in brick ballast** | Erection of RCC/PCC pole strut in brick ballast and ramming the foundation including excavation and refilling and secured with holding clamp nuts and bolts etc. as required. | Each | 2374.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.26** | | **Stay set complete (galvanized) 19mm. dia x 1.8 metre long stay rod** | Supplying and erection of stay set complete (galvanized) 19mm. dia x 1.8 metre long stay rod anchor plate of size 300 mm x 300 mm x7.5mm thimble stay clamp, turn buckle (19mm dia x 600mm), 7/4.00mm dia G.I. stay wire etc. in cement concrete M-10 (1 Cement: 3 Coarse sand: 6 graded stone aggregate 40mm nominal size) foundation including excavation and refilling etc. as required. | Each | 4600.00 |
| **42.27** | | **Stay set complete (galvanized) with 16mm dia x1.8 metre long stay** | Supplying and erection of stay set complete (galvanized) with 16mm dia x1.8 metre long stay anchor plate of size 200mm x 200mm x 6.0mm thimble stay clamp turn buckle 19mmx600mm, 7/3.15mm dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 grinded stone aggregate 40mm nominal size) foundation including excavation and refilling etc. as required. | Each | 3780.00 |
| **42.28** | | **Stay set complete with 19/20mm.dia 1.8 metre long stay rod** | Supplying and erection of stay set complete with 19/20mm.dia 1.8 metre long stay rod, turn buckle, anchor plate of size 300mm x 300mm x 7.5mm thimble stay clamp, bow tightener, 7/4.00 dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 granded stone aggregate 40mm nominal size) foundation including excavation and back refilling etc. as required. | Each | 4856.00 |
| **42.29** | | **Stay set complete with 16mm.dia 1.8 metre long stay rod** | Supplying and erection of stay set complete with 16mm.dia 1.8 metre long stay rod, turn buckle, anchor plate of size 200mm x 200mm x 6.0mm, stay clamp, bow tightener, 7/3.15 dia G.I. stay wire etc. in cement concrete M-10 (1 cement: 3 coarse and: 6 granded stone aggregate 40mm nominal size) foundation including excavation and back refilling etc. as required. | Each | 3600.00 |
| **42.30** | | **Erection of stay set complete in cement concrete M-10** | Erection of stay set complete in cement concrete M-10 (1 cement: 3 coarse sand: 6 graded stone aggregate 40mm nominal size) foundation including excavation and back refilling etc. as required. | Each | 2165.00 |
| **42.31** | | **Bow stay set complete (galvanized) with 19mm dia x 1.8m long stay rod** | Supplying and erection of bow stay set complete (galvanized) with 19mm dia x 1.8m long stay rod, anchor plate of size 300mmx300mm x 7.5mm, stay clamp, turn buckle 19mmx600mm, 7/4.00mm dia G.I. stay wire and bow tightener, etc. in cement concrete M-10 ( 1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx 50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | Each | 5033.00 |
| **42.32** | | **Bow stay set complete (galvanized) with 16mm dia. X 1.8m long stay rod** | Supplying and erection of bow stay set complete (galvanized) with 16mm dia. X 1.8m long stay rod, anchor plate of size 200mmx200mmx6.0mm, stay clamp, turn buckle 19mmx600mm, 7/3.15 mm dia G.I. Stay wire and bow tightener, etc. in cement concrete M-10 (1cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | Each | 4251.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.33** | | **Bow stay set complete with 19mm dia x 1.8m long stay rod** | Supplying and erection of bow stay set complete with 19mm dia x 1.8m long stay rod, anchor plate of size 300mmx300mm x 7.5mm, stay clamp, turn buckle 19mmx600mm, 7/4.00mm dia G.I. stay wire and bow tightener, etc. in cement concrete M-10 ( 1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx 50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | Each | 4947.00 |
| **42.34** | | **Bow stay set complete with 16mm dia. X 1.8m long stay rod** | Supplying and erection of bow stay set complete with 16mm dia. X 1.8m long stay rod, anchor plate of size 200mmx200mmx6.0mm thimble stay clamp, turn buckle 19mmx600mm, 7/3.15 mm dia G.I. Stay wire and bow tightener, etc. in cement concrete M-10 (1cement: 3 coarse sand: 6 graded stone aggregate 40mm nominal size) foundation including excavation and back refilling and also with 0.6m long brace of size 50mmx50mmx6mm angle iron with 50mm dia pulley fixed at one end of the brace as required. | Each | 4030.00 |
| **42.35** | | **50x65mm shackle insulator** | Supplying and erection of 50mm x 65mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | Each | 92.00 |
| **42.36** | | **75x90mm shackle insulator** | Supplying and erection of 75mm x 90mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | Each | 102.00 |
| **42.37** | | **100x110mm shackle insulator** | Supplying and erection of 100mmx110mm shackle insulator with G.I. bolts, nuts and straps etc. as required. | Each | 109.00 |
| **42.38** | | **Aluminium knob/bobbin** | Supplying and erection of aluminium knob/bobbin with nut, bolts & straps etc. as required. | Each | 119.00 |
| **42.39** | | **19/20mm dia. M.S. pipe 1 meter in length for lights** | Supplying and fixing of 19mm/20mm dia. M.S. pipe 1 meter in length for CFL/LED street light/bulkhead/water tight bracket including bending to shape and wiring with 1.5sqmm W.P./PVC wire etc. as required. | Each | 279.00 |
| **42.40** | | **19mm/20mm dia. G.I. pipe 2 meter long for lights** | Supplying and fixing of 19mm/20mm dia. G.I. pipe 2 meter long for CFL/LED street light/bulkhead/water tight bracket including bending to shape and wiring with 1.5sqmm W.P./PVC wire etc. as required. | Each | 542.00 |
| **42.41** | | **32mm dia. X 2.00 meter long G.I. pipe (medium class) bracket** | Supplying and fixing of 32mm dia. X 2.00 meter long G.I. pipe (medium class) bracket for mounting of fluorescent/S.V/M.H/L.E.D lamp street light fittings on poles including bending the pipe to the required shape and wiring with 1.5sq.mm W.P./PVC wire etc. as required. | Each | 791.00 |
| **42.42** | | **Angle/channel/flat iron fitting for overhead line & sub-station etc.** | Supply, fabrication and erection of angle/channel/flat iron fitting for overhead line & sub-station etc such as 'D' bracket, cross arms, top clamp, 'V' cross arms, back/support clamp or other similar work etc. including nut bolts of required size, making holes, fabrication, welding, cutting, etc. and painting with one coat of red oxide paint & two coat of aluminium paint as required as per specification. | Kg. | 105.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.43** | | **Labour charges for erection of angle/channel/flat iron fitting for overhead line & sub-station etc** | Labour charges for erection of angle/channel/flat iron fitting for overhead line & sub-station etc such as 'D' bracket, cross arms, top clamp, 'V' cross arms, back/support clamp or other similar work etc. including nut bolts of required size, as required as per specification. | Kg. | 23.00 |
| **42.44** | | **11KV porcelain pin insulator** | Supplying and erection of 11KV porcelain pin insulator complete with long steel head G.I. pin, nut, washer etc. as required. | Each | 375.00 |
| **42.45** | | **11KV porcelain disc insulator** | Supplying and erection of 11KV porcelain disc insulator for 11KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts washer etc. as required. | Each | 750.00 |
| **42.46** | | **33KV porcelain pin insulator** | Supplying and erection of 33KV porcelain pin insulator complete with long steel head G.I. pin, nut, washer etc. as required | Each | 788.00 |
| **42.47** | | **Set of three 11KV porcelain disc insulators for 33KV overhead lines** | Supplying and erection of a set of three 11KV porcelain disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | Each | 1928.00 |
| **42.48** | | **Set of four 11KV porcelain disc insulators for 33KV overhead lines** | Supplying and erection of a set of four 11KV porcelain disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | Each | 2639.00 |
| **42.49** | | **Labour, 11/33KV porcelain pin insulator** | Labour for erection of 11/33KV porcelain pin insulator complete with hardware as required. | Each | 169.00 |
| **42.50** | | **Labour, 11KV porcelain disc insulator for 11KV overhead lines** | Labour for erection of 11KV porcelain disc insulator for 11KV overhead lines complete with hardware as required. | Each | 169.00 |
| **42.51** | | **Labour, 11KV porcelain disc insulators for 33KV overhead lines** | Labour for erection of a set of three/ four 11KV porcelain disc insulators for 33KV overhead lines complete with hardware as required. | Each | 404.00 |
| **42.52** | | **11KV (5 KN) polymer pin insulator** | Supplying and erection of 11KV (5 KN) polymer pin insulator complete with long steel head G.I. pin, nut, washer etc. as required. | Each | 402.00 |
| **42.53** | | **11KV (45 KN) Polymer disc insulator for 11KV overhead lines** | Supplying and erection of 11KV (45 KN) Polymer disc insulator for 11KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts washer etc. as required. | Each | 502.00 |
| **42.54** | | **33KV (10 KN) polymer pin insulator** | Supplying and erection of 33KV (10 KN) polymer pin insulator complete with long steel head G.I. pin, nut, washer etc. as required | Each | 595.00 |
| **42.55** | | **33KV (45 KN) polymer disc insulators** | Supplying and erection of 33KV (45 KN) polymer disc insulators for 33KV overhead lines with galvanized insulator fittings, ball and socket type, and complete with galvanized strain clamp, bolts, nuts, washers etc. as required. | Each | 1018.00 |
| **42.56** | | **Labour, 11/33KV polymer pin insulator** | Labour for erection of 11/33KV polymer pin insulator complete with hardware as required. | Each | 160.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.57** | | **Labour, 11/33KV polymer disc insulators** | Labour for erection of 11/33KV polymer disc insulators for 11/33KV overhead lines complete with hardware as required. | Each | 259.00 |
| **42.58** | | **Single piece nonlinear resistor type lighting arrestor (set of 3 nos) for 3 wire, 11KV** | Supplying and erection of single piece nonlinear resistor type lighting arrestor (set of 3 nos) for 3 wire, 11KV overhead lines/sub-station with rated voltage of 9KV (rms) with a nominal discharge current rating of 5 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 3832.00 |
| **42.59** | | **Single piece non-linear resistor type lightning arrestor (set of 3 nos) for 3 wire, 33KV** | Supplying and erection of single piece non-linear resistor type lightning arrestor (set of 3 nos) for 3 wire, 33KV overhead lines/sub-station with rated voltage of 30KV (rms) with a nominal discharge current rating of 5 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 32662.00 |
| **42.60** | | **Single piece nonlinear resistor polymer type lighting arrestor (set of 3 nos) for 3 wire, 11KV** | Supplying and erection of single piece nonlinear resistor polymer type lighting arrestor (set of 3 nos) for 3 wire, 11KV overhead lines/sub-station with rated voltage of 9KV (rms) with a nominal discharge current rating of 10 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 3617.00 |
| **42.61** | | **Single piece non-linear resistor polymer type lightning arrestor (set of 3 nos) for 3 wire, 33KV** | Supplying and erection of single piece non-linear resistor polymer type lightning arrestor (set of 3 nos) for 3 wire, 33KV overhead lines/ sub-station with rated voltage of 27KV (rms) with a nominal discharge current rating of 10 KA and complete with galvanized clamping arrangement, G.I. bolts, nuts, washer etc. as required. | Set | 35697.00 |
| **42.62** | | **Outdoor H.T., AB switch assembly set** | Supplying installing, testing & commissioning of outdoor H.T., AB switch assembly set gang operated with brass contact parts, including required GI pipe operating rod, handle & locking arrangement on On-Off position conforming to IS complete with required material and installing on existing DP structure to complete the job as required as per specification.(Set of 3 nos.) | | |
| | 42.62.1 | 11 KV, 400Amp | 11 KV, 400Amp | set | 8989.00 |
| | 42.62.2 | 33KV, 600Amp | 33KV, 600Amp | set | 31525.00 |
| **42.63** | | **D.O. fuse** | Supplying installing, testing & commissioning D.O. fuse assembly with brass part contact for 33 or 11/0.4 KV DP Structure set of 3 Nos. with fuse barrel i/c required fuse element & other materials as per specification on existing D.P. structure as required. | | |
| | 42.63.1 | 11KV | 11KV | set | 4721.00 |
| | 42.63.2 | 33KV | 33KV | set | 16000.00 |
| **42.64** | | **Replacement of D.O. fuse barrel with brass parts** | Supplying & replacement of D.O. fuse barrel with brass parts for 33 or 11 KV i/c required fuse element according to load as per specification on existing D.P. structure as required. | | |
| | 42.64.1 | 11KV | 11KV | Each | 910.00 |
| | 42.64.2 | 33KV | 33KV | Each | 1895.00 |
| **42.65** | | **Replacement of D.O. fuse barrel for 33 or 11 KV** | Supplying & replacement of D.O. fuse barrel for 33 or 11 KV as per specification on existing D.P. structure as required. | | |
| | 42.65.1 | 11KV | 11KV | Each | 274.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.65.2 | 33KV | 33KV | Each | 314.00 |
| **42.66** | | **Replacement of D.O. fuse element** | Supplying & replacement of D.O. fuse element according to load for 33 or 11 KV as per specification on existing D.P. structure as required. | | |
| | 42.66.1 | 11KV | 11KV | Each | 102.00 |
| | 42.66.2 | 33KV | 33KV | Each | 119.00 |
| **42.67** | | **Anti-climbing device** | Supply & fixing anti climbing device with 2 ply G.I. barbed wire 1 Kg. per pole complete as per MPMKVVCO. Specification. | Each | 231.00 |
| **42.68** | | **G.I. earth coil of 4mm dia** | Supply, laying and fixing of G.I. earth coil of 4mm dia G.I. wire having 120 turns of nearly 50mm dia. and 3 mt. long tail in existing pit duly earthed with pole etc complete as required as per specification. | Each | 293.00 |
| **42.69** | | **Danger board** | Supplying and fixing 33/11/0.4 KV enamel coated danger board size 200x250mm with clamp on existing HT/LT structure/poles. | Each | 205.00 |
| **42.70** | | **Transformers, 11/0.4 KV 3 Phase** | Supplying, installing, testing and commissioning of 11/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, dehydrating silica gel breather on eye level, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, oil filling hole with plug, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification, hand operated off load tap changer switch above 100 KVA, oil conservator with drain valve, plug 63 KVA and above. | | |
| | | Level-2 | **Energy efficiency level -2 (New level -1)** | | |
| | 42.70.1 | Al. wound | **Aluminium wound** | | |
| | 42.70.1.1 | 25 KVA | 25 KVA | Each | 120963.00 |
| | 42.70.1.2 | 63 KVA | 63 KVA | Each | 228764.00 |
| | 42.70.1.3 | 100 KVA | 100 KVA | Each | 316509.00 |
| | 42.70.1.4 | 200 KVA | 200 KVA | Each | 607948.00 |
| | 42.70.2 | Cu. wound | **Copper wound** | | |
| | 42.70.2.1 | 315 KVA | 315 KVA | Each | 1096813.00 |
| | 42.70.2.2 | 500 KVA | 500 KVA | Each | 1472863.00 |
| | | Level-3 | **Energy efficiency level -3 (New level-2)** | | |
| | 42.70.3 | Al. wound | **Aluminium wound** | | |
| | 42.70.3.1 | 25 KVA | 25 KVA | Each | 169223.00 |
| | 42.70.3.2 | 63 KVA | 63 KVA | Each | 266369.00 |
| | 42.70.3.3 | 100 KVA | 100 KVA | Each | 348160.00 |
| | 42.70.3.4 | 200 KVA | 200 KVA | Each | 714495.00 |
| | 42.70.4 | Cu. wound | **Copper wound** | | |
| | 42.70.4.1 | 315 KVA | 315 KVA | Each | 1219029.00 |
| | 42.70.4.2 | 500 KVA | 500 KVA | Each | 1764301.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.71** | | **Transformers, 33/0.4 KV 3 Phase** | Supplying, installing, testing and commissioning of 33/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, hand operated off load Tap changer switch above 100 KVA, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, oil conservator with drain plug, oil filling hole with plug, dehydrating silica gel breather on eye level, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification. | | |
| | | Level-2 | **Energy efficiency level -2 (New level -1)** | | |
| | 42.71.1 | Cu. wound | **Copper wound** | | |
| | 42.71.1.1 | 100 KVA | 100 KVA | Each | 651820.00 |
| | 42.71.1.2 | 200 KVA | 200 KVA | Each | 842979.00 |
| | 42.71.1.3 | 315 KVA | 315 KVA | Each | 1153220.00 |
| | 42.71.1.4 | 500 KVA | 500 KVA | Each | 1595079.00 |
| | | Level-3 | **Energy efficiency level -3 (New level-2)** | | |
| | 42.71.2 | Cu. wound | **Copper wound** | | |
| | 42.71.2.1 | 100 KVA | 100 KVA | Each | 745833.00 |
| | 42.71.2.2 | 200 KVA | 200 KVA | Each | 968329.00 |
| | 42.71.2.3 | 315 KVA | 315 KVA | Each | 1407054.00 |
| | 42.71.2.4 | 500 KVA | 500 KVA | Each | 1783104.00 |
| **42.72** | | **Transformer, 33/0.4 KV 3 phase** | Supplying, installing, testing and commissioning of 33/0.4 KV 3 Phase 50 cycle oil immersed, naturally cooled, outdoor type transformer connected delta on H.T. side and star on L.T. side, hand operated off load Tap changer switch, rating and diagram plate, two earthing terminal, lifting lugs, oil level gauge, drain valve with plug, temperature not exceeding 50ºC on load, temperature dial gauge, Bucholz relay, oil conservator with drain plug, oil filling hole with plug, dehydrating silica gel breather on eye level, four unidirectional roller, arcing horns, explosion vent, terminal arrangement bushing on H.V. side and cable box on LV side, first filling of oil upto desired level and transformer installing on existing structure with all required materials arrangement as required ISI marked & as per IS specification. | | |
| | | Level-2 | **Energy efficiency level -2 (New level -1)** | | |
| | 42.72.1 | Cu. wound | **Copper wound** | | |
| | 42.72.1.1 | 630 KVA | 630 KVA | Each | 1723563.00 |
| | 42.72.1.2 | 1000 KVA | 1000 KVA | Each | 2130950.00 |
| | 42.72.1.3 | 1250 KVA | 1250 KVA | Each | 2312708.00 |
| | 42.72.1.4 | 1600 KVA | 1600 KVA | Each | 3002133.00 |
| | 42.72.1.5 | 2000 KVA | 2000 KVA | Each | 4255633.00 |
| | 42.72.1.6 | 2500 KVA | 2500 KVA | Each | 5007733.00 |
| | | Level-3 | **Energy efficiency level -3 (New level-2)** | | |
| | 42.72.2 | Cu. wound | **Copper wound** | | |
| | 42.72.2.1 | 630 KVA | 630 KVA | Each | 1848913.00 |
| | 42.72.2.2 | 1000 KVA | 1000 KVA | Each | 2344045.00 |
| | 42.72.2.3 | 1250 KVA | 1250 KVA | Each | 2550873.00 |
| | 42.72.2.4 | 1600 KVA | 1600 KVA | Each | 3315508.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.72.2.5 | 2000 KVA | 2000 KVA | Each | 5139350.00 |
| | 42.72.2.6 | 2500 KVA | 2500 KVA | Each | 5640750.00 |
| **42.73** | | **Shifting, erection, & commissioning of 100/200/315/500 KVA (11/0.433 KV) distribution transformer** | Shifting, erection, & commissioning of 25/63/100/200/315/500 KVA (11/0.433 KV) distribution transformer on existing structure including Bi-metallic clamp, ACSR/AAAC Conductor i/c colour code sleeve, for connection from H.T. Line to transformer as required as per specification. | Each | 10846.00 |
| **42.74** | | **Limit switch with accessories** | Supplying and fixing limit switch with accessories & arrangement on existing HT/LT structure/poles AB switch handle for inter locking between distribution panel main incoming LT switch gear/ system as required to complete the job. | Each | 4851.00 |
| **42.75** | | **Fencing of chain link mesh 100x100 mm, 12 SWG GI wire** | Supply & fixing fencing of chain link mesh 100 x 100 mm, 12 SWG GI wire for for D.P. structure electric sub-station switch yard including single post of angle iron size 50x50x5mm, 2.5 mt long should be provided upto 2.00 mt. apart or as required. The panel frame with required angle iron/ flat iron members/cross members including mesh welded in frame panel shall be erected between the posts. The angle iron post shall be erected in cement concrete M-15 (1:2:4) (20 mm graded metal) foundation, including excavation of pit and refilling the same. The fencing shall be provided with one entrance gate including locking arrangement as required as per specification. The height of fencing shall be 2 mt. above ground level. Fencing shall be painted with required shade. | sq.Mt | 1821.00 |
| **42.76** | | **Fencing of chain link mesh 100x100 mm, 12 SWG GI wire** | Supply & fixing fencing of chain link mesh 100 x 100 mm, 12 SWG GI wire for electric sub-station switch yard including single post of angle iron size 50x50x5mm, 2.5 mt long should be provided upto 2.00 mt. apart or as required with angle iron/ flat iron members /cross members including mesh fix on frame and erected between the post. The angle iron post shall be erected in cement concrete M-15 (1:2:4) (20 mm graded metal) foundation, cement concrete, excavation of pit and refilling the same. The fencing shall be provided with one entrance gate including locking arrangement as required as per specification. The height of fencing shall be 2 mt. above ground level. Fencing shall be painted with required shade. | sq.mt | 1491.00 |
| **42.77** | | **Rubber mat** | Supply of rubber mat 2 x 1.0Mtr, 1100V tested | Each | 1913.00 |
| **42.78** | | **Hand gloves 11KV** | Supply of hand gloves 11KV tested | Each | 1077.00 |
| **42.79** | | **Hand gloves 33KV** | Supply of hand gloves 33KV tested | Each | 1291.00 |
| **42.80** | | **Fiber discharge rod** | Supply of fiber discharge rod I/C connection copper lead tested | Each | 4042.00 |
| **42.81** | | **D.O. operating rod** | Supply of fiber D.O operating rod for 11/33KV tested | Each | 3854.00 |
| **42.82** | | **First aid box** | Supply of first aid box complete with medicine & bandage as per specification. | set | 1646.00 |
| **42.83** | | **Shock instruction chart** | Supply of shock instruction chart duly framed with glass, as reqd. | Each | 539.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.84** | | **Transformer breather with silica gel** | Supply and fixing transformer breather with silica gel suitable for 100 -500 KVA as reqd. | Each | 601.00 |
| **42.85** | | **New transformer oil** | Supply and filling of new transformer oil of desired break down value for topping as reqd. | Ltr. | 222.00 |
| **42.86** | | **Filtration of transformer oil** | Filtration of transformer oil for 33/0.4 KV, 11/0.4 KV 3 phase 50 cycle oil immersed, naturally cooled, distribution transformer upto desired break down value. | Ltr. | 30.00 |
| **42.87** | | **Construction of pedestal (plinth) for transformer** | Designing and construction of pedestal (plinth) for transformer of desired size including excavation in hard rock/soft rock, base concrete M-7.5 (1:4:8), 20mm metal and pedestal of RCC M-15 (1:2:4) including required tar steel reinforcement for required height as per IE rules from base concrete including shuttering and finishing with 10mm thick plaster in cement mortar 1:3 curing & white washing etc. complete as per specifications. | Cu. Mt. | 7379.00 |
| **42.88** | | **Spreading of metal dust, murum** | Collection and spreading of metal dust, murum and 40mm graded hard metal in sub-station yard or as required as mentioned below - | | |
| | 42.88.1 | Metal dust | Metal dust | Cu. Mt. | 1274.00 |
| | 42.88.2 | Murum | Murum | Cu. Mt. | 847.00 |
| | 42.88.3 | Metal 40 mm | Metal 40 mm | Cu. Mt. | 1478.00 |
| **42.89** | | **PVC insulated XLPE aerial bunch power cable** | Supplying & drawing/laying of PVC insulated XLPE aerial bunch power cable conductor H2/H4 grade aluminium solid/stranded conforming to IS-7098 Part-ll, Alu. alloy messenger of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| | 42.89.1 | **11KV grade** | **11KV grade arial bunch cable :-** | | |
| | 42.89.1.1 | 3x50 + 1x70 sq.mm. | 3 x50 + 1 x 70 sq.mm. | P.Mt. | 764.00 |
| | 42.89.1.2 | 3x70 + 1x70 sq.mm. | 3 x70 + 1 x 70 sq.mm. | P.Mt. | 887.00 |
| | 42.89.1.3 | 3x95 + 1x80 sq.mm. | 3 x95 + 1 x 80 sq.mm. | P.Mt. | 980.00 |
| | 42.89.1.4 | 3x150 + 1x120 sq.mm. | 3 x150 + 1 x 120 sq.mm. | P.Mt. | 1306.00 |
| | 42.89.1.5 | 3x150 + 1x120 sq.mm. | 4 x150 + 1 x 120 sq.mm. | P.Mt. | 1307.00 |
| | 42.89.1.6 | 50 sq.mm. | 50 sq.mm. | P.Mt. | 1549.00 |
| | 42.89.1.7 | 95 sq.mm. | 95 sq.mm. | P.Mt. | 2081.00 |
| | 42.89.1.8 | 150 sq.mm. | 150 sq.mm. | P.Mt. | 2394.00 |
| **42.90** | | **PVC insulated XLPE aerial bunch power cable** | Supplying & drawing/laying of PVC insulated XLPE aerial bunch power cable conductor H2/H4 grade aluminium solid/stranded conforming to IS-14255-1995, alu. alloy insulated messenger 1100 volts grade cable of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| | 42.90.1 | **Single core cable** | **Single core cable(insulated neutral messenger)** | | |
| | 42.90.1.1 | 2x25 sq.mm, 1x16 | 2x25 sq.mm, 1x16(insulated neutral messenger) | P.Mt. | 202.00 |
| | 42.90.1.2 | 2x35 sq.mm, 1x16 | 2x35 sq.mm, 1x16(insulated neutral messenger) | P.Mt. | 248.00 |
| | 42.90.2 | **Single core cable** | **Single core cable(insulated neutral messenger)** | | |
| | 42.90.2.1 | 3x25 sq.mm | 3x25 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 246.00 |
| | 42.90.2.2 | 3x35 sq.mm | 3x35 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 311.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.90.2.3 | 3x50 sq.mm | 3x50 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 321.00 |
| | 42.90.2.4 | 3x70 sq.mm | 3x70 sq.mm, 1x16(st. light), 1x35(insulated neutral messenger) | P.Mt. | 525.00 |
| | 42.90.2.5 | 3x95 sq.mm | 3x95 sq.mm, 1x16(st. light), 1x50(insulated neutral messenger) | P.Mt. | 647.00 |
| | 42.90.2.6 | 3x120 sq.mm | 3x120 sq.mm, 1x16(st. light), 1x70(insulated neutral messenger) | P.Mt. | 808.00 |
| | 42.90.2.7 | 3x150 sq.mm | 3x150 sq.mm, 1x35(st. light), 1x95(insulated neutral messenger) | P.Mt. | 922.00 |
| **42.91** | | **Aerial bunch cable accessories suspension clamp** | Supplying & fixing of aerial bunch cable accessories suspension clamp i/c required attachment for single core cable of approved make on pole with clamp, necessary material as per specification for required size. | Each | 427.00 |
| **42.92** | | **Aerial bunch cable accessories tension clamp (dead end)** | Supplying &fixing of aerial bunch cable accessories tension clamp (dead end) i/c required attachment for single core cable of approved make in air with clamp, necessary material as per specification for required size. | Each | 446.00 |
| **42.93** | | **Piercing connector** | Supplying & fixing of waterproof insulation piercing connector for aerial bunch cable i/c required attachment for single core cable of approved make in air with necessary material as per specification of required size as mentioned below – | | |
| | 42.93.1 | Main:16-95sq.mm | Main:16-95sq.mm, tap:1.5-50sq.mm | Each | 172.00 |
| | 42.93.2 | Main:50-150sq.mm | Main:50-150sq.mm, tap:6-50sq.mm | Each | 220.00 |
| **42.94** | | **Spring loaded busbar** | Supplying &fixing of spring loaded busbar screwless connector with enclosure of approved make including required clamp, nut, bolt, connection, testing as per specification to complete the job. | | |
| | 42.94.1 | **Single phase** | **Single phase** | | |
| | 42.94.1.1 | 1Ph, 1In+2Out | 1Ph, 1In+2Out | Each | 1972.00 |
| | 42.94.1.2 | 1Ph, 1In+3Out | 1Ph, 1In+3Out | Each | 2023.00 |
| | 42.94.1.3 | 1Ph, 1In+4Out | 1Ph, 1In+4Out | Each | 2989.00 |
| | 42.94.1.4 | 1Ph, 1In+5Out | 1Ph, 1In+5Out | Each | 3080.00 |
| | 42.94.1.5 | 1Ph, 1In+7Out | 1Ph, 1In+7Out | Each | 3226.00 |
| | 42.94.1.6 | 1Ph, 1In+8Out | 1Ph, 1In+8Out | Each | 3449.00 |
| | 42.94.2 | **Three Phase** | **Three Phase** | | |
| | 42.94.2.1 | 3Ph, 1In+2Out | 3Ph, 1In+2Out | Each | 4746.00 |
| | 42.94.2.2 | 3Ph, 1In+4Out | 3Ph, 1In+4Out | Each | 5839.00 |
| | 42.94.2.3 | 3Ph, 1In+6Out | 3Ph, 1In+6Out | Each | 7276.00 |
| | 42.94.2.4 | 3Ph, 1In+7Out | 3Ph, 1In+7Out | Each | 7404.00 |
| | 42.94.2.5 | 3Ph, 1In+8Out | 3Ph, 1In+8Out | Each | 7538.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.95** | | **12.5 meter high mast lighting** | Providing and erection on existing foundation12.5 meter high mast lighting system suitable to install 6 No. 2 x 400 w M.H/S.V/LED flood light fitting and control gear with integral Power tool, weight 340 kg. comprising of 2 section of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm, dia 100mm & 360mm for top and bottom respectively, stress fitting arrangement on site with 350mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 800mm at top and at base inside compartment with double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 520mm dia. and foundation bolts having 4 nos. bolts of 24mm dia, 750mm long (EN8 grade), Anchor plate 445 PCD, including accessories viz.<br><br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 250kg. load and upto 6 fittings symmetrically.<br>(2) Trailing copper cable 5x2.5 sq.mm, EPR insulated PCP sheathed.<br>(3) Double drum/350kg winch having gear 53:1, oil bath (SAE90/140) arrangement.<br>(4) 2 nos. stainless steel wire ropes 5mm dia (7/19) breaking load capacity 1450 kg.x 2.<br>(5) Integral power tool 3-phase, 0.75 HP 2m/min single speed.<br>(6) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 32 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(7) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without fixture and lamp) | Each | 205322.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.96** | | **16 meter high mast lighting** | Providing and erection on existing foundation 16 meter high mast lighting system suitable to install upto 8 No. 2 x 250/ 400 w M.H/S.V/LED flood light and control gear with integral power tool, weight 450 kg. comprising of 2 sections of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm & 4 mm, dia 150mm & 360mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 1200mm at top and at base inside compartment having double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 520mm dia. and foundation bolts having 6 nos. bolts of 24mm dia 750 mm long (EN.8 grade) anchor plate 445 PCD, including accessories viz.<br><br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 500kg. load and upto 8 fittings symmetrically.<br>(2) Trailing copper cable 5x2.5 sq.mm, PVC insulated PVC Sheathed<br>(3) Double drum 350Kg Winch having gear 53:1, oil bath (SAE90/140) arrangement<br>(4) 2 nos. Stainless steel wire ropes 5 mm dia (7/19) breaking load capacity 1450 kg x 2,<br>(5) Integral power tool 3-phase, 0.75HP 2m/min single speed .torque limiter upto 500 kg adjustable<br>(6) Single LED 300 Nos. aviation light<br>(7) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 32 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(8) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without fixture and lamp) | Each | 266303.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.97** | | **20 meter high mast lighting** | Providing and erection on existing foundation 20 meter high mast lighting system suitable to install upto 10 No. 2 x 250/ 400 w M.H/S.V/LED flood light and control gear with integral Power tool, weight 450 kg. comprising of 2 sections of hot dipped galvanized materials as per BSEN ISO 1461 thickness 3 mm & 4 mm, dia 150mm & 410mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 1200mm at top and at base inside compartment having double thickness 3 mm & 4 mm, dia 150mm & 410mm for top and bottom respectively, stress fitting arrangement on site with 500mm overlap dynamic loading to withstand max wind pressure as per -IS 875 part III, parameters for structural & foundation design must be taken from wind tunnel test. Lightning protection of GI single spike 1200mm at top and at base inside compartment having double internal lock with adequate size of MCB erected on PVC board complete with base plate of 25mm thick 570mm dia. and foundation bolts having 8 nos. bolts of 30mm dia 850 mm long (EN.8 grade) Anchor plate 490 PCD, including accessories viz.<br><br>(1) Lantern carriage of 50 NB ERW class-B, MS pipe covered with PVC sleeve suitable to carry 500kg. load and upto 10 fittings symmetrically.<br>(2) Trailing copper cable 5x4.0 sq.mm, PVC insulated PVC sheathed.<br>(3) Double drum 750Kg Winch having gear 53:1, oil bath (SAE90/140) arrangement.<br>(4) 2 nos. stainless steel wire ropes 6 mm dia (7/19) breaking load capacity 2350 kg. x 2,<br>(5) Integral power tool 3-phase, 1HP 2m/min single speed .torque limiter upto 500 kg adjustable<br>(6) 2 Nos. LED 300 nos. aviation light<br>(7) Feeder pillar fabricated out of 14 SWG CRCA sheet and comprise of incoming MCB 63 A TPN switch, HRC fuses, single dial timer, suitable size of contactors for lighting and power tool, 2 nos. outgoing, reversing switch for motor.<br>(8) Foundation drawing, test certificate and guarantee certificate shall be provided by manufacturer. (without fixture and lamp) | Each | 372547.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.98** | | **Octagonal poles** | Providing and erection of hot dipped galvanized octagonal poles in single section made from 3mm thick sheet having lockable weather proof flush door junction box, suitable size & type of foundation bolts 4 nos. 'J' type (EN8 grade) complete, erected in an approved manner on provided foundation. | | |
| | 42.98.1 | 3 meter | 3 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 28 kg ± 3% , 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | Each | 4691.00 |
| | 42.98.2 | 4 meter | 4 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 36 kg ± 3%, 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | Each | 5899.00 |
| | 42.98.3 | 5 meter | 5 meter height 130 X 70mm A/F, 200x200x12mm base plate having weight 44 kg ± 3%, 4 nos X 16 x 450mm bolt having weight 3.2 kg ± 3%. | Each | 6943.00 |
| | 42.98.4 | 6 meter | 6 meter height 130 X 70mm A/F, 220x220x12mm base plate having weight 52 kg ± 3%, 4 nos. X 20 x 600mm bolt having weight 6.7 kg ± 3%. | Each | 7956.00 |
| | 42.98.5 | 7 meter | 7 meter height, 130 X 70mm A/F, 220x220x12mm base plate having weight 60 kg ± 3%, 4 nos. X 20 x 700mm bolt having weight 7.7 kg ± 3%. | Each | 9863.00 |
| | 42.98.6 | 8 meter | 8 meter height, 135 X 70mm A/F, 225x225x16mm base plate having weight 70 kg ± 3%, 4 nos. X 20 x 750mm bolt having weight 8.2 kg ± 3%. | Each | 13415.00 |
| | 42.98.7 | 9 meter | 9 meter height, 155 X 70mm A/F, 250x250x16mm base plate having weight 87 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | Each | 15447.00 |
| | 42.98.8 | 10 meter | 10 meter height, 175 X 70mm A/F, 275x275x16mm base plate having weight 105 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | Each | 17114.00 |
| | 42.98.9 | 11 meter | 11 meter height, 175 X 70mm A/F, 275x275x16mm base plate having weight 115 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | Each | 18587.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.98.10 | 11 meter | 11 meter height, 210 X 90mm A/F, 300x300x16mm base plate having weight 141 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | Each | 21737.00 |
| | 42.98.11 | 12 meter | 12 meter height, 210 X 90mm A/F, 300x300x16mm base plate having weight 153 kg ± 3%, 4 nos. X 24 x 750mm bolt having weight 12 kg ± 3%. | Each | 24579.00 |
| **42.99** | | **Bakelite sheet MCBSP & stud type 4 way epoxy connector for decorative /octagonal pole box.** | Providing and fixing of bakelite sheet 10mm thick, one MCB 6Amp-32Amp SP & 16sq.mm stud type 4 way epoxy connector for decorative designer poles /octagonal pole box. | Each | 323.00 |
| **42.100** | | **Sword canopy type single arm GI bracket** | Providing and fixing of sword canopy type single arm GI bracket upto 2 meter Long for 70mm A/F top octagonal pole | Each | 1658.00 |
| **42.101** | | **Sword canopy type single arm GI bracket** | Providing and fixing of sword canopy type single arm GI bracket upto 2 meter Long for 90mm A/F top octagonal pole | Each | 2071.00 |
| **42.102** | | **Sword canopy type double arm GI bracket** | Providing and fixing of sword canopy type double arm GI bracket upto 2 meter Long for 70mm A/F top octagonal pole | Each | 2200.00 |
| **42.103** | | **Sword canopy type double arm GI bracket** | Providing and fixing of sword canopy type double arm GI bracket upto 2 meter Long for 90mm A/F top octagonal pole | Each | 3496.00 |
| **42.104** | | **Single arm standard bracket** | Single arm standard bracket | | |
| | 42.104.1 | | Providing and fixing of 1000mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | Each | 2059.00 |
| | 42.104.2 | | Providing and fixing of 1500mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | Each | 3084.00 |
| | 42.104.3 | | Providing and fixing of 2000mm long Single Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | Each | 4178.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.105** | | **Double arm standard bracket** | Double arm standard bracket | | |
| | 42.105.1 | | Providing and fixing of 1000mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4Nos 10mm dia Pinch bolts. | Each | 3695.00 |
| | 42.105.2 | | Providing and fixing of 1500mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts.. | Each | 4583.00 |
| | 42.105.3 | | Providing and fixing of 2000mm long Double Arm GI Hot Dip Galvanized Bracket with 42.4mm OD and 3.2mm thick Pipe. Canopy pipe should be 500mm with 4 Nos 10mm dia Pinch bolts. | Each | 5341.00 |
| **42.106** | | **Ring carriage dia** | Providing and fixing of ring carriage dia. not less than 750mm made of 50mm dia. GI pipe Light 'A' class for 70mm A/F top octagonal pole with required clamps, nut, bolt etc as required. | Each | 2468.00 |
| **42.107** | | **Decorative designer poles** | Providing and erection of black painted galvanized decorative designer poles in different section, top 60mm dia., bottom 140mm dia. with gold colour painted ornamental cast aluminium ring having lockable weather proof flush door junction box in bottom section, base plate dia. 290mm having 4 holes of 22mm dia. complete erected in an approved manner on existing foundation. Suitable size & type of foundation bolts 4 nos. X 16mm X 450mm 'J' type (EN8 grade) | | |
| | 42.107.1 | 3 meter | 3 meter Height | Each | 17737.00 |
| | 42.107.2 | 4 meter | 4 meter Height | Each | 19831.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 42.107.3 | 5 meter | 5 meter height | Each | 23654.00 |
| **42.108** | | **Casting of open raft shallow footing foundation for 12.5meter high mast** | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 12.5meter High Mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or Engineer-in-charge. | Each | 33771.00 |
| **42.109** | | **Casting of open raft shallow footing foundation for 16meter high mast** | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 16meter high mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or engineer-in-charge. | Each | 41931.00 |
| **42.110** | | **Casting of open raft shallow footing foundation for 20meter high mast** | Designing & casting of open raft shallow footing foundation with M-20 cement concrete suitable for 20meter high mast considering the safe soil bearing capacity at site as 10 T/sqm at 2meter depth including excavation, foundation nut bolts in an approved manner as per manufacturer drawing or engineer-in-charge. | Each | 42707.00 |
| **42.111** | | **Foundation for octagonal pole** | Designing and construction of footing foundation for octagonal pole of desired size including excavation in soil/soft rock, base concrete M-7.5 (1:4:8), 15mm thick and pedestal of RCC M-20 (1:1.5:3), required tar steel reinforcement for required size from base concrete, shuttering, curing etc. with existing foundation bolts complete as per specifications in an approved manner as per manufacturer drawing or engineer-in-charge. | Cu. Mt. | 6770.00 |
| **42.112** | | **Dismantling of existing pole accessories** | Dismantling of existing H- Beam/RS Joist/PCC/Rail pole i/c accessories such as cross arms, top clamp, V-cross arms, strut, stay & conductor/cable/guarding span complete. The material should be transport & handed over to MPMKVVCL/PWD store. | Each Pole | 1457.00 |
| **42.113** | | **KWh meter with LCD display (for energy metering) 5 - 30A , 1 phase 2 wire, 240 Volts** | Supply, installation, testing & commissioning of KWh meter with LCD display (for energy metering) 5 - 30A , 1 phase 2 wire, 240 Volts, accuracy class 1 duly tested from electricity board with uPVC BOX . | Each | 3732.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.114** | | **KWh meter with LCD display (for energy metering) 5 - 20A, 3 phase 4 wire** | Supply, installation, testing & commissioning of KWh meter with LCD display (for energy metering) 5 - 20A, 3 Phase 4 Wire, 440 Volts, accuracy class 1 duly tested from electricity board with uPVC BOX. | Each | 5675.00 |
| **42.115** | | **KWh meter with LCD display (for energy metering) 5 - 10A, CT operated, 3 phase 4 wire** | Supply, installation, testing & commissioning of KWh Meter with LCD display (For Energy Metering) 5 - 10A, CT Operated, 3 Phase 4 Wire, 440 Volts, and Accuracy Class 1 duly tested from electricity board with uPVC BOX. | Each | 8301.00 |
| **42.116** | | **LT trivector energy meter** | Supply, installation, testing & commissioning of LT trivector energy meter, DLMS compliant category-A (for substation/feeder metering) , 3 phase 4 wire, 440 Volts, 5 Amp, accuracy class 0.5S with GSM(GPRS compatible) modem suitable for required C.T. 5A duly tested from electricity board with uPVC box . | Each | 14882.00 |
| **42.117** | | **HT trivector energy meter** | Supply, installation, testing & commissioning of HT trivector energy meter, 3 phase 4 wire, 110 Volts, 5 Amp, accuracy class 0.5S with GSM (GPRS compatible) intelligent modem for consumer (store N forward type) suitable for required C.T. including uPVC box/sheet metal box with TTB duly tested from electricity board. | Each | 21686.00 |
| **42.118** | | **HT trivector energy meter** | Supply, installation, testing & commissioning of HT trivector energy meter, DLMS compliant category-A (for substation/feeder metering energy audit), 3 phase 4 wire, 110 Volts, 5 Amp, accuracy class 0.5S with GSM (GPRS compatible) modem suitable for required C.T. including uPVC box/sheet metal box with TTB duly tested from electricity board. | Each | 22454.00 |
| **42.119** | | **CT/PT ratio 300-150/5 Amp** | Supply, installation, testing & commissioning of HT 11 KV/110 V 3P4W CT/PT ratio 300-150/5 Amp outdoor oil immersed type duly tested from electricity board. | Each | 55469.00 |
| **42.120** | | **CT/PT ratio 400-200/5 Amp** | Supply, installation, testing & commissioning of HT 33 KV/110 V 3P4W CT/PT ratio 400-200/5 Amp outdoor oil immersed type duly tested from electricity board. | Each | 72425.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.121** | | **CT ratio 500-200/5 Amp** | Supply, installation, testing & commissioning of LT CT ratio 500-200/5 Amp epoxy coated Indoor Type duly tested from electricity board. | Each | 992.00 |
| **42.122** | | **Vacuum circuit breaker** | Supplying, installation, testing and commissioning of triple pole outdoor type 11KV/33KV vacuum circuit breaker with outdoor control & relay panel with numerical static relay for feeder protection (1F). The installation of VCB shall be carried out in accordance with IS: 3072. The VCB shall be mounted on suitable size duly painted MS channel/angle iron structure with nut bolts and washer. The structure shall be fixed on existing cement concrete foundation. The complete work should be done as per IS specification. | | |
| | 42.122.1 | 11 KV 630A VCB | 11 KV 630A VCB | Each | 190130.00 |
| | 42.122.2 | 33 KV, 1250A VCB | 33 KV, 1250A VCB | Each | 319644.00 |
| **42.123** | | **Single phase C.T./ P.T.** | Supplying, installation, testing and commissioning of single phase C.T./P.T. outdoor type 11KV/33KV with terminal box. The C.T./P.T. shall be mounted on existing structure fabricated with suitable size MS channel/angle iron structure with nut bolts and washer duly painted. The structure shall be fixed on existing cement concrete foundation as per IS specification as mentioned below. | | |
| | 42.123.1 | 11KV CT, 200-100/5 Amp | 11 KV CT outdoor type 200-100/5 Amp | Each | 13044.00 |
| | 42.123.2 | 33KV CT, 200-100/5 Amp | 33 KV CT outdoor type 200-100/5 Amp | Each | 18294.00 |
| **42.124** | | **Single phase C.T./ P.T.** | Supplying, installation, testing and commissioning of single phase C.T./P.T. outdoor type 11KV/33KV with terminal box. The C.T./P.T. shall be mounted on existing structure fabricated with suitable size MS channel/angle iron structure with nut bolts and washer duly painted. The structure shall be fixed on existing cement concrete foundation as per IS specification as mentioned below. | | |
| | 42.124.1 | 11 KV/110V, PTs | 11 KV/110V single phase PTs oil immersed | Each | 15989.00 |
| | 42.124.2 | 33 KV/110V, PTs | 33 KV/110V single phase PTs oil immersed | Each | 20488.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **42.125** | | **Indoor control relay panel** | Supplying, installation, testing and commissioning of indoor control relay panel shall consisting of ammeter, voltmeter, selector switch, auxiliary relays for anti-pumping device, trip and alarm of buchholz relay, oil and winding temperature circuit, electronic hooter, 230 V AC to 110 V DC power pack with charging equipment for closing, tripping and indication, triple pole IDMTL two O/C and one E/F delay with current setting range 50-200% and 10-50%, push button high speed trip relay etc. complete as per IS specification. | Each | 129039.00 |
| **42.126** | | **Sub-station** | **Compact outdoor sub-station :** | | |
| | 42.126.1 | **Compact outdoor sub-station comprising of HV compartment for 33/.433KV** | Design, supply, installation, testing and commissioning of compact outdoor sub-station comprising of HV compartment for 33/.433KV, DYn II, dry type resin cast transformer with standard accessories 1 no. of 33 KV, 630A, 25KA, 3 phase 50 HZ vacuum circuit breaker and 1 no. of 33 KV, 100A, 50KA, 3 phase 50 HZ Isolator (non - extensible RMU) connection on H.T side with single core H.T. cable and standard accessories, connection on LV side insulated aluminium busbar of required rating, compartment comprising of MCCB's 4 pole, and standard voltmeter 0 - 500 V with VSS, ammeter with ASS of required rating, LED indicator (RYB) 22.5mm dia. accessories as mentioned below :-<br><br>The incomer VCB and Isolator shall be interlocked mechanically and VCB with necessary interconnection wiring, feruled properly, earthing terminal, etc. The layout dimensional drawings, electrical wiring diagram and operation maintenance manuals in duplicate shall be supplied with VCB. The VCB/Isolator shall be supplied in conformity with relevant ISS & IEC with up to date amendments along with manufacturer's test certificate. The housing of compact outdoor sub - station base frame and detachable cover plates, walls, roof, doors and other components are made of 2 mm thick galvanized sheet steel with at least 20 & 40 µm zinc layer. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | Hinges, rivets are made of stainless steel. Wide door opening angle of 180 degrees, lock into top and bottom frame. Transformer roof has detachable facility to enable easy installation and maintenance without affecting other equipment's, MV & LV compartments roofs are separated from transformer roof. Roof slope for water runoff. air intake in bottom of doors for natural chimney effect and smooth air flow and sufficient self-cooling of transformer. One common earthing terminal. Type tested as per latest IEC standard 62271-202, IEC 62271-202 high-voltage/low-voltage prefabricated substation. MV & LV compartments have IP class - IP 54 & IP23 degree of protection for transformer compartment. The enclosure exterior shall be painted with epoxy based powder paint (colour RAL 7032) excluding platform, incoming/outgoing cable, earthing. | | |
| | 42.126.1.1 | **Cu. wound** | **Copper wound** | | |
| | 42.126.1.1.1 | 100KVA | 100KVA. 33/0.4KV O/G MCCB 4P - 2 Nos 160A 25KA | Each | 3418688.00 |
| | 42.126.1.1.2 | 200KVA | 200KVA, 33/0.4KV O/G MCCB 4P - 1 Nos 200A 25KA, 1 No 300A 36KA | Each | 3948566.00 |
| | 42.126.1.1.3 | 315KVA | 315KVA, 33/0.4KV O/G MCCB 4P - 1 Nos 630A 36KA, 1 No 300A 36KA | Each | 4230091.00 |

# CHAPTER - 43

# INTERCOM / EPABX SYSTEMS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **43.1** | | **EPABX system** | Supply, installation, testing and commissioning of EPABX system consist, no. of extension, no of junction, no. of expandable eport, operating console, current on max. load with ISDN compatible of approved make and labor as required as per specifications. | | |
| | 43.1.1 | 16 Nos. extensions | EPABX system consists of 16 nos. extensions, 8 nos of junction &upto 64 nos. of expandable eport, with one operating console, 1.35 Amp max. current | Each | 64894.00 |
| | 43.1.2 | 48 Nos. extensions | EPABX system consists of 48 nos. extensions, 8 nos of junction & upto 96 nos. of expandable eport, with one operating console, 2 Amp max. current | Each | 104928.00 |
| | 43.1.3 | 64 Nos. extensions | EPABX system consists of 64 nos. extensions, 16 nos of junction & upto 128 nos. of expandable eport, with one operating console, 3 Amp max. current | Each | 147635.00 |
| | 43.1.4 | 128 Nos. extensions | EPABX system consists of 128 nos. extensions, 24 nos of junction & upto 512 nos. of expandable eport, with one operating console, 11 Amp max. current | Each | 217082.00 |
| **43.2** | | **Float cum boost charger** | Supplying & fixing float cum boost charger for EPABX system of approved make with all necessary material as required as per specifications. | | |
| | 43.2.1 | 10 Amp 48 V | 10 Amp 48 V float cum boost charger | Each | 19316.00 |
| | 43.2.2 | 25 Amp 48 V | 25 Amp 48 V float cum boost charger | Each | 31701.00 |
| | 43.2.3 | 50 Amp 48 V | 50 Amp 48 V float cum boost charger | Each | 52170.00 |
| **43.3** | | **Essential additional accessories** | Supplying, fixing essential additional accessories to EPABX system of approved make with all necessary material as required as per specifications. | | |
| | 43.3.1 | EPABX E-1 card | EPABX E-1 card | Each | 28393.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 43.3.2 | R-2 TWT card | R-2 TWT card | Each | 14513.00 |
| | 43.3.3 | Hot standby card | Hot standby card | Each | 19936.00 |
| | 43.3.4 | EPABX-operator console | EPABX-operator console | Each | 6475.00 |
| | 43.3.5 | Digital phone | EPABX system digital phone with 2 line display (32 characters) with 8 programmable keys. | Each | 4264.00 |
| | 43.3.6 | Line card | EPABX system line card to extend the number of line by 8. | Each | 8965.00 |
| | 43.3.7 | Junction card | EPABX system junction card to extend the number of Index by 4. | Each | 5992.00 |
| | 43.3.8 | Junction card | EPABX system junction card to extend the number of junction by 8. | Each | 11643.00 |
| | 43.3.9 | Digital line card | EPABX system digital line card 8 line. | Each | 9015.00 |
| | 43.3.10 | CLI phone | 4 push button CLI phone. | Each | 717.00 |

# CHAPTER - 44

# LIFT/ELEVATORS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **44.1** | | **Lift/Elevators** | Supply, installation, testing, commissioning, and final testing of automatic lift complete in the shaft well and pit already constructed as per CPWD specification including automatic rescue device (ARD), All standard equipments, accessories and control equipments as per manufacturer's design and as per CPWD specification (Part III LIFTS) on turnkey basis, Conforming to NBC/statutory norms and fulfilling following requirements.<br>(i) **Type**: Passenger lift.<br>(ii) **Load**: As specified elsewhere from 4 passengers (272Kg) to 26 passengers (1768Kg) (for each person standard weight 68Kg.).<br>(iii) **Speed**: 1.00 metre/sec. or less as required.<br>(iv) **Control**: ACV3F, 32bit microprocessor integrated serial communication control system with RMS (remote monitoring system)/BMS (building monitoring system) supporting software.<br>(v) **Motor/machine**: Permanent magnetic synchronous gearless motor. | | |
| | | | (vi) Operation: Simplex collective selective with/without lift attendant.<br>(vii) Total rise: About 3 mtrs. (approx) two stops only.<br>(viii) Landings: All floors on same side.<br>(ix) Floors served: G+1.<br>(x) Signals: Digital car position indicator, car travel direction indicator inside car & at all landings, voice annunciator with suitable music shall be provided in the lift car.<br>(xi) Shaft size: As per manufacturer's design and as per CPWD specification (Part III LIFTS), Shaft to be constructed separately.<br>(xii) Guide rail: As per manufacturer's specification.<br>(xiii) Lift & car size: As per manufacturer's design and as per CPWD specification (Part III LIFTS).<br>(xiv) Car entrance: As per manufacturer's design and as per CPWD specification (Part III LIFTS).<br>(xv) Landing entrance: As per manufacturer's design and as per CPWD specification (Part III LIFTS). | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | (xvi) Car door: S.S. (stainless steel) hairline finish centre/telescopic opening door with vision panel.<br>(xvii) Landing door: S.S. (stainless steel) hairline finish centre/telescopic opening door with vision panel.<br>(xviii) Door operation: Automatic electric power operated door, full height infrared curtain with multiple criss/cross light beams.<br>(xix) Car interior: S.S. (stainless steel) panel with hair line finish on all sides.<br>(xx) Car floor: Anti skid PVC flooring on heavy duty M S (mild steel) platform.<br>(xxi) Car ceiling & lighting: S. S. (stainless steel) finish with aesthetic appearance with LED ceiling lights<br>(xxii) Carventilation: Pressure fan/blower fan (car fan with automatic sleep timer shall be provided).<br>(xxiii) Emergency lighting in car: Battery operated emergency lighting in the lift car.<br>(xxiv) Intercom system: Battery operated intercom system in lift car, machine room and ground | | |
| | | | floor/control room.<br>(xxv) Emergencyalarm: Battery operated alarm bell system in lift car & control room / ground floor. (xxvi) Position of machine room: Directly above the hoist-way.<br>(xxvii) Electric power supply: AC 415 V, 3 phases 50 Hz & AC 230 V 1phase 50Hz terminated in machine room.<br>(xxviii) Fire man switch : Fire man switch at machine room/terminal landing<br>(xxix) All car panel buttons and all floor switches must be with Braille language as per lift act.<br>(xxx) Audio visual indication in the lift car showing overloading shall be provided such that doors keep open till excess load is removed.<br>(xxxi) Spring buffer/hydraulic buffer shall be provided<br>(xxxi) Spring buffer/hydraulic buffer shall be provided<br>(xxxii) Mechanical over speed governor, door key holes in the floor doors shall be provided. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | (xxxiii) Lift machine hoisting arrangement in the lift machine room and monkey ladder for lift pit should be provided by the lift agency, along with the other steel structure works, foundations for the machines etc.<br>(xxxiv) In the hoist way fascia plate shall be provided.<br>(xxxv) Permanent wiring in lift machine room and lift well with proper no. of light points, with fixtures, exhaust fan, plug points and 3 phase 440V power supply shall be made in lift shaft by the department, machine room or at desired level.<br>(xxxvi) Car top safety barricade shall be provided.<br>(xxxvii) ARD (automatic rescue device): Solid state inbuilt automatic rescue device (ARD) automatically rescues passengers trapped in the lift car in between floors in the event of power failure. Automatic operation and immediate action in the event of mains failure, capable of moving the lift to the nearest landing, opens the automatic door of | | |
| | | | the lift car and floor. Sealed maintenance free battery backup with automatic charging unit and auto changeover device on mains failure. | | |
| | 44.1.1 | 4 Passenger | 4 Passenger (272Kg.) | Each | 1260000 |
| | 44.1.2 | 6 Passenger | 6 Passenger (408 Kg.) | Each | 1350000 |
| | 44.1.3 | 8 Passenger | 8 Passenger (544Kg.) | Each | 1485000 |
| | 44.1.4 | 10 Passenger | 10 Passenger (680Kg.) | Each | 1530000 |
| | 44.1.5 | 13 Passenger | 13 Passenger (884Kg.) | Each | 1710000 |
| | 44.1.6 | 15/16 Passenger | 15/16 Passenger (1088Kg.)/Stretcher Lift | Each | 1935000 |
| | 44.1.7 | 20 Passenger | 20 Passenger (1360Kg.) | Each | 2070000 |
| | 44.1.8 | 26 Passenger | 26 Passenger (1768Kg.)/Bed Lift | Each | 2205000 |
| **44.2** | | **Add extra for higher speed** | Add extra for higher speed to item 44.1 | | |
| | 44.2.1 | 1.5 metre/Sec | For higher speed 1.5 metre/Sec (For 4-10 Passengers) | Each | 81000 |
| | 44.2.2 | 1.5 metre/Sec | For higher speed 1.5 metre/Sec (For 13-26 Passengers) | Each | 99000 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 44.2.3 | 1.75 metre/Sec | For higher speed 1.75/2.00 metre/Sec (For 4-10 Passengers) | Each | 112500 |
| | 44.2.4 | 1.75 metre/Sec | For higher speed 1.75/2.00 metre/Sec (For 13-26 Passengers) | Each | 135000 |
| **44.3** | | **Add extra for every additional floor** | Add extra to item no. 44.1 for every additional floor beyond initial G+ 1 floor (additional stops beyond two stops) provides all necessary additional parts/equipments as per manufacturer's requirement. | | |
| | 44.3.1 | upto 9 stops | Cost for each additional floor : upto 9 stops (For 4-10 Passengers) | Each | 67500 |
| | 44.3.2 | upto 9 stops | Cost for each additional floor : upto 9 stops (For 13-26 Passengers) | Each | 85500 |
| | 44.3.3 | upto 16 stops | Above 9 stops but Max. upto 16 stops (For 4-10 Passengers) | Each | 76500 |
| | 44.3.4 | upto 16 stops | Above 9 stops but Max. upto 16 stops (For 13-26 Passengers) | Each | 135000 |
| **44.4** | | **Add extra for machine room less (MRL)** | Add extra to item no. 44.1for machine room less (MRL) lift in place of machine room directly above the hoist way. | | |
| | 44.4.1 | 4-13 Passenger | Extra cost for machine room less (4-13 Passenger) | Each | 135000 |
| | 44.4.2 | 15/16 & Above | Extra cost for machine room less (15/16 and above passenger capacity) | Each | 180000 |
| **44.5** | | **Add extra for group operation** | Add extra to item no. 44.1 for group operation | | |
| | 44.5.1 | Duplex | In group of two lift (Duplex). | Each | 72000 |
| | 44.5.2 | Triplex | In group of three lift (Triplex). | Each | 216000 |
| | 44.5.3 | Quadruplex | In group of four lift (Quadruplex) | Each | 270000 |
| **44.6** | | **Gong system** | Add extra to item no. 44.1 for hall lantern and arrival gong system. | | |
| | 44.6.1 | Extra cost/floor | Extra cost for per floor per lift | Each | 18000 |
| **44.7** | | **Moon rock finish** | Add extra to item no. 44.1 cost for moon rock finish S.S. (stainless steel) panel. | Each | 162000 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **44.8** | | **Toughened glass, on front door of car and each landing door** | Add extra to item no. 44.1 for transparent observation, toughened glass fixed to the structural steel frame of required size as per manufactures design on front door of car and each landing door in place of S.S. (stainless steel) panels including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.8.1 | For 4-10 passenger | For 4-10 passenger (for passenger car) | Each | 4500 |
| | 44.8.2 | For 4-10 passenger | For 4-10 passenger (for each landing door) | Each | 4500 |
| | 44.8.3 | For 13-26 passenger | For 13-26 passenger (for passenger car) | Each | 6300 |
| | 44.8.4 | For 13-26 passenger | For 13-26 passenger (for each landing door) | Each | 6300 |
| **44.9** | | **Toughened glass, on back side only** | Add extra to item no. 44.1 for transparent observation toughened glass fixed to the structural steel frame as per manufactures design on back side only in place of S.S. (stainless steel) panels including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.9.1 | For 4-10 passenger | For 4-10 passenger | Each | 67500 |
| | 44.9.2 | For 13-26 passenger | For 13-26 passenger | Each | 117000 |
| **44.10** | | **Toughened glass, on back side and front door of car** | Add extra to item no. 44.1 for transparent observation toughened glass fixed to the structural steel frame as per manufactures design on back side and front door of car and each landing door in place of S.S. (stainless steel) panels including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.10.1 | For 4-10 passenger | For 4-10 passenger | Each | 81000 |
| | 44.10.2 | For 13-26 passenger | For 13-26 passenger | Each | 135000 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **44.11** | | **Toughened glass, on all three sides and front door of car** | Add extra to item no. 44.1 for transparent observation toughened glass on all three sides and front door of car fixed to the structural steel frame as per manufactures design and each landing door in place of S.S. (stainless steel) panels including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.11.1 | For 4-10 passenger | For 4-10 passenger | Each | 135000 |
| | 44.11.2 | For 13-26 passenger | For 13-26 passenger | Each | 189000 |
| **44.12** | | **3 side toughened glass observation (pentagon shape) for capsule lift car** | Add extra to item no. 44.1 cost for 3 side toughened glass observation (pentagon shape) for capsule lift car including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.12.1 | For 10-13 passenger | For 10-13 passenger capacity lift car | Each | 216000 |
| | 44.12.2 | For 15/16 passenger | For 15/16 passenger capacity lift car | Each | 243000 |
| | 44.12.3 | For 20 passenger | For 20 passenger capacity lift car | Each | 270000 |
| | 44.12.4 | For 26 passenger | For 26 passenger capacity lift car | Each | 315000 |
| **44.13** | | **5 side toughened glass panoramic (hexagon shape) for capsule lift** | Add extra to item no. 44.1 cost for 5 side toughened glass panoramic (hexagon shape) for capsule lift car including cost of hold fastening of guide rails to a separately erected civil structure (steel or RCC columns) but excluding cost of civil structure. | | |
| | 44.13.1 | For 15/16 passenger | For 15/16 passenger capacity lift car | Each | 315000 |
| | 44.13.2 | For 20 passenger | For 20 passenger capacity lift car | Each | 360000 |
| | 44.13.3 | For 26 passenger | For 26 passenger capacity lift car | Each | 405000 |

# CHAPTER - 45

## MEDICAL GAS PIPE LINE SYSTEM (MGPS)

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **45.1** | | **Manifold system (excluding cylinders)** | Supply, installation, testing and commissioning of following gas manifold system (excluding cylinders). The Manifold is configured for designed numbers of cylinders and is suitable to withstand a pressure of 145 Kg/cm2, along with high-pressure copper annealed tail pipes with end brass adapter suitable for cylinders and manifold. Top frame comprising of high pressure copper pipes of size 1/2" NB x 15swg with high pressure brass fittings made of high tensile brass and connections through non-return valves, The design of middle and bottom frames is provided to fit both round and flat bottom cylinders safely. The manifold is tested (hydraulically) at 3500 psig.(It should be CE and ISO 13485 certified) | | |
| | 45.1.1 | Oxygen | Main manifold for Oxygen | Nos. | 4500.00 |
| | 45.1.2 | Emergency Oxygen | Emergency manifold for Oxygen | Nos. | 4500.00 |
| | 45.1.3 | Nitrous Oxide | Main manifold for Nitrous Oxide | Nos. | 4500.00 |
| | 45.1.4 | Emergency Nitrous Oxide | Emergency manifold for Nitrous Oxide | Nos. | 4500.00 |
| | 45.1.5 | Carbon dioxide | Main manifold for Carbon dioxide | Nos. | 4500.00 |
| | 45.1.6 | Emergency Carbon dioxide | Emergency manifold for Carbon dioxide. | Nos. | 4500.00 |
| **45.2** | | **Control panel for Oxygen** | Fully automatic three port control panel for Oxygen supply, installation, testing and commissioning of Automatic manifold control panel would provide a duty or standby gas supply from two cylinder banks with automatic changeover the manifold indicates local status and has a remote indicator facility. Automatic manifold control panel would be designed to supply the MGPS with a constant pressure supply via a control panel form two equal banks of cylinders and the changeover from the 'duty' to the 'standby' bank of cylinders would be automatic. Automatic changeover from the duty to the standby bank would occur at a cylinder pressure that will ensure the maximum usage of the contents of the duty bank. The changeover unit would be provided with monitoring to detect and display: duty bank operating, duty bank empty, standby bank operating, and standby bank below 10 % when | Nos. | 455000.00 |
| | | | the duty bank is empty, reserve bank low, pipeline pressure fault. There would be facility to connect liquid oxygen supply (i.e. three port control panel)<br>(It should be CE and ISO 13485 certified)<br>Fully automatic three port control panel for Oxygen<br>input pressure 150 kg/cm2<br>output pressure 4 kg/cm2<br>capacity: 1500 LPM<br>media : oxygen<br>It should be CE mark and with four digit number European notified body no. and country of origin certificate should be provided | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **45.3** | | **Fully-automatic control panel for Nitrous Oxide** | Fully-automatic control panel for Nitrous Oxide<br>input pressure 150 kg/cm2<br>output pressure 10 kg/cm2<br>capacity: 500 LPM<br>media : Nitrous Oxide<br>It should be CE mark and with four digit European notified body number and country of origin certificate should be provided | Nos. | 345800.00 |
| **45.4** | | **Emergency manifold line regulator** | Emergency manifold line regulator<br>Supply, installation, testing and commissioning of emergency manifold line regulator<br>input pressure 150 kg/cm2<br>working pressure 4 kg/cm2<br>It should be CE and ISO 13485 certified. | | |
| | 45.4.1 | oxygen | oxygen<br>input pressure 150 kg/cm2<br>working pressure 4 kg/cm2<br>Capacity : 1500 LPM | Nos. | 22000.00 |
| | 45.4.2 | Carbon dioxide | Carbon dioxide<br>Inlet pressure :150 kg/cm2<br>Output pressure 4 kg/cm2<br>including heater<br>capacity : 500 LPM | Nos. | 22000.00 |
| | 45.4.3 | Nitrous Oxide | Nitrous Oxide<br>including heater<br>capacity : 500 LPM | Nos. | 22000.00 |
| | 45.4.4 | Cylinder connecting pipe (pig tail) | Supply, installation, testing and commissioning of cylinder connecting pipe (pig tail) made up of flexible SS hose with safety cord 5/8 inch BSP gland nut and one side 1/2 inch BSP internal threads having one meter length. (Oxygen -24 , N2O - 10,CO2-4) | Nos. | 4000.00 |
| **45.5** | | **Gas outlet front loaded DIN /BS STD** | **Gas outlet front loaded DIN /BS STD:**<br>Supply, installation, testing and commissioning of DIN/British standard gas outlets are for use with medical gas system. The gas outlets completely comply with HTM2002 and C11 standard regulations and pass the test of leakage and connector stability. Medical gas outlet shall be gas specific for the services indicated and accept only corresponding DIN/British standard Quick connect adapter. Specifically, gas specific indexed to eliminate interchangeability of gas services. The gas outlet is offered in console type, wall type with box and hose barb type gas system. It should be CE and ISO 13485 certified.<br>DIN/BIS std. gas outlets feature and benefits:<br>* Comply with HTM2002 and C11 standard regulations<br>* Color coded according to ISO32 color<br>*100% pressure and leak tested<br>* Full range of mounting options<br>* Available in wall type with box, console type and hose barb type<br>* Gas specific adapters | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | **45.5.1** | **Quick connect adapter** | **DIN/British standard quick connect adapter (safety probe)** | | |
| | 45.5.1.1 | Oxygen | Oxygen | Nos. | 546.00 |
| | 45.5.1.2 | Nitrous Oxide | Nitrous Oxide | Nos. | 546.00 |
| | 45.5.1.3 | Medical Air | Medical Air (4 bar) | Nos. | 546.00 |
| | 45.5.1.4 | Surgical Air | Surgical Air ( 7 bar) | Nos. | 546.00 |
| | 45.5.1.5 | Vacuum | Vacuum | Nos. | 546.00 |
| | 45.5.1.6 | AGSS | AGSS | Nos. | 546.00 |
| | 45.5.1.7 | CO2 | CO2 | Nos. | 546.00 |
| | **45.5.2** | **Gas outlet** | **Gas outlet front loaded DIN /BS STD.** | | |
| | 45.52.1 | Oxygen | Oxygen | Nos. | 2730.00 |
| | 45.5.2.2 | Nitrous oxide | Nitrous oxide | Nos. | 2730.00 |
| | 45.5.2.3 | Medical Air | Medical Air (4 bar) | Nos. | 2730.00 |
| | 45.5.2.4 | Surgical Air | Surgical Air ( 7 bar) | Nos. | 2730.00 |
| | 45.5.2.5 | Vacuum | Vacuum | Nos. | 2730.00 |
| | 45.5.2.6 | AGSS | AGSS | Nos. | 2730.00 |
| | 45.5.2.7 | CO2 | CO2 | Nos. | 2730.00 |
| **45.6** | | **Copper pipes** | Supply, installation, testing and commissioning of copper pipe used would be half hard drawn, tempered seamless phosphorous deoxidized, non-arsenical and degreased confirming to BS EN: 13348. For copper and copper alloys seamless, round copper tubes for medical gases or vacuum.<br>The supply of pipes would be accompanied with manufactures test certificates for the physical properties of copper pipes and chemical composition. Copper pipe must have accepted third party inspection certificate for medical gas pipe line. | | |
| | 45.6.1 | 12 mm O.D. | Supply and installation of diameter 12 mm O.D. X 0.7 mm thick | Metre | 501.00 |
| | 45.6.2 | 15 mm O.D. | Supply and installation of diameter15 mm O.D. X 0.9 mm thick | Metre | 573.00 |
| | 45.6.3 | 22 mm O.D. | Supply and installation of diameter 22 mm O.D. X 0.9 mm thick | Metre | 865.00 |
| | 45.6.4 | 28 mm O.D. | Supply and installation of diameter 28 mm O.D. X 1.2 mm thick | Metre | 1001.00 |
| | 45.6.5 | 42 mm O.D. | Supply and installation of diameter 42 mm O.D. X 1.2 mm thick | Metre | 1775.00 |
| | 45.6.6 | 54 mm O.D. | Supply and installation of diameter 54 mm O.D. X 1.2 mm thick | Metre | 2548.00 |
| | 45.6.7 | 76 mm O.D. | Supply and installation of Diameter76 mm O.D. X 1.2 mm thick | Metre | 2958.00 |
| **45.7** | | **Back pressure compensated [BPC] Oxygen flow meter** | Supply, installation, testing and commissioning of back pressure compensated [BPC] Oxygen flow meter body should be made of brass material dully chrome plated and graduated from 0-15 L/m on an expanded lower scale for precision setting at low flow and 5-15 L/m for general use. The humidifier bottle should be made up of unbreakable polycarbonate/ polysulphone material and autoclavable at 121 degree centigrade temperature. It should be CE and ISO 13485 certified. | Nos. | 1550.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **45.8** | | **Digital area alarm** | Supply, installation, testing and commissioning of Area Pressure Alarm Panel would be wit digital display, closed circuit, self-monitoring type. The alarm would be designed to monitor the line pressure in piping systems. It should be CE and ISO 13485 Certified. with four digit notified no. | | |
| | 45.8.1 | 3 gas | 3 gas | Nos. | 24300.00 |
| | 45.8.2 | 5 gas | 5 gas | Nos. | 29700.00 |
| | 45.8.3 | 6 gas | 6 gas | Nos. | 31300.00 |
| **45.9** | | **Area valve box1** | Supply, installation, testing and commissioning of area valve box should be made up of M.S. body with powder coated. In side fitting copper pipe with on/off valve and pressure gauge. Lockable cover with breakable glass so that during normal operation access has been by key but during emergency operation, access by breaking the glass panel. The unit should provide a zone isolation facility. Gas flow direction should be indicated. It should be CE and ISO 13485 certified. | | |
| | 45.9.1 | 3 gas | 3 gas | Nos. | 12150.00 |
| | 45.9.2 | 5 gas | 5 gas | Nos. | 16000.00 |
| | 45.9.3 | 6 gas | 6 gas | Nos. | 19000.00 |
| **45.10** | | **Line shut off isolation valve** | Supply, installation, testing and commissioning of the ball valves [for dividing the pipelines into various zones for the case of maintenance] the internal ball with full borne is made of stainless steel with Teflon seats. These valves are leak proof and needs only quarter turn for its operation [either fully open or fully closed]. It should be CE and ISO 13485 certified. | | |
| | 45.10.1 | 15 mm O.D. | 15 mm O.D. | Nos. | 1500.00 |
| | 45.10.2 | 22 mm O.D. | 22 mm O.D. | Nos. | 3240.00 |
| | 45.10.3 | 28 mm O.D. | 28 mm O.D. | Nos. | 3510.00 |
| | 45.10.4 | 42 mm O.D. | 42 mm O.D. | Nos. | 4800.00 |
| **45.11** | | **Conversion kit for O.T. (3m Pipe)** | Conversion kit for O.T. (3m pipe) supply, installation, testing and commissioning of conversion kit. It should have safety plug on one end side and another side pin index according to specific gas and there should be standard 3mm long pipe in between those two ends. Color coded, and made of food grade material & suitable for working pressure of minimum 10 bar & Vacuum to 710 mm hg. One end is pin index and the respective end diameter indexing with safety plug. It should be CE and ISO 13485. certified Oxygen-8, Nitrous- 8, | Nos. | 1350.00 |
| **45.12** | | **High pressure tubing** | Supply, installation, testing and commissioning of color coded, and made of food grade material & suitable for working. It should be CE and ISO 13485 certified pressure of minimum 10 bar &vacuum to 710 mm hg. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 45.12.1 | Oxygen | Oxygen | Metre | 110.00 |
| | 45.12.2 | Nitrous Oxide | Nitrous Oxide | Metre | 110.00 |
| | 45.12.3 | Air | Air | Metre | 110.00 |
| | 45.12.4 | Vacuum | Vacuum | Metre | 110.00 |
| | 45.12.5 | CO2 | CO2 | Metre | 110.00 |
| **45.13** | | **Central compressed air system** | Supply, installation, testing and commissioning of oil free compressor having following system:<br>a. The medical air plant should be manufactured under an ISO 13485 qualify management system.<br>b. The Duplex medical air plant should be provided with 15 HP motor each.<br>Type of Compressor – Oil free air cooled compressor.<br>Motor – 15 HP, 50 Hz<br>No. of Stages – Two stages.<br>Piston Displacement in CFM – 36.66 CFM<br>Outlet Pressure – 8.5 Kg/Cm2<br>c. Duplex air compressor system suitable for both continuous and frequent start/stop operation<br>d. Two compressors shall be designated as one running and one standby operating in Duplex or Cascade mode.<br>e. compressor body should be of Aluminum alloy. | Complete job | 1865500 |
| | | | f. Duplex desiccant heatless dryer of 100CFM capacity shall be provided to reduce moisture content of the air.<br>g. 2 Nos. of Air receiver should be provided with the capacity of 500 liters.<br>h. Air Filtration system should be provided to supply medical quality It should be CE and ISO 13485 Certified air for each system (Running & Stand by)<br>i.duplex pressure reducing system of 4 bar and 7 bar each with same capacity in standby mode<br>Pre Filter –<br>Max Operating flow: 60 cfm<br>Max inlet pressure: 16 kg/cm2<br>Hsg. MOC – Aluminum die Casting<br>Initial Pressure Loss 0.1 (kg/cm2)<br>Element MOC- Borosilicate<br>End Connection – 1/2” BSP | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | After Filter –<br>Hsg. MOC – Aluminum die Casting<br>Initial Pressure Loss 0.1 (kg/cm2)<br>Element MOC- Borosilicate<br>End Connection – 1/2" BSP<br>i. Separate pressure reducing system should be provided for each system (Running & Stand by).<br>ii) Reciprocating Type Air Cooled Oil Free Air Compressors -2 Numbers<br>iii) Electrical Control Panel -1 Number<br>iv) Vertical Air storage tank with all accessories Capacity:500 L -2 Numbers<br>v) Air Dryer-100 CFM or as required Desiccant type -2 Numbers<br>vi) Air filters (3 Stage) - 1 micron, 0.1 micron, 0.01 micron with 1 inch end connection - 2 Numbers<br>vii) Air Regulator ( Pressure Reducing System )- 2 Numbers | | |
| **45.14** | | | Supply, installation, testing and commissioning of central vacuum pump<br>a. Vacuum Plant should be capable of running in Triplex mode depending on Hospital requirement (2 Running &1 as stand by.)<br>b. The medical plant should be manufactured under in ISO 13485 quality management system.<br>c. The vacuum plant should comprise of 3 nos of Motors of 10 HP Each.<br>e. The plant should be fitted with filters.<br>f. There should be a 2 No. of vacuum receiver of 2000 liters capacity.<br>g. one bacterial filter with secretion bottle having end connection of 2inch.<br>h. Electrical Control Panel | | |
| | 45.14.1 | | Vacuum pump with motor 10 HP<br>(Two Running & One stand by)<br>Electrical Control Panel<br>Vacuum reservoir Tank with safety accessories Capacity: 2000 L Filter with Secretion Tap Bottle | Nos. | 1164800.00 |
| | 45.14.2 | Ward vacuum jar | Ward vacuum jar with regulator capacity: 600 ml | Nos. | 1970.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **45.15** | | **Suction regulator** | a. The suction regulator should incorporate an anti-bacterial filter, an anti over flow safety device, wide membrane continuous suction controller and 600 ml polysulphone jar.<br>b. Should have vacuum gauge fitted with a protective bumper device.<br>c. Should have ON/OFF knob allowing quick readjustment of vacuum level.<br>d. Must have central knob with 0-1000mbar/kPa range.<br>e. Jar with lid should be unbreakable and auto cleavable upto 134 deg C must be transparent to ensure perfect sucked liquid visibility. It should be CE and ISO 13485 certified. | | |
| | 45.15.1 | Vacuum jar trolley | Vacuum jar trolley for O.T. | Nos. | 12285.00 |
| **45.16** | | AGSS system | Supply, installation, testing and commissioning of AGSS system should contain duplex system having capacity of 2500 LPM each. These are dedicated low velocity exhaust system for collection of excess gases from one or more ORs and discharged to the outdoors. The powerful side channel exhaust blower the unit very compact, reliable and creates sufficient negative pressure, so that cross-contamination does not occur in the other ORs connected to this system. It should consist of balancer/ receiver unit along with connecting hose for anesthesia machine to receiver and from outlet to receiver unit; AGSS hose assemblies, AGSS Remote Switch. AGSS central system is manufactured according to the regulations of EN ISO 7396-2d. It should be CE and ISO 13485 certified. | | |
| | 45.16.1 | Duplex AGSS unit | Duplex AGSS unit | Set | 560250.00 |
| | 45.16.2 | Balancer/receiver unit | Balancer/receiver unit along with connecting hose for anesthesia machine to receiver and from outlet to receiver unit | Set | 50000.00 |
| | 45.16.3 | hose assemblies | AGSS hose assemblies | Nos. | 12000.00 |
| | 45.16.4 | Remote switch | AGSS remote switch | Nos. | 10000.00 |
| | 45.16.5 | Outlets | Probe for AGSS outlets | Nos. | 700.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **45.17** | | **Bed head panels** | Supply, installation, testing and commissioning of bed head panels for ICU with electrical fitting (aluminum extruded with powder coating and length of 1500 mm, having provision for gas outlets & electrical switch & socket) It should be CE marked and ISO 13485 certified.lt should be efficient, safe &robust design in extruded aluminum section. lt should have the provision of gas outlets & electricals as per requirement. The smooth surfaces, with acceptable colour choice should have an integrated rail system to mount accessories. Entire pipe line should run in continuous horizontal panels with no break for each unit & length as per area where it has to be installed. | Nos. | 16000.00 |
| **45.18** | | **Medical Isolation Panel** | Supply,installation,testing and commissioning of fully automatic medical isolation panel for OT and ICU , isolatoted power distribution panel as per IEC 60364-7-710 and BIS SP 30-2011 part -3 section 4of medical establishments.medical isolaton transformer, earth fault location system and remote indicator panel should be CE certified imported from foriegn countries with 4 Digit number european notified body no. , country of origin mention on it and consists of following : | | |
| | **45.18.1** | **Medical Isolation Panel with automatic fault locator** | i) 1no. Of isolation transformer having primary voltage :230 volt AC and secondary voltage 230 volt AC as per IEC 60364-7-710 and IEC 61558-2-15 , with equipment supply voltage range 70 to 260 volt and frequency range 42 to 460 hz.<br>ii) 1 no. of 63 Amp 10KA Dp, ELCB,30 ma leakage current on primary side and 1 no. 63 Amp ,10KA,DP,MCB on secondary side of isolation transformer.<br>iii) panel must have provision for two different power supply at primary side of transformer and system should be designed and installed to facilitate the automatic changeover from main distributin system to electrical safety power source feeding essential loads as per IEC 60364-5-536-2.2.4.<br>iv) 12 no. of 20Amp ,10KA , DP,MCB for outgoings.<br>v) Isolation transformer shall be provided with overload and over temperature monitoring alarm through PTC. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | vi) enviroment qualification : Ambient temp: 10 deg celcius to 55 deg celcius. relative humidity : 95% non condensing.<br>vii) 12no. of core current transformer having response sensitivity 0.2mAmp.<br>viii) 1 set of exhaust fan and filter.<br>ix) Insulation Monitor shall be as per IEC 61557-8.<br>x) 1 no. of earth fault locator system as per IEC 61557-9 with injection less than 1mAmp.<br>xi) integrated digital voltmeter and ammeter and provison of LED indicator for mains supply and UPS supply.<br>xii) power consumption should be less than 14VA,maximum leakage current capacitance shold be less than 5 micro farad , measuring injected current less than 50micro Amp and noise level less than 28 db. | | |
| | | | xiii) 1no. of load current measuring transformer.<br>xiv) Panel should be capable of detecting all combination of capacitive and restive faults,including balanced , unbalanced and hybrid faults. A self test and self calibration function is included. The above work will be housed in 16 gauge MS enclosure floor/wall mounted duly coated with approved colour shade, cable entry from top,prewired ,earthing,etc. required to complete the job in 450W*400D*1900H mm dimension. Submitting of SLD of panel to Engineer-in-charge before commencement of work . | | |
| | 45.18.1.1 | 10 KVA | 10 KVA | Each | 1250000.00 |
| | 45.18.1.2 | 8 KVA | 8 KVA | Each | 1184000.00 |

# CHAPTER -46

## AIR CONDITIONER HVAC

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **46.1** | | **Dismantling& shifting of window air conditioning unit** | Dismantling & shifting of window air conditioning unit 1.0TR/ 1.5TR/2TR capacity excluding fascia and cutting, welding complete work as required within town. | Each | 1155.00 |
| **46.2** | | **Dismantling& shifting** | Dismantling &shifting, installation, testing & commissioning of split air conditioning unit 1.0TR/1.5TR/2TRcapacity including indoor / outdoor unit complete as required within town. | Each | 1455.00 |
| **46.3** | | **Suction/ discharge copper pipe line in split AC** | Supply, installation, testing & commissioning of suction/discharge copper pipe line in split air conditioning unit 1.0/1.5 TR/2 TR/3 TR capacity including brazing, nitrile rubber insulation 6/9mm thick class 'O', etc complete as required. | | |
| | 46.3.1 | Copper tube 1/4" | Copper tube 1/4" | P. Mt | 437.00 |
| | 46.3.2 | Copper tube 3/8" | Copper tube 3/8" | P. Mt | 550.00 |
| | 46.3.3 | Copper tube 1/2" | Copper tube 1/2" | P. Mt | 657.00 |
| | 46.3.4 | Copper tube 5/8" | Copper tube 5/8" | P. Mt | 815.00 |
| **46.4** | | **Chemical servicing** | Chemical servicing of window/split air conditioning unit 1TR/1.5TR/2TR/3TR capacity including cleaning of cooling & condensing units, filters, grills etc & all other parts oiling & greasing of fan motor, testing of A.C unit etc. | Each | 1073.00 |
| **46.5** | | **Gas charging** | Gas charging (R410A/Eco Friendly Green Gas) of window/split air conditioning unit including flushing of cooling & condensing units with nitrogen gas & rectification of leakage, replacement of filter, flair nut etc. if any. | | |
| | 46.5.1 | 1.0 TR | 1.0 TR | Each | 1629.00 |
| | 46.5.2 | 1.5 TR | 1.5 TR | Each | 1836.00 |
| | 49.5.3 | 2.0 TR | 2.0 TR | Each | 2042.00 |
| | 46.5.4 | 3.0 TR | 3.0 TR | Each | 3264.00 |
| **46.6** | | **Gas charging of refrigerator** | Gas charging of refrigerator upto 300 ltr capacity including flushing of cooling & condensing units with nitrogen gas & rectification of leakage, replacement of filter, etc. if any. | Each | 1290.00 |
| **46.7** | | **Thermostat switch** | Replacement & connection of thermostat switch of required size & capacity for Refrigerator upto 300 ltr capacity including all required material to complete the job. | Each | 1684.00 |
| **46.8** | | **Starting relay** | Replacement & connection of starting relay required size & capacity for Refrigerator upto 300 ltr capacity including all required material to complete the job. | Each | 939.00 |

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| **46.9** | | **Comprehensive maintenance and servicing** | Comprehensive maintenance and servicing of air-conditioners of following type and capacity including replacement with new compressor, and related parts such as capacitors, relay, electronic controls, PCB, display, thermostat, remote control, and related accessories, rewinding of fan/blower motor, flushing of cooling & condensing units with nitrogen gas, required refrigerant gas charging & rectification of leakage, replacement of filter, flair nut etc. if any. | | |
| | **46.9.1** | **Window type AC** | **Window type air conditioners :-** | | |
| | 46.9.1.1 | 1.5 TR | 1.5 TR | Each/year | 2358.00 |
| | **46.9.2** | **Split type** | **Split type wall mounted air conditioners single phase** | | |
| | 46.9.2.1 | 1.5 TR | 1.5 TR | Each/year | 3273.00 |
| | 46.9.2.2 | 2 TR | 2 TR | Each/year | 4513.00 |
| | **46.9.3** | **Split type cassette/floor type** | **Split type cassette/floor type air conditioners single/three phase** | | |
| | 46.9.3.1 | 1.5 TR | 1.5 TR | Each/year | 4502.00 |
| | 46.9.3.2 | 2 TR | 2 TR | Each/year | 4912.00 |
| | 46.9.3.3 | 3 TR | 3 TR | Each/year | 6640.00 |
| | **46.9.4** | **Ductable AC** | **Ductable air conditioners three phase** | | |
| | 46.9.4.1 | 5.5 TR | 5.5 TR | Each/year | 15770.00 |
| | 46.9.4.2 | 7.5 TR | 7.5 TR | Each/year | 21518.00 |
| | 46.9.4.3 | 8.5 TR | 8.5 TR | Each/year | 24807.00 |
| | 46.9.4.4 | 8.75 TR | 8.75 TR | Each/year | 37632.00 |
| **46.10** | | **Comprehensive maintenance of single/double door refrigerator** | Comprehensive maintenance of single/double door refrigerator upto 300 ltr capacity including replacement of compressor, and related parts such as condenser, relay, electronic controls, display, thermostat, and related accessories, rewinding of motor, flushing of cooling & condensing units with nitrogen gas, required refrigerant gas charging & rectification of leakage, replacement of lamp, etc. if any. | Each/year | 3399.00 |

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| **46.11** | | **SPLIT AC (Inverter model, Heating & Cooling)** | SPLIT AC (inverter model) :- Supply, installation, testing and commissioning of Split type wall mounted air conditioning unit of following capacity including indoor, outdoor unit fitted with inverter based rotary compressor, copper condenser and first charge refrigerant R410A. The condensing unit working range should be capable for providing cooling between +/-5 deg. C to 50 deg. C & heating within 0 deg. C to 15 deg. C including connection of copper pipe of required size upto 3 mtr., insulation, cordless remote control, auto air swing louvers, wall stand for outdoor unit, connection copper lead wire, condensed water outlet pipe 2 mtr. complete etc as required. Outdoor/indoor units should be of same make (brand). | | |
| | 46.11.1 | 1.0 TR , 3 star | 1.0 TR star rating 3 star | Each | 43340.00 |
| | 46.11.2 | 1.0 TR , 5 star | 1.0 TR star rating 5 star | Each | 50932.00 |
| | 46.11.3 | 1.5 TR, 3 star | 1.5 TR star rating 3 star | Each | 51364.00 |
| | 49.11.4 | 1.5 TR, 5 star | 1.5 TR star rating 5 star | Each | 59142.00 |
| | 46.11.5 | 1.8 /2 TR, 3 star | 1.8/2 TR star rating 3 star | Each | 65939.00 |
| | 46.11.6 | 1.8 /2 TR, 5 star | 1.8/2 TR star rating 5 star | Each | 75826.00 |
| | **46.11.2** | **SPLIT AC (inverter model, Cooling only)** | **SPLIT AC (inverter model, Cooling only)** | | |
| | 46.11.2.1 | 1.0 TR , 3 star | 1.0 TR star rating 3 star | Each | 37800.00 |
| | 46.11.2.2 | 1.0 TR , 5 star | 1.0 TR star rating 5 star | Each | 45000.00 |
| | 46.11.2.3 | 1.5 TR, 3 star | 1.5 TR star rating 3 star | Each | 46000.00 |
| | 46.11.2.4 | 1.5 TR, 5 star | 1.5 TR star rating 5 star | Each | 52800.00 |
| | 46.11.2.5 | 1.8 /2 TR, 3 star | 1.8/2 TR star rating 3 star | Each | 59500.00 |
| | 46.11.2.6 | 1.8 /2 TR, 5 star | 1.8/2 TR star rating 5 star | Each | 67800.00 |
| **46.12** | | **Split AC (non inverter)** | Split AC (non inverter) :- Supply installation testing and commissioning of split type wall mounted air conditioning unit of following capacity including indoor, outdoor unit fitted with rotary compressor, copper condenser and first charge refrigerant R410A. The condensing unit working range should be capable of providing cooling between +/-5deg.C to 50 deg. C including connection of copper pipe of required size upto 3 mtr., insulation, cordless remote control, auto air swing louvers, wall stand for outdoor unit, connection copper lead wire, condense water outlet pipe 2 mtr. complete etc as required. Outdoor/indoor units should be of same make (brand). | | |
| | 46.12.1 | 1.0 TR , 3 star | 1.0 TR star rating 3 star | Each | 33039.00 |
| | 46.12.2 | 1.5 TR, 3 star | 1.5 TR star rating 3 star | Each | 39036.00 |
| | 46.12.3 | 1.8/2 TR, 3 star | 1.8/2 TR star rating 3 star | Each | 53029.00 |

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| **46.13** | | **Cassette AC** | Cassette AC :- Supply installation testing and commissioning of split type ceiling cassette Slim line/floor standing air conditioning unit of capacity including indoor, outdoor unit fitted with rotary compressor, copper condenser and first charge refrigerant R410A/ecofriendly green gas only, cordless remote control, auto air swing louvers, connection copper lead wire between outdoor and indoor unit, connection copper lead wire upto 1.5 mtr., excluding connection copper pipe of required size, insulation, floor stand for outdoor unit, condense water outlet complete etc as required. Outdoor/indoor units should be of same make (brand). | | |
| | **46.13.1** | **Ceiling suspended/mounted cassette type** | **Ceiling suspended/mounted cassette type 4 way unit with inbuilt drain pump & decorative panels.** | | |
| | 46.13.1.1 | 1.8 TR/2.0 TR single phase | 1.8 TR/2.0 TR single phase | Each | 86198.00 |
| | 46.13.1.2 | 2.4 TR single phase | 2.4 TR single phase | Each | 101281.00 |
| | 46.13.1.3 | 2.7/3.0 TR three phase | 2.7/3.0 TR three phase | Each | 125091.00 |
| | 46.13.1.4 | 3.4/3.5 TR three phase | 3.4/3.5 TR three phase | Each | 141939.00 |
| | 46.13.1.5 | 3.8/4 TR three phase | 3.8/4 TR three phase | Each | 147387.00 |
| | 46.13.2 | Floor standing/slim line | Floor standing/slim line | | |
| | 46.13.2.1 | 2.0 TR/2.4 TR single phase | 2.0 TR/2.4 TR single phase | Each | 101887.00 |
| | 46.13.2.2 | 3.0TR/3.3 TR three phase | 3.0TR/3.3 TR three phase | Each | 112984.00 |
| | 46.13.2.3 | 3.8TR/4.0 TR three phase | 3.8TR/4.0 TR three phase | Each | 119239.00 |
| | **46.13.3** | **AC installation, wall mounted** | AC installation: - Installation of split type wall mounted air conditioner indoor, outdoor unit fitted with rotary compressor, copper condenser and first charge refrigerant R410A, including connection of copper pipe and material supplied with complete unit etc as required. (excluding cost of copper pipe) | Each | 1500.00 |
| | **46.13.4** | **AC installation, split type ceiling cassette air conditioning** | AC installation: - Installation of split type ceiling cassette air conditioning unit including indoor, outdoor unit fitted with rotary compressor, condenser and first charge refrigerant R410A/ecofriendly green gas only including connection of copper pipe and material supplied with complete unit etc as required.(excluding cost of copper pipe) | Each | 5000.00 |

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| | **46.13.5** | **AC installation, split type slim line /floor standing air conditioning unit** | AC Installation: - Installation of split type slim line/floor standing air conditioning unit including indoor, outdoor unit fitted with rotary compressor, condenser and first charge refrigerant R410A/ecofriendly green gas only including connection of copper pipe and material supplied with complete unit etc as required.(excluding cost of copper pipe) | Each | 8000.00 |
| **46.14** | | Ductable AC | Ductable AC :- Supply installation testing and commissioning of, air cooled ductable air conditioning unit of following capacity with scroll compressor of approved make suitable for operation on 3 phase 415 V, 50 Hz AC supply. Comprising of copper condenser, and aluminum fins, and first charge refrigerant. The indoor comprising of fan, fan motor, copper tube, aluminum finned cooling coil, drain pan, filter, controls, safety with ecofriendly gas R410/R410A excluding connection copper pipe of required size, insulation, stand for outdoor unit, connection copper lead wire, condensed water outlet pipe complete etc as required. Outdoor/indoor units should be of same make (brand). | | |
| | 46.14.1 | 5.0-5.5 TR three phase | 5.0 - 5.5 TR three phase | Each | 127411.00 |
| | 46.14.2 | 8.75-8.5 TR three phase | 8.75 - 8.5 TR three phase | Each | 170936.00 |
| | 46.14.3 | 10.0-11.0 TR three phase | 10.0 - 11.0 TR three phase | Each | 238073.00 |
| | 46.14.4 | 16.5-18.0 TR three phase | 16.5 - 18.0 TR three phase | Each | 338443.00 |
| | 46.14.5 | 20.8-22.0 TR three phase | 20.8 - 22.0 TR three phase | Each | 471859.00 |

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| **46.15** | | **Package AC** | Package AC: - Supply installation testing and commissioning of, air cooled package air conditioning unit of following capacity with scroll compressor of approved make suitable for operation on 3 phases 415 V, 50 Hz AC supply. Comprising of copper condenser, with aluminums fins, and first charge refrigerant. The indoor comprising of fan, fan motor, copper tube, aluminum finned cooling coil, drain pan, filter, controls, safety with ecofriendly gas R410 excluding connection copper pipe of required size, insulation, stand for outdoor unit, connection copper lead wire, condensed water outlet pipe complete etc as required. Outdoor/indoor units should be of same make (brand). | | |
| | 46.15.1 | 5.5 TR - 6.0 TR three phase | 5.5 TR - 6.0 TR three phase | Each | 149405.00 |
| | 46.15.2 | 8.3 TR - 9.0 TR three phase | 8.3 TR - 9.0 TR three phase | Each | 198074.00 |
| | 46.15.3 | 11.0 TR - 12.0 TR three phase | 11.0 TR-12.0 TR three phase | Each | 248464.00 |
| | 46.15.4 | 16.5 TR - 18.0 TR three phase | 16.5 TR-18.0 TR three phase | Each | 376982.00 |
| | 46.15.5 | 22.0 TR three phase | 22.0 TR three phase | Each | 423433.00 |
| **46.16** | | **AC installation, air cooled ductable air conditioning unit** | AC installation :- Installation, air cooled ductable air conditioning unit of following capacity with scroll compressor suitable for operation on 3 phase 415 V, 50 Hz AC supply. The indoor/outdoor comprising of condenser, fan, fan motor, cooling coil, drain pan with filter and first charge refrigerant ecofriendly gas including connection of copper pipe and material supplied with complete unit etc as required. | | |
| | 46.16.1 | 5.5 TR | 5.5 TR | Each | 5500.00 |
| | 46.16.2 | 8.75/8.5 TR | 8.75/8.5 TR | Each | 8500.00 |
| | 46.16.3 | 11.0 TR | 11.0 TR | Each | 11000.00 |
| | 46.16.4 | 17.0/16.7 TR | 17.0/16.7 TR | Each | 34000.00 |
| | 46.16.5 | 22.0/20.8 TR | 22.0/20.8 TR | Each | 44000.00 |
| **46.17** | | **AC installation, air cooled package air conditioning unit** | AC installation: - Installation, air cooled package air conditioning unit of following capacity with scroll compressor suitable for operation on 3 phase 415 V, 50 Hz AC supply. The indoor/outdoor comprising of condenser, fan, fan motor, cooling coil, drain pan with filter and first charge refrigerant eco friendly gas including connection of copper pipe and material supplied with complete unit etc as required. | | |
| | 46.17.1 | 5.5 TR/6.0 TR | 5.5 TR/6.0 TR | Each | 7000.00 |
| | 46.17.2 | 8.75 TR/9.0 TR | 8.75 TR/9.0 TR | Each | 9000.00 |
| | 46.17.3 | 11.0 TR /12.0 TR | 11.0 TR /12.0 TR | Each | 11000.00 |
| | 46.17.4 | 17.0 TR /18.0 TR | 17.0 TR /18.0 TR | Each | 34000.00 |
| | 46.17.5 | 22.0 TR | 22.0 TR | Each | 44000.00 |

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| **46.18** | | **VRV/VRF outdoor AC unit** | VRV/VRF outdoor AC unit:- Supply installation testing and commissioning of VRV /VRF /MRV outdoor units with following multiple inverter based scroll type compressors, electronic expansion valve, R 410A/ecofriendly green refrigerant, cooling & heating mode both options, special pre-coated fin type heat exchanger, low noise direct driven propeller condenser fans in each module with top discharge including internal wiring, microprocessor controls, safety. The condensing unit should be capable of providing cooling between +/- 5deg. C. to 48 deg. C & heating within 0 deg. C to 15 deg. C, COP not less than 3.3. Each system should have the provision to combine maximum 3 nos compressor in each outdoor unit, every single outdoor unit should not have more than maximum Capacity 20 HP (3 phase) each compressor complete as per specification. Outdoor/indoor units should be of same make (brand). (Single phase/three phase) | | |
| | 46.18.1 | 6 HP | 6 HP | Each | 204128.00 |
| | 46.18.2 | 8 HP | 8 HP | Each | 222389.00 |
| | 46.18.3 | 10 HP | 10 HP | Each | 275889.00 |
| | 46.18.4 | 12 HP | 12 HP | Each | 309153.00 |
| | 46.18.5 | 14 HP | 14 HP | Each | 351203.00 |
| | 46.18.6 | 16 HP | 16 HP | Each | 399471.00 |
| | 46.18.7 | 18 HP | 18 HP | Each | 438757.00 |
| | 46.18.8 | 20 HP | 20 HP | Each | 483570.00 |
| | 46.18.9 | 22 HP | 22 HP | Each | 591261.00 |
| | 46.18.10 | 24 HP | 24 HP | Each | 624525.00 |
| | 46.18.11 | 26 HP | 26 HP | Each | 666575.00 |
| | 46.18.12 | 28 HP | 28 HP | Each | 714843.00 |
| | 46.18.13 | 30 HP | 30 HP | Each | 754129.00 |
| | 46.18.14 | 32 HP | 32 HP | Each | 805161.00 |
| | 46.18.15 | 34 HP | 34 HP | Each | 844447.00 |
| | 46.18.16 | 36 HP | 36 HP | Each | 889260.00 |
| | 46.18.17 | 38 HP | 38 HP | Each | 928546.00 |
| | 46.18.18 | 40 HP | 40 HP | Each | 973359.00 |
| | 46.18.19 | 42 HP | 42 HP | Each | 1069106.00 |
| | 46.18.20 | 44 HP | 44 HP | Each | 1113919.00 |
| | 46.18.21 | 46 HP | 46 HP | Each | 1162187.00 |
| | 46.18.22 | 48 HP | 48 HP | Each | 1210456.00 |
| | 46.18.23 | 50 HP | 50 HP | Each | 1249741.00 |

| | | | | | |
|---|---|---|---|---|---|
| | 46.18.24 | 52 HP | 52 HP | Each | 1289027.00 |
| | 46.18.25 | 54 HP | 54 HP | Each | 1328313.00 |
| | 46.18.26 | 56 HP | 56 HP | Each | 1373126.00 |
| | 46.18.27 | 58 HP | 58 HP | Each | 1417940.00 |
| | 46.18.28 | 60 HP | 60 HP | Each | 1462753.00 |
| **46.19** | | **VRF/VRV indoor ACunit (wall mounted)** | VRF/VRV indoor AC unit (wall mounted):- Supply installation testing and commissioning of wall mounted indoor split unit comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, and temperature & on/off display with provision of discharge angle setting by wireless remote controller including internal wiring, controls complete as per specification. (Single phase/three phase) | | |
| | 46.19.1 | 0.65TR | 0.65TR | Each | 39439.00 |
| | 46.19.2 | 0.83 TR | 0.83TR | Each | 40057.00 |
| | 46.19.3 | 1.05 TR | 1.05TR | Each | 40777.00 |
| | 46.19.4 | 1.33 TR | 1.33TR | Each | 43350.00 |
| | 46.19.5 | 1.65 TR | 1.65TR | Each | 44482.00 |
| | 46.19.6 | 2.08TR | 2.08TR | Each | 46025.00 |
| **46.20** | | **VRF/VRV indoor AC unit (ceiling mounted 4 way)** | VRF/VRV indoor AC unit (ceiling mounted 4 way):- Supply installation testing and commissioning of 4 way round flow ceiling suspended/mounted cassette type unit comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature &on/off display with inbuilt drain pump & decorative panels including internal wiring, controls. Dimension in terms of length & width should not be more than 950mmX 950mm with wireless remote controller. (Single phase/three phase) | | |
| | 46.20.1 | 0.83 TR | 0.83 TR | Each | 54319.00 |
| | 46.20.2 | 1.05 TR | 1.05 TR | Each | 56227.00 |
| | 46.20.3 | 1.33 TR | 1.33 TR | Each | 58758.00 |
| | 46.20.4 | 1.65 TR | 1.65 TR | Each | 60373.00 |
| | 46.20.5 | 2.08 TR | 2.08 TR | Each | 61785.00 |
| | 46.20.6 | 2.64 TR | 2.64 TR | Each | 67939.00 |
| | 46.20.7 | 3.30 TR | 3.30 TR | Each | 69251.00 |
| | 46.20.8 | 4.13 TR | 4.13 TR | Each | 70562.00 |
| | 46.20.9 | 4.50 TR | 4.50 TR | Each | 73690.00 |

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| **46.21** | | **VRF/VRV indoor Ac unit (ceiling mounted compact 4 way)** | VRF/VRV indoor AC unit (ceiling mounted compact 4 way):- Supply installation testing and commissioning of 4 way round flow ceiling suspended/mounted compact cassette type unit comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature &on/off display with inbuilt drain pump & decorative panels including internal wiring, controls. Dimension in terms of length & width should not be more that 750mmX 750mm with wireless remote controller. (Single phase/three phase) | | |
| | 46.21.1 | 0.65 TR | 0.65 TR | Each | 52064.00 |
| | 46.21.2 | 0.83 TR | 0.83 TR | Each | 55367.00 |
| | 46.21.3 | 1.05 TR | 1.05 TR | Each | 56283.00 |
| | 46.21.4 | 1.33 TR | 1.33 TR | Each | 59884.00 |
| | 46.21.5 | 1.65 TR | 1.65 TR | Each | 61119.00 |
| **46.22** | | **VRF/VRV indoor AC unit (ceiling mounted 2 way)** | VRF/VRV indoor AC unit (ceiling mounted 2 way):- Supply installation testing and commissioning of 2 way flow ceiling suspended/mounted cassette type unit comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature &on/off display with inbuilt drain pump & decorative panels including internal wiring, controls. Dimension in terms of length & width should not be more that 950mmX 950mm with wireless remote controller. (Single phase/three phase) | | |
| | 46.22.1 | 0.65 TR | 0.65 TR | Each | 68323.00 |
| | 46.22.2 | 0.83 TR | 0.83 TR | Each | 78819.00 |
| | 46.22.3 | 1.05 TR | 1.05 TR | Each | 80465.00 |
| | 46.22.4 | 1.33 TR | 1.33 TR | Each | 85714.00 |
| | 46.22.5 | 1.65 TR | 1.65 TR | Each | 87875.00 |
| | 46.22.6 | 2.08 TR | 2.08 TR | Each | 90241.00 |
| | 46.22.7 | 2.64 TR | 2.64 TR | Each | 99709.00 |
| | 46.22.8 | 4.13 TR | 4.13 TR | Each | 103413.00 |
| **46.23** | | **VRF/VRV indoor AC unit (ceiling mounted one way)** | VRF/VRV indoor AC unit (ceiling mounted one way) :- Supply installation testing and commissioning of corner cassette type unit (one way) comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature &on/off display with inbuilt drain pump & decorative panels including internal wiring, controls with wireless remote controller. (Single phase/three phase) | | |

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|---|---|---|---|---|---|
| | 46.23.1 | 0.65 TR | 0.65 TR | Each | 71453.00 |
| | 46.23.2 | 0.83 TR | 0.83 TR | Each | 72791.00 |
| | 46.23.3 | 1.05 TR | 1.05 TR | Each | 63941.00 |
| | 46.23.4 | 1.33 TR | 1.33 TR | Each | 79171.00 |
| | 46.23.5 | 1.65 TR | 1.65 TR | Each | 82258.00 |
| | 46.23.6 | 2.08 TR | 2.08 TR | Each | 83081.00 |
| **46.24** | | **VRF/VRV indoor AC unit (below ceiling mounted)** | VRF/VRV indoor AC unit (below ceiling mounted) :- Supply installation testing and commissioning of ceiling suspended (below ceiling) cassette type comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature &on/off display with inbuilt drain pump & decorative panels including internal wiring, controls with cordless remote & connection unit. (Single phase/three phase) | | |
| | 46.24.1 | 2.36 TR | 2.36 TR | Each | 137886.00 |
| | 46.24.2 | 3.30 TR | 3.30 TR | Each | 140665.00 |
| **46.25** | | **VRF/VRV indoor AC unit (ceiling mounted duct - low silhouette)** | VRF/VRV indoor AC unit (ceiling mounted duct - low silhouette):- Supply installation testing and commissioning of ceiling mounted duct-low silhouette type comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, with inbuilt drain pump including internal wiring, controls with cordless remote. (Single phase/three phase) | | |
| | 46.25.1 | 0.65 TR | 0.65 TR | Each | 43728.00 |
| | 46.25.2 | 0.83 TR | 0.83 TR | Each | 44963.00 |
| | 46.25.3 | 1.05 TR | 1.05 TR | Each | 45683.00 |
| | 46.25.4 | 1.33 TR | 1.33 TR | Each | 48462.00 |
| | 46.25.5 | 1.65 TR | 1.65 TR | Each | 49697.00 |
| | 46.25.6 | 2.08 TR | 2.08 TR | Each | 50726.00 |
| **46.26** | | **VRF/VRV indoor AC unit (mid - static ductable** | VRF/VRV indoor AC unit (mid - static ductable) :- Supply installation testing and commissioning of mid static ductable type comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, with inbuilt drain pump including internal wiring, controls with cordless remotes. (Single phase/three phase) | | |
| | 46.26.1 | 1.33 TR | 1.33 TR | Each | 51696.00 |
| | 46.26.2 | 1.65TR | 1.65 TR | Each | 52503.00 |
| | 46.26.3 | 2.08 TR | 2.08TR | Each | 55530.00 |
| | 46.26.4 | 2.64 TR | 2.64TR | Each | 63298.00 |

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|---|---|---|---|---|---|
| | 46.26.5 | 3.30 TR | 3.30 TR | Each | 66729.00 |
| **46.27** | | **VRF/VRV indoor AC unit (high static ductable)** | VRF/VRV indoor AC unit (high static ductable) :- Supply, installation, testing and commissioning of high static ductable type comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, with inbuilt drain pump including internal wiring, controls with cordless remote. (Single phase/three phase) | | |
| | 46.27.1 | 0.65 TR | 0.65 TR | Each | 47661.00 |
| | 46.27.2 | 0.83 TR | 0.83 TR | Each | 48367.00 |
| | 46.27.3 | 1.05 TR | 1.05 TR | Each | 49376.00 |
| | 46.27.4 | 1.33 TR | 1.33 TR | Each | 56741.00 |
| | 46.27.5 | 1.65 TR | 1.65 TR | Each | 58557.00 |
| | 46.27.6 | 2.08 TR | 2.08 TR | Each | 61079.00 |
| | 46.27.7 | 2.64 TR | 2.64 TR | Each | 69856.00 |
| | 46.27.8 | 3.30 TR | 3.30 TR | Each | 73690.00 |
| | 46.27.9 | 4.13 TR | 4.13 TR | Each | 77322.00 |
| | 46.27.10 | 4.75 TR | 4.75 TR | Each | 91042.00 |
| | 46.27.11 | 5.4 TR | 5.4 TR | Each | 129884.00 |
| | 46.27.12 | 6.54 TR | 6.54 TR | Each | 136744.00 |
| | 46.27.13 | 8.19 TR | 8.19 TR | Each | 144008.00 |
| **46.28** | | **VRF/VRV indoor AC unit (floor standing package)** | VRF/VRV indoor AC unit (floor standing package):- Supply, installation, testing and commissioning of floor standing package type unit comprising of copper evaporator, electronic expansion valve, blower, motor, filter pad, auto air swing louvers, temperature & ON/OFF display with inbuilt drain pump & decorative panels including internal wiring, controls with wireless remote controller.(Single phase/three phase) | | |
| | 46.28.1 | 4.13 TR | 4.13 TR | Each | 196483.00 |
| | 46.28.2 | 6.54 TR | 6.54 TR | Each | 209535.00 |
| | 46.28.3 | 8.19 TR | 8.19 TR | Each | 222192.00 |
| | 46.28.4 | 13.12 TR | 13.12 TR | Each | 337619.00 |
| | 46.28.5 | 16.5 TR | 16.5 TR | Each | 413666.00 |
| **46.29** | | **VRF/VRV indoor AC unit** | VRF/VRV indoor AC unit (surface ceiling suspended compact):- Surface ceiling suspended type compact with cordless remote. (Single phase/three phase) | | |
| | 46.29.1 | 1.05 TR | 1.05 TR | Each | 54002.00 |
| | 46.29.2 | 2.08 TR | 2.08 TR | Each | 60631.00 |
| | 46.29.3 | 3.30TR | 3.30 TR | Each | 67629.00 |

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| **46.30** | | **Copper piping** | Copper piping: - Supply, installation, testing and commissioning of interconnecting copper refrigerant pipe work (19mm/13mm thick) duly insulated with elastomeric nitrile rubber type tubular insulation between indoor & outdoor units as per specifications. And exposed piping should be wrapped with glass woven cloth and lag coating for vapour barrier and weather protection. Joints shall be covered with nitrile rubber tape of 3 mm thick complete as per specification. | | |
| | 46.30.1 | 44 mm OD | 44 mm O.D. wall thickness ≥ 1.4mm (insulation – 19 mm thick) | Rmt | 2965.00 |
| | 46.30.2 | 41.1 mm OD | 41.1 mm O.D. wall thickness ≥ 1.4mm (insulation – 19 mm thick) | Rmt | 2379.00 |
| | 46.30.3 | 38 mm OD | 38 mm O.D. wall thickness ≥ 1.4mm (insulation – 19 mm thick) | Rmt | 1977.00 |
| | 46.30.4 | 34.9 mm OD | 34.9 mm O.D. wall thickness ≥ 1.4mm (insulation – 19 mm thick) | Rmt | 1821.00 |
| | 46.30.5 | 28.6 mm OD | 28.6 mm O.D. wall thickness ≥ 1.2mm (insulation – 19 mm thick) | Rmt | 1390.00 |
| | 46.30.6 | 25 mm OD | 25 mm O.D. wall thickness ≥ 1.2mm (insulation – 19 mm thick) | Rmt | 1249.00 |
| | 46.30.7 | 22.2 mm OD | 22.2 mm O.D. wall thickness ≥ 1.0mm (insulation – 19 mm thick) | Rmt | 975.00 |
| | 46.30.8 | 19.1 mm OD | 19.1 mm O.D. wall thickness ≥ 1.0mm (insulation – 13 mm thick) | Rmt | 889.00 |
| | 46.30.9 | 15.9 mm OD | 15.9 mm O.D. wall thickness ≥ 0.8mm (insulation – 13 mm thick) | Rmt | 874.00 |
| | 46.30.10 | 12.7 mm OD | 12.7 mm O.D. wall thickness ≥ 0.8mm (insulation – 13 mm thick) | Rmt | 689.00 |
| | 46.30.11 | 9.5 mm OD | 9.5 mm O.D. wall thickness ≥ 0.8mm (insulation – 13 mm thick) | Rmt | 588.00 |
| | 46.30.12 | 6.4 mm OD | 6.4 mm O.D. wall thickness ≥ 0.8mm (insulation – 13 mm thick) | Rmt | 475.00 |
| **46.31** | | **Drain piping** | Drain piping :- Providing and fixing of CPVC, 6Kg/cm2 drain water piping with fittings, support and insulated with 6 mm thick closed cell elastomeric nitrile rubber insulation complete as per specification. | | |
| | 46.31.1 | 20 mm dia | 20 mm dia ID | Rmt | 171.00 |
| | 46.31.2 | 25 mm dia | 25 mm dia ID | Rmt | 202.00 |
| | 46.31.3 | 32 mm dia | 32 mm dia ID | Rmt | 297.00 |
| | 46.31.4 | 40 mm dia | 40 mm dia ID | Rmt | 354.00 |
| **46.32** | | **Flexible pre** | Flexible pre insulated duct: - Supply and fixing of aluminum foil flexible pre-insulated duct in accordance with the approved specifications. | | |
| | 46.32.1 | 100 mm dia | 100 mm dia | Rmt | 360.00 |
| | 46.32.2 | 150 mm dia | 150 mm dia | Rmt | 448.00 |
| | 46.32.3 | 200 mm dia | 200 mm dia | Rmt | 547.00 |
| **46.33** | | **PVC fresh air pipe fan** | PVC fresh air pipe fan: -Supply and fixing of PVC fresh air pipe fan complete as per specification. | | |
| | 46.33.1 | 100 mm dia | 100 mm dia | Rmt | 235.00 |
| | 46.33.2 | 150 mm dia | 150 mm dia | Rmt | 276.00 |
| | 46.33.3 | 200 mm dia | 200 mm dia | Rmt | 344.00 |
| | 46.33.4 | 250 mm dia | 250 mm dia | Rmt | 412.00 |

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| | 46.33.5 | Installation of wall mounted indoor split unit | VRF/VRV indoor AC unit installation charges: - Installation of wall mounted indoor split unit including labor for internal wiring, controls complete as per specification. (Single phase/three phase) | Each | 1500.00 |
| | 46.33.6 | Installation of cassette unit 4/2/1 way | VRF/VRV indoor AC unit installation charges: - Installation of cassette unit 4/2/1 way flow ceiling suspended/mounted cassette type unit & decorative panels including labor for internal wiring, controls with wireless remote controller. (Single phase/three phase) | Each | 4000.00 |
| | 46.33.7 | Installation of low static ductable ceiling mounted duct | VRF/VRV indoor AC unit installation charges: - Installation of low static ductable ceiling mounted duct-low silhouette type including labor for internal wiring, controls with cordless remote. (Single phase/three phase) | Each | 4000.00 |
| | 46.33.8 | Installation of mid static ductable ceiling mounted type | VRF/VRV indoor AC unit installation charges:- Installation of mid static ductable ceiling mounted type including labor for internal wiring, controls with cordless remote. (Single phase/three phase) | Each | 4000.00 |
| | 46.33.9 | Installation of high static ductable ceiling mounted type | VRF/VRV indoor AC unit installation charges:- Installation of high static ductable ceiling mounted type including labor for internal wiring, controls with cordless remote. (Single phase/three phase) | Each | 8500.00 |
| | 46.33.10 | Installation of floor standing package type unit | VRF/VRV indoor AC unit installation charges: - Installation of floor standing package type unit including labor for internal wiring, controls with wireless remote controller.(Single phase/three phase) | Each | 16000.00 |
| | 46.33.11 | Installation of VRF/VRV/ MRV outdoor unit | VRF/VRV outdoor AC unit installation charges: - Installation of VRF/VRV/ MRV outdoor unit is including labor for internal wiring, controls complete as per specification. (single phase/three phase) | Per HP | 1400.00 |
| **46.34** | | **Refrigerant** | Refrigerant:- Supply and additional refrigerant charge of refrigerant R410/410A & equivalent Eco friendly green gas to compensate the additional piping done at site for all systems including testing commissioning as required, | Kg | 1437.00 |
| **46.35** | | **Y joints** | Y joints:- Supply, installation, testing & commissioning of ref nut/Y-joints brass for indoor and outdoor units of suitable size as per requirement, | Each | 5500.00 |
| **46.36** | | **Ducting** | Ducting:- Supply and fixing of site fabricated GI sheet metal ducts in accordance with the approved shop drawings and as required as per specifications. GI sheet will be as per IS-277. All necessary GI /MS supports using thread rods (as directed). Angle, flat, and anchor bolts, food grade neoprene gaskets, food grade adhesive included in the scope. | | |
| | 46.36.1 | 24G | 24G | Sq.mtr | 1291.00 |
| | 46.36.2 | 22G | 22G | Sq.mtr | 1473.00 |
| | 46.36.3 | 20G | 20G | Sq.mtr | 1603.00 |
| | 46.36.4 | 18G | 18G | Sq.mtr | 1742.00 |

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| **46.37** | | **Thermal insulation** | Thermal insulation :- Supplying and fixing of following thickness of mat finish closed cell nitrile rubber (class "1") insulation on existing duct after applying two coats of cold setting adhesive complete as per specifications and as required. | | |
| | 46.37.1 | 25 mm thick | 25 mm thick | Sq.mtr | 1148.00 |
| | 46.37.2 | 19 mm thick | 19 mm thick | Sq.mtr | 960.00 |
| | 46.37.3 | 12/13 mm thick | 12/13 mm thick | Sq.mtr | 691.00 |
| **46.38** | | **Internal duct acoustic insulation** | Internal duct acoustic insulation: - Supply, installation, testing and commissioning of internal duct acoustic insulation with open cell structured class "O" fire retardant nitrile rubber with density 140 - 180 kg/m³ and specs as per RISAC chart specification sheet inside of the AHU supply air ducting up to 4 mtr. length from AHU mouth as per the specifications, it should have antimicrobial properties. | | |
| | 46.38.1 | 6 mm thick | 6 mm thick | Sq mtr | 893.00 |
| | 46.38.2 | 10 mm thick | 10 mm thick | Sq mtr | 989.00 |
| **46.39** | | | **Grills and diffusers** | | |
| | 46.39.1 | Al. air grills | Supplying & fixing of powder coated extruded aluminum air grills without volume control damper complete as required. | Sq.mtr | 9565.00 |
| | 46.39.2 | Coller damper | Supplying & fixing of coller damper complete as required. | Sq.mtr | 10483.00 |
| | 46.39.3 | Al. square diffuser | Supplying & fixing of powder coated extruded aluminum square diffuser without volume control damper complete as required. | Sq.mtr | 11910.00 |
| | 46.39.4 | Al. square diffuser | Supplying & fixing of powder coated extruded aluminum square diffuser with volume control damper complete as required. | Sq.mtr | 14067.00 |
| | 46.39.5 | G.I duct | Supplying & fixing of G.I duct volume control damper | Sq.mtr | 10100.00 |
| **46.40** | | **Flexible connections** | Flexible connections :- Providing & fixing of flexible canvas connection between indoor unit and duct to avoid the vibration transfer | Each | 1794.00 |
| **46.41** | | **Control cable** | Controlcable: - Supply and laying, testing & commissioning of following sizes of copper conductor control cables on existing tray including fixing clamps & proper connections to equipment etc. complete as required. | | |
| | 46.41.1 | 2Cx0.75 sq.mm | 2C x 0.75 sq.mm | Rmt | 64.00 |
| | 46.41.2 | 4Cx1.5 sq.mm | 4Cx1.5 sq.mm | Rmt | 140.00 |
| | 46.41.3 | 4Cx2.5 sq.mm | 4Cx 2.5 sq.mm | Rmt | 214.00 |
| | 46.41.4 | 4Cx4 sq.mm | 4Cx 4 sq.mm | Rmt | 350.00 |
| | 46.41.5 | 4Cx6 sq.mm | 4Cx 6 sq.mm | Rmt | 473.00 |
| | 46.41.6 | 4Cx10 sq.mm | 4Cx 10 sq.mm | Rmt | 899.00 |
| **46.42** | | **Communication cable** | Controlcable: - Supply and laying, testing & commissioning of following sizes of shielded copper conductor Communication cables on existing tray including fixing clamps & proper connections to equipment etc. complete as required. | | |
| | 46.42.1 | 2Cx1.5 sq.mm | 2Cx1.5 sq.mm | Rmt | 255.00 |

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| **46.43** | | **Outdoor stand fabrication** | Outdoor stand fabrication :- Supply, fabrication and erection of angle/channel/flat iron stand, hanging arrangement for air conditioning machines & ducts etc, back /support clamps or other similar work etc. including nut bolts of required size, making holes, fabrication, welding, cutting, etc. and painting with one coat of red oxide paint & two coat of enamel paint as required as per specification. | Kg | 85.00 |
| **46.44** | | **Fresh air in line fan** | Fresh air in line fan: - Supply, installation, testing and commissioning of fresh air inline fan. | | |
| | 46.44.1 | 100 CFM | 100 CFM (static pressure upto 20 mm) | Each | 5473.00 |
| | 46.44.2 | 150 CFM | 150 CFM (static pressure upto 20 mm) | Each | 5473.00 |
| | 46.44.3 | 200 CFM | 200 CFM (static pressure upto 20 mm) | Each | 6478.00 |
| | 46.44.4 | 250 CFM | 250 CFM (static pressure upto 20 mm) | Each | 6478.00 |
| | 46.44.5 | 300 CFM | 300 CFM (static pressure upto 20 mm) | Each | 9492.00 |
| | 46.44.6 | 500 CFM | 500 CFM (static pressure upto 20 mm) | Each | 12507.00 |
| | 46.44.7 | 700 CFM | 700 CFM (static pressure upto 20 mm) | Each | 15415.00 |
| | 46.44.8 | 900 CFM | 900 CFM (static pressure upto 20 mm) | Each | 18482.00 |
| **46.45** | | **Installation fresh air in line fan** | Installation fresh air in line fan: - Installation of fresh air inline fan including internal wiring, controls complete as per specification as mentioned below :- | | |
| | 46.45.1 | Upto 300 CFM | Upto 300 CFM | Each | 1500.00 |
| | 46.45.2 | Above 300 CFM | Above 300 CFM to 900 CFM | Each | 2500.00 |
| **46.46** | | **External duct insulation** | External duct insulation :- Supplying & fixing of following thickness of mat finish closed cell nitrile rubber (class "O") insulation along with company fitted weather proof UV protected glass wool cloth for protection of weather conditions and mechanical damages to insulation on existing duct after applying two coats of cold setting adhesive complete as per the specifications as required. | | |
| | 46.46.1 | 19 mm thick | 19 mm thick | Sq mtr | 2243.00 |
| | 46.46.2 | 12/13 mm thick | 12/13 mm thick | Sq mtr | 1973.00 |
| **46.47** | | **Fire damper** | Fire damper: - Supply, installation, testing and commissioning of fire dampers in supply air duct/return air duct/main branch of required size as and where required i/c control wiring. The motorized damper shall open/close on power failure and restore. The system shall also function on fire signals accordingly complete as required as per specification. | | |
| | 46.47.1 | Fire damper | Fire damper | Sq mtr | 7184.00 |
| | 46.47.2 | Actuators | Actuators | Each | 6536.00 |
| **46.48** | | **AHU kit** | AHU kit: - Supply installation testing and commissioning of AHU kit to integrate with AHUs to various combination of static and CFM requirement shall be connected to IDUs of VRF complete as required as per specification. | | |
| | 46.48.1 | 5 TR to 15TR | 5 TR to 15TR | Each | 42515.00 |

# CHAPTER - 47
# UNINTERRUPTED POWER SUPPLY (UPS)

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **47.1** | | **UPS (1 phase input & 1 phase output)** | Supply, installation, testing and commissioning of UPS with isolation transformer & battery for backup, minimum load power factor 0.8,power factor 0.99, having OVCD (over voltage cut-off device), having CVCF (constant voltage constant frequency, bypass parameters configurable, 2 stage charging(constant current/float charge), having temperature compensated charger, Fan speed control, digital signal processor controller, total harmonic distortion (current)-THDi (<10%@50% R Load, Input 1 phase voltage 230v, voltage range 110-300v, Output voltage (220AC/230AC/240AC ± 1%), Total harmonic distortion Voltage (THDv- <3% linear load, <5% nonlinear load, 90 % AC/AC efficiency, Over load capacity(105-110%: 3min, 111-130%: 30Sec), operating temperature with full load 0-40°C, communication features (RS232, USB com port, Intelligent slot(optional). UPS installation with all required material arrangements as required as per IS specification. | | |
| | **47.1.1** | **1 KVA 1Ph** | **1 KVA (1 phase input & 1 phase output)** | | |
| | 47.1.1.1 | 30 min backup | For 30 minute backup, min. 800 VAH | Each | 38416.00 |
| | 47.1.1.2 | 60 min backup | For 60 minute backup, min. 1600 VAH | Each | 49618.00 |
| | 47.1.1.3 | 120 min backup | For 120 minute backup, min. 3200 VAH | Each | 66978.00 |
| | **47.1.2** | **2 KVA 1Ph** | **2 KVA (1 phase input & 1 phase output)** | | |
| | 47.1.2.1 | 30 min backup | For 30 minute backup, min. 1600 VAH | Each | 73606.00 |
| | 47.1.2.2 | 60 min backup | For 60 minute backup, min. 3200 VAH | Each | 87271.00 |
| | 47.1.2.3 | 120 min backup | For 120 minute backup, min. 6400 VAH | Each | 117652.00 |
| | **47.1.3** | **3 KVA 1Ph** | **3 KVA (1 phase input & 1 phase output)** | | |
| | 47.1.3.1 | 30 min backup | For 30 minute backup, min. 2400 VAH | Each | 76832.00 |
| | 47.1.3.2 | 60 min backup | For 60 minute backup, min. 4800 VAH | Each | 102403.00 |
| | 47.1.3.3 | 120 min backup | For 120 minute backup, min. 9600 VAH | Each | 144983.00 |
| | **47.1.4** | **5 KVA 1Ph** | **5 KVA (1 phase input & 1 phase output)** | | |
| | 47.1.4.1 | 30 min backup | For 30 minute backup, min. 4000 VAH | Each | 138062.00 |
| | 47.1.4.2 | 60 min backup | For 60 minute backup, min. 8000 VAH | Each | 165628.00 |
| | 47.1.4.3 | 120 min backup | For 120 minute backup, min. 16000 VAH | Each | 269555.00 |
| **47.2** | | **UPS (3 phase input & 1 phase output)** | Supply, installation, testing and commissioning of UPS with isolation transformer & battery for backup, having three phase input with ground (R-Y-B-N-G), | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | Voltage range 190v AC-478v AC (based on load %), line low transformer (0-50% :190v/51-100% : 305v ± 3% VAC), line low comeback +25 - 35 VAC ( ± 3%), line high transfer 478 VAC (± 3%), line high comeback 461 VAC (± 3%), frequency range 46 - 54 Hz, power factor 0.99, Total harmonic distortion (current)-THDi (<5% with full load, having output voltage 220 VAC/230 VAC/240 VAC, voltage regulation ± 1%, frequency 50 Hz ± 0.1 Hz, Synchronization range 46 -54 Hz, voltage distortion ≤ 2 % (linear load) & ≤ 5 % (nonlinear load), having pure sine wave output waveform, crest factor 3 : 1, efficiency (AC -AC) 94%, power factor 0.9, having inbuilt isolation transformer, over load 101 - 109% : 5 Min, 110 - 129% : 1 Min, 130 - 150% : 10 sec, having SMF/VRLA battery type, transfer time line mode to battery mode 0 ms, inverter to bypass, ECO mode 0 ms, bypass to inverter 0 ms, ECO to inverter mode < 10 ms, when bypass voltage of frequency is out of range, having features Eco mode, Fan speed control, EPO function, frequency converter mode & parallel function (optional) up to 4 units. Operating temperature 0 - 45°C, noise level < 55 db @ 1 meter having display & indication features i.e. user friendly dot matrix LCD display, input : voltage & frequency, battery : voltage & level in %, internal DC bus voltage z frequency, current, power& load level in %), UPS status (operating mode/warning/fault/codes (fault & warnings), UPS settings (output voltage/ frequency/ bypass enable/disable/special function., UPS/CVCF/ECO.<br>LED status (normal mode/load on battery/load on bypass/system fault), Audible alarm (main failure alarm, low battery alarm, UPS warning, overload, fault & bypass mode etc.), protections (advance electronic protection for device, safety backed with MCBs, fast acting fuses, high speed pulse blanking, electronic overvoltage/under voltage), connections (terminal block for input/bypass/battery & output), static bypass, communication interface RS 232/USB port for software interface, intelligent slot for SNMP (optional). UPS installation with all required material arrangements as required as per IS specification. | | |
| | **47.2.1** | **7.5 KVA 3Ph** | **7.5 KVA (3 phase input & 1 phase output)** | | |
| | 47.2.1.1 | 30 min backup | For 30 minute backup, min. 6000 VAH | Each | 203164.00 |
| | 47.2.1.2 | 60 min backup | For 60 minute backup, min. 12000 VAH | Each | 252195.00 |
| | 47.2.1.3 | 120 min backup | For 120 minute backup, min. 24000 VAH | Each | 365976.00 |
| | **47.2.2** | **10 KVA 3Ph** | **10 KVA (3 phase input & 1 phase output)** | | |
| | 47.2.2.1 | 30 min backup | For 30 minute backup, min. 8000 VAH | Each | 222284.00 |
| | 47.2.2.2 | 60 min backup | For 60 minute backup, min. 16000 VAH | Each | 326094.00 |
| | 47.2.2.3 | 120 min backup | For 120 minute backup, min. 32000 VAH | Each | 487968.00 |
| | **47.2.3** | **15 KVA 3Ph** | **15 KVA (3 phase input & 1 phase output)** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 47.2.3.1 | 30 min backup | For 30 minute backup, min. 12000 VAH | Each | 367149.00 |
| | 47.2.3.2 | 60 min backup | For 60 minute backup, min. 24000 VAH | Each | 525504.00 |
| | 47.2.3.3 | 120 min backup | For 120 minute backup, min. 48000 VAH | Each | 787083.00 |
| | **47.2.4** | **20 KVA 3Ph** | **20 KVA (3 phase input & 1 phase output)** | | |
| | 47.2.4.1 | 30 min backup | For 30 minute backup, min. 16000 VAH | Each | 470373.00 |
| | 47.2.4.2 | 60 min backup | For 60 minute backup, min. 32000 VAH | Each | 602922.00 |
| | 47.2.4.3 | 120 min backup | For 120 minute backup, min. 64000 VAH | Each | 909075.00 |
| **47.3** | | **UPS (3 phase input & 3 phase output)** | Supply, installation, testing and commissioning of UPS with isolation transformer & battery for backup, having three phases input three phase output. Online double conversion VFI-SS-111 technology, with sinusoidal wave form, architecture (standalone of distributed parallel up to 8 units), having input characteristics i.e. input voltage (380v, 400v, 415v, 3Ph+N+PE), input frequency (45 -65 Hz), input voltage range (Ph - Ph) : (half load 208v - 467v/full load 312v - 467v), total harmonic distortion of input current < 3% (at full linear load), compatibility with diesel generators configurable for synchronization between the input and output frequencies even for high frequency variations, input power factor > 0.99, having output characteristics i.e. output voltage (380v, 400v, 415v, 3Ph+N (adjustable from front panel), efficiency up to 96%, efficiency in ECO mode up to 98.5%, output frequency (nominal) 50 /60 Hz 0.01% free run (adjustable from front panel), crest factor 3 : 1, total harmonic distortion of output voltage < 2 % (at full linear load), output power factor 0.9, output voltage tolerance ± 1%, built in automatic and maintenance bypass, inbuilt isolation transformer, battery type (VRLA - AGM maintenance free). Communication & management i.e. LCD display (touch screen, LED bar status, live synoptic view for real time), communication ports (RS 232, genset, programmable 4 relay contacts, modbus), Audible alarm (acoustic alarms and warnings), Net interface slot (SNMP card (optional)), having emergency power off (EPO) & remote management. Ambient conditions i.e. operating temperature °C (0°C - 40°C), relative humidity % (20% - 95% not condensing), protection index (IP20), noise at 1 m dBA < 55. UPS installation with all required material arrangements as required as per IS specification. | | |
| | **47.3.1** | **10 KVA 3Ph** | **10 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.1.2 | 60 min backup | For 60 minute backup, min. 16000 VAH | Each | 442808.00 |
| | **47.3.2** | **15 KVA 3Ph** | **15 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.2.1 | 30 min backup | For 30 minute backup, min. 12000 VAH | Each | 487968.00 |
| | 47.3.2.2 | 60 min backup | For 60 minute backup, min. 24000 VAH | Each | 557175.00 |
| | **47.3.3** | **20 KVA 3Ph** | **20 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.3.1 | 30 min backup | For 30 minute backup, min. 16000 VAH | Each | 525504.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 47.3.3.2 | 60 min backup | For 60 minute backup, min. 32000 VAH | Each | 702627.00 |
| | **47.3.4** | **30 KVA 3Ph** | **30 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.4.1 | 30 min backup | For 30 minute backup, min. 24000 VAH | Each | 609960.00 |
| | 47.3.4.2 | 60 min backup | For 60 minute backup, min. 48000 VAH | Each | 826965.00 |
| | **47.3.5** | **40 KVA 3Ph** | **40 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.5.1 | 30 min backup | For 30 minute backup, min. 30000 VAH | Each | 941919.00 |
| | 47.3.5.2 | 60 min backup | For 60 minute backup, min. 60000 VAH | Each | 1309068.00 |
| | **47.3.6** | **60 KVA 3Ph** | **60 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.6.1 | 15 min backup | For 15 minute backup, min. 24000 VAH | Each | 1112004.00 |
| | 47.3.6.2 | 30 min backup | For 30 minute backup, min. 48000 VAH | Each | 1296165.00 |
| | **47.3.7** | **80 KVA 3Ph** | **80 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.7.1 | 15 min backup | For 15 minute backup, min. 30000 VAH | Each | 1290300.00 |
| | 47.3.7.2 | 30 min backup | For 30 minute backup, min. 60000 VAH | Each | 1649238.00 |
| | **47.3.8** | **100 KVA 3Ph** | **100 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.8.1 | 15 min backup | For 15 minute backup, min. 40000 VAH | Each | 1847475.00 |
| | 47.3.8.2 | 30 min backup | For 30 minute backup, min. 80000 VAH | Each | 2322540.00 |
| | **47.3.9** | **120 KVA 3Ph** | **120 KVA (3 phase input & 3 phase output)** | | |
| | 47.3.9.1 | 15 min backup | For 15 minute backup, min. 48000 VAH | Each | 2334270.00 |
| | 47.3.9.2 | 30 min backup | For 30 minute backup, min. 96000 VAH | Each | 2873850.00 |
| **47.4** | | Inverter | Supply, installation, testing and commissioning of Pure sine wave Inverter , Display Indications (Status, Fault conditions and Battery Charging) ,Standard Voltage range from 100 V~ 300 V , High Peak Current Handling Capability & High Voltage Surge Protection excluding battery and wiring for connections. | | |
| | 47.4.1 | 700-850 VA | 700-850 VA | Each | 6680.00 |
| | 47.4.2 | 1-1.05 KVA | 1-1.05 KVA | Each | 7791.00 |
| | 47.4.3 | 1.2-1.5 KVA | 1.2-1.5 KVA | Each | 11660.00 |
| | 47.4.4 | 2-2.5 KVA | 2-2.5 KVA | Each | 15778.00 |
| | 47.4.5 | 3-3.5 KVA | 3-3.5 KVA | Each | 25105.00 |
| **47.5** | | Battery | Supply, installation, testing and commissioning of 12/24 V Battery system in series/parallel combinations or as single as per requirement including interconnetion wiring, lugs and other required material as required with maunting stand/trolley, warranty period: Minimum 3 year | | |
| | 47.5.1 | 110/120 AH, Tall Tubular | 110/120 AH, Tall Tubular | Each | 14691.00 |
| | 47.5.2 | 150/160 AH, Tall Tubular | 150/160 AH, Tall Tubular | Each | 16825.00 |
| | 47.5.3 | 180/200 AH, Tall Tubular | 180/200 AH, Tall Tubular | Each | 20125.00 |
| | 47.5.4 | 26 AH, Maintenance free | 26 AH, Maintenance free | Each | 4377.00 |
| | 47.5.5 | 65 AH, Maintenance free | 65 AH, Maintenance free | Each | 9085.00 |

# CHAPTER 48
# DIESEL GENERATING SET

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **48.1** | | **DG set, Water cooled** | Supply, installation, testing and commissioning of 'Silent Type' diesel generating set comprising with diesel engine water cooled, Synchronous alternator, AMF control panel single/three phase, acoustic enclosure, battery with charger and related accessories including incoming cable 10 mtr, exhaust pipe 'B' class with insulation as per CPWD specification and CPCB - IV Norms. Note:-<br>1. Two No Cu plate (600mmx600mmx3mm) earthing and Two No GI plate (600mmx600mmx6mm) earthing with GI/Cu Strip to be done for each installation as per specification and will be separtely paid. 2. Suitable size of PCC/RCC plateform to be made seprately as per requirement and instruction of Enguineer-in-charge. | | |
| | 48.1.1 | 1 Phase, 7.5 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set single phase, water cooled with AMF control panel, **7.5 KVA, 230V.** | Each | 546726.00 |
| | 48.1.2 | 3 Phase, 7.5 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel**, 7.5 KVA, 415V**. | Each | 545376.00 |
| | 48.1.3 | 3 Phase, 15 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **15 KVA, 415V**. | Each | 638767.00 |
| | 48.1.4 | 3 Phase, 20 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **20 KVA, 415V.** | Each | 713129.00 |
| | 48.1.5 | 3 Phase, 25 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **25 KVA, 415V.** | Each | 883438.00 |
| | 48.1.6 | 3 Phase, 30 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **30 KVA, 415V.** | Each | 974177.00 |
| | 48.1.7 | 3 Phase, 40 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **40 KVA, 415V**. | Each | 1088696.00 |
| | 48.1.8 | 3 Phase, 50 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **50 KVA, 415V.** | Each | 1204061.00 |
| | 48.1.9 | 3 Phase, 62.5 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **58.5/62.5 KVA, 415V**. | Each | 1214206.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 48.1.10 | 3 Phase, 75 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **75 KVA, 415V**. | Each | 1486101.00 |
| | 48.1.11 | 3 Phase, 82.5 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **82.5 KVA, 415V**. | Each | 1753503.00 |
| | 48.1.12 | 3 Phase, 100 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **100 KVA, 415V**. With electronic governor. | Each | 2032070.00 |
| | 48.1.13 | 3 Phase, 125 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **125 KVA, 415V**. With electronic governor. | Each | 2158580.00 |
| | 48.1.14 | 3 Phase, 150/160 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **150/160 KVA, 415V**.With electronic governor. | Each | 2746784.00 |
| | 48.1.15 | 3 Phase, 180 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **180 KVA, 415V.** With electronic governor. | Each | 3131615.00 |
| | 48.1.16 | 3 Phase, 200 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **200 KVA, 415V**. With electronic governor. | Each | 3434792.00 |
| | 48.1.17 | 3 Phase, 250 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **250 KVA, 415V.** With electronic governor. | Each | 3983017.00 |
| | 48.1.18 | 3 Phase, 300/320 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel**, 300/320 KVA, 415V**. With electronic governor. | Each | 5209096.00 |
| | 48.1.19 | 3 Phase, 400 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **400 KVA, 415V**. With electronic governor. | Each | 6668331.00 |
| | 48.1.20 | 3 Phase, 500 KVA, Water cooled | Supply, installation, testing and commissioning of diesel generating set three phase, water cooled with AMF control panel, **500 KVA, 415V**. With electronic governor. | Each | 8109464.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **48.2** | | Diesel Generator Set, Air cooled | Supply, installation, testing and commissioning of 'Silent Type' diesel generating set comprising with diesel engine **air cooled,** Synchronous alternator, AMF control panel single/three phase, acoustic enclosure, battery with charger and related accessories including incoming cable 10 mtr, exhaust pipe 'B' class with insulation as per CPWD specification and CPCB - IV Norms<br>Note:-<br>1. Two No Cu plate (600mmx600mmx3mm) earthing and Two No GI plate (600mmx600mmx6mm) earthing with GI/Cu Strip to be done for each installation as per specification and will be separtely paid.<br>2. Suitable size of PCC/RCC plateform to be made seprately as per requirement and instruction of Enguineer-in-charge. | | |
| | 48.2.1 | 1 Phase, 7.5 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set single phase, air cooled with AMF control panel**, 7.5 KVA, 230V**. | Each | 406549.00 |
| | 48.2.2 | 3 Phase, 7.5 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **7.5 KVA, 415V.** | Each | 406549.00 |
| | 48.2.3 | 3 Phase, 15 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **15 KVA, 415V**. | Each | 582728.00 |
| | 48.2.4 | 3 Phase, 25 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **25 KVA, 415V**. | Each | 739753.00 |
| | 48.2.5 | 3 Phase, 30 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **30 KVA, 415V.** | Each | 808644.00 |
| | 48.2.6 | 3 Phase, 40 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **40 KVA, 415V**. | Each | 917656.00 |
| | 48.2.7 | 3 Phase, 50 KVA, Air cooled | Supply, installation, testing and commissioning of diesel generating set three phase, air cooled with AMF control panel, **50 KVA, 415V.** | Each | 1074226.00 |
| | 48.2.8 | 3 Phase, 62.5 KVA, Air cooled | Supply, installation, testing and commissioning of Diesel Generating Set Three Phase,air Cooled with AMF Control Panel, 62.5 KVA, 415V . | Each | 1086610.00 |

# CHAPTER - 49

# FIRE DETECTION AND ALARM SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **49.1** | | **2 Loop microprocessor** | Supply, installation, testing and commissioning of 2 loop microprocessor based intelligent and electronically addressable, modular, loop capacity of 159 detectors. Fire alarm control panel with 5.7 inch touch screen LCD display, multiple access levels, 10,000 event history logs in the nonvolatile memory (EEPROM). The panel should be modular microprocessor based in nature. The panel shall have 240V AC power supply, automatic battery charger, 24 V, ceiled lead acid maintenance free batteries sufficient for 24 hours normal working and then be capable of operating the system for 30 mints during emergency condition. The panel shall be EN 54/UL certification. | Each | 188534.00 |
| **49.2** | | **4 Loop microprocessor** | Supply Installation, testing & commissioning of 4 loop microprocessor based intelligent and electronically addressable, modular, with networking card & printer port- loops panel (expandable) with each loop capacity of 159 detectors. Fire alarm control panel with 5.7 inch touch screen LCD display, multiple access levels 10,000 event history logs in the nonvolatile memory (EEPROM). The panel should be modular microprocessor based in nature and should be expandable from single loop to up to 32 loops. The panel shall have 240V AC power supply, automatic battery charger, 24 V, ceiled lead acid maintenance free batteries sufficient for 24 hours normal working and then be capable of operating the system for 30 minutes during emergency condition. Provision for integration to BMS back net software. The panel shall be EN 54/UL certification. | Each | 489527.00 |
| **49.3** | | **Loop card** | Supply installation, testing and commissioning of loop card LSN 0300 A LSN module. | Each | 47162.00 |
| **49.4** | | **Active and networkable repeater panel** | Supply, installation, testing and commissioning of active and networkable repeater panel with LCD display complete system function keys for programming and control like system reset, alarm acknowledgement, alarm silence, trouble acknowledgement UL/EN54 certified. | Each | 140353.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **49.5** | | **Addressable–optical detector** | Supply, installation, testing and commissioning of addressable–optical detector with inbuilt isolator, automatic addressing shall be UL/EN54 certified. | Each | 4929.00 |
| **49.6** | | **Addressable–optical thermal detector** | Supply, installation, testing and commissioning of addressable–optical thermal detector with inbuilt isolator, automatic addressing shall be UL/EN54 certified | Each | 5654.00 |
| **49.7** | | **Addressable– automatic dual LED smoke detector** | Supply, installation, testing and commissioning of addressable– automatic dual LED smoke detector with two integrated light-scattering smoke sensors using LEDs with different colors/wavelengths (blue/infrared) to avoid false alarm & should comply to test fire 1, Inbuilt Isolator, automatic addressing shall be UL/EN54 certified | Each | 6300.00 |
| **49.8** | | **Addressable– automatic dual LED multi detector (Thermal Sensor)** | Supply, installation, testing and commissioning of addressable– automatic dual LED multi detector with two integrated light-scattering smoke sensors using LEDs with different colors/wavelengths (blue/infrared) to avoid false alarm & with thermal sensor, inbuilt isolator, automatic addressing shall be UL/EN54 certified | Each | 8485.00 |
| **49.9** | | **Addressable– automatic dual LED multi detector (Thermal & Chemical sensor)** | Supply, installation, testing and commissioning of addressable– automatic dual LED multi detector with two integrated light-scattering smoke sensors using LEDs with different colors/wavelengths (blue/infrared) to avoid false alarm & with thermal and chemical sensor, inbuilt isolator, automatic addressing shall be UL/EN54 certified | Each | 13670.00 |
| **49.10** | | **Indoor addressable manual break glass unit** | Supply, installation, testing and commissioning of indoor addressable manual break glass unit (double action) with inbuilt isolators, with flexible network structures & necessary fixing arrangements with key complete as required shall be UL/EN54 certified | Each | 5224.00 |
| **49.11** | | **Outdoor addressable manual break glass unit** | Supply, installation, testing and commissioning of outdoor addressable manual break glass unit (double action) with inbuilt isolators, with flexible network structures & necessary fixing arrangements with key complete as required, shall be UL/EN54 certified | Each | 10703.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **49.12** | | **Indoor standalone loop powered detector base sounder** | Supply, installation, testing and commissioning of Indoor Standalone Loop Powered detector base sounder with inbuilt isolators & with 32 different tone variants selection options & adjustable sound pressure by 5 levels, the sound pressure 92.1 dB(A), shall be UL/EN54 certified | Each | 8749.00 |
| **49.13** | | **Outdoor standalone loop powered detector base** | Supply, installation, testing and commissioning of outdoor standalone loop powered detector base sounder with inbuilt isolators & with 32 different tone variants selection options & adjustable sound pressure by 5 levels, the sound pressure 92.1 dB(A), shall be UL/EN54 certified | Each | 11577.00 |
| **49.14** | | **Strobe for base** | Supply, installation, testing and commissioning of strobe for base sounder red with inbuilt isolators, shall be UL/EN54 certified | Each | 7807.00 |
| **49.15** | | **Beam detector** | Supply, installation, testing and commissioning of beam detector including mounting base along with interface module to direct connect to the loop, having a coverage area with a distance of min. 10 m and max. 100 m with extra width of up to 14 m ,operating temperature range: -20 C to +55 C, response sensitivity of 40% UL/EN54 certified | Each | 99921.00 |
| **49.16** | | **Addressable control relay module** | Supply, installation, testing and commissioning of addressable control relay module with inbuilt isolators & with flexible network structures with a maximum switching current of 1A @ 30V DC, shall be UL/EN54 certified | Each | 18038.00 |
| **49.17** | | **Addressable output module** | Supply, installation, testing and commissioning of addressable output module with inbuilt isolators, shall be UL/EN54 certified | Each | 20351.00 |
| **49.18** | | **Interface module** | Supply, installation, testing and commissioning of 1output+1input interface module with inbuilt isolator UL/EN54certified | Each | 19114.00 |
| **49.19** | | **Response indicator** | Supply, installation, testing and commissioning of response Indicator shall be UL/EN54certified | Each | 267.00 |

# CHAPTER - 50
# CONVENTIONAL FIRE ALARM SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **50.1** | | **Fire panel** | Fire panel : Providing, fixing, testing & commissioning microprocessor based main fire alarm panel and indicating panel solid state modular card type pulsar, timer for dual stage alarm facility with indicators, standby battery ,complies to IS 2189:1999.2 x 16 character backlit LCD display lamp test features. Complete with integral power supply and battery charger for SMF battery. NO/NC contacts for both fire and fault. Switches to enable on board operation. LCD and LED Indication for open, short circuit, Isolate, fire in each zone. reset, lamp test, silence, isolate, short, open, fire continuous buzzer for fault and intermittent buzzer for fire battery charger box, operating voltage 220 Volt mains A.C. input/24V D.C. output etc. complete in all respect. | | |
| | 50.1.1 | 2 zone | 2 zone fire panel | Each | 13665.00 |
| | 50.1.2 | 4 zone | 4 zone fire panel | Each | 22991.00 |
| | 50.1.3 | 8 zone | 8 zone fire panel | Each | 44123.00 |
| | 50.1.4 | 12 zone | 12 zone fire panel | Each | 53164.00 |
| | 50.1.5 | 16 zone | 16 zone fire panel | Each | 65534.00 |
| | 50.1.6 | 24 zone | 24 zone fire panel | Each | 87910.00 |
| **50.2** | | **Smoke detector (ionization type)** | Smoke detector (ionization type) : Providing &fixing of ionization type smoke detectors as per IS: 11360-1985 of dual chamber, 12/24 volt DC, ionization source-less than one micro curie maximum Americium 241, twin visual alarm indicator (LED's) "Blink" in stand by and "steady" in alarm complete in all respects. | Each | 3726.00 |
| **50.3** | | **Smoke detector (optical type)** | Smoke detector (optical type) : Providing &fixing of optical type smoke detector as per IS:11360-1985 of photo-optic sensing chamber, 12/24 volt DC, ionization source-less than one micro curie maximum Americium 241, twin visual alarm indicator (LED's) "Blink" in stand by and "steady" in alarm complete in all respects. | Each | 3675.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **50.4** | | **Heat detector** | Heat detector : providing & fixing of heat detector based on heat rate-of- rise/fixed temperature with working voltage 9-33V DC twin visual alarm indicator (LED's) "Blink" in stand by and "steady" in alarm complete in all respects. | Each | 2962.00 |
| **50.5** | | **Beam detector** | Beam detector: Providing & fixing of beam detector including mounting base along with Interface module to direct connect to the loop, having a coverage area with a distance of min. 10 m and max. 100 m with extra width of up to 14 m, operating temperature range: -20 C to +55 C, response sensitivity of 40% | Each | 86612.00 |
| **50.6** | | **Response indicator** | R.I. : Providing & fixing of response indicator | Each | 228.00 |
| **50.7** | | **Manual call point** | Manual call point: Providing & fixing of manual calls point flush or surface mounting. Front protection glass cover. Unbreakable glass ABS plastic body. | Each | 716.00 |
| **50.8** | | **Internal hooter** | Internal hooter: Providing &fixing of internal hooter with working AC Voltage 220V A.C./24V D.C. current consumption, sound output 120 DB, material M.S. sheet. | Each | 794.00 |
| **50.9** | | **Industrial hooter** | Industrial hooter: Providing & fixing of industrial hooter speaker horn type ABS body with working AC Voltage 220V A.C./24V D.C. current consumption, sound output 120 DB, material M.S. sheet. | Each | 2008.00 |

# CHAPTER - 51

## AUTOMATIC FIRE DETECTION AND SUPPRESSION SYSTEM

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **51.1** | | **Automatic linear pneumatic tube detection** | Supply, fixing, testing and commissioning of automatic linear pneumatic tube detection based NOVEC 1230/System for electrical panels consisting of the following components: | | |
| | 51.1.1 | 2.5 LB capacity NOVEC 1230, DLP assembly | 2.5 LB capacity NOVEC 1230, DLP assembly with automatic valve, push in connector for tube, 2.5 LB NOVEC 1230 gas, mounting bracket, end of line adopter and low pressure switch for monitoring system activation. | No. | 96782.00 |
| | 51.1.2 | 5 LB capacity NOVEC 1230, DLP assembly | 5 LB capacity NOVEC 1230, DLP assembly with automatic valve, push in connector for tube, 5 LB NOVEC 1230 gas, mounting bracket, end of line adopter and low pressure switch for monitoring system activation. | No. | 147202.00 |
| | 51.1.3 | 10 LB capacity NOVEC 1230, DLP assembly | 10 LB capacity NOVEC 1230, DLP assembly with automatic valve, push in connector for tube, 10 LB NOVEC 1230 gas, mounting bracket, end of line adopter and low pressure switch for monitoring system activation. | No. | 237615.00 |
| | 51.1.4 | Heat detection tube | Linear pneumatic heat detection tube with all necessary fittings & supports. | Metre | 2606.00 |
| | 51.1.5 | Master control unit | Master control unit with audio visual alarm with wiring to make complete system operational. The control panel should have provision for integration with fire alarm/SCADA/BMS system. | No. | 7099.00 |

# CHAPTER - 52

# ENERGY SAVING AND PROTECTION SOLUTIONS FOR BUILDINGS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.1** | | **Light dependent relay (LDR)** | Light dependent relay (LDR)– Pole mounting for outdoor lights | | |
| | 52.1.1 | Pole mounting for outdoor lights | Supply, installation, testing and commissioning of 230 V AC operated LDR shall have an operating range of 96…253V AC with 1 NO, 16 A, AgSnO2 contact material with starting current of 120A and light dependent relay with a lux sensitivity adjustment of 1 to 80 lux shall operate on 230 V AC with an operating voltage range of 0.8 to 1.1 times that of nominal voltage. The contacts shall be rated for 16 Amp, cadmium free along with the cadmium free light sensor and shall operate at an ambient temperature of -30 deg C to + 70 deg C. The electrical service life of LDR shall be 1 lakh switching cycles at full load. The dielectric strength between open contacts should be 1000 V AC and conducted disturbance immunity i.e. surge (1.2/50μs) on L and N (differential mode) of 4 kV. The LDR shall have necessary approvals like CE, GOST and IMQ and shall be IP 54 protected and able to switch phase and neutral lines for LED lamp load of 400 Watt with light feedback compensation. | Each | 7139.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.2** | | **Light dependent relay (LDR)** | Light dependent relay (LDR) – Panel mounting for lighting control | | |
| | 52.2.1 | Panel mounting for lighting control | Supply, installation, testing and commissioning of 230 V AC operated LDRs shall have an operating range of 96…253V AC with 1 NO, 16 A, AgSnO2 contact material with starting current of 120A and light dependent relay with a lux sensitivity adjustment of 1 to 1000 lux shall operate on 230 V AC with an operating voltage range of 0.8 to 1.1 times that of nominal voltage. The changeover contacts shall be rated for 16 Amp, cadmium free along with the cadmium free light sensor and shall operate at an ambient temperature of -20 deg C to + 50 deg C. The electrical service life of LDR shall be 100X103 with dielectric strength between open contacts of 1000 V AC and conducted disturbance immunity i.e. surge (1.2/50μs) on L and N (differential mode) of 4 kV. The LDR shall have necessary approvals like CE, GOST and IMQ and shall be lighting sensor IP 54 and LDR IP 20 protected. able to switch phase lines for LED lamp Load of 400 Watt with light feedback compensation and zero hysteresis for energy saving. | Each | 8824.00 |
| **52.3** | | **Astro time switch** | Astro time switch – For lights on-off based on sunrise and sunset timing | | |
| | | | Supply, installation, testing and commissioning of astro-time switch based on astro program i.e. calculation of sunrise and sunset times through date, time and Indian PIN codes. It shall have an option of astro ON period override by time switch and Offset function which allows programming of switching times offset from the astronomic time | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.3.1 | For lights on-off based on sunrise and sunset timing | . The 230 V AC operated astro time switch shall have an operating range of 96…253V AC with 1 NO, 16 A, AgSnO2 contact material with starting current of 120A and 1 CO cadmium free contacts, it shall have LCD status indication with back light display along with internal easily replaceable battery on the front side for set-up and programming. The electrical life at rated load AC1 shall be 1 lakh cycles and shall work at an ambient temperature of -20 to +50 deg C. It shall have a dielectric strength between supply and open contacts of 4000V AC, between open contacts 1000V AC respectively and rated impulse voltage (between supply and contacts) of 6kV and rated impulse voltage (between open contacts) of 1.5kV. The mounting shall be on 35 mm DIN rail (EN 60715) and shall have necessary approvals like CE, IMQ and EAC. | Each | 9524.00 |
| **52.4** | | **PIR movement detector** | PIR movement detector–For corridors, common rooms, stair cases, store room, washroom | | |
| | 52.4.1 | For corridors, common rooms, stair cases, store room, washroom | Supply, installation, testing and commissioning of 230 V AC operated movement detector shall have an operating range of 96…253V AC with 1 NO, 10 A, AgSnO2 contact material with starting current of 100Aand shall have universal mounting position to permit the selection of any area of survey with IP 40 protection. It shall have an ambient temperature range of -30 to +50 deg C. The movement detector shall have a dielectric strength between open contacts of 1000V AC with an electrical life at rated load AC1 of 1 lakh cycles with necessary approvals like CE, EAC and IMQ | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.4.1.1 | PIR movement detector for stair case | PIR movement detector for stair case: - To be of dual mounted type (wall and surface mounted) with the angle of survey shall be 110 degree and depth of field shall be 10 metre, it shall have ambient light intervention threshold of 5…350lux. The movement detector shall have a sensing area of dia 8 metre X 5 metre when ceiling mounted at a height of 2.8 metre. With adjustable light ON time after last detection of 10 sec…12 min, | Each | 6685.00 |
| | 52.4.1.2 | PIR movement detector for washroom | PIR movement detector for washroom: - Ceiling and surface mounted angle of survey shall be 360 degree it shall have ambient light intervention threshold of 5…350 lux. The movement detector shall have a sensing area of dia 8X 5metreat a height of 2.8metre. With adjustable light ON time after last detection of 10 sec…12 min | Each | 6181.00 |
| | 52.4.1.3 | PIR movement detector for corridor | PIR movement detector for corridor:- Ceiling and surface angle of survey shall be 360 degree it shall have ambient light intervention threshold of 1…500 lux. The movement detector shall have a sensing area of 30 X 4 metre when ceiling mounted at a height of 3metre. With adjustable light ON time after last detection of 12 sec…35 min, | Each | 9793.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.5** | | **PIR movement and presence detector** | PIR movement and presence detector | | |
| | 52.5.1 | PIR Movement and Presence Detector | Supply, installation, testing and commissioning of 230 V AC operated movement detector shall have an operating range of 96…253V AC with 1 NO, 10 A, AgSnO2 contact material with starting current of 100Aand shall have universal mounting position to permit the selection of any area of survey with IP 40 protection. It shall have an ambient temperature range of -30 to +50 deg C. The movement detector shall have a dielectric strength between open contacts of 1000V AC with an electrical life at rated load AC1 of 1 lakh cycles with necessary approvals like CE and TUV | | |
| | 52.5.1.1 | PIR movement and presence detector | PIR movement and presence detector:- Office and meeting room it shall have ambient light intervention threshold of 1…500 lux and motion sensing area dia of 8 metre and presence sensing area dia of 4 metre when ceiling mounted at a height of 3metre. With adjustable light ON time after last detection of 12 sec…35 min. | Each | 10248.00 |
| | 52.5.1.2 | PIR movement and presence detector | PIR movement and presence detector:- Conference hall of high ceiling it shall have ambient light intervention threshold of 5…350 lux and motion sensing area dia of 12 X 8 metre when ceiling mounted at a height of upto 6 metre. With adjustable light ON time after last detection of 30 sec…35 min, | Each | 9524.00 |
| **52.6** | | **Modular contactor for the lighting feeder panel** | Modular contactor for the lighting feeder panel | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.6.1 | Modular contactor for the lighting feeder panel | Supply, installation, testing and commissioning of 25A rating four pole contactor, designed specific contact material AgSnO2 for lamp loads and AC load for high starting current. It shall work at an ambient temperature of -20 to +50 deg C and auto-On-Off selector version. It shall have 3 Kv short circuit protections. The mounting shall be on 35 mm DIN rail (EN 60715) | Each | 4181.00 |
| **52.7** | | | **Surge overvoltage protection device** | | |
| | 52.7.1 | Surge overvoltage protection device | Supply, installation, testing and commissioning of SOVP device should have single high capacity varistor in the L-N plug and encapsulated spark gap for N-PE. L-N SOVP should have a rated voltage of minimum 320VAC to handle Indian power supply conditions. SOVP should have visual indication on replaceable varistor and spark gap modules as well as a potential free contact on the socket for remote indication. SOVP device should be wired up as per TT configuration of wiring in accordance with EN 61643-11 standards. Both the L-N and N-PE, SPDs should be rated for handling 30KA of nominal 8/20 μs surge current. | | |
| | 52.7.1.1 | SOVP for high mast light | SOVP for high mast light- Type 1+2 pluggable SPD 3+1, rating L-N/N-PE Uc 275/ 255 V AC, limp 12.5/50 KA, Imax 60/100 KA, Up 1.2/1.5 KV | Each | 32428.00 |
| | 52.7.1.2 | SOVP for pole light sub feeder panels | SOVP for pole light sub feeder panels- Type 2 pluggable SPD 3+1,rating L-N/N-PE Uc 255/320 V ln30/60KA KA, Imax 30Ka/60 KA, Up 1.5KV | Each | 12897.00 |
| | 52.7.1.3 | SOVP for pole light | SOVP for pole light - Type 2 pluggable SPD 1+1, rating L-N/N-PE Uc 230/400 V ln 5 KA, Imax 10 KA, Up 1.2KV. | Each | 4534.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.8** | | **Three phase voltage monitoring relay** | Three phase voltage monitoring relay:– For protection of sensitive equipment against under/over voltages which are caused due to voltage fluctuations, neutral loss, phase reversals at the end of the utility transformers etc | | |
| | 52.8.1 | Electronic voltage monitoring relays | Supply, installation, testing and commissioning of electronic voltage monitoring relays for three phase applications shall be multifunctional type providing the flexibility of monitoring under voltage, over voltage, voltage fault memory, window mode, phase rotation and phase loss. The relay shall be based on positively safe logic. All the functions and values shall be adjusted by selector and trimmer on front side. With a supply voltage of 220 V AC-510 V AC and 6 Amp 1 CO cadmium free contact with a dielectric strength between supply and contacts of 2500 V AC and between open contacts 1000 V AC, it shall have colored LED's for visual indication and shall comply to EN standards EN 6100-4-2/3/4/5/6/11. The switch-off delay time shall be selectable in a range of 0.5 to 0 sec. The electrical life at rated load AC1 shall be 60X10^3 cycles and shall work at an ambient temperature of -20 to +60 deg C. The mounting shall be on 35 mm DIN rail (EN 60715) and shall have necessary approvals like CE and GOST. | Each | 13361.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.9** | | **Current monitoring relay** | Current monitoring relay:– For checking the health of the crucial motors, pumps, etc as an advance warning system | | |
| | 52.9.1 | Programmable universal (AC/DC) current monitoring relay | Supply, installation, testing and commissioning of programmable universal (AC/DC) current monitoring relay shall be usable with current transformer for detecting upper and lower value, upper set point minus hysteresis range (5-50%) for switch ON, lower set point plus hysteresis range (5-50%) for switch ON and fault memory. It shall be immune to supply interruptions of < 200 ms with detection levels of AC (0.1…10) A at transducer to 600A/ DC (0….10) A. The switch off/reaction/start delay shall be for (0.1…12)s/0.35s/(0.1..20)s respectively. The current monitoring relay shall have insulation according to EN 61810-1 with electrical isolation between supply to measuring circuits and shall comply to EMC specifications as per EN 31000-4-2/3/4/5/6 and EN 55022. It shall work at a supply voltage of 230 V AC with and electrical life at rated load AC1 shall be 1 lakh cycles and shall work at an ambient temperature of -20 to +55 deg C. The mounting shall be on 35 mm DIN rail (EN 60715) and shall have necessary approvals like CE and GOST. | Each | 16047.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **52.10** | | **Room thermostat** | Room thermostat – For controlling/monitoring room temperature | | |
| | 52.10.1 | Touch screen programmable room thermostat | Supply, installation, testing and commissioning of touch screen programmable room thermostat shall be operated at 3V DC (2 batteries 1.5VDC AAA) shall have a temperature setting range of (+5...+37) °C. It shall be available in daily/weekly programmable versions and shall have calendar with automatic leap year function with day light saving updates. It shall have 2 level of security- simple touch screen blocking or full 4 digit PIN Lock and shall be IP 20 protected and functions like frost protection, automatic control, manual control, holiday program and pump anti-seizure function with a facility for video and audio confirmation of key and function entry. With a rated power of 5Amp at 230V AC, it shall have 1CO contact with on/off hysteresis of (0.1-0.9) K. It should with weekly programming facility. | Each | 11038.00 |
| **52.11** | | **Lightning &surge protection** | Lightning & surge protection for solar plants on the building roof for KW plants:- Type 1 + 2 SPD | | |
| | 52.11.1 | Surge arrester for protection of DC side(1,000V) of systems | Supply, installation, testing and commissioning of surge arrester for protection of DC side(1,000V) of systems in photovoltaic applications against overvoltage caused by direct lightning strike – Iimp 12.5 KA 10/350µs and induced overvoltage's upto Imax 30 KA, 8/20 µs and a protection level of 1.8 KV. Short circuit withstand current of 125 A. Has Visual indication of Healthy/Replace status onboard and remote, with Replaceable modules and Complies with prEN 50539-11:2010 | Each | 50529.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.11.2 | For smaller plants: type 2 SPD | For smaller plants: Type 2 SPD | | |
| | 52.11.2.1 | Surge arrester for protection of DC side(420V to 1,200V) | Supply, installation, testing and commissioning of surge arrester for protection of DC side(420V to 1,200V) of systems in photovoltaic applications upto Imax 30 KA, 8/20 µs and a protection level of 2 KV. Short circuit withstand current of 125 A. Has visual indication of healthy/replace status onboard and remote, with replaceable modules and complies with EN 50539-11:2010 | Each | 21724.00 |
| **52.12** | | **Temperature &humidity monitoring and controlling** | Temperature &humidity monitoring and controlling solutions for electrical panels and rising mains | | |
| | 52.12.1 | Humidity and temperature monitor | Supply, installation, testing and commissioning of humidity and temperature monitor should be small size to get mounted on a DIN rail inside the electrical panels and distribution panels. Operating voltage is 230V AC with LED status indicator lights. Should have relay output to switch on the space heater which should be DIN rail mountable and encapsulated with fire retardant housing for touch proof and accident proof operations. Switching point for temperature and relative humidity should be adjustable. The electrical life shall be rated for minimum 1 lakh cycles | Each | 8158.00 |
| **52.13** | | **Panel thermostat** | **Panel thermostat** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.13.1 | Thermostat | Supply, installation, testing and commissioning of 250 V AC operated panel thermostat shall have 1 NO in heating control and 1 NC in ventilation control , 10A, AgNi contact material and shall have wide temperature setting range -20 to +40 & +0 to +60deg C in heating/ventilation control. It shall have very small and compact size 17.5 mm wide and thermostat have snap action thermostatic bimetal sensor .The electrical life at rated load AC1 shall be 1 lakh cycles and shall work at an ambient temperature of -45 to +80 deg C. The mounting shall be on 35 mm DIN rail (EN 60715) and shall have necessary approvals like CE and GOST and EAC. | Each | 2490.00 |
| **52.14** | | **Filter fan** | **Filter fan-** | | |
| | 52.14.1 | Filter fan | Supply, installation, testing and commissioning of 230 v Ac operated filter fan shall have air volume 14 to 470 m3/h with exhaust filter installed in cabinet and air volume 24 to 630 m3/h with free flow. It shall have power consumption is 4 to 130 W and minimal depth with in enclosure and very low acoustic noise. The filter material consists of a synthetic fiber progressive construction which is moisture resistant to 100 degree C and shall work at an ambient temperature of -10 to +70 deg C. The filter fan shall have necessary approvals like CE, EAC and UL and shall be IP 54 protected. The filter mats are to filter class EU3 accordingly DIN24185 and have an average filtering degree of 80 to 90 % and the filter fan shall have 50,000 hr life at 40 degree C. | Each | 9440.00 |
| **52.15** | | **Panel heater** | **Panel heater-** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 52.15.1 | Panel heater | Supply, installation, testing and commissioning of 110 to 230 V AC operated panel heater shall have an operating range of 88…253V AC. The panel heater shall have heating power of 100W. It shall have PTC resistor, self-regulating heating system and shall work at an ambient temperature of -45 to +50 deg C. The mounting clip shall be on 35 mm DIN rail (EN 60715) and shall have necessary approvals like CE, UL and should be touch proof. | Each | 9812.00 |
| **52.16** | | **Water level control relay** | **Water level control relay – reliable solution for tank water level control** | | |
| | 52.16.1 | Water level control relay | Supply, installation, testing and commissioning of 230V AC operated water level control relay should ee able to connect three level sensors to detect the water level in the tank. The relay should be Ip 20 rated with 1 no, 16 A, AgCdO contact material. It shall have an ambient temperature range of -20 to +60 Deg C and have led indicator. The reinforced insulation level must be 6 KV - 1.2/50µs between: supply and contacts, electrodes and supply and contacts and electrodes. It should be able to control about a single level or between min./max. limits with adjustable sensitivity range. The float switch float switch of 8 ampere contact rating should be with 2 watertight chambers, counterweight (300gm) with TUV approved cable length of 10 meters including grip. | Each | 31945.00 |

# CHAPTER - 53

## AUDIO SPEAKER, AMPLIFIER

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **53.1** | | **Digital voice alarm controller** | 8 zone IP based digital voice alarm controller | | |
| | 53.1.1 | Digital voice alarm controller expandable upto 128 zones. | Supply, installation, testing and commissioning of 8 zone IP based, digital voice alarm controller expandable upto 128 zones. The voice alarm controller should be able to connect directly over ethernet without any additional hardware or convertor. It should have functions like the audio playing, zone control, fault monitoring, log recording, volume control and amplifier switchover. The voice alarm controller should also have the following functions:<br>. 100 - 255 priorities<br>. Time schedule broadcasts<br>. Capable to Connect PTT<br>. Capable of amplifier redundancy<br>. 8 trigger inputs/outputs | Each | 482820.00 |
| **53.2** | | **Networkable touch screen/keypad paging station** | **Networkable touch screen/keypad paging station** | | |
| | 53.2.1 | Networkable touch screen/Keypad paging station for selection of zones | Supply, installation, testing and commissioning of IP based networkable touch screen/ Keypad paging station for selection of zones, supervision of system status, setting of scheduled broadcasts with the following functions:<br>. Built-in monitoring loudspeaker<br>. Goose-neck microphone<br>. LCD touch screen/Keypad display<br>. Audio input | Each | 132683.00 |
| **53.3** | | Amplifiers | Amplifiers | | |
| | 53.3.1 | 4X125W class-D amplifier with 4 independent channels | Supply, installation, testing and commissioning of 4X125W class-D amplifier with 4 independent channels and can provide automatic re-settable over current, overload, overheating, overvoltage, under-voltage and DC protection | Each | 329017.00 |
| | 53.3.2 | 2X250W class-D amplifier with 2 independent channels | Supply, installation, testing and commissioning of 2X250W class-D amplifier with 2 independent channels and can provide automatic re-settable over current, overload, overheating, overvoltage, under-voltage and DC protection | Each | 258405.00 |
| | 53.3.3 | 1X500W class-D amplifier with 1 independent channel | Supply, installation, testing and commissioning of 1X500W class-D amplifier with 1 independent channel and can provide automatic re-settable over current, overload, overheating, overvoltage, under-voltage and DC protection | Each | 217261.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 53.3.4 | 240W power amplifier with 100,70V speaker OP | Supply, installation, testing and commissioning of 240W power amplifier with 100,70V speaker op, combination XLR/RCA line in line out connectors. Speaker OP short circuit protection.(with require mounting arrangement) | Each | 53845.00 |
| | 53.3.5 | 480w power amplifier with 100,70V speaker OP | Supply, installation, testing and commissioning of 480W power amplifier with 100,70V speaker OP, combination XLR/RCA line in line out connectors. Speaker OP short circuit protection.(with require mounting arrangement) | Each | 63172.00 |
| | 53.3.6 | 1000W power amplifier | Supply, installation, testing and commissioning of 1000W power amplifier with following features:- Rated output wattage - 1000W, amplifier output 100V, 70V and 4-16 ohms transformer isolated speaker outputs, Supply input 0.775 v (0dB) unbalanced, 10k ohms, XLR and phone jack frequency response 50 - 18,000 Hz, +/- 1.5 dB ,S/N ratio> 105 dB | Each | 171261.00 |
| | 53.3.7 | Power Amplifier with 2 x 600W | Supply, installation, testing and commissioning of Network Based Digital DriveCore Dual Channel Power Amplifier with 2 x 600W; built-in DSP, Frequency Response (8Ω, 20 Hz - 20 kHz) +/-0.5dB, THD (full rated power, 20 Hz - 20 kHz) 0.35%, etc. | Each | 253752.00 |
| | 53.3.8 | Power Amplifier with 2 x 1200W | Supply, installation, testing and commissioning of Network Based Digital DriveCore Dual Channel Power Amplifier with 2 x 1200W; built-in DSP, Frequency Response (8Ω, 20 Hz - 20 kHz) +/-0.25dB, THD (full rated power, 20 Hz - 20 kHz) 0.35%, etc. | Each | 285923.00 |
| **53.4** | | Mixer | Mixer | | |
| | 53.4.1 | 8-Channel Mixer | Supply, installation, testing and commissioning of 8-Channel Mixer with Low-cost, high-performance Lexicon effects mixers. 8+2 and 12+2 channel frame sizes. Built-in 24 bit Lexicon digital effects processor. 32 FX settings | Each | 63800.00 |
| | 53.4.2 | 12-Channel Mixer | Supply, installation, testing and commissioning of12-Channel Mixer with Low-cost, high-performance Lexicon effects mixers. 8+2 and 12+2 channel frame sizes. Built-in 24 bit Lexicon digital effects processor. 32 FX settings | Each | 70125.00 |
| | 53.4.3 | . Multi-Channel Analog Mixing Console | Supply, installation, testing and commissioning of Multi-Channel Analog Mixing Console with 8 mono input channels and 4 stereo input channels. Frequency Response : Mic/Line Input to any output, 20 Hz-20 kHz<1 dB. CMRR Typical @ max gain @ 1 kHz>80 dB | Each | 105593.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 53.4.4 | Multi-Channel Analog Mixing Console | Supply, installation, testing and commissioning of Multi-Channel Analog Mixing Console with 16 mono input channels and 4 stereo input channels. Frequency Response : Mic/Line Input to any output, 20 Hz-20 kHz<1 dB. CMRR Typical @ max gain @ 1 kHz>80 dB | Each | 132405.00 |
| **53.5** | | Speakers | Speakers | | |
| | 53.5.1 | 6W ceiling speaker with max SPL1M/1W 96dB | Supply, installation, testing and commissioning of 6W ceiling speaker with max SPL1M/1W 96dB. Frequency response of 80Hz-20KHz with a dispersion angle of 160 deg. The speaker should have tapings at 6W/3W/1.5W (with require mounting arrangement) | Each | 1703.00 |
| | 53.5.2 | 20W ceiling speaker with max SPL1M/1W 90dB | Supply, installation, testing and commissioning of 20W ceiling speaker with max SPL1M/1W 90dB. Frequency response of 100Hz-20KHz with a dispersion angle of 120 deg. The speaker should have tapings at 20W/10W (With require mounting arrangement) | Each | 13206.00 |
| | 53.5.3 | 6W wall mount with max SPL1M/1W 96dB | Supply, installation, testing and commissioning of 6W wall mount with max SPL1M/1W 96dB. Frequency response of 160Hz-20KHz. The speaker should have tapings at 6W/3W (with require mounting arrangement) | Each | 2530.00 |
| | 53.5.4 | 10W wall mount with max SPL1M/1W 92dB | Supply, installation, testing and commissioning of 10W wall mount with max SPL1M/1W 92dB. Frequency response of 150Hz-17KHz with a dispersion angle of 170 deg. The speaker should have tapings at 10W/5W (with require mounting arrangement) | Each | 6836.00 |
| | 53.5.5 | 40W wall mount with max SPL1M/1W 86dB | Supply, installation, testing and commissioning of 40W wall mount with max SPL1M/1W 86dB. Frequency response of 100Hz-20KHz with a dispersion angle of 160 deg. The speaker should have tapings at 40W/20W/10W (with require mounting arrangement) | Each | 17361.00 |
| | 53.5.6 | 15W horn speaker (IP66) with max SPL1M/1W 110dB | Supply, installation, testing and commissioning of 15W horn speaker (IP66) with max SPL1M/1W 110dB. Frequency response of 500Hz-5KHz with a dispersion angle of 140 deg. The speaker should have tapings at 15W/7.5W | Each | 8538.00 |
| | 53.5.7 | 30W horn speaker (IP66-ABS) with max SPL1M/1W 104dB | Supply, installation, testing and commissioning of 30W horn speaker (IP66-ABS) with max SPL1M/1W 104dB. Frequency response of 300Hz-10KHz. The speaker should have tapings at 30/15W.(with require mounting arrangement) | Each | 14619.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 53.5.8 | 20W column speaker with max SPL1M/1W 94dB | Supply, installation, testing and commissioning of 20W column speaker with max SPL1M/1W 94dB. Frequency response of 300Hz-15KHz with a dispersion angle of 160 deg. The speaker should have tapings at 20 W / 10 W / 5 W / 2.5 W (With require mounting arrangement) | Each | 27863.00 |
| | 53.5.9 | Ultra Compact 2-way Loudspeaker 150W | Supply, installation, testing and commissioning of SPEAKER : Ultra Compact 2-way Loudspeaker with 1 x 5.25" LF with Frequency Range (-10dB) 80 Hz -20 kHz, Frequency Response (±3 dB) : 90 Hz -18 kHz, Coverage Pattern 90° x 90°, Impedance : 16 ohms, System Power Rating : 150 W / 300 W / 600 W (Continuous/Program/Peak) (IEC rated) (With require mounting arrangement) | Each | 69340.00 |
| | 53.5.10 | Two-Way Stage Monitor and Loudspeaker System 300W | Supply, installation, testing and commissioning of SPEAKER with 12" Two-Way Stage Monitor and Loudspeaker System Frequency Response (±3 dB) 62 Hz – 19kHz Coverage Pattern 90° x 50°nominal System Sensitivity 95 dB SPL (1w@1m) Power Rating 300 W / 600 W / 1200 W (Continuous/Program/Peak) (With require mounting arrangement) | Each | 87007.00 |
| | 53.5.11 | 2-way Loudspeaker 375W | SPEAKER : Compact 2-way Loudspeaker with 2 x 8" LF with Frequency Range (-10dB) : 53 Hz - 20 kHz, Frequency Response (±3 dB) : 65 Hz – 20 kHz, Coverage Pattern : /26: 120° x 60°/95: 90° x 50°, Impedance 8 ohms, System Power Rating : 375 W / 750 W / 1500 W (Continuous / Program /Peak) (IEC rated). (With require mounting arrangement) | Each | 189890.00 |
| | 53.5.12 | Line Array Column Loudspeaker 150W | Supply, installation, testing and commissioning of SPEAKER : Constant Beamwidth Technology Line Array Column Loudspeaker with 8 x 2" Full Range Drivers Frequency Range (-10dB) : 80 Hz – 20 kHz, Coverage Angle Vertical : 20°; Horizontal : 150°, Sensitivity : 93 dB , Impedance : 8 ohms, Power Rating 150W (600W peak), 2 hrs; 100 W (400W peak), 100 hrs (With require mounting arrangement) | Each | 78901.00 |
| | 53.5.13 | Line Array Column Loudspeaker 325W | Supply, installation, testing and commissioning of SPEAKER : Constant Beamwidth Technology Line Array Column Loudspeaker with 16 x 2" full Range Drivers with Frequency Range (-10dB) : 80 Hz – 20 kHz; Coverage Angle Vertical: Switchable, Horizontal: 150°(±20°), Sensitivity : 96 dB, Impedance : 8 ohms, Power Rating : 325 W (1300W peak), 2 hrs; 200 W (800W peak), 100 hrs (With require mounting arrangement) | Each | 158929.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 53.5.14 | Two-Way Powered Line Array Loudspeaker 800W | SPEAKER : 12 in. Two-Way Powered Line Array Loudspeaker System with Frequency Range (-10dB) : 57 Hz - 20 kHz, Frequency Response 75 Hz - 20 kHz (+/- 3db), Coverage Pattern : 100 x 15 nominal, Sensitivity Passive: 95 dB SPL, Nominal Impedance : 8 Ohms, Power Rating : 800 W / 1600 W / 3200 W (Program / Peak / Continuous). (With require mounting arrangement) | Each | 313108.00 |
| | 53.5.15 | Compact Line Array Loudspeaker 390W | Supply, installation, testing and commissioning of SPEAKER : Dual 8-inch Compact Line Array Loudspeaker with Frequency Range (-10dB) 50Hz - 19kHz, Coverage Pattern 110-degrees nominal (300Hz - 18kHz), Max SPL 139 dB, System Input Power Rating :- LF : 600W Continuous (IEC/100 hour); MF/HF : 390W Continuous (IEC/100 hour), Nominal Impedance :- LF = 8 ohms, MHF = 8 ohms. (With require mounting arrangement) | Each | 639531.00 |
| **53.6** | | | Conference System | | |
| | 53.6.1 | Delegate Unit | Supply, installation, testing and commissioning of Conference Delegate Unit Nominal Acoustic Input at the microphone 85dB SPL Frequency Response 400Hz to 10Khz. Length of the microphone from the base of the stem 488mm or more. | Each | 15111.00 |
| | 53.6.2 | Chairman Unit | Supply, installation, testing and commissioning of Conference Chairman unit Nominal Acoustic Input 85dB SPL Frequency Response 400Hz to 10kHz. Length of the microphone from the base of the stem 488mm or more. | Each | 16847.00 |
| | 53.6.3 | Conference Control unit | Supply, installation, testing and commissioning of Conference Control unit Frequency response 200 Hz to 12 kHz Input Impedance 4.7 kohm Rated output Impedance 1 kohm Equaliser Center frequencies of filters 250Hz,2kHz and 8kHz Mic input sensitivity 1 mV (+1/-3dB) Aux input sensitivity 100mV (+1/-3dB) | Each | 37466.00 |
| | 53.6.4 | Extension Cable | Supply, installation, testing and commissioning of Extension Cable 10 Mtrs. For connection of additional length other than provided in delegate /chairman unit | Each | 2515.00 |
| **53.7** | | | Microphones | | |
| | 53.7.1 | Vocal Microphone : | Supply, installation, testing and commissioning of Professional Dyanamic Vocal Microphone Audio frequency bandwidth: 70 to 20000 Hz Sensitivity: 2.6 mV/Pa Electrical impedance: 600 Ohms Recommended load impedance: 2000 Ohms Polar Pattern: Supercardioid | Each | 16252.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 53.7.2 | Instrumental Microphone : | Supply, installation, testing and commissioning of Professional instrumental microphone Audio frequency bandwidth 50 - 20000 Hz Sensitivity at 1000 Hz2.5 mV/Pa (–52 dBV re 1 V/Pa) Max. SPL for 1 % / 3 % THD147 dB SPL / 156 dB SPL Electrical impedance at 1000 Hz≥ 600 Ohms Recommended load impedance≥ 2000 Ohms Polar Pattern Cardioid | Each | 21413.00 |
| | 53.7.3 | Gooseneck Microphone: | Supply, installation, testing and commissioning of Professional tabletop microphone Audio frequency bandwidth70 - 18000 Hz Electrical impedance600 Ohms Recommended load impedance2000 Ohms Voltage9 to 52 V Current3 mA | Each | 49784.00 |
| | 53.7.4 | Wireless Handheld Microphone: | Supply, installation, testing and commissioning of Wireless microphone system Microphone capsule: Patented dynamic D5 capsule Polar pattern: supercardioid Audio bandwidth: 70 – 20 000 Hz (+/- 3dB) Dimensions: 237 x 51mm | Each | 49419.00 |
| | 53.7.5 | Lapel Wireless Microphone: | Supply, installation, testing and commissioning of WIRELESS BODYPACK TRANSMITTER Type: Three pin mini XLR connector (female)Maximum audio level: 12dB Rated Impedance: 1 Ohms Audio bandwidth: 40 – 20 000 Hz (+/- 3dB) Dimensions: 60 x 75,5 x 30 mm | Each | 52796.00 |
| **53.8** | | HDMI Transmitter | Supply, installation, testing and commissioning of Transmitter : HDMI HDBaseT-Lite Extender with PoH (4K @ 40 mtr.) (HDBaseT Class B) extends an HDMI signal up to 70m from the HDBaseT source using one Cat 5e/6/6a cable. Video :- Input Interfaces : 1 x HDMI Type A Female (Black); Impedance : 100 Ω; Max. Distance : 3 m, Max. Data Rate : 10.2Gbps (3.4Gbps per lane); Max. Pixel Clock : 340 MHz; Compliance : HDMI (3D, Deep Color, 4K), HDCP2.2 Compatible, Consumer Electronics Control (CEC)Audio Input : 1 x HDMI Type A Female | Each | 52066.00 |
| **53.9** | | HDMI Receiver | Supply, installation, testing and commissioning of Receiver : HDMI HDBaseT-Lite Extender with PoH (4K @ 40 mtr.) (HDBaseT Class B) extends an HDMI signal up to 70m from the HDBaseT source using one Cat 5e/6/6a cable. Video :- Output Interfaces : 1 x HDMI Type A Female (Black); Impedance: 100 Ω; Max. Distance : 3 m, Max. Data Rate : 10.2Gbps (3.4Gbps per lane); Max. Pixel Clock : 340 MHz; Compliance : HDMI (3D, Deep Color, 4K), HDCP2.2 Compatible, Consumer Electronics Control (CEC), Audio Output : 1 x HDMI Type A Female (Black). | Each | 52066.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **53.10** | | HDMI Transmitter | Supply, installation, testing and commissioning of Transmitter : HDMI HDBaseT-Lite Extender (4K@100m) (HDBaseT Class A) guarantees the optimum 4K HDMI signal transmission across up to100 m via a single Cat 5e/6/6a. Video :- Input Interfaces : 1 x HDMI Type A Female (Black); Impedance : 100 Ω; Max. Distance : 3 m, Max. Data Rate : 10.2Gbps (3.4Gbps per lane); Max. Pixel Clock : 340 MHz; Compliance : HDMI (3D, Deep Color, 4K), HDCP2.2 Compatible, Consumer Electronics Control (CEC)Audio Input : 1 x HDMI Type A Female | Each | 69662.00 |
| **53.11** | | HDMI Receiver | Supply, installation, testing and commissioning of Receiver : HDMI HDBaseT-Lite Extender (4K@100m) (HDBaseT Class A) guarantees the optimum 4K HDMI signal transmission across up to100 m via a single Cat 5e/6/6a. Video :- Output Interfaces : 1 x HDMI Type A Female (Black); Impedance: 100 Ω; Max. Distance : 3 m, Max. Data Rate : 10.2Gbps (3.4Gbps per lane); Max. Pixel Clock : 340 MHz; Compliance : HDMI (3D, Deep Color, 4K), HDCP2.2 Compatible, Consumer Electronics Control (CEC), Audio Output : 1 x HDMI Type A Female (Black). | Each | 75950.00 |
| **53.12** | | 4 x 2 True 4K Presentation Matrix Switch | Supply, installation, testing and commissioning of 4 x 2 True 4K Presentation Matrix Switch - Analog/Digital AV Matrix Switching – switch among 3 HDMI and 1 VGA inputs to 2 HDMI outputs and support of audio embedding/de-embedding, Scaler – supports video upscaling to 4K for display A and downscaling to 1080p for display B. Multiple control options – flexible control via front-panel pushbuttons, IR remote control, RS-232, Contact Closure, and web-based GUI through Ethernet. | Each | 202937.00 |

# CHAPTER - 54

# CLOSED CIRCUIT TELEVISION (CCTV)

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **54.1** | | **2 MP IP IR eyeball camera** | **2 MP IP IR eyeball camera with H.264 video compression** | | |
| | 54.1.1 | IP dome camera | Supply, Installation testing and commissioning of the IP dome camera shall be equipped with a 1/2.8 2 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 3.6 mm fixed lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.1 lux @F1.2, AGC ON, 0 lux with IR video compression :- H.264, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, encode mode:- CBR/VBR (freely adjustable in CBR, 5 levels in VBR). UL listed | Each | 10221.00 |
| **54.2** | | **2 MP IP IR bullet camera** | **2 MP IP IR bullet camera with H.264 Video compression** | | |
| | 54.2.1 | IP bullet camera | Supply, Installation testing and commissioning of the IP bullet camera shall be equipped with a 1/2.8 2 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 3.6 mm fixed lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.1 lux @F1.2, AGC ON, 0 lux with IR video compression :- H.264, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, encode mode:- CBR/VBR (freely adjustable in CBR, 5 levels in VBR). | Each | 12480.00 |
| **54.3** | | **2 MP IP IR eyeball camera** | **2 MP IP IR eyeball camera with H.265 video compression** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.3.1 | IP mini dome camera | Supply, installation testing and commissioning of the IP mini dome camera shall be equipped with a 1/3″ 2 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 2.8 mm fixed lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, UL listed | Each | 15506.00 |
| **54.4** | | **2 MP IP IR bullet camera** | **2 MP IP IR bullet camera with H.265 video compression** | | |
| | 54.4.1 | IP bullet camera | Supply, installation testing and commissioning of the IP bullet camera shall be equipped with a 1/2.9" 2MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 3.6 mm fixed lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (Color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, UL listed | Each | 15868.00 |
| **54.5** | | **2MP IP varifocal MFZ dome camera** | **2MP IP varifocal MFZ dome camera** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.5.1 | IP dome camera | Supply, Installation testing and commissioning of the IP dome camera shall be equipped with a 1/2.8 2 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 2.8 - 12 mm verifocal lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.1 lux @F1.2, AGC ON, 0 lux with IR video compression :- H.264, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, encode mode:- CBR/VBR (freely adjustable in CBR, 5 levels in VBR). | Each | 16280.00 |
| **54.6** | | **2MP IP varifocal MFZ bullet camera** | **2MP IP varifocal MFZ bullet camera** | | |
| | 54.6.1 | IP bullet camera | Supply, installation testing and commissioning of the IP bullet camera shall be equipped with a 1/2.8 2 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps) images, a 2.8 - 12 mm Verifocal lens, and a waterproof (IP66) enclosure. As a true day/night solution, the camera shall use Smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. Illumination:-0.1 lux @F1.2, AGC ON, 0 lux with IR video compression :- H.264, noise reduction: 3DNR,backlight compensation :- BLC/HLC/DWDR, encode mode:- CBR/VBR (freely adjustable in CBR, 5 levels in VBR). | Each | 17224.00 |
| **54.7** | | **4MP IP fixed rugged dome cam.** | **4MP IP fixed rugged dome camera** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.7.1 | IP rugged dome camera | Supply, installation testing and commissioning of the IP rugged dome camera shall be equipped with a 1/3″ 4 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps)/2688 × 1520 images, a 2.8 mm fixed lens, and a waterproof (IP66) IK10 impact-resistant camera housing, as a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- True WDR 120 DB, angle of view 104°/87° (H), 57°/48° (V) ,UL listed | Each | 19553.00 |
| **54.8** | | **4MP IP fixed IR bullet camera** | **4MP IP fixed IR bullet camera** | | |
| | 54.8.1 | IP bullet camera | Supply, installation testing and commissioning of the IP bullet camera shall be equipped with a 1/3" 4MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps)/2688 × 1520 (20fps) images, a 3.6 mm fixed lens, and a waterproof (IP66) IK10 impact-resistant camera housing, As a true day/night solution, the camera shall use Smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (Color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- True WDR 120DB, angle of view :-104°/87° (H), 57°/48° (V),UL listed | Each | 19309.00 |
| **54.9** | | **4MP IP varifocal MFZ dome camera 2.7-13.5 MM** | **4MP IP varifocal MFZ dome camera 2.7-13.5 MM** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.9.1 | IP rugged dome camera | Supply, installation testing and commissioning of the IP rugged dome camera shall be equipped with a 1/3″ 4 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps)//2688 × 1520 (20fps) images, 2.7 -13.5 mm, MFZ lens, digital zoom 16x and a waterproof (IP66) IK10 impact-resistant camera housing, As a true day/night solution, the camera shall use smart IR technology and provide up to 145 ft (45 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- True WDR 120 DB, angle of view 104°/87° (H), 57°/48° (V) ,UL listed | Each | 29009.00 |
| **54.10** | | **4MP IP varifocal MFZ IR bullet camera** | **4MP IP varifocal MFZ IR bullet camera** | | |
| | 54.10.1 | IP bullet camera | Supply, installation testing and commissioning of the IP bullet camera shall be equipped with a 1/3″ 4 MP progressive scan CMOS imager to capture full HD 1080p (25/30 fps)/2688 × 1520 (20fps), images, 2.7 -13.5 mm, MFZ lens, digital zoom 16x and a waterproof (IP66) IK10 impact-resistant camera housing, as a true day/night solution, the camera shall use smart IR technology and provide up to 164ft (50 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 3DNR,backlight compensation :- True WDR 120 DB, angle of view104°/87° (H), 57°/48° (V) ,UL listed | Each | 29192.00 |
| **54.11** | | **8 MP IP IR fixed lens dome camera** | **8 MP IP IR fixed lens dome camera** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.11.1 | IP mini dome camera | Supply, installation testing and commissioning of the IP mini dome camera shall be equipped with a 1/2.5″ 8 MP progressive scan CMOS imager to capture ultra HD 4K/8 MP resolution at 15 fps or3 MP at 25/30 fps, a 4.0 mm fixed lens, and a waterproof (IP66) IK10 impact-resistant camera housing, as a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. Illumination:-0.1 lux (color)/0 lux (B/W) @ F1.6 (with IR LEDs on)video compression :- H.265 , noise reduction: 3DNR, backlight compensation :- DWDR, angle of view horizontal: 88° vertical: 48°,UL listed | Each | 38926.00 |
| **54.12** | | **8 MP IP IR fixed lens bullet camera** | **8 MP IP IR fixed lens bullet camera** | | |
| | 54.12.1 | IP bullet camera | Supply, installation testing and commissioning of the IP bullet camera shall be equipped with a 1/2.5″ 8 MP progressive scan CMOS imager to capture ultra HD 4K/8 MP resolution at 15 fps or3 MP at 25/30 fps, a 4.0 mm fixed lens, and a waterproof (IP66) IK10 impact-resistant camera housing, as a true day/night solution, the camera shall use smart IR technology and provide up to 98 ft (30 m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:0.07 lux (color)/0 lux (B/W) @ F2.0 (with IR LEDs on)) video compression :- H.265 , noise reduction: 3DNR,Backlight Compensation :- DWDR, angle of view horizontal: 88° vertical: 48°,UL listed | Each | 39785.00 |
| **54.13** | | **2 MP IP 25X zoom PTZ camera** | **2 MP IP 25X zoom PTZ camera** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.13.1 | 2MP IP PTZ camera | Supply, installation testing and commissioning of 2MP IP PTZ camera shall be equipped with a 1/2.8" exmor CMOS imager to capture full HD 1080p (50/60 fps)/ 25x zoom images, 4.8 mm-120mm, and a waterproof (IP66) IK10 impact-resistant camera housing, as a true day/night solution, the camera shall use Smart IR technology and provide up to 328 ft (100m) of IR illumination. The camera shall accept PoE (802.3af) or 12 V DC power input. min. illumination:-0.08 lux color @ F2.0 (color, 1/3s, 30 IRE),0 lux B/W with IR LEDs on @ F2.0) video compression :- H.265 +, noise reduction: 2D/3D,backlight compensation :- True WDR 120 DB, built-in analytics, including face detection, automatically restores to previous PTZ and lens poSITC ion after power failure preset speed pan: Up to 240°/s, tilt: Up to 200°/s, angle of view H: 59.2° – 2.4° ,UL listed | Each | 86237.00 |
| **54.14** | | **4MP IP varifocal MFZ dome camera IP67** | **4MP IP varifocal MFZ dome camera IP67** | | |
| | 54.14.1 | Outdoor WDR IR rugged IP dome cameras | Supply, installation testing and commissioning of outdoor WDR IR rugged IP dome cameras shall be equipped with 1/2.7" or 1/2.8" 2 MP progressive scan imager or 1/3" 4 MP progressive scan imager to capture 1080p resolution at 25/30 fps or 4 MP at 20 fps images, 2.7–12 mm or 7–22 mm, F1.4, motorized focus/zoom len, H.264 and MJPEG codec, triple stream support, 50 m (150') IR distance, depending on scene reflectance, waterproof (IP67) and IK10 vandal resistant camera housing, ONVIF profile S and G compliant , built-in PoE (Power over Ethernet) eliminates, separate power supply and associated wiring, 24 V AC/12 V DC inputs where PoE power is unavailable, supports up to 128 GB micro SDHC (class 10) card for local video storage when network is interrupted. Card not included. | Each | 46383.00 |
| **54.15** | | **4MP IP varifocal MFZ bullet camera IP67** | **4MP IP varifocal MFZ bullet camera IP67** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.15.1 | Outdoor WDR IR rugged IP bullet cameras | Supply, installation testing and commissioning of Outdoor WDR IR rugged IP bullet cameras shall be equipped with : 1/2.7" 2 MP progressive CMOS imager to capture 1080p resolution at 25/30 fps or 4 MP at 20 fps images, 1: 2.7-12 mm, MFZ, F1.4, H.264 and MJPEG codec, triple stream support, 60 m (150') IR distance, depending on scene reflectance, waterproof (IP67) and IK10 vandal resistant camera housing, ONVIF profile S and G compliant,, built-in PoE (Power over Ethernet) eliminates, separate power supply and associated wiring, 24 V AC/12 V DC inputs where PoE power is unavailable, Supports up to 128 GB micro SDHC (class 10) card for local video storage when network is interrupted. Card not included, True WDR: up to 140 dB | Each | 49281.00 |
| **54.16** | | **2 MP IP 30X zoom PTZ camera** | **2 MP IP 30X zoom PTZ camera** | | |
| | 54.16.1 | Outdoor PTZ IR ultra low light IP cameras | Supply, installation testing and commissioning of outdoor PTZ IR ultra low light IP cameras with full HD 1080p resolution, 25/30 fps image with a 1/1.9 in. 2 MP sensor, true WDR 120 dB ensures glare-free images, 6.0-180 mm, F1.6-F4.8, 30x optical zoom lens with 360° continuous rotation for the flexibility to clearly view the target area with desired width and depth, H.264 and MJPEG codec, triple stream support, IR LEDs provide up to 200 m, waterproof (IP67) and IK10 vandal resistant camera housing, ONVIF profile S compliant supports up to 128 GB micro SDHC (Class 10) card for local video storage when network is interrupted. | Each | 207766.00 |
| **54.17** | | **4 channel NVR with 1 SATA** | **4 channel NVR with 1 SATA** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.17.1 | 4CH network video recorder | Supply, installation testing and commissioning of 4CH network video recorder with real-time recording: up to 4 MP resolution live view, preview and playback/up to 4 channels at 1080p,withinbuilt PoE, up to 1TB supports complete with UL certification | Each | 41629.00 |
| **54.18** | | **8 channel NVR with 1 SATA** | **8 channel NVR with 1 SATA** | | |
| | 54.18.1 | 8CHnetwork video recorder | Supply, installation testing and commissioning of 8CHnetwork video recorder with real-time recording: up to 4 MP resolution live view, preview and playback/up to 8 channels at 1080p,withinbuilt PoE, up to 1TB supports complete with UL certification | Each | 51795.00 |
| **54.19** | | **16 channel NVR with 1 SATA** | **16 channel NVR with 1 SATA** | | |
| | 54.19.1 | 16ch network video recorder | Supply, installation testing and commissioning of 16ch network video recorder with real-time recording: up to 4 MP resolution live view, preview and playback/up to 16 channels at 1080p,with inbuilt PoE, up to 1TB supports complete with UL certification | Each | 68855.00 |
| **54.20** | | **8 channel NVR with 2 SATA** | **8 channel NVR with 2 SATA** | | |
| | 54.20.1 | 8 channel NVR | Supply, installation testing and commissioning of 8 channel NVR. With 1080p real-time live view H.265/H.264MJPEG dual codec decoding up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 2 SATA HDDs up to 16TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications (a flashing light, bell, or siren) complete with UL certification | Each | 66722.00 |
| **54.21** | | **16 channel NVR with 2 SATA** | **16 channel NVR with 2 SATA** | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 54.21.1 | 16 channel NVR with 1080p | Supply, installation testing and commissioning of 16 channel NVR with 1080p real-time live view H.265/H.264MJPEG dual codec decoding up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 2 SATA HDDs up to 16TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 84721.00 |
| **54.22** | | **16 channel NVR with 4 SATA** | **16 channel NVR with 4 SATA** | | |
| | 54.22.1 | 16 channel NVR With 1080p | Supply, installation testing and commissioning of 16 channel NVR with 1080p real-time live view H.265/H.264MJPEG dual codec decoding Up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 4 SATA HDDs up to 32TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 114098.00 |
| **54.23** | | **32 channel NVR with 4 SATA** | **32 channel NVR with 4 SATA** | | |
| | 54.23.1 | 32 channel NVR with 1080p | Supply, installation testing and commissioning of 32 channel NVR with 1080p real-time live view H.265/H.264 MJPEG dual codec decoding Up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 4 SATA HDDs up to 32TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 140084.00 |
| **54.24** | | **64 channel NVR with 4 SATA** | **64 channel NVR with 4 SATA** | | |
| | 54.24.1 | 64 channel NVR with 1080p | Supply, installation testing and commissioning of 64 channel NVR with 1080p real-time live view H.265/H.264 MJPEG dual codec decoding Up to 12 MP resolution preview &playback max 320 MBPS incoming bandwidth support 4 SATA HDDs up to 32TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 209793.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **54.25** | | **32 channel NVR with 8 SATA** | **32 channel NVR with 8 SATA** | | |
| | 54.25.1 | 32 channel NVR with 1080p | Supply, installation testing and commissioning of 32 channel NVR with 1080p real-time live view H.265/H.264 MJPEG dual codec decoding Up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 8 SATA HDDs up to 64TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 222294.00 |
| **54.26** | | **64 channel NVR with 8 SATA** | **64 channel NVR with 8 SATA** | | |
| | 54.26.1 | 64 channel NVR With 1080p | Supply, Installation testing and commissioning of 64 channel NVR With 1080p real-time live view H.265/H.264 MJPEG dual codec decoding up to 12 MP resolution preview & playback max 320 MBPS incoming bandwidth support 8 SATA HDDs up to 64TB, supports fisheye video de-warping in local and web user interface, supports visual or auditory notifications(a flashing light, bell, or siren) complete with UL certification | Each | 261535.00 |
| **54.27** | | **8 port PoE switch with 2 SFP ports** | **8 port PoE switch with 2 SFP ports** | | |
| | 54.27.1 | 8 port PoE switch with 2 SFP ports | Supply, installation testing and commissioning of 8 port PoE switch with 2 SFP ports, having security features like port security supports 64 MACs per port, auto surveillance VLAN, loopback detection automatically disables a port when a loop is detected, cable diagnostics allows administrators to determine cable status, UTP cat. 5, cat. 5e (100 m max.), full/half-duplex for 10/100 MBPS, full-duplex for 1000 MBPS, auto MDI/MDIX adjustment for all twisted-pair ports, switching capacity, 40 GBPS, maximum 64 bytes packet forwarding rate, 14.88 MBPS, ports 1 to 8 compliant with 802.3at | Each | 23024.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **54.28** | | **24 port PoE switch with 4 SFP ports** | **24 port PoE switch with 4 SFP ports** | | |
| | 54.28.1 | 24 port PoE switch with 4 SFP ports | Supply, installation testing and commissioning of 24 port PoE switch with 4 SFP ports, having security features like port security supports 64 MACs per port, auto surveillance VLAN, loopback detection automatically disables a port when a loop is detected, cable diagnostics allows administrators to determine cable status, UTP Cat. 5, Cat. 5e (100 m max.), full/half-duplex for 10/100 MBPS, full-duplex for 1000 MBPS, auto MDI/MDIX adjustment for all twisted-pair ports, switching capacity, 56 GBPS, maximum 64 bytes packet forwarding rate, 41.7mbps, Ports 1 to 24 compliant with 802.3 at | Each | 51329.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **54.29** | | **Workstation with client license** | **Workstation with client license** | | |
| | 54.29.1 | Work station | The work station shall have the following specifications:- Processor Intel @ i5 750 2.66 GHz with internal memory of 4 GB RAM. Operating system window 7professional 64-bit (WOW 64mode) or window 7 professional 32-bit NVIDIA 1 GB independent graphic card. Resolution support for cameras 2560x1600 display resolution capable and direct X true colour 32 bit. Dual link VDI outputs. Client software shall support Dual monitors each displaying 16 cameras simultaneously. A maximum of 32 cameras viewing per workstation at a time. Programmable to sequence. | Each | 231656.00 |
| **54.30** | | **Hard disk** | **Hard disk** | | |
| | 54.30.1 | 2 TB | Providing &fixing of 2 TB hard disk | Each | 6414.00 |
| | 54.30.2 | 4 TB | Providing &fixing of 4 TB hard disk | Each | 10825.00 |

# CHAPTER - 55

## SOLAR WATER HEATER AND SOLAR PLANT

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 55.1 | | **Evacuated tube collectors solar water heating system** | Supply, installation, testing and commissioning of evacuated tube collectors solar water heating system having water circulation natural thermo syphon suitable for inlet water with chloride hardness up to 600 ppm & TDS up to 1000 ppm comprising of evacuated tube collectors with double walled vacuum tube made from hard borosilicate glass able to resist impact from hail stone up to 30 mm in diameter, of length 1800mm having ID/OD 48/58mm, thickness 1.6mm.heavy duty HR black CGO-4 steel of 2 mm thick horizontal cylindrical hot water tank with dished end capsule shape including access doors having PUF/rock wool insulation density 42 Kg/cu. mt., 50mm thick having nonstick lining inside the tank and UV stabilized polyester precoated G.I. outer cover. Stand assembly for tank having polyester powder coated G.I. folded angles and electrical back up 2 KW. Heat loss < 0.70 W/ (m 2˚C ), The system should be as per latest specifications of IREDA/BIS/MNRE norms and valid test reports issued by authorized test centers confirming to MNRE guidelines for following capacities :- | | |
| | 55.1.1 | 100 LPD | Rated capacity 100 LPD, storage tank 100 Ltr | Each | 39817.00 |
| | 55.1.2 | 200 LPD | Rated capacity 200 LPD, storage tank 200 Ltr | Each | 58093.00 |
| | 55.1.3 | 500 LPD | Rated capacity 500 LPD, storage tank 500 Ltr | Each | 121889.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **55.2** | | **Flat plate collector solar water heating system** | Supply, installation, testing and commissioning of flat plate collector solar water heating system having water circulation natural thermo syphon suitable for inlet water with chloride hardness up to 600 ppm & TDS up to 1000 ppm comprising of Copper collector box, coated with aluminum outer casing, absorber coating with black chrome selective coating, header &riser electrolytic copper tube OD 24.7/12.7mm, absorber fin electrolytic copper sheet with special shape grove & embossing including ultrasonic bonding between absorber fins and riser tubes, copper sheet high selective coated, connections with four brass flanges 62mm outer diameter. Aluminum housing frame 2094x1018x90mm, overall Area 2.13 sq.mm. with cream white ultraviolet resistant pure polyester powder coating, extruded aluminum frame with fully welded joints for glass retainer serrated aluminum sections fitted with SS 304 screws. Rear panel/bottom sheet aluminum sheet riveted & joints are made leak proof by using silicon sealant. Reflector sheet highly reflective thin aluminum foil. Gaskets &grommets EPDM rubber suitable for high temperature, nut bolts & screws stainless steel (S.S.-304).<br><br>Specially designed swivel hinges & sockets to adjust the collector inclination. Transparent tempered 4 mm low iron solar glass cover, light transmittance of glass 85%. Mineral wool insulation 50 mm bottoms& 25 mm side, gross density 48 kg / cu mtr. Hot water storage tank of length 1800mm having ID/OD 48/58mm, thickness 1.6mm.heavy duty HR black CGO-4 steel of 2 mm thick horizontal cylindrical hot water tank with dished end capsule shape with access doors having PUF/rock wool insulation density 42 Kg/ cu. mt., 50mm thick having nonstick lining inside the tank and UV stabilized polyester precoated G.I. outer cover. Stand assembly for tank having polyester powder coated G.I. folded angles and electrical back up 2 KW heat loss < 0.70 W/ (m 2 ˚ C ) and minimum maintenance, The system should be as per latest specifications of IREDA/BIS/ MNRE norms and valid test reports issued by authorized test centers confirming to MNRE guidelines for following capacities :- | | |
| | 55.2.1 | 100 LPD | Rated capacity 100 LPD, storage tank 100 Ltr | Each | 54436.00 |
| | 55.2.2 | 200 LPD | Rated capacity 200 LPD, storage tank 200 Ltr | Each | 99043.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 55.2.3 | 500 LPD | Rated capacity 500 LPD, storage tank 500 Ltr | Each | 162601.00 |
| **55.3** | | **Water pipe line of chlorinated polyvinyl chloride (CPVC)** | Supply, installation, testing and commissioning of water pipe line of chlorinated polyvinyl chloride (CPVC) SDR - 11 grade (IS 14930 Part I marked) for mentioned below size inclusive of CPVC plain and brass threaded all necessary required fittings such as elbow, tee, union, nipple, reducer, tank nipple, FABT, MABT, solvent, labor for fixing, etc. to complete the job as per specification. | | |
| | 55.3.1 | 15.00 mm OD | 15.00 mm OD | P Mtr. | 188.00 |
| | 55.3.2 | 20.00 mm OD | 20.00 mm OD | P Mtr. | 210.00 |
| | 55.3.3 | 25.00 mm OD | 25.00 mm OD | P Mtr. | 227.00 |
| | 55.3.4 | 32.00 mm OD | 32.00 mm OD | P Mtr. | 288.00 |
| | 55.3.5 | 40.00 mm OD | 40.00 mm OD | P Mtr. | 402.00 |
| **55.4** | | **Ball valve (brass)** | Supply, installation, testing and commissioning of ball valve (brass) high or low pressure with plastic floats inclusive of all necessary required material, labor for mentioned below size. | | |
| | 55.4.1 | 15.00 mm OD | 15.00 mm OD | Each | 583.00 |
| | 55.4.2 | 20.00 mm OD | 20.00 mm OD | Each | 714.00 |
| | 55.4.3 | 25.00 mm OD | 25.00 mm OD | Each | 906.00 |
| **55.5** | | **Gun metal gate valve** | Supply, installation, testing and commissioning of gun metal gate valve with brass lever handle ISI inclusive of all necessary required material, labor for mentioned below size. | | |
| | 55.5.1 | 15.00 mm | 15.00 mm | Each | 1102.00 |
| | 55.5.2 | 20.00 mm | 20.00 mm | Each | 1308.00 |
| | 55.5.3 | 25.00 mm | 25.00 mm | Each | 2043.00 |
| | 55.5.4 | 32.00 mm | 32.00 mm | Each | 2829.00 |
| | 55.5.5 | 40.00 mm | 40.00 mm | Each | 5753.00 |
| **55.6** | | **CPB Bib cock** | Supply, installation, testing and commissioning of CPB Bib cock inclusive of all necessary required material, labor for mentioned below size. | | |
| | 55.6.1 | 15.00 mm | 15.00 mm | Each | 1394.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 55.7 | | **On grid Solar Photo voltaic Power Plant** | Supply, Installation, Testing and Commissioning of on grid Solar Photo voltaic Power Plant conforming to MNRE specifications as amended, consisting of Mono/Poly Crystalline silicon solar cells, net metering facility, necessary protections, earthing, mountedon Aluminium/GI structure of suitable strength with following components complete as required:-<br><br>a) Solar Photo voltaic Module of capacity 330Wp or aove, manufactured in India, conforming to IS14286/IEC61215 ,IS/IEC61730-Part-1, IS/IEC61730-Part-2. Solar Photovoltaic Module conversion efficiency shall not be less than 16.5%. PV modules used in solar power plants/systems must be warranted for their out put peak watt capacity, which should not be less than 90%at the end of 10 years and 80%at the end of 25 years.<br>b) Power Conditioning Unit (PCU)of 350-800V DC Input voltage range and 400 VAC, three phase, 4wire, 50Hz+/-2.5Hz, output voltage suitable to generate AC Power with efficiency not less than 97%, total harmonic distortion less than 3% and suitable for ambient temperature from 0 to 50 degree C. The PCU shall adjust the voltage and frequency level to suit the Grid Voltage Frequency.<br>c) Data Monitoring System complete with accessories. d)Fixing of Array junction box & Main junction box with IP65 protection and termination arrangement for incoming and outgoing cable along with glands, lugs and other accessories etc. as required. e) Lightning and surge voltage protection. f)Connections & Interconnections by supplying & fixing required size XLPE insulated copper conductor 1.1kV grade armoured power and control cables between solar modules, main power cable to grid supply PCU unit along with supplying & fixing of necessary channel/conduit lugs and other accessories etc | KWp | 74750.00 |

## CHAPTER - 56

## FACADE LIGHT

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **56.1** | | **Surface wall mounted 9W LED** | Supply and fixing of IP 65, Surface wall mounted 9W LED up/downlighter decorative wall light fixture with anti glare beam angles 24deg - 36 deg as required. Fixture should be Aluminium with light reflectance surface, SS clamps, screws as required. CRI80+ NON DIM with led power supply built in color tolerance: <MacAdam 4 SDCM LED. | Each | 14635.00 |
| **56.2** | | **Surface/recessed ceiling mounted LED downlighter** | Supply and fixing of surface/recessed ceiling mounted LED downlighter fixture. Fixture should be Aluminium die cast, ADC 12, cable gland, SS clamps, screws as required. Glare quotient to be minimum & black body surface for glare control, should have honeycomb louvers for anti-glare, minimum 100 lm/watt, 8deg-15deg as required. working temperature-20°C +45°C IP Rating IP65 (not for permanent immersion) insulation class III. Choice of three colour temperatures as standard 2700K-5700K and delivers 90CRI for excellent colour rendering, and with binning-free LEDs carefully selected from two-step Macadam's ellipse. Fittings smart shield protected against polarity inversion, Protection class IP 65, safety class III. Safety glass, clear. Reflector made of pure anodised aluminium. | | |
| | 56.2.1 | 20W LED | 20W LED Downlighter with honeycomb louvers | Each | 21952.00 |
| | 56.2.2 | 30W LED | 30W LED Downlighter with honeycomb louvers | Each | 26831.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **56.3** | | **Surface/recessed ceiling mounted 15W deep LED** | Supply and fixing of IP65 of surface/recessed ceiling mounted 15W deep LED downlighter fixture. Fixture should be Aluminium die cast, ADC 12, cable gland, SS clamps, screws as required. Glare quotient to be minimum & black body surface for glare control, should have honeycomb louvers for anti-glare, 24deg-48 deg as required. working temperature-20°C +45°C IP rating IP65 (not for permanent immersion) insulation class III. Choice of three colour temperatures as standard 2700K-5700K and delivers 90CRI for excellent colour rendering, and with binning-free LEDs carefully selected from two-step Macadam's ellipse. fittings smart shield protected against polarity inversion, Protection class IP 65, safety class III. Safety glass, clear. Reflector made of pure anodised aluminium. | Each | 14635.00 |
| **56.4** | | **Surface ground mounted 3-6W small LED projector** | Supply and fixing of IP 65 surface ground mounted 3-6W small LED projector fixture with snoot & Honeycomb louver, product fitted with 700ma 8deg-24deg as required. 3000K-4000K CRI80+ NON DIM with led power supply built in color tolerance: <MacAdam 4 SDCM LED mounting AR111 similar optics PC lens. Fixture should be Aluminium die cast, SS clamps, screws as required. The finish should be stainless steel. | Each | 12196.00 |
| **56.5** | | **Recessed ground mounted deep LED uplighter** | Supply and fixing of recessed ground mounted deep LED uplighter fixture with Honeycomb Louver. protected Uno PATHFINDER IP66 2700K-5500K 8deg-24deg as required. stainless steel body hard anodized aluminium finishes stainless steel & IK rating 08. Installation housing box power supply cables35 cm NS20N PCP 2x0,5 mm2 power supply 24V DC power consumption max 9 Watt lumen output 240-680lm working temperature-20°C +45°C IP rating IP66 (not for permanent immersion) insulation class III Weight upto 500gm. and delivers 90CRI for excellent colour rendering, and with binning-free LEDs carefully selected from two-step Macadam's ellipse. PATHFINDER is smart shield protected against polarity inversion, an integrated super fast diode prevents accidental damage to the luminaire during installation on due to reverse polarity. Smart shield protects against 'hot-plugging'. Fully integrated spike-reduce on protects the constant- current driver and prevents damage to the LED die surface. With housing box included and LED driver in IP66 casing box included. | | |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 56.5.1 | 3W recess deep | 3W recess deep uplight | Each | 14025.00 |
| | 56.5.2 | 6W recess deep | 6W recess deep uplight | Each | 15245.00 |
| | 56.5.3 | 9W recess deep | 9W recess deep uplight | Each | 17684.00 |
| **56.6** | | **Instant outline direct view diffuse** | Supply and fixing of IP66 protected aluminium instant outline direct view diffuse 3000K-5000K with back profile with integrated cable alley for cable management with integrated IP rated cable connectors hidden within profile for additional protection from weather. Cross section of profile Silver anodized surface mounted aluminum profile 25 mm (W) x 20 mm (D) (minimum), with opal semi translucent sealed encapsulation, protection grade IP66, Minimum 85 lm/watt, DC 24 V, CRI>80Ra, standard length 2000 mm. excl. converter and end caps. The support is made of an anodized extruded aluminum frame which makes it possible to first fix SS the wall/ceiling. | | |
| | 56.6.1 | 5W/meter | 5W per meter | Meter | 4269.00 |
| | 56.6.2 | 10W/meter | 10W per meter | Meter | 6098.00 |
| | 56.6.3 | 15W/meter | 15W per meter | Meter | 7317.00 |
| **56.7** | | **Instant outline 10W/pm flexible LED direct view diffuse** | Supply and fixing of IP66 protected aluminum instant outline 10W/pm flexible LED direct view diffuse 3000-5000K/RGB with back profile with integrated cable alley for cable management with integrated IP rated cable connectors hidden within profile for additional protection from weather. Cross section of profile Silver anodized surface mounted aluminum profile 35 mm (W) x 35 mm (D) OR 50 mm (W) x 50 mm (D) (minimum), with opal semi translucent sealed encapsulation, protection grade IP66, DC 24 V, 750 lm/m, 10 W/m, CRI>80Ra , standard length 2000 mm. Excl. converter and end caps. The support is made of an anodized extruded aluminum frame which makes it possible to first fix SS the wall/ceiling. | Meter | 12196.00 |
| | 56.7.1 | RGB DMX controller | RGB DMX controller having 24V/48V DC input maximum 300W. | Each | 60979.00 |
| | 56.7.2 | RGB DMX SLAVE | RGB DMX SLAVE having 24V/48V DC input. | Each | 42685.00 |
| **56.8** | | **Long linear LED uplighter extrusion fixture** | Supply and fixing of recessed/surface ground mounted 3000K-5500K, 10deg-15deg as required 1m long linear LED uplighter extrusion fixture with VISOR & Honeycomb Louver with & IK rating 06, IP 65 cable gland, SS Clamps, screws as required with swivel mounting. | | |
| | 56.8.1 | 10W | 10W | Each | 14635.00 |
| | 56.8.2 | 20W | 20W | Each | 21952.00 |
| | 56.8.3 | 30W | 30W | Each | 29270.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **56.9** | | **Asymmetric LED step light** | Supply and fixing of decorative recessed wall mounted 6W Asymmetric LED step light 4"x4" 15-18 deg luminary Aluminium die cast, ADC 12, IP 65 cable gland, SS clamps, screws as required | Each | 10976.00 |
| **56.10** | | **20W LED with snoot** | Supply and fixing of IP 65 surface ground mounted product fitted with 700ma 20W LED with snoot 8deg-24deg 3000K-4000K CRI80+ nondimmable led power supply built in Color tolerance: <MacAdam 4 SDCM LED similar optics PC lens. Fixture should be IK rating 07, Aluminium die cast, SS clamps, screws as required. The finish should be stainless steel. | Each | 21952.00 |
| **56.11** | | **Surface wall mounted 6W 4 way grazer** | Supply and fixing IP65 surface wall mounted 6W 4 way grazer 3"x3" LED fixture fitted with 700ma 8deg-15 deg 3000K-4000K . Luminary should be Aluminium die cast, CRI 80+ non-dimmable with led power supply , < 4 Macadam SDCM LED. | Each | 17684.00 |
| **56.12** | | **Recessed ground mounted deep LED marker fixture** | Supply and fixing of recessed ground mounted deep LED marker fixture with diffused light 3000K-5000K, CRI>80, 45deg-60deg stainless steel 1W-3W, bodyhard anodized aluminium finish, stainless steel & IK rating 08. working temperature -20°C +45°C IP rating IP66 (not for permanent immersion) insulation class III weight upto 100gm, protected against polarity inversion, fully integrated spike-reduce on protects the constant current driver and prevents damage to the LED die surface, with housing box included and LED driver in IP66 casing box included. | Each | 9147.00 |
| **56.13** | | **Decorative surface wall mounted LED** | Supply and fixing of decorative surface wall mounted LED 3"X3" fixture fitted with 700ma LED, 10deg-15deg 3000K-4000K . Fixture should be Aluminium die cast, CRI 80+, IP65, non-dimmable led power supply , < 4 Macadam SDCM LED with 10 deg/15 deg optics PMMA. | | |
| | 56.13.1 | 3W LED | 3W LED | Each | 7927.00 |
| | 56.13.2 | 6W LED | 6W LED | Each | 9757.00 |
| | 56.13.3 | 9W LED | 9W LED | Each | 10976.00 |
| **56.14** | | **11W LED bollard** | Supply and fixing of IP 65 surface ground mounted luminary fitted with 700ma 11W LED bollard with 8deg-24deg 3000K-4000K CRI80+ nondimmable LED power supply built in Color tolerance <MacAdam 4 SDCM LED similar Optics PC lens. Fixture should be IK rating 07, Aluminium die cast, SS clamps, screws as required. The finish should be stainless steel. | Each | 21952.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **56.15** | | **100W LED pole light** | Supply and fixing of IP 65 surface ground mounted product fitted with 700ma 100W LED pole light with 36deg-60deg 3000K-4000K CRI80+ nondimmable LED power supply built in Color tolerance: <MacAdam 4 SDCM LED similar optics PC lens. Fixture should be IK rating 07, Aluminium die cast, SS clamps, screws as required. pole length as per site requirement. | Each | 42685.00 |
| **56.16** | | **LED RGB/RGBW flood lighting luminaire** | Supply and fixing of LED RGB/RGBW flood lighting luminaire for facade lighting with Die-cast aluminium housing, powder-coated finish, of following wattage. Luminaire shall be capable of producing dynamic color changing light for with 16 million colors by DMX/Ethernet based control. The beam angle of the fitting shall be narrow, medium, wide or asymmetric as per the requirement. Luminaire shall be complete with driver and capable of operating at line voltage without any separate power supply from 100-270 V AC, 50 Hz with power factor > 0.9 . Fixture shall be suitable to operate at an ambient temperature range of -10°C to +50°C and shall be IP 65 and IK 06 rated. Lifetime shall be at least L70:50000 burning hours at 50°C. Luminaire shall be conforming to BIS (IS 10322) and shall be complete with all necessary accessories required for proper working of fixture including weather proof connection cables, water proof connectors etc. as required. | | |
| | 56.16.1 | 600W/580W | 600W/580W | Each | 254678.00 |
| | 56.16.2 | 300W/280W | 300W/280W | Each | 122840.00 |
| | 56.16.3 | 250W | 250W | Each | 105392.00 |
| | 56.16.4 | 140W | 140W | Each | 65775.00 |
| | 56.16.5 | 130W | 130W | Each | 59964.00 |
| | 56.16.6 | 110W/115W | 110W/115W | Each | 50022.00 |
| | 56.16.7 | 90W | 90W | Each | 45997.00 |
| | 56.16.8 | 75W | 75W | Each | 42972.00 |
| | 56.16.9 | 55W | 55W | Each | 37709.00 |
| | 56.16.10 | 45W/40W | 45W/40W | Each | 32652.00 |
| | 56.16.11 | 30W | 30W | Each | 17553.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **56.17** | | **LED RGB/RGBW Linear Wall Washer luminaire** | Supply and fixing of LED RGB/RGBW Linear Wall Washer luminaire for Facade lighting with aluminium extruded housing, of following wattage. Luminaire shall be capable of producing dynamic color changing light with 16 million colors by DMX/Ethernet based control. The beam angle of the fitting shall be narrow, medium, wide or asymmetric as per the requirement. Luminaire shall be complete with driver and capable of operating at line voltage without any separate power supply from 100 to 270 V AC, 50 Hz, power factor > 0.9 . Fixture shall be suitable to operate at ambient temperature range of -10°C to +50°C and shall be IP 65 and IK06 rated. Lifetime should be at least L70:50000 burning hours at 50°C. Luminaire shall be conforming to BIS and shall be complete with all necessary accessories required for proper working of fixture including weather proof connection cables, water proof connectors etc. as required. | | |
| | 56.17.1 | 80W | 80W | Each | 52219.00 |
| | 56.17.2 | 70W | 70W | Each | 47435.00 |
| | 56.17.3 | 60W | 60W | Each | 40365.00 |
| | 56.17.4 | 40W | 40W | Each | 34240.00 |
| | 56.17.5 | 25W | 25W | Each | 20280.00 |

Note: As regards to color temperature and beam angle the approval of Engineer In-charge shall have to be obtained mandatory before procurement of luminaries, So as to meet the design parameters.

## CHAPTER - 57

## ELECTRICAL/ELECTRONIC EQUIPMENTS

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **57.1** | | Refrigerator | Supply, installation, testing & commissioning of Refrigerator : minimum 3 Star rating, Smart Inverter Compressor,single/multy door, Smart Connect, With Base Stand/Drawer and other material as per required, Warranty: Minimum 1 year comprehensive warranty on complete product and a 10 years warranty on the compressor | | |
| | 57.1.1 | | 60-80 Ltr. Capacity | Each | 10174.00 |
| | 57.1.2 | | 160-180 Ltr. Capacity | Each | 16122.00 |
| | 57.1.3 | | 220 Ltr. Capacity | Each | 21957.00 |
| | 57.1.4 | | 240 Ltr. Capacity | Each | 25798.00 |
| | 57.1.5 | | 300 Ltr. Capacity | Each | 39816.00 |
| | 57.1.6 | | 700 Ltr. Capacity | Each | 146034.00 |
| **57.2** | | LED TV | Supply, installation, testing & commissioning of L.E.D. Television (4K L.E.D. Display O.S. Webos /Android or equivalent, minimum 50 Hz refresh rate Min. 20 W speaker, Wifi/bluetooth enabled, minimum 2 HDMI & 1USB Port, i/c remote and wall/table top accessories) | | |
| | 57.2.1 | | 32” | Each | 20257.00 |
| | 57.2.2 | | 42”/43” | Each | 32365.00 |
| | 57.2.3 | | 49”/50” | Each | 46494.00 |
| | 57.2.4 | | 55” | Each | 50997.00 |
| | 57.2.5 | | 65” | Each | 73740.00 |
| | 57.2.6 | | 75’/76” | Each | 131294.00 |
| | 57.2.7 | | 85”/86” | Each | 292184.00 |
| **57.3** | | Drinking Water Cooler | Supply, installation, testing & commissioning of Water Cooler with Eco Friendly Refrigerant, IN-SITU PUF Insulation, Anti-corrosive Pre-coated Body with storage of required capacity | | |
| | 57.3.1 | 20 LITRE | WATER COOLER 20 LITRE | Each | 25595.00 |
| | 57.3.2 | 40 LITRE | WATER COOLER 40 LITRE | Each | 30815.00 |
| | 57.3.3 | 80 LITRE | WATER COOLER 80 LITRE | Each | 32525.00 |
| | 57.3.4 | 120 LITRE | WATER COOLER 120 LITRE | Each | 43928.00 |
| | 57.3.5 | 150 LITRE | WATER COOLER 150 LITRE | Each | 51801.00 |
| **57.4** | | Domestic water purifier | Supply, installation, testing & commissioning of Domestic water purifier having RO+UV+MTDS with one number prefilter having minimum water purifying capacity of 16 Ltrs per Hour storage . Installation Type- Wall Mounted/Table Top, suitable for hardness upto 2000mg/L operating input voltage 150-250V AC/50Hz. | Each | 24840.00 |
| **57.5** | | AMC charges of Domestic Water purifier | AMC charges of Domestic Water purifier having RO+UV+MTDS with minimum water purifying capacity of 16 Ltrs per hour. AMC includes replacement/cleaning all parts/filters of Water purifier and cleaning of prefilter as required | Each | 4945.00 |
| **57.6** | | 50 LPH RO Plant | Supply, installation, testing & commissioning of 50 LPH RO Plant with sediment filtration having minimum purifying capacity of 50LPH per hour. Suitable for reducing hardness, TDS, heavy metal contaminants, and pathogenic micro organism, operating input voltage 230V AC/50 Hz | Each | 53406.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| **57.7** | | AMC of 50 LPH RO Plant | AMC of 50 LPH RO Plant having minimum purifying capacity of 50LPH per hour. AMC includes replacement/cleaning all parts/filters/membrane of plant and cleaning of prefilter as required. | Each | 20700.00 |
| **57.8** | | Room Air Heater | Supply, installation, testing & commissioning of Room Heater (Air Blower Type) with over heat protection, Variable Heat Settings and inbuilt thermostat. | | |
| | 57.8.1 | | (1000/2000 WATTS) | Each | 2573.00 |
| **57.9** | | Hot water storage water heater | Supplying and fixing of hot water storage water heatervertical with required safety, ABS body including connection pipe, foundation bolt and nut, connection pipe, stop cock, testing etc. to complete the job. | | |
| | 57.9.1 | 15 Ltr. | 15 Ltr. (5 Star rating- standing losses ≤0.419 kWh/24 hrs/45°C) | Each | 14322.00 |
| | 57.9.2 | 25 Ltr. | 25 Ltr. (5 Star rating- standing losses ≤0.511 kWh/24 hrs/45°C) | Each | 15885.00 |
| | 57.9.3 | 35 Ltr. | 35 Ltr. (5 Star rating- standing losses ≤0.584 kWh/24 hrs/45°C) | Each | 18791.00 |
| **57.10** | | Interactive Panel | Supply, installation, testing and commissioning of Interactive Panel having Resolution 3840(H) ×2160 (V), Display Color 1.07Billon Colors, Min. Brightness 400cd/m2, LCD (with LED backlit), View Angle R/L 178 (Min.), U/D 178 (Min.), speaker output 20W × 2, HDMI Input 2, HDMI Output 1, Multi-Media USB 4 (2 Front 3.0, 1x3.0 rear, 1x2.0 rear), RS232 - 01, RJ-45-Yes(LAN) / Ethernet, Touch input 02, Bluetooth BT 5.0, WIFI 6, VGA In 1, Line Out 1, MIC 1, DP 1.2 1, Cover Glass anti glare 9 Mohs, Touch Resolution 32768x32768, Touch Accuracy 1mm, having Annotation Over Any Source, Response Time <5ms, having Palm Detection, Pen and Touch Differentiation, Touch - 20 points (Drawing, Annotation, Painting, Eraser, Highlighter, PDF Viewer, PPT Viewer, Browser, Movies & Image Viewer, External Source annotation), Files like PDF or objects can be saved in a private folder, 4 Drawing through split screens / Export and Import of files from mails / pen-drive or internet / Multimedia feature, Compatibility Windows 11/ Linux/Android /Chrome, OS Version ANDROID 11.0 , Processor A55 Quadcore 1.55Ghz, Flash 32 GB EMMC5, RAM 4 GB DDR4 or equivalent latest software version and specification, having accessories- User manual, wireless key board and mouse, remote control, power cable, touch cables, HDMI cable, Touch pens (2 stylus, 1 extendable),VESA Standard (Wall mount) / 3 Year warranty. | | |
| | 57.10.1 | 65 Inch | 65 Inch Interactive Panel | Each | 150080.00 |
| | 57.10.2 | 75 Inch | 75 Inch Interactive Panel | Each | 170363.00 |
| | 57.10.3 | 85/86 Inch | 85/86 Inch Interactive Panel | Each | 497840.00 |
| **57.11** | | OPS | Supply, installation, testing and commissioning of OPS (Open pluggable specificatio) with Core i5 Processor, 512 GB SSD, 8 GB RAM, Latest Windows. | Each | 61730.00 |
| **57.12** | | PVC Body Desert coolers/ Tower Cooler / Window cooler | Supplying of PVC Body Desert coolers/ Tower Cooler / Window cooler operating at 230 V and 50 Hz Supply, with submersible pump, Fan motor with regulator, woodwool wall pads/Honeycomb type wall pad, without stand. | | |
| | 57.12.1 | Window | Window Coolers with woodwool wall pads/Honeycomb type wall pad | | |
| | 57.12.1.1 | 1500 cubic metre/hr | Manul operating type with Tank capacity 40-45ltrs, Air delivery 1500 cubic metre/hr | Each | 9946.00 |

| Item No. | Sub Item No. | Item Name | Description of Item | Unit | Rate (in Rs) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | 57.12.1.2 | 2000cubic metre/hr | Manul operating type with Tank capacity 46-50 ltrs, Air delivery 2000cubic metre/hr | Each | 10599.00 |
| | 57.12.1.3 | 2000 cubic metre/hr | Manul operating type with Tank capacity 51-55 ltrs, Air delivery 2000 cubic metre/hr | Each | 11284.00 |
| | 57.12.2 | Tower Coolers | Tower Coolers with woodwool wall pads/Honeycomb type wall pad | | |
| | 57.12.2.1 | 1200-1500 cubic metre/hr | Manual operating type with tank capacity 20-28 ltrs , Air delivery 1200-1500 cubic metre/hr | Each | 11467.00 |
| | 57.12.2.2 | 1200-1500 cubic metre/hr | Remote operating type with tank capacity 20-28 ltrs, Air delivery 1200-1500 cubic metre/hr | Each | 13573.00 |
| | 57.12.2.3 | 1200-1500 cubic metre/hr | Remote operating type with tank capacity 35-40 ltrs, Air delivery 1200-1500 cubic metre/hr | Each | 14468.00 |
| | 57.12.2.4 | 1200-1500 cubic metre/hr | Remote operating type with tank capacity 50-55 ltrs, Air delivery 1200-1500 cubic metre/hr | Each | 15730.00 |
| | 57.12.3 | Desert Coolers | Desert Coolers Honey comb type | | |
| | 57.12.3.1 | 3000 cubic metre/hr | Manual operating type with tank capacity 50-60 ltrs, Air delivery 3000 cubic metre/hr | Each | 15010.00 |
| | 57.12.3.2 | 3000-3500 cubic metre/hr | Manual operating type with tank capacity 61-70 ltrs, Air delivery 3000-3500 cubic metre/hr | Each | 16694.00 |
| | 57.12.3.3 | 3500-4000 cubic metre/hr | Manual operating type with tank capacity 71-80 ltrs, Air delivery 3500-4000 cubic metre/hr | Each | 17573.00 |
| | 57.12.3.4 | 4000-5000 cubic metre/hr | Manual operating type with tank capacity 81-90 ltrs, Air delivery 4000-5000 cubic metre/hr | Each | 18836.00 |
| | 57.12.3.5 | 4000-5000 cubic metre/hr | Manual operating type with tank capacity 91-110 ltrs, Air delivery 4000-5000 cubic metre/hr | Each | 18939.00 |

## M.P.P.W.D. SCHEDULE OF TESTING CHARGES (2023)

### For National Accreditation Board for Testing and Calibration Laboratories (NABL)

| S.No | Material Of Test | Specified Test Performed | Reference Of Test Method | Quantity Of Sample Required | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | PVC insulated unsheathed and sheathed cables/cords with flexible conductor for rated voltages up to and including 1100 Volts as per IS 694 - 2010 | Acceptance tests | | | |
| | | A) For single core | | | |
| | | i)Unsheathed | IS 694 - 2010 | 16 metre | 3975 |
| | | ii) Sheathed | IS 694 - 2010 | 16 metre | 6075 |
| | | iii)FRLS unsheathed | IS 694 - 2010 | 16 metre | 5175 |
| | | iv) FRLS sheathed | IS 694 - 2010 | 16 metre | 7275 |
| | | v)FRLSH unsheathed | IS 694 - 2010 | 16 metre | 5850 |
| | | vi) FRLSH sheathed | IS 694 - 2010 | 16 metre | 7770 |
| | | B)For each additional core | IS 694 - 2010 | 16 metre | 1180 |
| 2 | PVC insulated (heavy duty) electric cables, for working voltages up to and including 1100V as per IS 1554 (Part-1) : 1988 | Acceptance tests | | | |
| | | i) Unarmored without FRLS | IS 1554 (Part- 1) : 1988 | 7 metre | 4500 |
| | | ii)Armored with FRLS | IS 1554 (Part- 1) : 1988 | 7 metre | 5680 |
| | | A)For each additional core | IS 1554 (Part- 1) : 1988 | 7 metre | 1180 |
| | | B)For optional test | IS 1554 (Part- 1) : 1988 | 7 metre | 800 |
| 3 | XLPE insulated power cable confirming toIS 7098 (Part-1): 1988 upto and including 1100 Volts with aluminum stranded /solid conductor. | Acceptance tests | | | |
| | | i)Unarmored without FRLS | IS 7098 (Part- 1): 1988 | 7 metre | 4500 |
| | | ii)Armored without FRLS | IS 7098 (Part- 1): 1988 | 7 metre | 4500 |
| | | iii) Unarmored without FRLS | IS 7098 (Part- 1): 1988 | 7 metre | 8900 |
| | | iv)Armored with FRLS | IS 7098 (Part- 1): 1988 | 7 metre | 10800 |
| | | a)For each additional core | IS 7098 (Part- 1): 1988 | 7 metre | 1180 |
| | | b)For optional test | IS 7098 (Part- 1): 1988 | 7 metre | 800 |
| 4 | Aerial bunched cable (ABC) confirming to IS 14255: 1955 upto and including 1100 Volts. | Acceptance tests | | | |
| | | a) XLPE insulated each aluminium conductor core | IS 14255 : 1995 | 7 metre | 4500 |
| | | b)For each additional core | IS 14255 : 1995 | 7 metre | 1180 |
| | | c)For optional test | IS 14255 : 1995 | 7 metre | 800 |
| 5 | Aluminum conductors for overhead transmission purposes aluminum stranded conductors as per IS 398 (Part- 1) : 1996 | Acceptance tests | | | |
| | | a)Aluminium strand upto 7 wires | IS 398 (Part- 1) : 1996 | 1 metre | 3300 |
| | | b)For each additional strand | IS 398 (Part- 1) : 1996 | 1 metre | 500 |
| 6 | Aluminum conductors galvanized steel reinforced as per IS 398 (Part- 2) : 1996 | Acceptance tests | | | |
| | | a)Aluminium strand upto 7 wires | IS 398 (Part- 2) : 1996 | 1 metre | 3300 |
| | | b)For each additional strand | IS 398 (Part- 2) : 1996 | 1 metre | 500 |
| 7 | Aluminum alloy stranded conductors (aluminum-magnesium- silicon type) as per IS 398 (Part- 4) : 1994 | Acceptance tests | | | |
| | | a)Aluminium strand upto 7 wires | IS 398 (Part- 4) : 1994 | 1 metre | 3300 |
| | | b)For each additional strand | IS 398 (Part- 4) : 1994 | 1 metre | 500 |

| S.No | Material Of Test | Specified Test Performed | Reference Of Test Method | Quantity Of Sample Required | Testing Charges Per Test (Rs.) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 8 | Aluminum conductors galvanized steel reinforced- For extra high voltage as per IS 398 (Part- 5) : 1992 | Acceptance tests | | | |
| | | a)Aluminium strand upto 7 wires | IS 398 (Part- 5) : 1992 | 1 metre | 3300 |
| | | b)For each additional strand | IS 398 (Part- 5) : 1992 | 1 metre | 500 |
| 9 | Copper analysis as per IS 191 : 2007 | Copper purity tests as per electrolysis method | | | |
| | | a) Copper purity | IS 191 : 2007 | 0.5 metre of cable sample | 900 |
| 10 | Rigid PVC conduit pipe as per IS 9537 (Part- 3) : 1983 | Acceptance tests | | | |
| | | a) Conduit pipe of PVC material | IS 9537 (Part- 3) : 1983 | 6 metre for each size | 5500 |

Note: - Frequency of test to be decided by Engineer in charge.


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